Table 8-11. Summary of Computations for Ground Station Passes for LEO Circular Orbits. Example computations are based on the following: Circular orbit at H = 1,000 km, i = 50 deg, orbit pole at 40 deg N, 150 deg E, ground station in Phoenix at 33.5 deg N, 248 deg E, and minimum allowable elevation angle ε_{min} = 5 deg. "same" indicates that either the formula or value listed is the same for both approximations.

Parameter	"Nonrotating Earth"			"Rotating Earth"		
	Formula	Eq.	Ex.	Formula	Eq.	Ex.
Approximation	Great circle centered on instantaneous orbit pole			Small circle centered on instantaneous rotation axis		
Accuracy	Evaluation point only			Point plus 1st and 2nd derivatives of path and velocity		
Earth ang radius, ρ	$\sin\rho = R_E/(R_E+H)$	8-26	59.8 deg	same		same
Period, P	$P = 2\pi\sqrt{a^3/\mu}$	9-4b	105 min	$P = 2\pi\sqrt{a^3/\mu}$	9-4b	same
Max nadir ang, η_{max}	$\sin\eta_{max} = \sin\rho\cos\varepsilon_{min}$	8-64	59.4 deg	same		same
Max Earth cen ang, λ_{max}	$\lambda_{max} = 90\text{ deg} - \varepsilon_{min} - \eta_{max}$	8-65	25.6 deg	same		same
Max distance, D_{max}	$D_{max} = R_E(\sin\lambda_{max}/\sin\eta_{max})$	8-66	3,194 km	same		same
Min Earth cen ang, λ_{min}	$\sin\lambda_{min} = \sin lat_{pole}\sin lat_{gs} + \cos lat_{pole}\cos lat_{gs}\cos(\Delta long)$	8-71	15.4 deg	$\lambda_{min} = \lvert\rho_S - \gamma\rvert$	8-84	13.7 deg
Min nadir ang, η_{min}	$\tan\eta_{min} = (\sin\rho\sin\lambda_{min})/(1-\sin\rho\cos\lambda_{min})$	8-72	54.1 deg	same		51.9 deg
Max elev ang, ε_{max}	$\varepsilon_{max} = 90\text{ deg} - \lambda_{min} - \eta_{min}$	8-73	20.5 deg	same		24.4 deg
Min dist, D_{min}	$D_{min} = R_E(\sin\lambda_{min}/\sin\eta_{min})$	8-74	2,095 km	same		1,914 km
Max ang rate, $\dot{\theta}_{max}$	$\dot{\theta}_{max} = [(360\text{ deg})(R_E+H)]/(P D_{min})$	8-75	12.1 deg/min	$\dot{\theta}_{max} = [(2\pi)(R_E+H)]/(PD_{min})$	8-75	13.2 deg/min
Az range, $\Delta\phi$	$\cos(\Delta\phi/2) = (\tan\lambda_{min}/\tan\lambda_{max})$	8-76	109.5 deg	$\cos(\Delta\phi/2) = \dfrac{\cos\gamma\cos\lambda_{max} - \cos\rho_s}{\sin\gamma\sin\lambda_{max}}$	8-86	118.9 deg
Az of center of pass, θ_{center}	$\cos\theta_{center} = \dfrac{\sin\lambda_{min}\sin lat_{gs} - \sin lat_{pole}}{\cos\lambda_{min}\cos lat_{gs}}$	8-78	128.1 deg	same Eq. substituting $\sin lat_E$ for $\sin lat_{pole}$	8-77 8-78	125.1 deg
Time in view, T	$T = (P/180\text{ deg})\,\text{acos}(\cos\lambda_{max}/\cos\lambda_{min})$	8-77	12.05 min	$T = W/(60\ \omega_E)$ $\cos(W/2) = \dfrac{\cos\lambda_{max} - \cos\rho_s\cos\gamma}{\sin\rho_s\sin\gamma}$	8-85a 8-85b	13.34 min

where as usual, α and δ are the longitude and latitude of the instantaneous rotation axis and the ground station. Having computed ρ_S and γ, the off ground track angle, or minimum Earth central angle, λ_{min}, is given by:

$$\lambda_{min} = |\rho_S - \gamma| \tag{8-84}$$

as can be seen from Fig. 8-38. Also from the figure, we can immediately determine both the azimuth range as seen from the ground station, $\Delta\phi$, and the arc through which the satellite moves, W, as measured about the instantaneous rotation axis. By the definition of the rotation rate about E, this lets us determine immediately the time in view, T. Specifically:

$$T = W/\omega_E \tag{8-85a}$$

where:

$$\cos(W/2) = \frac{\cos\lambda_{max} - \cos\rho_s\cos\gamma}{\sin\rho_s\sin\gamma} \tag{8-85b}$$

and

$$\cos(\Delta\phi/2) = \frac{\cos\gamma\cos\lambda_{max} - \cos\rho_s}{\sin\gamma\sin\lambda_{max}} \tag{8-86}$$

These equations allow us to obtain a substantially more precise yet still analytic estimate of the ground station pass parameters. The revised equations and an example of their application are provided in Table 8-11. It is important to recognize the basis for this revised formulation. Previously we represented the ground track by a great arc through the subsatellite point, but not quite tangent to the actual ground track. We are now representing it by a small circle which is both tangent to and has the same curvature and rotation velocity as the real dual-axis spiral ground track.

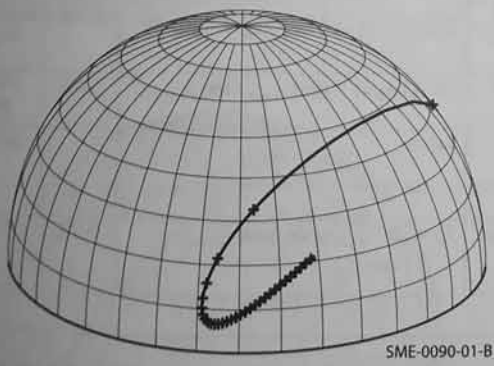
8.4.2 Elliptical Orbits

The computation of viewing parameters for satellites in elliptical orbits are conceptually similar but computationally more complex than for circular orbits. At perigee, the access area is small, and the satellite is moving

rapidly. At apogee, the access area is large, and the satellite moves more slowly. Consequently, substantially larger areas of the Earth will see the satellite at apogee for longer periods of time. Figure 8-39 shows the apparent motion as seen from the Earth for a spacecraft in a geosynchronous transfer orbit. The parameters for any specific ground station pass will be typically done by simulation because of the complexity of the analytic formulation. However, we can apply the elliptical orbit parameters defined in Sec. 9.3.2 and the coverage equations from Table 8-11 to compute the extrema at apogee and perigee and also representative parameters for other ground station passes for which the parameters



(A) On the Earth's Surface



(B) On the Ground Station Celestial Sphere

Fig. 8-39. Apparent Motion of a Satellite in a 200 km × 26,000 km GEO Transfer Orbit.

are known. These computations using the previous equations are shown in Table 8-12 for a satellite in a Molniya orbit and ground station passes at apogee and at perigee.

8.5 Satellite Relative Motion

Thus far, we have been concerned primarily with the motion of satellites relative to the surface of the Earth, or the apparent motion as seen from Earth. However, as illustrated in Fig. 8-40, the motion of satellites relative to each other is also of interest. This is important for constellation analysis and design, operations planning and analysis, and intersatellite communications, such as data transfer between satellites or low-Earth orbit communications constellations.

What does the motion of one spacecraft look like as seen from another? In some sense, the answer is easy. We integrate the equations of motion for both the target and the observer, difference the two, plot the results against the background of fixed stars, and the computer will have done all the work for us. This is the traditional approach. It works well for detailed analysis in an operations environment but is entirely unsuccessful at providing the physical insight and broad overview critical for conceptual mission design, constellation design and analysis, mission and operations planning, or the analysis of alternative approaches. Consequently, we are interested here not in a precise numerical answer but in broad characteristics and general limits on the motion, so that we can understand how to design, plan, and analyze the process of spacecraft interacting with each other.



Fig. 8-40. INTELSAT Communications Satellite Seen from the Space Shuttle Orbiter. Because the satellite is about to be captured, the orbits have been nearly perfectly matched and the two vehicles are moving very slowly with respect to each other. In most cases, satellites cross paths moving at relative velocities on the order of 7 km/sec or 30 times the speed of normal commercial jet airliners. (Photo courtesy Johnson Space Center.)

Because the relative motion of spacecraft, particularly on a large scale, is normally done by computer simulation, there have been very few analytical studies of the characteristics of this motion. The one exception to this is the case of two spacecraft orbiting very near each other. Here the equations of motion can be considerably simplified and are reduced to what are frequently called *Hill's equations* or the *Clohessy-Wiltshire equations*. These are summarized in OCDM Sec. 10.2.3.

The relative motion of satellites naturally breaks down into three broad areas. First, there is the large-scale relative motion due to individual satellites not being in the same orbit. This creates the gross motion of one satellite relative to another that is needed for constellation design, coverage



Table 8-12. Summary of Computations for Ground Station Passes for Elliptical Orbits. "At Perigee" refers to a ground station pass centered on perigee and uses the subscript P. "At Apogee" refers to a ground station pass centered on apogee and uses the subscript A. Because of the large arc at apogee that can see the ground, the time in view computation there is valid only for small eccentricities. Sample calculations are based on a satellite in an elliptical orbit at 63.4 deg inclination, 500 km perigee, and 40,000 km apogee. The orbit pole is assumed to be at 26.6 deg N, 150.0 deg E. The ground station is at 33.5 deg N, 248.0 deg E. All formulas are based on the "non-rotating Earth" approximation.

Parameter	At Perigee			At Apogee		
	Formula	Eq.	Ex.	Formula	Eq.	Ex.
Earth ang radius, ρ	$\sin \rho_{max} = R_E / (R_E + H_P)$	8-26	68.0 deg	$\sin \rho_{min} = R_E / (R_E + H_A)$	8-25	7.90 deg
Semimajor axis, a	$a = R_E + (H_A + H_P)/2$	8-23	26,628 km	$a = R_E + (H_A + H_P)/2$	8-23	26,628 km
Period, P	$P = 2\pi\sqrt{a^3 / \mu}$	9-4b	721 min	$P = 2\pi\sqrt{a^3 / \mu}$	9-4b	721 min
Mean motion, n	$n = (180 / \pi)\sqrt{\mu / a^3}$	9-17	0.008 32 deg/s	same		same
Orbital angular velocity, ω	$\omega_{max} = na/(R_E + H_P)$ $\omega_E = n(\sin i / \cos \delta_E)$	8-80	0.032 2 deg/s	$\omega_{min} = na/(R_E + H_A)$ $\omega_E = n(\sin i / \cos \delta_E)$	8-80	0.004 78 deg/s
Max nadir angle, η_{max}	$\sin \eta_{maxP} = \sin \rho_{max} \cos \varepsilon_{min}$	8-95	67.5 deg	$\sin \eta_{maxA} = \sin \rho_{min} \cos \varepsilon_{min}$	8-95	7.87 deg
Max Earth central angle, λ_{max}	$\lambda_{maxP} = 90 \text{ deg} - \varepsilon_{min} - \eta_{maxP}$	8-65	17.5 deg	$\lambda_{maxA} = 90 \text{ deg} - \varepsilon_{min} - \eta_{maxA}$	8-64	77.1 deg
Max distance, D_{max}	$D_{maxP} = R_E(\sin \lambda_{maxP} / \sin \eta_{maxP})$	8-66	2,078 km	$D_{maxA} = R_E(\sin \lambda_{maxA} / \sin \eta_{maxA})$	8-65	45,385 km
Min Earth central angle, λ_{min}	$\sin \lambda_{min} = \sin lat_{pole} \sin lat_{gs} + \cos lat_{pole} \cos lat_{gs} \cos(\Delta\text{long})$	8-71	8.2 deg	same		same
Min nadir angle, η_{min}	$\tan \eta_{minP} = \dfrac{\sin \rho_{max} \sin \lambda_{min}}{1 - \sin \rho_{max} \cos \lambda_{min}}$	8-72	58.2 deg	$\tan \eta_{minA} = \dfrac{\sin \rho_{min} \sin \lambda_{min}}{1 - \sin \rho_{min} \cos \lambda_{min}}$	8-71	1.3 deg
Max elevation angle, ε_{max}	$\varepsilon_{maxP} = 90 \text{ deg} - \lambda_{min} - \eta_{minP}$	8-73	23.5 deg	$\varepsilon_{maxA} = 90 \text{ deg} - \lambda_{min} - \eta_{minA}$	8-72	80.5 deg
Min distance, D_{min}	$D_{minP} = R_E(\sin \lambda_{min} / \sin \eta_{minP})$	8-74	1,075 km	$D_{minA} = R_E(\sin \lambda_{min} / \sin \eta_{minA})$	8-73	40,076 km
Max angular rate, $\dot{\theta}_{max}$	$\dot{\theta}_{max} = \omega_{max}(H_P + R_E) / D_{minP}$	8-74	0.206 deg/s	$\dot{\theta}_{max} = \omega_{min}(H_A + R_E) / D_{minA}$	8-74	0.005 53 deg/s
Azimuth range, $\Delta\phi$	$\cos(D\phi_P/2) = (\tan \lambda_{min} / \tan \lambda_{maxP})$	8-75	125.4 deg	$\cos(\Delta\phi_A/2) = (\tan \lambda_{min} / \tan \lambda_{maxA})$	8-75	176.2 deg
Azimuth of center of pass, ϕ_{center}	$\cos \phi_{center} = \dfrac{\sin \lambda_{min} \sin lat_{gs} - \sin lat_{pole}}{\cos \lambda_{min} \cos lat_{gs}}$	8-77 8-75	116.5 deg	same		same
Time in view, T	$T_P = (2/\omega_{max}) \operatorname{acos}(\cos \lambda_{maxP} / \cos \lambda_{min})$	8-76	982 sec.	$T_A = (2/\omega_{min}) \operatorname{acos}(\cos \lambda_{maxA} / \cos \lambda_{min})$	8-76	N/A*

* At apogee, the approximation given is valid only for low eccentricities.

computations, or how to fill in holes in coverage. Superimposed on this is the small-scale relative motion due to varying orbit perturbations on each satellite and individual satellites not being in precisely the intended orbit. This produces almost a "Brownian motion" within the constellation that causes each satellite to move around relative to its ideal location. This small-scale motion, discussed in detail in OCDM Chap. 10, impacts the size of the station-keeping box that we can create in a constellation and, therefore, impacts coverage and crosslink communications. Finally, in addition to how satellites actually move, we are interested in how they appear to move as seen from an observer satellite. This is important for understanding how to design and point antennas, lasers, or cameras for communications or observations.

For the large-scale relative motion of co-altitude satellites in circular orbits there are two key parameters, as illustrated in Fig. 8-41. The *relative inclination*, i_R, is the angle at which the orbits of the two satellites intersect. This is **not** the same as the difference between the inclinations of the two orbits since orbits at the same inclination but different nodes will intersect at non-zero angles, as shown in Fig. 8-42. The second parameter is the *relative phase*, ϕ_R, which is the angular separation between the two satellites at the time they pass through one another's orbit plane. As can be seen from Fig. 8-41, this will occur four times per orbit—once when satellite 1 crosses the plane of satellite 2 and later when satellite 2 crosses the plane of satellite 1. It then occurs twice more at the other intersection of the orbits, 180 deg away.

Table 8-12, Fig. 8-40◀, Eq. 8-86◀

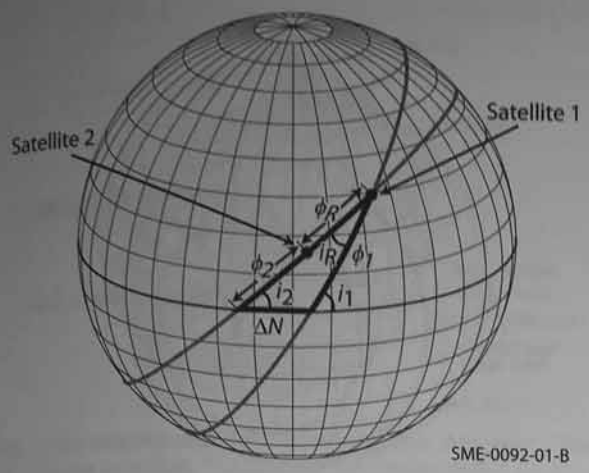


Fig. 8-41. The Relative Motion of Co-Altitude Satellites in Circular Orbits Depends on Only Two Parameters—the Relative Inclination, i_R, and the Relative Phase, ϕ_R.

The relative phase and relative inclination fully define the relative motion between the satellites. This is perhaps the most important message of this section:

> **The relative motion of co-altitude satellites in circular orbits depends only on the relative phase and relative inclination.**

It does not depend on the specific inclination or on the satellite's being at a common inclination. Once we have calculated i_R and ϕ_R, we can ignore the actual inclinations and use these parameters to calculate all of the characteristics of the relative motion of the satellites.

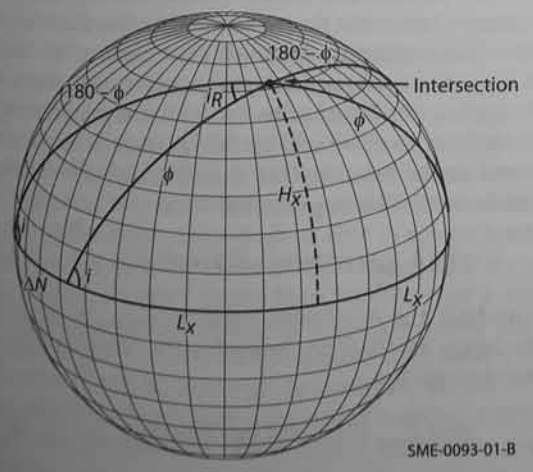


Fig. 8-42. Intersection of Two Orbits with the Same Inclination. The intersection point will be midway between the pairs of ascending and descending nodes. For one satellite the arc length, ϕ, from the ascending node to the intersection point will be less than 90 deg. For the other it will be 180 deg − ϕ.

In most constellations, the actual inclination of any two satellites is the same, i.e., $i_1 = i_2 = i$. (See Fig. 8-42.) In this case, there are particularly simple expressions for i_R and ϕ_R:

$$\cos i_R = \cos^2 i + \sin^2 i \cos\Delta N \tag{8-87a}$$

$$\phi_R = (T_2 - T_1)n + \Delta\phi \tag{8-87b}$$

where

$$\Delta\phi = 180 \text{ deg} - 2\phi \tag{8-87c}$$

$$\tan\phi = \frac{\tan(90 \text{ deg} - \Delta N/2)}{\cos i} \tag{8-87d}$$

ΔN is the angular separation along the equator between the two ascending nodes, $T_2 - T_1$ is the time between satellites 1 and 2 crossing their respective ascending nodes, n is the mean motion or angular frequency of both satellites, ϕ is defined on Fig. 8-42, and $\Delta\phi$ is an intermediate variable that represents the difference in arc length from the point where the two orbits intersect to their respective ascending nodes.

In terms of classical orbit elements, Eq. (8-87b) can be replaced by:

$$\phi_R = L_{02} - L_{01} + \Delta\phi \tag{8-88a}$$

where the *spacecraft longitude*, L_{0_1} is defined by:

$$L_{0_i} \equiv M_{0_i} + \omega_i \tag{8-88b}$$

where ω_i is the argument of perigee and M_{0_i} is the anomaly at a common epoch for the two orbits. For nearly circular orbits ω_i and M_{0_i} will not be well-defined but L_{0_i} will be.*

In the more general case in which the inclinations of the two satellites are different ($i_1 \neq i_2$), the relative inclination and relative phase are somewhat more complex:

$$\cos i_R = \cos i_1 \cos i_2 + \sin i_1 \sin i_2 \cos\Delta N \tag{8-89a}$$

$$\phi_R = (T_2 - T_1)n + \Delta\phi \tag{8-89b}$$

where

$$\Delta\phi = \phi_2 + \phi_R - \phi_1 \tag{8-89c}$$

$$\cos\phi_1 = \left[\frac{\cos i_2 - \cos i_R \cos i_1}{\sin i_R \sin i_2}\right] \tag{8-89d}$$

$$\cos\phi_2 + \cos\phi_R = -\left[\frac{\cos i_1 - \cos i_R \cos i_2}{\sin i_R \sin i_2}\right] \tag{8-89e}$$

Here ϕ_1 is the arc length from the ascending node of satellite 1 to the intersection of the orbits and ϕ_2 is the arc length from the ascending node of satellite 2 to where satellite 2 will be when satellite 1 is at the intersection of the orbits.

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* If the orbit is circular, we simply use the spacecraft longitude, L_{0_i}, with respect to node as the defining element.

We can use the above equations to determine the motion of a satellite relative to any arbitrarily defined *base satellite*. Any satellite in a constellation may be chosen as the base satellite. If the constellation is highly symmetric, it will not matter which satellite is chosen as the base.

To simplify the analysis, we use the *relative motion reference frame* defined in Fig. 8-43. This is an Earth centered frame of reference, co-rotating with the base



Fig. 8-43. Coordinate Frame of Reference Used for Relative Motion Analysis.

satellite, with the equator of the frame aligned with the orbit plane and the coordinate pole at the orbit pole. In comparing the motion of two satellites, it doesn't matter which is the base satellite. Changing the base satellite simply flops the relative motion pattern to the other side and reverses the sequence of motion, leaving the pattern unchanged. The purpose of using the relative motion reference frame is to remove the rotation of the base satellite relative to the Earth and concentrate on the motion relative to the satellites themselves. (For most constellation designs we want complete coverage of the Earth so our location with respect to the surface doesn't matter. However, we can easily go back to motion over the surface of the Earth if this becomes important.) Depending on what is convenient, we can think of the surface of the Earth-centered sphere containing the base satellite as representing either the direction as seen from the center of the Earth or the sphere with radius equal to the common altitude, i.e. the sphere which contains the actual motion of the satellites.

For circular co-altitude orbits, the relative motion of any satellite with respect to the other is an *analemma* or figure eight in the relative motion frame. As shown in Fig. 8-44 the half height of the analemma equals the relative inclination and the distance along the equator from the base satellite equals the relative phase. Thus, if the orbit planes remain the same but the satellites change position within the orbit, then the relative motion analem-

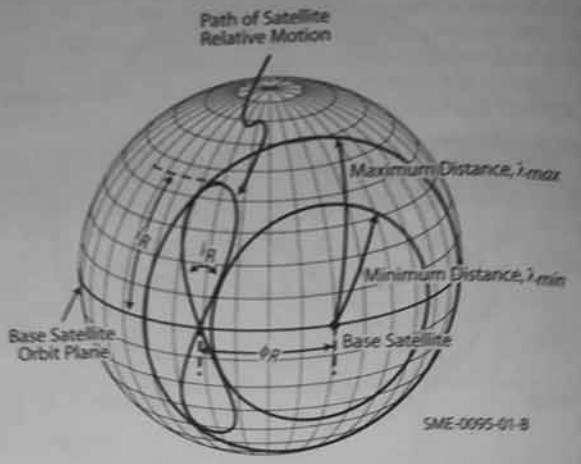


Fig. 8-44. The Relative Motion of Two Co-Altitude Satellites in Circular Orbits as Seen in the Relative Motion Frame. Here the motion is simply an analemma offset from the base satellite by the relative phase and with height equal to the relative inclination.

ma will remain unchanged, but will be located at a different distance, ϕ_R. The equations of motion for the analemma are given most easily in parametric form:

$$\sin \delta = \sin i_R \sin nt \tag{8-90}$$

$$\alpha = nt - \text{atan}\,(\cos i_R \tan nt) \tag{8-91}$$

where t is the time, n is the angular frequency of both satellites, δ is the elevation component in the relative motion frame, and α is the azimuthal component, relative to the point where the analemma crosses the equator (i.e., α and δ are the azimuth and elevation coordinators in Fig. 8-44). At the orbit plane, the left and right sides of the analemma cross at an angle equal to the relative inclination. Consequently, the angle that each curve makes with respect to the base orbit plane as it crosses this plane is 90 deg $- i_R/2$. The minimum angular separation, λ_{min}, from the base satellite will be the radius of a small circle centered on the base satellite and tangent to the inside of the analemma. This is given by:

$$\sin(\lambda_{min}/2) = \sin(\phi_R/2)\cos(i_R/2) \tag{8-92}$$

Similarly the maximum distance, λ_{max}, is the radius of the larger small circle tangent to the outside of the analemma given by:

$$\cos(\lambda_{max}/2) = \cos(\phi_R/2)\cos(i_R/2) \tag{8-93}$$

The shape of the relative motion analemma is a one parameter family of curves depending only on the relative inclination between the two orbits. A few representative analemmas are shown in Fig. 8-45. Recall that the height of the analemma equals the relative inclination. If the two satellites are in the same orbit plane, the height will be 0 and the analemma reduces to a point offset from the base satellite by an amount equal to the spacing be-

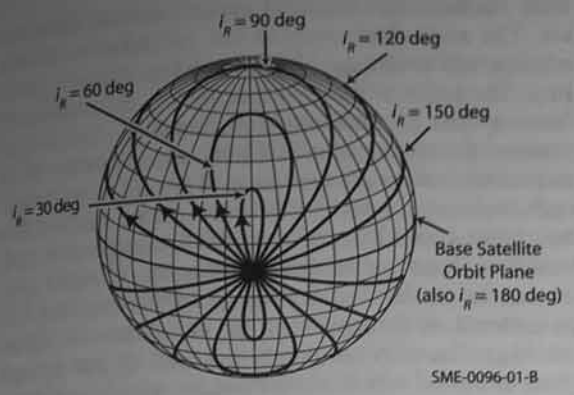


Fig. 8-45. Representative Relative Motion Analemma for Co-altitude Satellites. The shape of the relative motion analemma for co-altitude satellites depends only on the relative inclination, i_R, which is listed on each curve. i_R = 180 deg is a satellite in the same orbit as the base satellite, but moving in the opposite direction.

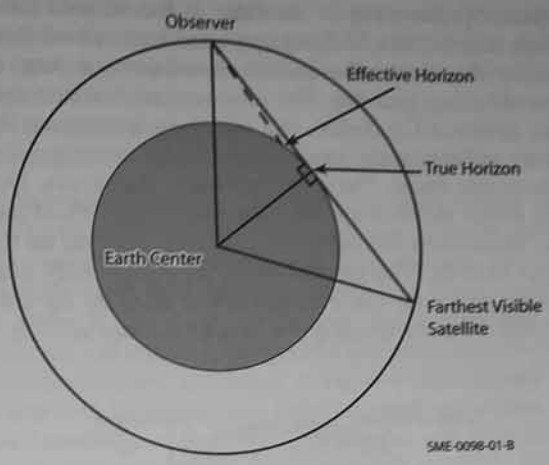


Fig. 8-46. Determining Regions of Intersatellite Visibility. Whenever two satellites are within twice the maximum Earth central angle (true horizon), they will be able to see each other.

tween the two satellites. For very small values of the relative inclination, the analemma is nearly a vertical line with very little sideways motion. As the analemma gets larger, it also gets "fatter" and covers a larger azimuth range. If the two orbits are at right angles to each other, the analemma crosses the equator at 45 deg and just reaches the pole of the coordinate system. When the relative inclination is larger than 90 deg, the two satellites are traveling in opposite directions and the analemma sweeps through all azimuth angles. Finally, when the relative inclination becomes 180 deg, the analemma reduces to a great circle along the coordinate equator, with the two satellites moving in the same plane but in opposite directions, such that they pass each other twice per orbit.

We can use the relative motion analemma to determine when 2 satellites can see each other. Ignoring atmospheric refraction and absorption, satellites will be able to just see each other when the line joining them is tangent to the surface of the Earth. As shown in Fig. 8-46, this will occur at twice the maximum Earth central angle seen by a single satellite. This is another way of saying that whenever two satellites can see the same point on the surface of the Earth, they must be able to see each other. Thus, on our relative motion reference frame we can draw a circle centered on the base satellite with a radius equal to twice the Earth central angle corresponding to the true horizon. Whenever the second satellite is within this circle, it will be above the horizon and visible to the base satellite. Whenever it is outside this circle, it will be below the horizon and the satellites cannot see each other.

As shown in Fig. 8-47, we can extend this concept to determine the nadir angle, η, as seen from either satellite to the other. For any co-altitude satellite at Earth central angle, λ, the nadir angle from each satellite to the other is given by:

$$\eta = 90 \text{ deg} - \lambda / 2 \qquad (8\text{-}94)$$

As must be the case, all co-altitude satellites will be seen within 90 deg of nadir.

In addition to the relative motion of the two satellites, we are also interested in the apparent motion, that is, how does one satellite move in the spacecraft-centered celestial sphere of the other satellite. This problem is similar in many respects to transforming between geometry on the surface of the Earth and geometry as seen by the satellite, with similar results. Like the Earth projection

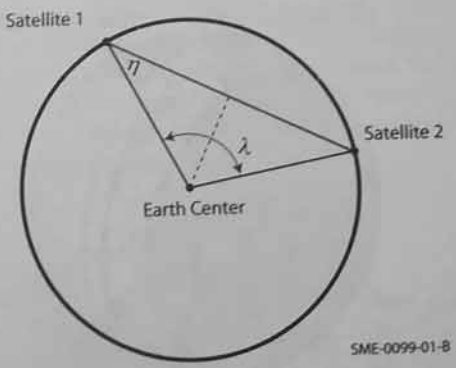
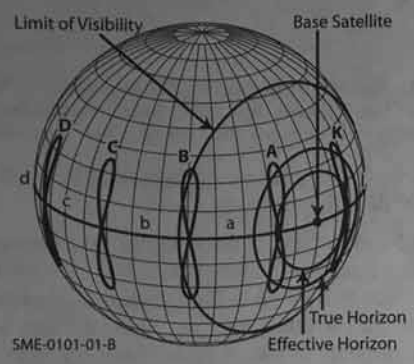


Fig. 8-47. Relationship Between Geometry as Seen from the Satellite and as Seen from the Earth's Center.

problem, azimuths measured relative to the base satellite don't change. They are the same in Earth-centered and spacecraft-centered coordinates. However, the Earth central angle, measured in the relative motion frame, is transformed into the nadir angle measured in the spacecraft frame according to Eq. (8-94). This allows us to transform any of the relative motion analemmas into the apparent motion, as seen from the base satellite.

Fig. 8-48 shows this relative motion and apparent motion for a series of satellites in a plane at a relative inclination of 25 deg with respect to the base satellite's orbit

plane. We have put 11 satellites in the inclined plane such that they are 32.7 deg apart and have spaced them so that the relative phase of the closest one is as large as possible, i.e., 16.4 deg. The relative motion analemmas are shown is Fig. 8-48A. All of the analemmas are the same and are simply spaced out along the equator of the coordinate frame. The visibility limit is also shown. Figure 8-48B shows the same analemmas transformed into the spacecraft-centered celestial sphere, centered on the base satellite. The general effect is to bend each analemma about the nadir direction and elongate the analemma for those which are nearest to the base satellite.



(A) In Relative Motion Frame (Earth at the Center).



(B) On Spacecraft-Centered Celestial Sphere
(Base Satellite at the Center).

Fig. 8-48. Relative Motion vs. Apparent Motion. Satellites A to K are assumed to be in a common plane at i_R = 25 deg with respect to the base satellite orbit plane. See text for discussion.

To get a better feel for the relative motion, we can think of the background sphere in Fig. 8-48A as the sphere containing the constellation of circular orbits and rotating such that the "base satellite" remains fixed on the x axis. The orbit plane of the base satellite is the equator of the plot and the orbit pole is the pole of the plot. The 10 other satellites in the orbit plane of the base satellite are the dots, a through i, spread out along the equator. The next plane over in our constellation is at a relative inclination, i_R, of 25 deg and contains 11 satellites. The paths of these 11 satellites in this coordinate frame are the 11 analemmas, A through K. The nearest plane on the other side would have a similar series of 11 analemmas, although the phase of the satellites within each analemma would be different. Planes that are further away would have a larger relative inclination and, therefore, larger analemmas. Nonetheless, all of the relative motion analemmas for all of the other satellites will be centered on the equator in the plot. While the figure would get "busy" if we tried to plot them all, this process gives us a good way to understand the relative motion of all of the other satellites in the constellation.

In Fig. 8-48B, we are looking at the motion as seen by an observer sitting on the base satellite. The other satellites in the same plane are simply at fixed locations in front or behind. The closest satellites (a and l) are always visible and the rest are always invisible. The closest satellite in the neighboring plane (A or K), oscillates back and forth across the base orbit plane moving mostly parallel to the horizon. The furthest satellite in this adjacent plane (F), moves in a very small apparent analemma but is never visible because it is always behind the Earth's disk, as are the 3 satellites on either side of it. Thus, of the 11 satellites in the adjacent plane, 2 are always visible, 7 are always invisible, and 2 are invisible part of the time and visible, but very near the horizon, most of the time.

From the plot, we can determine a number of **general characteristics of the apparent motion of satellites:**

- All co-altitude satellites are within the hemisphere centered on nadir (i.e., below the *horizontal plane* which is the great circle perpendicular to nadir and is the limit of apparent satellite positions when they are very close to the base satellite).
- Satellites near the base satellite will appear near the horizontal plane; those far from the base satellite will appear near nadir and will be obscured by the Earth disk.
- The visible motion is confined to an annulus or band between the horizontal plane and the Earth's horizon. Any satellite which is over a location on the Earth less than twice as far away as the true geometric horizon will be visible and in this band.
- Generally, the motion is back and forth, approximately parallel to the horizon; the center of this back and forth motion is the base satellite orbit plane.

As an example of this motion, satellite A is the one in the neighboring plane closest to the base satellite. In the relative motion frame (Fig. 8-48A), the analemma is near the base satellite and is well within the limit of visibility at all times. In the apparent motion frame (Fig. 8-48B), the motion is basically horizontal, moving back and forth

across the base satellite's orbit plane with relatively little elevation change.

Satellite B is the next closest satellite in the adjacent plane. Here the satellite is principally within the limit of visibility but goes outside of that limit for brief periods. In the apparent motion frame, this means that the satellite lies very near the Earth's horizon. It is above the horizon most of the time, and for brief periods dips below the horizon and becomes invisible. The remaining satellites remain below the horizon at all times and are invisible to the base satellite. The apparent motion is still an analemma but is obscured by the disk of the Earth. Combining Eqs. (8-92), (8-93), and (8-94), we can determine the maximum and minimum nadir angles for any other satellite as follows:

$$\sin \eta_{min} = \cos(\lambda_{max} / 2) = \cos(\phi_R / 2) \cos (i_R / 2) \quad (8\text{-}95)$$

$$\cos \eta_{max} = \sin(\lambda_{min} / 2) = \sin(\phi_R / 2) \cos (i_R / 2) \quad (8\text{-}96)$$

In addition, we can determine the following expression for the maximum azimuth angle, α_{max}, as seen from the base satellite as:

$$\tan \eta_{max} = \tan (i_R / \sin \phi_R) \quad (8\text{-}97)$$

Finally, Table 8-13 summarizes the results of our assessment of the large scale relative motion for co-altitude satellites. For a discussion of the small scale relative motion see OCDM, Sec. 10.1.2.

Table 8-13. Results of Large-Scale Relative Motion for Co-altitude Satellites.

- For co-altitude satellites in circular orbits, the large-scale relative motion depends only on the relative inclination and relative phase of the satellites
- The general motion is an analemma centered on the base satellite orbit plane
  - Height is equal to relative inclination
  - Position of central point is equal to the relative phase
- The apparent motion as seen from one of the satellites "bends" the analemma about the nadir direction
  - Visible motion is confined to a band between the Earth's horizon and the horizontal plane perpendicular to nadir
  - Generally, the motion is back-and-forth, approximately parallel to the horizon, and centered on the orbit plane
  - Any satellite will be visible when it is over a location on the Earth less than twice as far as the true (geometric) horizon

8.6 Mapping and Pointing Budgets

As discussed in Sec. 1.3, most space missions are not concerned with orbit, attitude, and timing per se, but rather with the more fundamental aspects of mapping and pointing. In this context, *pointing* means orienting the spacecraft, camera, sensor, or antenna to a target having specific geographic coordinates or inertial direction. *Mapping* is determining the coordinates of the *look-point* of a camera, sensor, or antenna. Satellites used only for communications will generally require only pointing. Satellites having some type of viewing instrument, such as weather, ground surveillance, or Earth resource satellites, will ordinarily require both pointing (point the instrument at Los Angeles) and mapping (determine the geographic location of the feature in pixel 1096).

In order to create a mapping and pointing budget, we must first understand the sources of error in determining the mapping and pointing, and then evaluate how to quantify these error sources, and how the errors should be added to create the final budget. The components of pointing and mapping budgets for an Earth-oriented observation mission are shown in Fig. 8-49 and defined in Table 8-13.* Mapping and pointing errors are related to the knowledge of the spacecraft position and attitude in space. However, even if these are known precisely, other errors will be present. For example, an error in the observation time will result in an error of the computed location of the target, because the target frame of reference moves relative to the spacecraft. A target fixed on the Earth's equator will rotate with the Earth at 464 m/s. A 10 sec error in the observation time would produce an error of 5 km in the computed geographic location of the target. Errors due to the target altitude come about because of the projection of the direction of the target onto a location on the surface of the Earth. For example, if the spacecraft has an elevation angle of 45 deg as seen from the target, and if the assumed altitude of the target is off by 1 km, then there will be a 1 km error in "placing" the target on the geographic surface of the Earth. Since the roughness of the surface of the Earth has a variation of several kilometers relative to a perfect spheroid, this implies that observations must be made nearly vertically, or if we wish geographic accuracies much greater than this, the target altitudes must be well known.

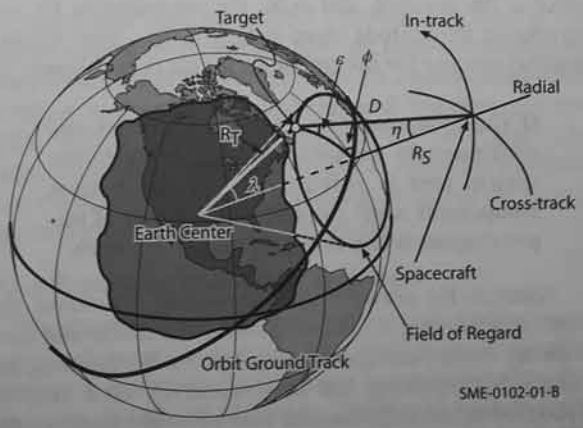


Fig. 8-49. Definition of Pointing and Mapping Error Components for an Earth-oriented Observation Mission.

* Errors associated with an inertially pointed mission are similar, but do not include any of the error sources associated with projecting the position onto the Earth.

Table 8-14. Sources of Pointing and Mapping Errors. Sensing axis orientation errors include errors in (1) attitude determination, (2) instrument mounting, and (3) stability for mapping control for pointing.

Spacecraft Position Errors:		
Δl	In- or along-track	Displacement along the spacecraft's velocity vector
ΔC	Cross-track	Displacement normal to the spacecraft's orbit plane
ΔR_S	Radial	Displacement toward the center of the Earth (nadir)
Sensing Axis Orientation Errors		
$\Delta\eta$	Elevation	Error in angle from nadir to sensing axis
$\Delta\phi$	Azimuth	Error in rotation of the sensing axis about nadir
Other Errors:		
ΔR_T	Target altitude	Uncertainty in the altitude of the observed object
ΔT	Clock error	Uncertainty in the real observation time (results in) uncertainty in the rotational position of the Earth)

Pointing errors differ from mapping errors in the way they include inaccuracies in attitude control and angular motion. The pointing error must include the entire control error for the spacecraft. On the other hand, the only control-related component of the mapping error is the angular motion during the exposure or observation time. This short term jitter results in a blurring of the look-point of the instrument or antenna. Except for this blur, it is attitude knowledge, rather than attitude control that is important for observation missions and therefore mapping. We may achieve an accuracy goal for either pointing or mapping in many ways. For example, we may know the position of the spacecraft precisely, and the attitude only poorly. Or we may choose to allow a larger error in the position, and make the requirements for determining the attitude more stringent. The ideal mathematical process for minimizing the cost is as follows:

> **If the components of an error budget are continuously variable then the minimum system cost is obtained by adjusting each component until the differential cost of improvement is the same for all components.**

Assume, for example, that we wish to minimize the cost of achieving a given pointing accuracy, and that with the current design, we can improve the pointing accuracy by adjusting the orbit accuracy at a cost of $100,000/mrad. Also assume that we could improve the pointing accuracy by improving the attitude accuracy at a cost of only $50,000/mrad. In this case, we should adjust our error budgets so as to be less stringent in terms of orbit accuracy and more stringent in terms of attitude accuracy until the differential cost of improvement for both was $75,000/mrad. In a sense, we are selling orbit accuracy for $100,000/mrad, in order to buy attitude accuracy for only $50,000/mrad. Continuing to do this so long as one component costs more than another in terms of improvement allows us to continually reduce the cost of achieving a given level of performance.

In practice the above process seldom works. Ordinarily, we cannot improve accuracy continuously. Normally, we have some components for which very little adjustment in accuracy is possible, and others for which there are very large steps in both performance and cost as we change the methods or techniques that we are using. Even if we can make nearly continuous adjustments, we seldom know precisely how much money it costs to achieve a given level of performance. In practice, the mission designer attempts to balance the components, often by relying on experience and intuition as much as analysis. In any case, the overall goal remains correct. We should try to balance the error budget so the incremental cost of improvement in any of the components results in approximately comparable cost.

Table 8-15. Construction of Mapping and Pointing Error Budgets. This process is essentially the same for all types of spacecraft error budgets.

Optimal Solution (How It Should be Done)
Adjust each error component until the differential cost of improvement ($ per milliradian or $ per km) is the same for each
Doesn't work in practice because performance improvements are not continuously variable.
Practical Solution (How it Works in Practice)
Allocate the budget equally among all components
Divide the components into three categories A. Those allowing very little adjustment (e.g., position) B. Those that easily meet the allocation (e.g., time) C. Those allowing increased accuracy at increased cost (e.g., attitude)
Adjust both the requirement and category (C) components to meet the cost vs. performance trade

A more practical technique for creating an error budget is summarized in Table 8-15. We begin the process by writing down all of the components of the budget. We then spread the total available budget equally among all of the components. For example, if there are five error sources relevant to the problem, then if we are going to add the errors linearly we would assign each source 1/5 of the error and if we are going to use a root sum square then we would assign each source the square root of 1/5 of the error. This provides a starting point for allocating the errors. Our next step is to look at the range of options available to us and divide the error sources into the following categories:

A. Those allowing very little adjustment
B. Those easily meeting their error allocation
C. Those allowing increased accuracy at increased cost

Table 8-16. Mapping and Pointing Error Formulas. ε is the elevation angle of the spacecraft as seen from the target, *lat* is the latitude of the target, ϕ is the target azimuth relative to the ground track, λ is the Earth central angle from the target to the satellite, D is the distance from the satellite to the target, R_T is the distance from the Earth's center to the target (typically ~ R_E, the Earth's radius), and R_S is the distance from the Earth's center to the satellite. See Fig. 8-49.

Error Source	Error Magnitude (units)	Magnitude of Mapping Error (km)	Magnitude of Pointing Error (rad)	Direction of Error
Attitude Errors:(1)				
Azimuth	$\Delta\phi$ (rad)	$\Delta\phi D \sin\eta$	$\Delta\phi \sin\eta$	Azimuthal
Nadir Angle	$\Delta\eta$ (rad)	$\Delta\eta$ D / sin ε	$\Delta\eta$	Toward nadir
Position Errors:				
In-Track	ΔI (km)	$\Delta I\,(R_T/R_S)\cos H$(2)	$(\Delta I/D)\sin Y_I$(5)	Parallel to ground track
Cross-Track	ΔC (km)	$\Delta C\,(R_T/R_S)\cos G$ (3)	$(\Delta C/D)\sin Y_C$(6)	Perpendicular to ground track
Radial	ΔR_S (km)	$\Delta R_S \sin\eta / \sin\varepsilon$	$(\Delta R_S/D)\sin\eta$	Toward nadir
Other Errors:				
Target Altitude	ΔR_T (km)	$\Delta R_T / \tan\varepsilon$	—	Toward nadir
S/C Clock	ΔT (s)	$\Delta T\,V_e \cos(lat)$ (4)	$\Delta T(V_e/D)\cos(lat) \cdot \sin J$ (7)	Parallel to Earth's equator

Notes:
(1) Includes attitude determination error, instrument mounting error, stability over exposure time (mapping only), and control error (pointing only). The formulas given assume that the attitude is measured with respect to the Earth.
(2) $\sin H = \sin\lambda \sin\phi$.
(3) $\sin G = \sin\lambda \cos\phi$.
(4) V_e = 464 m/s (Earth rotation velocity at the equator).
(5) $\cos Y_I = \cos\phi \sin\eta$.
(6) $\cos Y_C = \sin\phi \sin\eta$.
(7) $\cos J = \cos\phi_E \cos\varepsilon$, where ϕ_E = azimuth relative to East.

For example, determining the position of the spacecraft is a normal spacecraft operational function, done either by the ground station or by GPS. Whichever method is chosen, the level of accuracy is essentially fixed and cannot be adjusted without a dramatically high cost. (Launching a new and more accuracy GPS constellation is not a practical solution for most missions). Consequently, we assign these errors to category A and accept whatever error is inherent in the process. Timing errors frequently fall into category B, since modern spacecraft clocks can easily maintain an accuracy of tens of milliseconds which represents a small motion on the part of the spacecraft. Therefore, the error associated with the timing will typically be much smaller than the allocation and we will accept the more accurate values in order to allow a looser budget in another category. Attitude determination errors ordinarily fall into category C. Here, we could use a gravity-gradient stabilized system at very low cost, but accurate to only a few degrees, or a horizon sensor system, accurate to 0.05–0.10 deg at moderate cost, or an expensive star sensor system, accurate to better than 0.01 deg.

This process not only allows us to balance cost between appropriate components, but also to go back to the mission definition process and find the real requirements. For example, achieving a mapping accuracy of 100 m on the ground might triple the cost of the space mission, by requiring highly accurate attitude determination, a new system for orbit determination, and a detailed list of target altitudes. Reducing this accuracy requirement to 500 m might lower the cost enough to make the mission possible within established budget constraints. This is an example of trading on mission requirements as discussed in more detail in Chaps. 3 and 6. Requirements trading is extremely important to a cost-effective mission, but we often omit this step in the normal process of defining and allocating mission requirements. However, it is this evaluation of the cost of achieving a particular level of performance that allows technical information and technical decision-making to flow back into the requirements definition process.

To carry out the trade process defined above, we need to know how an error in each of the components listed in Table 8-14 relates to the overall mapping and pointing errors. Table 8-16 gives the formulas relating the errors in each of the seven basic components to the overall mapping and pointing errors. The notation here is the same as in Fig. 8-49. For any specific mission conditions, these formulas relate the errors in the fundamental components to the resulting pointing and mapping accuracies. This set of formulas provides the basic algebraic information which we use to transform attitude, position, and other error sources into specific mapping and pointing requirements for a given mission.

8.6.1 Representative Mapping Budget

To illustrate the process of constructing detailed mapping and pointing budgets, we will look at specific numerical examples of a moderate accuracy satellite, flying at an altitude of 1,000 km. The Excel model on the website allows you to transform this example to the parameters of your particular mission.

The errors associated with mapping, depend strongly on how close to the horizon we choose to work. Working in a very small region directly under the spacecraft gives excellent mapping accuracy and resolution, but provides

Table 8-17. Representative Mapping and Pointing Error Budgets. See Figs. 8-50 and 8-51 for plots.

Source	Error in Source	Error Budgets: Mapping Error (km) ε = 10 deg	Mapping Error (km) ε = 30 deg	Pointing Error (deg) ε = 10 deg	Pointing Error (deg) ε = 30 deg
Attitude Errors:					
Azimuth	0.06 deg	2.46	1.33	0.051	0.045
Nadir Angle	0.03 deg	8.33	1.78	0.030	0.030
Position Errors:					
In-Track	0.2 km	0.17	0.17	0.002	0.005
Cross-Track	0.2 km	0.16	0.17	0.004	0.007
Radial	0.1 km	0.49	0.15	0.002	0.003
Other Errors:					
Target Altitude	1 km	5.67	1.73	—	—
S/C Clock	0.5 sec	0.23	0.23	0.005	0.008
Root Sum Square		10.39	2.84	0.060	0.055

very poor coverage. On the other hand, working near the horizon provides very wide coverage, but poor mapping accuracy. Thus, we must trade resolution and mapping accuracy for coverage. The mapping accuracy for a particular mission will depend on the spacecraft's elevation angle at the edge of the coverage region. In almost all cases the mapping accuracy would be much better looking straight down, and the limiting accuracy will be closest to the horizon. To see how this works in practice, Table 8-17 provides representative mapping and pointing error budgets which are then plotted for an altitude of 1,000 km in Fig. 8-50 as a function of spacecraft elevation angle as seen from the ground, using the equations listed in Table 8-16. The total mapping error is the RSS of the individual components. Generally, uncertainty in target altitude and attitude determination contribute the most to the errors in mapping accuracy. In most cases, improving the other factors will have only a second-order effect. Consequently, determining the target altitude and spacecraft attitude are high priorities in assessing a mission's mapping performance and cost.

The uncertainty in target altitude typically contributes most to determining a geographic location on the Earth. The oblateness of the Earth has the largest effect on target altitude. It causes a variation in distance from the center of the Earth of approximately 25 km between the poles and the equator. However, we can account for this factor analytically at a very low cost, so it does not usually contribute to the error. The next plateau is for airplanes, clouds or other atmospheric features. The uncertainty in target altitude for these will typically be 10 km or larger, unless we have some a priori estimate of the altitude. For features fixed on the Earth's surface, the uncertainty in target altitude reduces to approximately 1 km, unless data analysis includes a detailed map of target altitudes. Figure 8-50 incorporates this 1 km error in target altitude as the dominant source of error. To do significantly better would require one of two options. First, the spacecraft could work only very near nadir and therefore have very poor coverage. Alternatively, it could include the elevation of the target region as a part of the data reduction process, therefore requiring a large database and making the data processing more complex.

8.6.2 Representative Pointing Budget

Unlike mapping, pointing depends only weakly on the spacecraft's elevation angle. This can be seen in Fig. 8-51, which is based on the same parameters as Fig. 8-50. For missions which require only pointing, working in a region near the horizon is almost as easy as pointing to a target or ground antenna at nadir. In this case the working limit on the spacecraft's elevation angle will depend on other factors, such as the transmission of the atmosphere for a selected wavelength, or possible obstruction by local geography. For example, ground stations ordinarily limit their work to approximately 5 deg above the horizon because of the reduced transmission of the atmosphere at lower elevation angles.

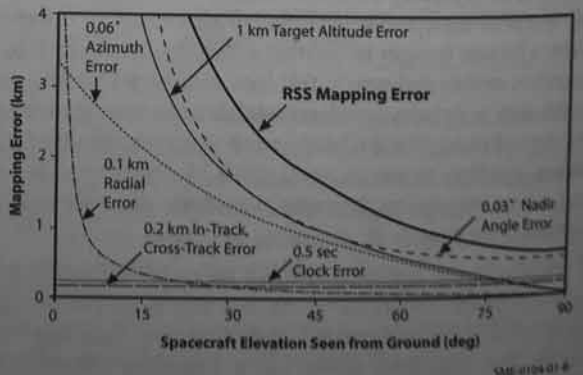


Fig. 8-50. Mapping Error as a Function of Elevation Angle for a Spacecraft at 1,000 km. Errors are from Table 8-17.

Pointing requirements normally arise from the spacecraft's housekeeping function, or from the need to point a particular instrument or antenna toward a ground tar-

get. Housekeeping requirements, such as solar array pointing and orbit maneuvers ordinarily demand pointing accuracies of 0.25–1 deg. Consequently, for most missions, the need to point the mission sensor or antenna is more important. Here again, two cases exist. If we wish to point the sensor at a single target, then we will generally try to point the center of the sensor at the target. By doing so, we establish a pointing requirement to place the target within the field of view. If the payload sensor's field of view is four times the 3σ pointing error, then the target will lie within the field of view with a 6σ probability, or virtual certainty. For example, if a sensor has a 1 deg square field of view, an overall pointing requirement of 0.25 deg will assure that the target will be within the field of view on essentially every observation.

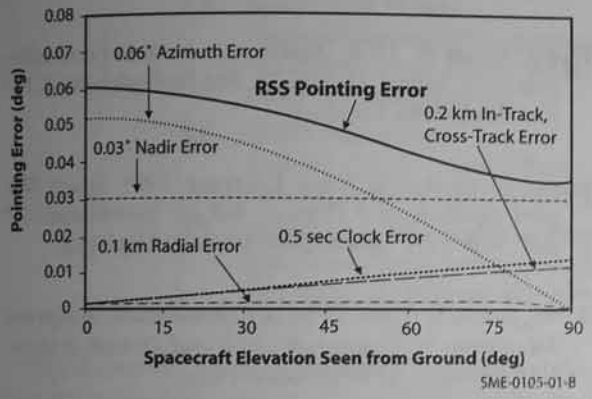


Fig. 8-51. Pointing Error as a Function of Elevation Angle for a Spacecraft at 1,000 km Altitude. Compare with Fig. 8-50. Note that mapping error is very sensitive to elevation angle while pointing error is not.

In pointing, we may also want to eliminate overlapping coverage. For example, if we wish to take a series of pictures, we must overlap the pictures by more than the pointing error to ensure continuous coverage of the ground. This requirement in turn implies that the spacing between pictures must equal the field of view size, less the pointing error. With a large pointing error, we must accept having fewer pictures in a given time, and increased resource cost in terms of time, power, and data rate for a given level of coverage. It is common to have a pointing requirement of 10%–20% of the field of view diameter. Driving the pointing under 10% of the field of view diameter will only slightly improve overall coverage. On the other hand, a pointing error worse than 20% of the field of view size can require substantial overlap, thus significantly diminishing the system's coverage and resource utilization.

8.6.3 Minimizing Cost

The first approach for minimizing cost is to eliminate entirely elements from the overall budget, thereby improving performance at essentially no cost. The key issue here is to look at the budget components and see which ones could be eliminated. For example, consider the pointing budget for a spacecraft whose goal is to remain nadir pointed. We can do this by using either star sensors or Earth sensors. However, star sensors are sensing the orientation with respect to inertial space, and therefore require an additional coordinate transformation which includes uncertainties in the position of the spacecraft in order to determine where nadir is and, therefore, remain nadir pointed. This is eliminated by using an Earth sensor to provide measurements directly with respect to the item we would like to be pointed toward. This implies that I could end up with a system that achieves lower accuracy using the high-cost star sensors than using low-cost Earth sensors. In general, I can maximize accuracy and minimize cost by using sensors that detect directly the object that I am interested in measurements with respect to.

The second principal way to reduce cost is to look at the overall problem of system requirements specification. If you do not meet the initial requirement, then the classic response is to go back and tighten up on the specifications. However, in a cost-conscious system, this is the wrong answer. The process needs to involve an iterative trade between what we want (the initial requirement) and what we can afford (what the engineering budget comes back with). In the end, the organization putting up the money must come up with a decision on how to proceed and whether or not meeting the accuracy objectives is worth the differential cost of doing so.

A key issue is to look for creative mechanisms for defining accuracy that may be able to meet the needs of the end user while still holding down cost. For example, as discussed above, accuracy is a strong function of elevation angle for a mapping error budget. This results in a very undesirable trade. If we increase the elevation angle to improve the accuracy, we will dramatically reduce coverage, which could drive up the number of spacecraft that are required. On the other hand, working at low elevation angles to achieve wide coverage with a small number of spacecraft provides extremely poor mapping accuracy at the edges of the swath. In cases such as this, a possible alternative is a "flexi-spec" in which different accuracies are required at different elevation angles. For example, in examining Fig. 8-50, we might set a mapping requirement of 3 km at an elevation angle of 25 deg, and 1 km at an elevation angle of 60 deg. This means that the user will get very wide and frequent coverage, with limited accuracy, and much higher accuracy, but at a less frequent rate, from the narrow swath. Certainly, this is less desirable than achieving high accuracy at all times. However, we may be able to reduce the number of spacecraft required by a factor of two or three and, therefore, dramatically drive down the cost of the system, if we can find a way to make use of poorer accuracy data at a higher frequency, and more accurate data at a lower frequency. This is representative of the types of trade that need to be made in modern systems, in order to drive down the overall system cost and maximize the performance.

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9 Orbits and Astrodynamics

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9.1 Keplerian Orbits
Kepler's First Law; Kepler's Second Law; Kepler's Third Law
Vis Viva Equation; Keplerian Orbit Elements and Terminology

9.2 Orbits of the Moon and Planets

9.3 Spacecraft Orbit Terminology

9.4 Orbit Perturbations, Geopotential Models, and Satellite Decay
Nonspherical Mass Distribution; Third Body Interactions;
Solar Radiation Pressure; Atmospheric Drag and Satellite Decay

9.5 Specialized Orbits
Geosynchronous Orbit; Repeating Ground Track Orbits;
Sun Synchronous Orbits; Molniya Orbits; Lagrange Point Orbits;
Other Specialized Orbits

9.6 Orbit Maneuvers
Transfer Orbits; Plane Change and Phase Shifts;
Planetary Assist Trajectories; Spacecraft Disposal

9.7 Summary—The Rules of Practical Astrodynamics
Important Characteristics of Spacecraft Orbits;
Ten Rules of Practical Astrodynamics

This chapter is a general introduction to orbits and defines the terminology used in orbit analysis and mission planning. It provides physical motivation for the principal effects important to orbit and mission design, formulas or approximations for all of the basic astronautical computations, and formulas needed for orbit and constellation design and the computation of ΔV budgets. As listed in the annotated bibliography on the book website, many books in celestial mechanics and orbit analysis are widely available. Those which provide the most comprehensive mathematical background are Battin [1999] and Seidelmann [2006]. High accuracy coordinate and time systems are very well explained by both Seidelmann [2006] and Kaplan [2005]. The best current book on orbits and astrodynamics, including both derivations of most of the equations presented here and free orbit software, is Vallado [2007]. The problem of selecting an orbit for a particular space mission (vs. the mathematical properties of an orbit described in this section and by the above references) is discussed in Chap. 10 and in more detail in Wertz [2009].

The *orbit* or *trajectory* is the path of a spacecraft or natural body through space. We will use orbit and trajectory interchangeably, although some authors prefer to use *orbit* to mean only a closed circular or elliptical path and *trajectory* to refer to other shapes such as hyperbola or paths which intersect the surface of the Earth. *Astrodynamics* is the mathematical analysis of orbits and their properties. An orbit is ordinarily specified by a *state vector* which can be the position and velocity of the spacecraft at some specified time or *epoch*, or a number of other equivalent forms described in Sec. 9.1.5. In principle, the state vector at any one time allows us to predict the position and velocity of the spacecraft at all future times. The list of successive positions of a satellite or planet is known as its *ephemeris* (plural *ephemerides*). Planetary ephemerides are published annually by the Government Printing Office. They are either in tabular form, such as *The Astronomical Almanac*, or more typically in machine readable form.

A *Keplerian orbit* is one in which gravity is the only force; the central body is spherically symmetric; the central body's mass is much greater than that of the satellite; and the central body and satellite are the only two objects in the system. Although this appears to be a large number of constraints, Keplerian orbits provide a remarkably good approximation for most spacecraft motion. A *perturbation* is a deviation from Keplerian motion of which there are two types. *Secular perturbations* are those for which the effects build up over time. *Cyclic perturbations* are periodic such that the effects cancel after one cycle or orbit.

Generally, this chapter addresses astrodynamics from a mission analysis and design perspective such that we can understand the underlying physics and make use of approximations and top-level formulas to evaluate most orbit effects. The detailed mathematical basis of astrodynamics is available from most volumes listed in the bibliography.

9.1 Keplerian Orbits

Predicting the motion of the Sun, Moon, and planets was a major part of the scientific revolution of the 16th and 17th centuries. Galileo's discovery of the satellites of Jupiter in 1610 provided a break with Aristotelian science and a strong argument for Copernicus' heliocentric theory. Danish astronomer Tycho Brahe determined the positions of the planets to about 1 minute of arc (1/60 deg) and the length of the year to about 1 sec with the unaided eye. Brahe's German assistant, Johannes Kepler, used these precise observations to derive empirically the rules of planetary motion which would later be justified by Newton.

It was the search for the underlying cause of the motion of celestial objects that motivated much of Newton's development of mechanics. In 1665, he determined that if gravity were an inverse square force it could account both for objects falling at the Earth's surface and for the motion of the Moon. However, he did not publish the detailed theory until 1687 in *Philosophiae Naturalis Principia Mathematica*. A major cause of this 22-year delay was Newton's inability to show that spherically symmetric objects (e.g., the Earth) behave gravitationally as though all the mass were concentrated at the center, the proof of which required development of the calculus. Having achieved this, Newton was able to combine his second law of motion, $\mathbf{F} = m\mathbf{a} \equiv m\ddot{\mathbf{r}}$, with his law of gravitation, $\mathbf{F} = -(G(Mm)/r^3)\mathbf{r}$, to obtain the two-body equation of motion which closely approximates the motion of spacecraft and planets:

$$\ddot{\mathbf{r}} = -\left(GM/r^3\right)\mathbf{r} \equiv -\left(\mu/r^3\right)\mathbf{r} \qquad (9\text{-}1)$$

where **F** is the force between two objects of mass m and M, **r** is the vector between them, **a** is the acceleration of the small body, G is Newton's constant of gravitation, and $\mu \equiv GM$ is the measure of the gravitational effect of the large body. Accurate orbit work includes the effect of the non-spherical symmetry of the Earth, perturbations due to other bodies, and non-gravitational forces—but nearly all of the basic foundations of orbit theory are direct extrapolations of Newton's work as foreseen by Newton himself.

Using gravitational theory and his laws of mechanics, Newton was able to derive Kepler's three laws of planetary motion. These laws apply to any two point masses (or spherically symmetric objects) moving under their mutual gravitational attraction. Kepler's laws in the form derived by Newton are:

- **Kepler's First Law:** *If two objects in space interact gravitationally, each will describe an orbit that is a conic section with the center of mass at one focus. If the bodies are permanently associated, their orbits will be ellipses; if not, their orbits will be hyperbolas*
- **Kepler's Second Law:** *If two objects in space interact gravitationally (whether or not they move in closed elliptical orbits), a line joining them sweeps out equal areas in equal intervals of time.*
- **Kepler's Third Law:** *If two objects in space revolve around each other due to their mutual gravitational attraction, the sum of their masses multiplied by the square of their period of mutual revolution is proportional to the cube of the mean distance between them; that is,*

Consequences of the Equation for Orbital Motion. Although Eq. (9-1) is a relatively simple set of equations, they provide very interesting consequences for satellite motion. First, notice that the right-hand side of Eq. (9-1) does not contain any properties of the spacecraft (m canceled in the two preceding equations). This implies that the acceleration and subsequent motion of objects in space does not depend on any physical properties of the objects themselves, so long as we ignore non-gravitational forces. A feather follows the same orbit as a bowling ball or a large spacecraft. Consequently, when the astronaut lets go of their pen, it follows the same trajectory that the spacecraft is following and appears to float next to the astronaut because they are both falling together with the same acceleration. This is the phenomenon often called *weightlessness* or *zero g*. However, gravity has not gone away and objects are certainly not weightless. An object sitting on a tall platform attached to the Earth, but at the altitude of the Space Shuttle, would weigh about 95% of what it does on the surface of the Earth. Gravity is very strong and objects are falling very rapidly. It's just that they are falling together, such that the motion relative to each other is very slow. The result is the astronaut and the pen floating gently at 7 km/s relative to the Earth's surface.

There is a second more subtle implication of these equations. In Newton's second law, m is the *inertial mass* which is the property of an object which resists acceleration or gives it momentum. A feather or ping pong ball has relatively little inertial mass whereas a freight train has a great deal. On the other hand, the m in the law of gravitation is the *gravitational mass*, which is a measure of the gravitational charge or how strong gravity acts. This is equivalent to the electrical charge on an object which determines how strong the electrical forces are. Objects with a similar mass can have very different electrical charges and, consequently, behave very differently in the presence of electrical fields. For some reason, this isn't true of gravity. Objects with the same inertial mass all have the same gravitational charge and all of them behave the same in the presence of a gravitational field. It is this very strange equivalence of gravitational and inertial mass that led Einstein, in the early 20th century, to develop the theory of relativity which implies, in part, that gravity is more a property of space than a property of objects, as electrical forces are. The theory of relativity plays a very minor role in the orbits of planets (particularly Mercury) and spacecraft designed specifically to test gravitational theories. For all but remarkably demanding applications, relativistic effects may be totally ignored in spacecraft calculations. Nonetheless, it is interesting that the seeds of relativity lie in the very foundations of Newtonian mechanics.

$$(m+M)\,P^2 = \frac{4\pi^2}{G}\,a^3 \qquad (9\text{-}2)$$

where P is their mutual period of revolution, a is the mean distance between them, m and M are the two masses, and G is Newton's gravitational constant.

The more massive of the two objects, M, is called the *primary* and the other, m, is called the *secondary*, or *satellite*. The *barycenter* is the location of the center of mass between the two objects. Kepler's empirical relations presented in two works in 1609 and 1619 were essentially the same as those derived by Newton, except that the constant of proportionality in the third law was obtained empirically and the shape of the orbits specified in the first law was an ellipse (one of 4 possible conic sections) because his experience was limited to planets.

In 1673, Christian Huygens introduced the quantity $(1/2)\,mV^2$ which he called the *vis viva* or "living force," to explain the motion of the compound pendulum. This concept was further developed by Gottfried Leibnitz in terms of "living" and "dead" forces which later came to be known as kinetic and potential energy. The application of this theory to celestial mechanics led to the fourth fundamental relationship for two objects rotating under their mutual gravitational attraction, the *vis viva equation*:

$$\mathcal{E} = \frac{V^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} \qquad (9\text{-}3)$$

where $\mathcal{E}$ is the total *specific energy* (i.e., energy per unit mass), and $V^2/2$ and $-\mu/r$ are the kinetic energy and potential energy, respectively, of the satellite, r is the instantaneous separation of the objects, and V is the magnitude of the relative velocity.

For some time there was bitter controversy between the followers of Huygens and Leibnitz, who believed that $\mathbf{F}\Delta x = (1/2)mV^2$ was the correct measure of the effect of a force, **F**, and the followers of Galileo and Newton, who believed that $\mathbf{F}\Delta t = \Delta(mV)$ was the proper measure. The controversy was resolved in 1743 when Jean D'Alembert published his *Traité de Dynamique*, which showed that both measures were correct and that they were not equivalent. (For a discussion of this controversy see, for example, Girvin [1948] or Dugas [1988].)

9.1.1 Kepler's First Law

Recall from the introduction that a *Keplerian orbit* is one in which gravity is the only force; the central body is spherically symmetric, the central body's mass is much greater than that of the satellite; and the central body and the satellite are the only two objects in the system. Kepler's first law states that the orbits of celestial objects are *conic sections*, i.e., figures produced by the intersection of a plane and a cone (Fig. 9-1 and Table 9-1) or any quadratic function in a plane. If the objects are permanently associated, this figure will be an *ellipse*, as

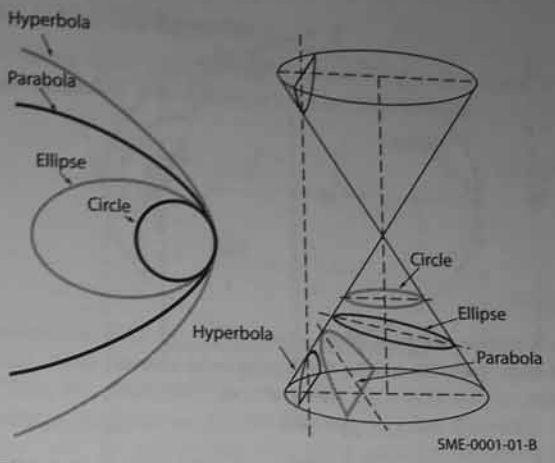


Fig. 9-1. The 4 Conic Sections Result from the Intersection of a Plane and a Right Circular Cone. Two special cases occur when the angle between the plane and axis of the cone is either 90 deg (resulting in a circle) or equal to the angular radius of the cone (resulting in a parabola).

shown in Fig. 9-2. Geometrically, an ellipse is defined by two points known as *foci*; the ellipse is then the locus of all points such that the sum of the distances from each point on the ellipse to the two foci is $2a$, where a is called the *semimajor axis* and is half the long axis or *major axis* of the ellipse. The semimajor axis is also the mean distance between either focus and points on the ellipse and is often listed this way in tables of orbit parameters. In Fig. 9-2, the quantity c is half the distance between the foci, and the *semiminor axis*, b, is half the short axis or *minor axis* of the ellipse. One of the foci is the center of mass or *barycenter* of the two objects; the other focus is of only geometric interest and is an empty point in space. Finally for some computations it is convenient to define the *semiparameter*, p, which is the distance from the focus to the orbit measured perpendicular to the major axis.

Table 9-1. Orbit Properties for the 4 Conic Sections. In practice only elliptical and hyperbolic orbits occur. Circular and parabolic orbits are special cases, but are convenient approximations for many orbits.

Conic	Total Energy, $\mathcal{E}$	Semimajor Axis, a	Eccentricity, e
Circle	< 0	= radius (> 0)	$= 0$
Ellipse	< 0	> 0	$0 < e < 1$
Parabola	$= 0$	∞	$= 1$
Hyperbola	> 0	< 0	> 1

The shape of an ellipse is uniquely specified by a single parameter, such as the ratio of the semimajor and semiminor axes. The parameter normally used to specify this shape is the *eccentricity*, e, defined as the ratio $c/a = (a^2 - b^2)^{1/2}/a$. The eccentricity also serves as a convenient ratio to parameterize all the conic sections as listed in Table 9-1. Specifically, $e = 0$ for a circle; $0 < e < 1$

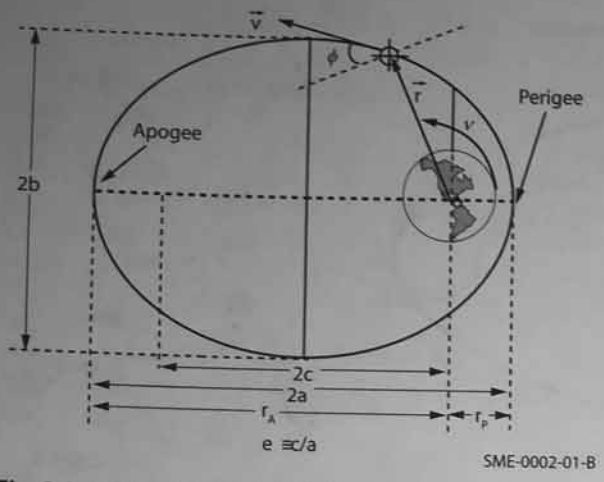


Fig. 9-2. Geometry of an Elliptical Orbit with Eccentricity $e \equiv c/a = 0.6$. Both the satellite and the primary go around their mutual center of mass. However, for most practical problems (except double stars), this center of mass is nearly the same as the center of mass of the primary.

for an ellipse; $e = 1$ for a *parabola*; and $e > 1$ for a *hyperbola*. In the last case, the nearest point on the curve to the focus is between the focus and the center of the two branches of the hyperbola. These 4 classes of curves are illustrated in Fig. 9-3 and their properties are summarized at the end of the section.*

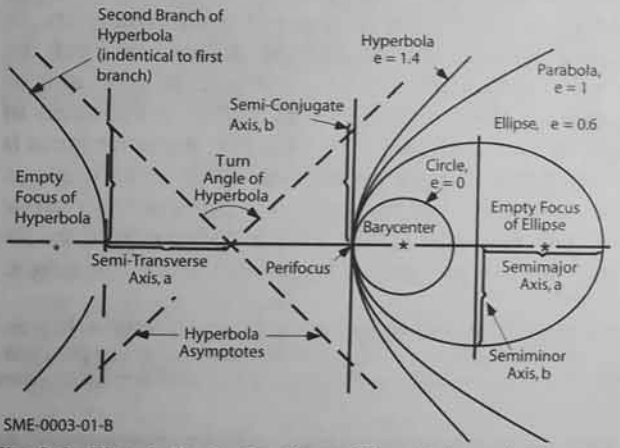


Fig. 9-3. The 4 Conic Sections. The circle and ellipse are closed "bound" orbits which continuously repeat. The parabola and hyperbola are open, "unbound" orbits such that the satellite passes perifocus only once and then recedes to infinity.

Both the circle and parabola represent special cases of the infinite range of possible eccentricities and therefore will never occur in nature. Orbits of objects which are gravitationally bound will be elliptical and orbits of objects which are not bound will be hyperbolic. Thus, an object approaching a planet from "infinity," such as a spacecraft approaching Mars, travels on a hyperbolic trajectory relative to the planet and will swing past the planet and recede to infinity, unless some non-gravitational force (a rocket firing or a collision with the planet) intervenes. Similarly, a rocket with insufficient energy to escape a planet must travel in an elliptical orbit in the absence of non-gravitational forces. Because the ellipse is a closed curve, the rocket will eventually return to the point in space at which the engine last fired.

9.1.2 Kepler's Second Law

As shown in Fig. 9-4, Kepler's second law is a restatement of the conservation of angular momentum. The angular momentum is proportional to the magnitude of the radius vector, r, multiplied by the perpendicular component of the velocity, $V_\perp$. In any infinitesimal time interval δt, the area swept out by a line joining the barycenter and the satellite will be $(1/2)mV_\perp r$. Hence, the area swept out per unit time is proportional to the angular momentum per unit mass which is a constant.

9.1.3 Kepler's Third Law

Kepler's third law applies only to elliptical orbits and relates the orbital period to the semimajor axis. In the case of Earth satellites, and very nearly in the case of the planets orbiting the Sun, we may ignore the mass of the secondary and write:

$$a^3 = \left[G(M+m)/4\pi^2\right]P^2 \approx \left(GM/4\pi^2\right)P^2 \equiv \left(\mu/4\pi^2\right)P^2 \quad \text{or} \tag{9-4a}$$

$$P = 2\pi\sqrt{a^3/\mu} \tag{9-4b}$$

Table 9-2. Values of $\mu \equiv GM$ for the Earth, Sun, Moon, and Mars. (See App. B for other values.)

Central Body	μ (m^3/s^2)	$\mu/4\pi^2$ (m^3/s^2)
Earth	$3.986\,004\,356 \times 10^{14}$	$1.009\,666\,698 \times 10^{13}$
Moon	$4.902\,800\,15 \times 10^{12}$	$1.241\,893\,786 \times 10^{11}$
Earth & Moon	$4.035\,032\,357\,5 \times 10^{14}$	$1.022\,085\,636 \times 10^{13}$
Mars	$4.282\,837\,522 \times 10^{13}$	$1.084\,855\,418 \times 10^{12}$
Sun	$1.327\,124\,400\,41 \times 10^{20}$	$3.361\,645\,377 \times 10^{18}$

The values of $\mu \equiv GM$ for the Earth, Moon, Sun, and Mars are given in Table 9-2 and for the major objects in the Solar System in App. D. Note that μ can be measured with considerable precision by astronomical observations. However, the values of M are limited by the accuracy of G to about 0.06%. (This is the most poorly known of the fundamental physical constants.) Therefore, the use of G is normally avoided and calculations are best

* Alternatively, we may define a conic section as the locus of all points which maintains a fixed ratio between the distance to the focus and the perpendicular distance to a fixed line called the *directrix*. The directrix is perpendicular to the major axis of any conic section. The ratio of the distance to the focus and to the directrix is the *eccentricity*, e.

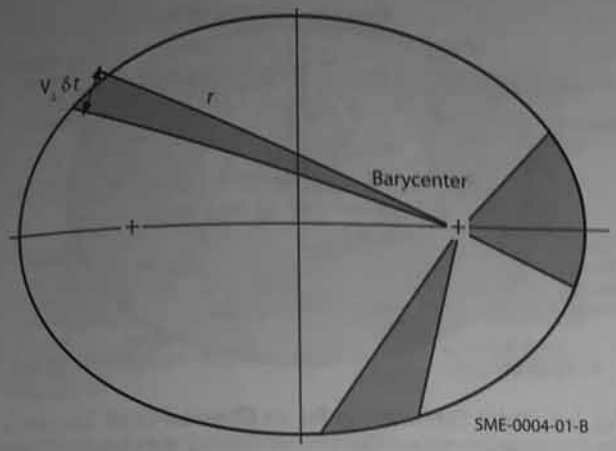


Fig. 9-4. Kepler's Second Law. Because the shaded areas are equal the time required for the satellite to sweep out each area is equal. The area swept out is directly proportional to both the time interval and the satellite's angular momentum.

done in terms of μ and the ratio of the masses of Solar System objects.

As long as the mass of the secondary is small, such that Eq. (9-4a) holds, then the constant of proportionality in Kepler's third law may be evaluated directly from observing orbiting objects. For example, the *astronomical unit*, or *AU*, is a unit of length equal to the semimajor axis of the Earth's orbit about the Sun; thus, in units of years and AU, $\mu/4\pi^2 = 1$ for the Sun and, therefore, $a^3 = P^2$ in units of astronomical units and years for the planets and any other satellites of the Sun. Similarly, for satellites of the Earth, Moon, or planets at different altitudes, the ratio of the periods may be determined from the expression

$$P = P_0 \left(\frac{a}{a_0} \right)^{3/2} \tag{9-5a}$$

where P_0 and a_0 are any known period and semimajor axis. For Earth satellites:

$$P = 1.658\,669 \times 10^{-4} \left(6{,}378.136\ 6 + H \right)^{3/2} \quad H \text{ in km} \tag{9-5b}$$

$$P = 4.180\,432 \times 10^{-4} \left(3{,}443.918\ 3 + H \right)^{3/2} \quad H \text{ in NMi} \tag{9-5c}$$

where P is in minutes and H is the mean altitude.

9.1.4 Vis Viva Equation

If we again assume that the mass of the secondary is small, the *vis viva* equation may be rewritten as

$$V^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right) \tag{9-6a}$$

$$\frac{1}{2} V^2 - \frac{GM}{r} = -\frac{\mu}{2a} \equiv \varepsilon \tag{9-6b}$$

where the *total specific energy*, ε, is the total energy per unit mass (kinetic plus potential) of the orbiting object. Thus, the semimajor axis is a function **only** of the total energy. Because the potential energy is a function only of position, the semimajor axis for a satellite launched at any point in space will be a function only of the launch velocity and not the direction of launch. (The shape and orientation of the orbit will, of course, depend on the launch direction). The orbit is hyperbolic if $\varepsilon > 0$ and elliptical if $\varepsilon < 0$. The special case between these, zero total energy, is an orbit with infinite semimajor axis. The velocity in this case is called the *parabolic velocity*, or *velocity of escape*, V_e. At any distance, R, from the center of a spherically symmetric object we have:

$$V_e \equiv \sqrt{(2\mu)/R} \equiv \sqrt{(2GM)/R} \tag{9-7}$$

A satellite launched with this velocity in any direction will not return, assuming that there are no other forces and the satellite doesn't run into the planet.

The vis viva equation may be used to obtain two other velocities of particular interest. If $R = a$, then

$$V_c = \sqrt{\mu/R} \tag{9-8}$$

is the *circular velocity*, or the velocity needed for circular orbit of radius R. Finally,

$$V_h \equiv \sqrt{2\varepsilon} = \sqrt{V^2 - 2\mu/R} \tag{9-9}$$

is the *hyperbolic velocity*, or the velocity of an object infinitely far away from the primary. Here V is the instantaneous velocity in the hyperbolic orbit at an arbitrary distance R from the center of the massive object. Values of the circular velocity and escape velocity for the Earth, Moon, Mars, and the Sun are shown in Table 9-3.

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Table 9-3. Values of the Circular Velocity and Escape Velocity for the Earth, Sun, Moon, and Mars. Constants are evaluated at the surface, except for the last row. (See website or App. C.1 for other values.)

Central Body	Equatorial Radius (km)	Circular Velocity (km/s)	Circular Period (min)	Escape Velocity (km/s)
Earth	6,378.136 6	7.905	84.49	11.180
Moon	1,738	1.680	108.36	2.375
Mars	3,390	3.551	100.19	5.021
Sun (at surface)	695,990	436.7	166.91	617.5
Sun (at Earth's orbit)	1.496×10^8	29.78	5.260×10^5	42.12

Table 9-3, Fig. 9-4, Eq. 9-9

9.1.5 Keplerian Orbit Elements and Terminology

For a purely Keplerian orbit, if we know the position and velocity at any given instant, we can integrate the equations of motion to determine the position and velocity at all future times. Consequently, a Keplerian orbit can be fully specified by giving the three components of the position and three components of the velocity at any instant. This numerical specification of an orbit is called the *orbit elements*. The position and velocity at any instant is convenient for computer applications but provide relatively little insight into the fundamental characteristics of the orbit. A convenient set for conceptualization are the classical or *Keplerian elements* described below. (Unfortunately, in many cases, these are inconvenient for numerical computation.) The information needed to fully specify a Keplerian orbit is:

- The orbit size and shape (2 parameters)
- The orientation of the orbit plane in space (2 parameters)
- The rotational orientation of the semimajor axis within that plane (1 parameter)
- Where the satellite is on the orbit (1 parameter)

In addition to the orbit elements, we need to know the value of $\mu = GM$ for the central body in order to calculate the motion of the satellite over time.

Orbit Size and Shape. For either hyperbolic or elliptical orbits, *perifocus* is the point on the orbit where the secondary is closest to the center of mass (Fig. 9-5). The *periapsis* or *perifocal distance* is the distance between the center of mass and the perifocus. This is equal to $a-c = a(1 - e)$ for an elliptical orbit of semimajor axis, a, and eccentricity, e. Unfortunately, the terminology here is both well established and awkward because different words are used for the point of closest approach to different primaries. Thus, we have *perihelion* (closest approach to the Sun), *perigee* (closest approach to the Earth), *pericynthiane* or *perilune* (closest approach to the Moon), *perijove* (closest approach to Jupiter) and even *periastron* (closest approach of two stars in a binary pair).

Perihelion and perifocal distance are measured from the center of mass, but *perigee height*, frequently shortened to "perigee" in common usage, is measured from the surface of the Earth. (See Fig. 9-6.) This terminology arises because we are interested primarily in the height above the surface for low-altitude spacecraft. The most unambiguous procedure is to use perigee height or perigee altitude whenever the distance is being measured from the surface;* however, this is frequently not done.

* Throughout this book when discussing distances relative to the Earth, we use "height" exclusively for distances measured from the Earth's surface; e.g., apogee height, perigee height, or height of the atmosphere.

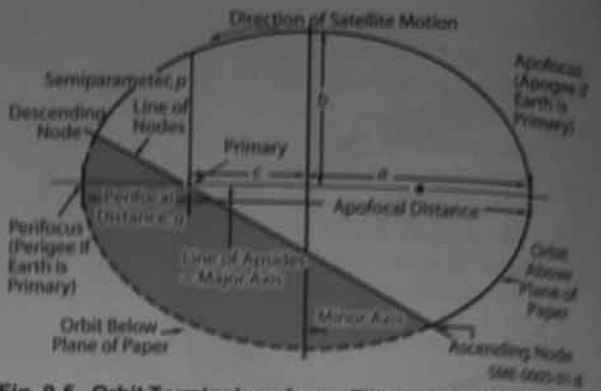


Fig. 9-5. Orbit Terminology for an Elliptical Orbit. The orbit is tilted, or inclined, with respect to the plane of the paper such that the dashed segment is below the paper which is assumed to be the reference plane.

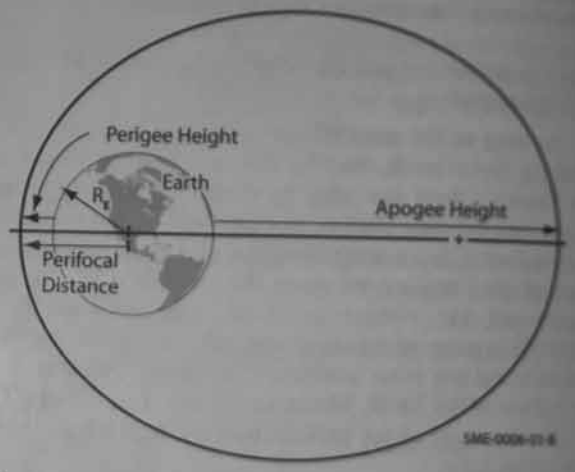


Fig. 9-6. Definition of Perigee Height and Apogee Height.

In elliptical orbits, the most distant point from the primary is called *apofocus*; the *apoapsis* or *apofocal distance* is $a+c = a(1+e)$. Again, the words *aphelion*, *apolune*, and *apogee* or *apogee height* are used, the latter being measured from the Earth's surface. The straight line connecting apogee, perigee, and the two foci is called the *line of apsides*, which is the same as the major axis of the ellipse. If, h_p, h_A, and R_E are the perigee height, apogee height, and radius of the Earth, respectively, then for an Earth satellite, the semimajor axis is:

$$a = R_E + (h_p + h_A)/2 \qquad (9\text{-}10)$$

The size and shape of any Keplerian orbit can be defined equivalently by either the semimajor axis and eccentricity or the apogee height and perigee height. Both methods are in common use depending on the application.

Orientation of the Orbit Plane. As shown in Fig. 9-7, the *inclination*, i, is the angle between the orbit plane and a reference plane which also contains the center of mass. The most commonly used reference planes are the *equatorial plane* (the plane of the Earth's equator) for Earth's satellites and the *ecliptic* (the plane of the Earth's orbit about the Sun) for interplanetary orbits.

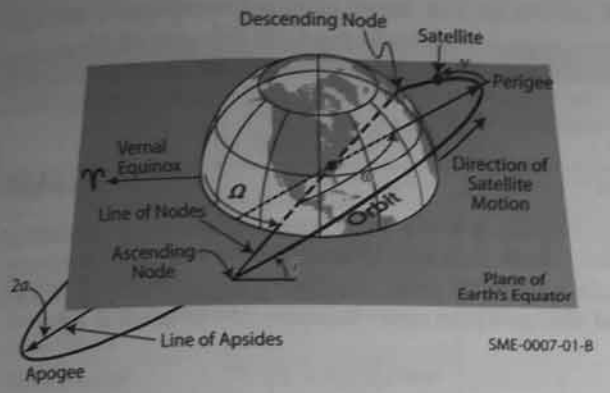


Fig. 9-7. Keplerian Orbit Elements. ♈ marks the direction of the vernal equinox. The line of nodes is the intersection between the equatorial plane and the orbit plane. Ω is measured in the equatorial plane, and ω is measured in the orbit plane.

Both the rotation of the Earth on its axis and the revolution of the Earth around the Sun are in a counterclockwise direction as seen from the North Pole. Most satellites travel in this same direction and are said to be in a *prograde* orbit which has an inclination between 0 and 90 deg. Satellites traveling in a direction opposite the rotation of the Earth are in a *retrograde* orbit and have inclinations between 90 deg and 180 deg.

The intersection of the orbit plane and the equatorial plane through the center of mass is the *line of nodes*. For an Earth satellite the *ascending node* is the point in its orbit where the satellite crosses the equator going from south to north. The *descending node* is the point where it crosses the equator going from north to south.

To fully define the orbit plane, we need to specify both its inclination and the orientation of the line of nodes around the equator. Because the Keplerian orbit is approximately fixed in inertial space,* we need to define this rotational orientation with respect to inertial space rather than with respect to the surface of the Earth. The reference point that is ordinarily used for inertial space is the ascending node of the Earth's orbit about the Sun. This is the location of the Sun in the sky on the first day of spring. It is called the *vernal equinox* or *first point of Aries*. This is the zero point for *right ascension* in the sky which is the equivalent of longitude measured on the surface of the Earth. (See Table 8-4 in Sec. 8.1.2) Consequently, the orientation of the ascending node on the sky is defined by the *right ascension of the ascending node*, Ω (often called RAAN in computer code), which is the angle in the equatorial plane measured eastward from the vernal equinox to the ascending node of the orbit. At times, it is also convenient to specify the *longitude of the ascending node* measured on the surface of the Earth from the Greenwich meridian to the ascending node. However, this parameter is changing continuously as the Earth rotates underneath the orbit.

The vernal equinox is universally used as the zero point for coordinates in inertial space, i.e., relative to the "fixed" stars. Unfortunately, this is an exceptionally inconvenient reference because it is moving very slowly through the sky due to the precession of the Earth's axis as described in Sec. 9.2. The motion is remarkably slow with a period of approximately 26,000 years. Nonetheless, it implies that all inertial coordinate systems have a date or *epoch* attached to them. Modern almanacs list the locations of stars in *2000 coordinates*, which means that they are expressed in a coordinate system in which the vernal equinox has been precessed to January 0 of the year 2000.

Equally unfortunately, nearly all satellite orbit systems use what are called *true-of-date coordinates* in which the vernal equinox is precessed to the time of specification of the orbit called the *orbit epoch*. This correction to the fundamental coordinate system is extremely small and is easily done by standard computer routines. However, it means that it is very difficult to verify orbit computations with simple hand calculators because there are always small differences in the answer due simply to differences in the coordinate system specification. The principal merit of this approach is that it provides continued employment for astrodynamicists.

Orientation of the Orbit within the Plane. Having specified the orientation of the orbit plane, we now need to specify the rotational orientation of the major axis (line of apsides) within that plane. This is normally done by defining the *argument of perigee*, ω, which is the angle at the center of mass of the Earth measured in the orbit plane in the direction of the satellite's motion from the ascending node to perigee. (See Fig. 9-7.)

Position of the Satellite within the Orbit. Finally, we need some mechanism to specify where the satellite is in its orbit. The *true anomaly*, ν, is the angle measured at the center of mass between perigee and the satellite. This gives us a series of three angular measurements. The right ascension of the ascending node is measured eastward from the vernal equinox to the ascending node of the orbit. The argument of perigee is then measured from the ascending node in the direction of the motion of the satellite to perigee, and, finally, the true anomaly is measured in the direction of motion from perigee to the location of the satellite.

Unfortunately, the true anomaly is difficult to calculate. Consequently, those who were first studying the mathematics of orbital motion introduced the *mean anomaly*, M, as 360 $(\Delta t/P)$ degrees where P is the orbital period and Δt is the time since perigee passage. Thus, $M = \nu$ for a satellite in a perfectly circular orbit. The mean anomaly at any time is a trivial calculation and of no physical interest. The quantity of real interest is the true anomaly which is difficult to calculate. The *eccentric*

Table 9-3◀, Fig. 9-7, Eq. 9-10◀

* Friction with the Earth's surface drags the atmosphere around as the Earth rotates on its axis. This friction is negligible in orbit. Thus, a satellite orbit remains approximately fixed in inertial space (except for orbit perturbations described in Sec. 9.4) as the Earth rotates once per day underneath the orbit. This causes satellites not in an equatorial orbit to view most of the Earth's surface twice daily—once on the upward portion of the orbit and once on the downward portion.

anomaly, E, was introduced as an intermediate variable relating the two.* The mean and eccentric anomalies are related by *Kepler's equation* (not related to Kepler's laws):

$$M = E - e\sin E \tag{9-11}$$

where e is the eccentricity. E is then related to ν by *Gauss' equation*:

$$\tan\left(\frac{\nu}{2}\right) = \left(\frac{1+e}{1-e}\right)^{1/2}\tan\left(\frac{E}{2}\right) \tag{9-12a}$$

or

$$\tan\left(\frac{E}{2}\right) = \left(\frac{1-e}{1+e}\right)^{1/2}\tan\left(\frac{\nu}{2}\right) \tag{9-12b}$$

$E/2$ and $\nu/2$ are used because these quantities are always in the same quadrant.

The above approach was used for historical computations. In modern computer programs, the true anomaly is determined by simply integrating the equations of motion and taking into account other forces, as well as the purely central body forces which produce Keplerian motion. Alternatively, for a Keplerian orbit the mean anomaly and true anomaly can be determined by the following recursive formula:†

$$E_{i+1} = E_i + \frac{M - E_i + e\sin E_i}{1 - e\cos E_i} \tag{9-13}$$

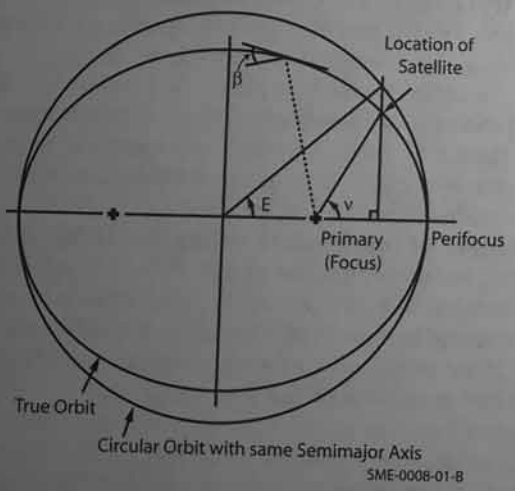


Fig. 9-8. Definition of the True Anomaly, ν, and Eccentric Anomaly, E. The *flight path angle*, β, is the angle between the velocity vector and the perpendicular to the radius vector.

For small eccentricities, ν may be expressed directly as a function of M by expanding in a power series in e to yield:

$$\nu = M + 2e\sin M + \frac{5}{4}e^2\sin 2M - \frac{1}{4}e^3\sin M + \frac{13}{12}e^2\sin 3M + \mathcal{O}\left(e^4\right)^{\ddagger} \tag{9-14}$$

Finally, it is often convenient to discuss variations in the orbital elements in terms of the *mean angular rate*, often called the *mean motion*, n, which is simply the rate of change of the mean anomaly. If n is in rad/s, then

$$n \equiv \mathrm{d}M/\mathrm{d}t = 2\pi/P \tag{9-15}$$

$$M = M_0 + n(t - t_0) \tag{9-16}$$

where P is the period, and M_0 is the mean anomaly at epoch t_0. From Eq. (9-4a), we have

$$n = \sqrt{\mu/a^3} \tag{9-17}$$

where $\mu \equiv GM$ and a is the semimajor axis. Thus, n and M are easily determined from fundamental orbit parameters and can then be used as the basis for computing the true anomaly and the variations in the elements due to orbit perturbations as discussed in Sec. 9.4.

Summary. The elements of an orbit are the parameters needed to fully specify the motion of a satellite. Table 9-4 and Figs. 9-7 and 9-8 show the classical elements or Keplerian elements for an Earth satellite (planetary elements are slightly different and are defined in Sec. 9.2). The semimajor axis and eccentricity define the size and shape of the orbit; the inclination and right ascension of the ascending node define the orbit plane. The argument of perigee defines the rotation of the orbit within the plane, and finally, the true anomaly or mean anomaly specifies the position of the satellite in its orbit at the epoch time at which the orbit is specified.

Unfortunately, the Keplerian elements are poorly defined under conditions that commonly occur for Earth satellites. For near circular orbits, perigee will be very poorly defined and consequently, both the argument of perigee and true anomaly will not be well-defined. In this case, the *longitude of the satellite* measured from the ascending node is often used. If the orbit plane is near the equatorial plane, then the ascending node will also be poorly defined and, in this case, the basic angle will be measured from the vernal equinox. For a hyperbolic orbit, the period is undefined and is replaced by the *aerial velocity* or area per unit time swept out by a line joining the satellite and the planet (i.e., the angular momentum per unit mass). The equations for this are in Table 9-4.

In practice, a variety of orbit perturbations, described in Sec. 9.4, cause continuing changes and oscillations in the orbit elements. The *mean orbit elements* given in

---

* E is defined as the angle measured at the center of the orbit between perigee and the projection of the satellite perpendicular to the major axis onto a circular orbit with the same semimajor axis as shown in Fig. 9-8.

† This convenient formula is from David Diaz, Microcosm.

‡ $\mathcal{O}$ in Eq. (9-14) and subsequent equations stands for "of order" and refers to subsequent terms of expansion. Thus, Eq. (9-14) includes all terms of order e^3 or lower and excludes terms of order e^4 and higher.

Table 9-4. Properties of Keplerian Orbits. See Fig. 9-2, 9-3, 9-5, 9-6, and 9-8 for illustration of the variables.

Quantity	Circle	Ellipse	Parabola	Hyperbola
Defining Parameters	a = semimajor axis = radius	a = semimajor axis b = semiminor axis	p = semi-latus rectum q = perifocal distance	a = semi-transverse axis ($a < 0$) b = semi-conjugate axis
Parametric Equation	$x^2 + y^2 = a^2$	$x^2/a^2 + y^2/b^2 = 1$	$x^2 = 4qy$	$x^2/a^2 - y^2/b^2 = 1$
Eccentricity, e	$e = 0$	$e = \sqrt{a^2 - b^2}/a \quad 0 < e < 1$	$e = 1$	$e = \sqrt{a^2 - b^2}/a \quad e > 1$
Perifocal Distance, q	$q = a$	$q = a(1 - e)$	$q = p/2$	$q = a(1 - e)$
Velocity, V, at Distance, r, from Focus	$V^2 = \mu/r$	$V^2 = \mu(2/r - 1/a)$	$V^2 = 2\mu/r$	$V^2 = \mu(2/r - 1/a)$
Total Energy Per Unit Mass, ε	$\varepsilon = -\mu/2a < 0$	$\varepsilon = -\mu/2a < 0$	$\varepsilon = 0$	$\varepsilon = -\mu/2a > 0$
Mean Angular Motion, n	$n = \sqrt{\mu/a^3}$	$n = \sqrt{\mu/a^3}$	$n = 2\sqrt{\mu/p^3}$	$n = \sqrt{\mu/(-a)^3}$
Period, P	$P = 2\pi/n$	$P = 2\pi/n$	$P = \infty$	$P = \infty$
Anomaly	$\nu = M = E$	Eccentric anomaly, E $\tan\frac{\nu}{2} = \left(\frac{1+e}{1-e}\right)^{1/2} \tan\left(\frac{E}{2}\right)$	Parabolic anomaly, D $\tan\frac{\nu}{2} = D/\sqrt{2q}$	Hyperbolic anomaly, F $\tan\frac{\nu}{2} = \left(\frac{1+e}{1-e}\right)^{1/2} \tanh\left(\frac{F}{2}\right)$
Mean Anomaly, M	$M = M_0 + nt$	$M = E - e \sin E$	$M = qD + (D^3/6)$	$M = (e \sinh F) - F$
Distance from Focus, $r = q(1 + e)/(1 + e\cos\nu)$	$r = a$	$r = a(1 - e\cos E)$	$r = q + (D^2/2)$	$r = a(1 - e\cosh F)$
$r\,dr/dt \equiv r\dot{r}$	0	$r\dot{r} = e\sqrt{a\mu}\sin E$	$r\dot{r} = \sqrt{\mu}\,D$	$r\dot{r} = e\sqrt{(-a)\mu}\sinh F$
Areal Velocity, $\frac{dA}{dt} = \frac{1}{2}r^2\frac{d\nu}{dt}$	$\frac{dA}{dt} = \frac{1}{2}\sqrt{a\mu}$	$\frac{dA}{dt} = \frac{1}{2}\sqrt{a\mu(1 - e^2)}$	$\frac{dA}{dt} = \frac{1}{2}\sqrt{\frac{\mu q}{2}}$	$\frac{dA}{dt} = \frac{1}{2}\sqrt{a\mu(1 - e^2)}$ © 2011 Microcosm, Inc.

Note: $\mu = GM$ is the gravitational constant of the central body; ν is the true anomaly, and $M \equiv n(t - T)$ is the mean anomaly, where t is the time of observation, T is the time of perifocal passage, and n is the mean angular motion. See App. C for additional formulas and a discussion and listing of terminology and notation.

most general purpose tables define the average motion over some span of time. For more precise calculations, it is preferable to use *osculating elements* which are the elements of a true Keplerian orbit instantaneously tangent to the real orbit. Thus, the *osculating elements* fluctuate continuously as various forces (e.g., atmospheric drag and perturbations from other planets) alter the simple shape defined by Kepler's laws.

Keplerian orbits are not sufficiently accurate for spacecraft ephemerides or for calculations which require a precise knowledge of the spacecraft's position or velocity; however, they are accurate enough to estimate overall mission characteristics in most regions of space. The normal procedure for adding additional detail to orbit analysis is to treat orbits as Keplerian with additional perturbations produced by any of the various interactions which may be important. Approximations for the most important perturbations are discussed in Sec. 9.4.

9.2 Orbits of the Moon and Planets

Orbits of the planets and other natural objects in the Solar System exhibit considerable regularity. With the exception of cometary orbits, they are all nearly circular, coplanar, and regularly spaced.

Although perturbations due to third bodies (mostly Jupiter) must be taken into account for computing accurate planetary ephemerides, the description of planetary orbits in terms of Keplerian elements describes most orbital characteristics quite well. The position of the center of mass of the Solar System relative to the Sun depends on the position of all the planets but is typically somewhat below the surface of the Sun in the general direction of Jupiter. For most practical purposes we can regard the Sun as the center of mass of the Solar System and, therefore, as at one focus for all planetary orbits.

To define the Keplerian elements for the orbits of either planets or interplanetary spacecraft we must establish a reference plane for the solar system. The standard plane chosen for this is the *ecliptic* or the plane of the Earth's orbit about the Sun. This plane is inclined to the Earth's equatorial plane by 23.44 deg, an angle known as the *obliquity of the ecliptic*. The intersection of the plane of the ecliptic and the plane of the Earth's equator define two opposite directions in space known as the *vernal* and *autumnal equinoxes* represented by the symbols, ♈ and ♎, respectively. The vernal equinox is the direction of the Sun (viewed from the center of the Earth) as it crosses the equatorial plane from south to north on the first day of spring. It serves as the reference direction for coordi-

Table 9-4, Fig. 9-8◀, Eq. 9-17◀

nate systems using either the equatorial or ecliptic plane. Perturbative forces on the Earth cause the rotational axis of the Earth to move in a cone of 23.44 deg radius about a vector perpendicular to the ecliptic plane. This *precession of the equinoxes* has a period of 25,700 years. This implies that all celestial or "inertial" coordinate systems will have a date attached to them called the *epoch* for the coordinate system.

Because the Earth's orbit is not perfectly Keplerian, and because of the drift of the vernal equinox, the orbital period of the Earth about the Sun (and similarly the periods of the other planets) depends on how it is measured. The *sidereal year*, about 365.26 days, is the period of revolution of the Earth relative to the fixed stars. The *tropical year* is the Earth's period relative to the vernal equinox, and is about 20 minutes shorter than the sidereal year. This is the basis of the civil calendar, since for calendar purposes we are interested in the seasons, which are determined by the position of the Sun relative to the Earth's equatorial plane. Finally, the *anomalistic year*, 5 minutes longer than the sidereal year, is the period of the Earth relative to perihelion. Recall that *perihelion* is the perifocal point when the Earth is closest to the Sun. The drift in the inertial position of perihelion is due to perturbations from the other planets.

The orbital elements of the planets and other Solar System objects are analogous to the Earth satellite elements defined in Table 9-4 with the Earth's equatorial plane replaced by the ecliptic plane.* Thus the semimajor axis and eccentricity retain the same definitions. The inclination of planetary orbits is measured relative to the ecliptic so the inclination of the Earth's orbit is zero. The *longitude of the ascending node*, Ω, is the angle from the vernal equinox to the ascending node of the planet's orbit measured eastward (i.e., in the direction of motion of the Earth in its orbit) along the ecliptic plane. The *argument of perihelion*, ω, is the angle from the ascending node to perihelion measured along the planet's orbit in the direction of its motion. In some tables, ω is replaced by the *longitude of perihelion*, $\tilde{\omega}$, defined as $\Omega + \omega$. Note that this is not a true angular measure because Ω and ω are measured in different planes. Finally, the mean anomaly of satellite orbits is replaced by the *time of perihelion passage*, which is one of the times, usually the most recent, when the planet was at perihelion. Numerical values for the planetary orbital elements are given in App. C.

Planetary orbits within the Solar System are fairly uniform in both shape and orientation; with the exception of Pluto and Mercury, the orbital inclinations are all less than 3.5 deg and the eccentricities are less than 0.1. The semimajor axes of the planetary orbits are also nearly regular and are approximately given by an empirical relationship known as *Bode's Law* in which the semimajor axes of the planets and asteroid belt in AU are approximately 0.4, 0.7, 1.0, 1.6, 2.8, 5.2, and so on.†

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For both interplanetary spacecraft and for determining the brightness of the planets as seen by Earth-orbiting spacecraft, we are interested in the orientation of the planets relative to the Earth and Sun, as well as the orientation relative to the fixed stars. The various geometrical orientations of the planets relative to the observer and the Sun are called *planetary configurations*, or *aspects*, and are discussed on the book website.

A critical characteristic for interplanetary flight is the length of time it takes for planets to return to the same relative orientation, known as the *synodic period*, S, which is more formally defined as the interval between successive oppositions of a superior planet or successive inferior conjunctions of an inferior planet. The relation between the synodic period and the sidereal period, P, relative to the fixed stars is shown in Fig. 9-9. In this example, the innermost planet has made one and one-third revolutions in the period of time that the outer planet has made one-third of a revolution.

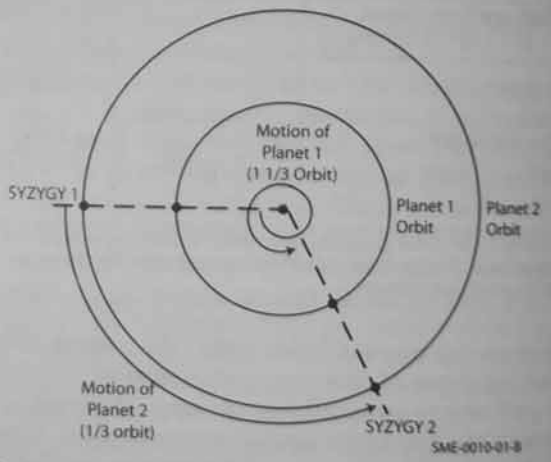


Fig. 9-9. Determining the Synodic Period or Period with Respect to the Earth.

If P_{inner} and P_{outer} are the periods of the inner and outer planets, respectively, then in general:

$$S^{-1} = P_{inner}^{-1} - P_{outer}^{-1} \qquad \text{or} \qquad (9\text{-}18a)$$

$$S = \frac{P_{outer} P_{inner}}{P_{outer} - P_{inner}} \qquad (9\text{-}18b)$$

For an observer on the Earth, if S and P are both expressed in years, then for superior planets the synodic period is given by:

* Note that *longitude* in the Solar System is measured in the ecliptic plane from the vernal equinox and is not the same as longitude on the surface of the Earth or right ascension in the sky which is measured along the celestial equator.

† Bode's Law states that the semimajor axis of the planets is given approximately by the empirical expression $a = 0.4 + 0.3 \times 2^n$, where n is the series of integers, 0, 1, 2, ... Bode's Law did reasonably well at predicting the orbit of Uranus, which had not yet been discovered, but didn't do well for Neptune or Pluto.

$$S_{superior}^{-1} = 1 - P_{outer}^{-1} \quad \text{or} \quad (9\text{-}19a)$$

$$S_{superior} = P_{outer}/(P_{outer} - 1) \quad (9\text{-}19b)$$

and for an inferior planet by:

$$S_{inferior}^{-1} = P_{inner}^{-1} - 1 \quad \text{or} \quad (9\text{-}20a)$$

$$S_{inferior} = P_{inner}/(1 - P_{inner}) \quad (9\text{-}20b)$$

For planets at very different distances from the Sun, the synodic period will be slightly longer than the period of the inner planet. For planets which are close together the synodic period will become longer since the two planets revolve about the Sun at nearly the same rates and, therefore, pass each other very slowly. For example, oppositions of Mars occur approximately every 780 days. Because planetary orbits are not circular, the actual synodic periods vary by several weeks. Synodic periods of the various planets are shown in Table 9-5. Specific oppositions for Mars are shown in Table 9-6 along with the associated launch opportunities.

Planetary configurations are important for interplanetary flight as well as for observations because they define the opportunities for planetary travel. For example, as discussed in Sec. 9.6.1, trips to Mars along a minimum energy trajectory will leave the Earth about 97 days before opposition, and will arrive at Mars about 162 days after opposition (see Table 9-6), although various factors may cause the actual flight times, (particularly the arrival), to vary by several weeks. Because an opposition of Mars occurred on March 17, 1997, we would expect flights that left Earth on about December 10, 1996, to arrive at Mars about August 26, 1997. The two spacecraft flown during this launch opportunity, Mars Pathfinder and Mars Global Surveyor, were launched on December 4, 1996, and November 7, 1996, and arrived at Mars on July 4, 1997, and September 11, 1997, respectively. Similarly, because of the Mars opposition on October 13, 2020, we can expect flights to Mars at that time to leave Earth about July 8, 2020, and arrive at Mars about March 24, 2021.

Table 9-5. Synodic Periods of the Planets.

Planet	Synodic Period (Days)	1st Opposition after 2000
Mercury	115.9132	February 12, 2001
Venus	583.9178	March 29, 2001
Mars	779.9317	June 13, 2001
Jupiter	398.8854	January 1, 2002
Saturn	378.0914	December 3, 2001
Uranus	369.6563	August 15, 2001
Neptune	367.4861	June 30, 2001
Pluto*	366.7326	June 2, 2001

*Pluto is officially a *dwarf planet* as discussed in App. D.

The orbits of the natural satellites in the Solar System are generally less uniform than the orbits of the planets primarily because perturbations cause substantial variations in satellite orbits. For example, the perigee location for the Moon makes one complete revolution about the Moon's orbit in only 8.85 years, and the line of nodes rotates fully around the orbit in 18.6 years. Thus, in analogy with the various types of years, the month is defined in several ways, depending on the measurement reference. For most purposes the fundamental intervals are the *sidereal month*, relative to the fixed stars, about 27.32 days, the *synodic month* from new moon to new moon, about 29.53 days and the *nodical* or *draconic month* between successive ascending nodes, about 27.21 days. Other periods are listed in App. C.

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Table 9-6. Earth-Mars Oppositions and Launch Opportunities, 1994–2035. Launch opportunities assume a Hohmann Transfer (see Sec. 9.6.1).

Launch Opportunity	Opposition	Arrival Date	Launch Opportunity	Opposition	Arrival Date
11/7/94	2/12/95	7/24/95	2/15/16	5/22/16	10/31/16
12/9/96	3/16/97	8/25/97	4/20/18	7/26/18	1/4/19
1/16/99	4/23/99	10/2/99	7/8/20	10/13/20	3/24/21
3/8/01	6/13/01	11/22/01	9/1/22	12/7/22	5/18/23
5/23/03	8/28/03	2/6/04	10/10/24	1/15/25	6/26/25
8/2/05	11/7/05	4/18/06	11/14/26	2/19/27	7/31/27
9/17/07	12/23/07	6/2/08	12/18/28	3/25/29	9/3/29
10/24/09	1/29/10	7/10/10	1/25/31	5/2/31	10/11/31
11/27/11	3/3/12	8/12/12	3/22/33	6/27/33	12/6/33
12/31/13	4/7/14	9/16/14	6/10/35	9/15/35	4/24/36

Table 9-6, Fig. 9-9◀, Eq. 9-20b

Table 9-7. Orbit Classification by Distance from the Earth. This classification is convenient since it represents approximately the difficulty of getting there and, therefore, the cost and time. See text for additional discussion of each.

Name	Location	Uses	Examples
LEO (Low-Earth Orbit)	< 3,000 km (most < 900 km)	All applications (cheapest to get to)	Space Telescope, ISS, LandSat, Iridium
MEO (Medium-Earth Orbit)	3,000 km to GEO	Communications, navigation, some observation	GPS
HEO (Highly Elliptical Orbit), also called Molniya Orbits	Typically perigee in LEO and apogee near GEO	Communications	Russian Comm Satellites
GEO (Geosynchronous)	35,856 km	Communications, weather	TDRS, Intelsat, DBS (radio, TV), BrazilSat
Super-Synchronous	Above GEO, below the Moon	Limited	Vela
Lunar and Lagrange Point	At or near Moon distance (350,000 km)	Science, potentially manufacturing	Apollo, Lunar Orbiter, Lunar Prospector
Interplanetary, Deep Space	Beyond the Moon, within the Solar System	Exploration	Viking, Mars Pathfinder, Galileo, Mars Rovers
Interstellar	Outside the Solar System	Exploration	Pioneer 10,11

9.3 Spacecraft Orbit Terminology

Objects launched into space are categorized by their orbit. *Ballistic missiles, sounding rockets* and *suborbital vehicles* travel in elliptical orbits which intersect the surface of the Earth. This is frequently called a *ballistic trajectory* because it is also the path of a baseball, bullet, or cannonball. The ballistic missile and sounding rocket are distinguished by their functions. The missile is used to strike some specific target, whereas the sounding rocket or suborbital vehicle is used to make measurements in or above the Earth's atmosphere. The sounding rocket may either impact the surface, burn up in the atmosphere, or be recovered by parachute.

The semimajor axis of a satellite orbit must be at least as large as the radius of the planet, whereas the semimajor axis for a sounding rocket or suborbital vehicle may be as small as approximately half the radius of the planet.* Because the total energy of a spacecraft depends only on the semimajor axis, the energy of a sounding rocket is normally, though not necessarily, much less than that of an Earth satellite. However, sounding rockets and ballistic missiles frequently reach altitudes well above those of low-Earth satellites because they travel in very elongated elliptical orbits.

Any object which travels in an elliptical orbit around a planet is called a *satellite* of that planet. As shown in Table 9-7, Earth satellites are typically classified by their distance. The principal differences between the orbit altitudes is the energy or velocity required to achieve them and the radiation level in that environment. Most Earth satellites are in *low-Earth orbit*, or *LEO*, below the Van Allen radiation belts, which start somewhat above 1,000 km. This orbit is both easy to get to and has a much lower radiation density than higher orbits and, therefore, has the potential for more economical satellites. The next most common is *geosynchronous orbit* or *GEO*, at an altitude of 35,786 km at which the satellite's orbit period is just equal to the Earth's rotation period so the satellite remains approximately fixed over a location on the Earth's equator. This is the single most used satellite orbit. Between LEO and GEO is a broad regime known as *medium-Earth orbit* or *MEO*, which typically contains relatively few satellites because of high levels of radiation in the Earth's Van Allen belt. A few satellites are in *super-synchronous orbits* above GEO, but below the Moon. Finally, a few Earth satellites are at approximately the distance of the Moon, in what are called *Lagrange point orbits*. As discussed further in Sec. 9.5, the *Lagrange points* are five points relative to the Earth and Moon at which the satellite will maintain the same orientation relative to the Earth-Moon system.

If the velocity of an object is greater than the escape velocity of a planet it will be an *interplanetary probe*, or deep space spacecraft traveling in a hyperbolic trajectory relative to the planet and, after it has left the vicinity of the planet, traveling in an elliptical orbit about the Sun. Finally, if an object attains a velocity greater than the Sun's escape velocity, it will be an *interstellar probe*. Pioneer 10, swinging past Jupiter in December 1973, gained sufficient energy in the encounter, as described in Sec. 9.5, to become the first man-made interstellar probe.

All known satellites or probes are assigned an international designation by the World Warning Agency on behalf of the Committee on Space Research, COSPAR, of the United Nations. These designations are of the form 2015-27c, where the first number is the year of launch, the second number is a sequential numbering of launches in that year, and the letter identifies each of the separate objects launched by a single vehicle. Thus, 1997–069c

* Assume all the mass of the Earth is concentrated at its center and a high platform is built to the former location of New York City. An object dropped from the platform will *not* go all the way to the former location of Australia, but will swing very rapidly around the central mass (with perigee essentially at the former center of the Earth) and return to apogee at the platform tip. [Use Eq. (9-5a) with $V = 0$.] Therefore, the semimajor axis will be about half the radius of the Earth and the total energy will be a factor of two less than that for a circular orbit at the Earth's surface.

was the third of five Iridium satellites orbited on the 69th launch of 1997 (a Delta II launched from Vandenberg Air Force Base on Nov. 9, 1997).

In addition to the international designation, most satellites are assigned a name by the launching agency. For NASA, spacecraft in a series are given a letter designation prior to launch, which is changed to a number after a successful launch. Thus, the second Synchronous Meteorological Satellite was SMS-B prior to launch, and SMS-2 after being successfully orbited. Because of launch failures or out of sequence launches, the lettering and numbering schemes do not always follow the pattern A = 1, B = 2, and so forth. For example, in the Interplanetary Monitoring Platform series, IMP-B failed on launch, IMP-E was put into a lunar orbit and given another name and IMP-H and -I were launched in reverse order. Thus IMP-I became IMP-6 and IMP-H became IMP-7. The satellites may also be assigned names in different series; IMP-6 was also Explorer-43 and IMP-7 was Explorer-47.

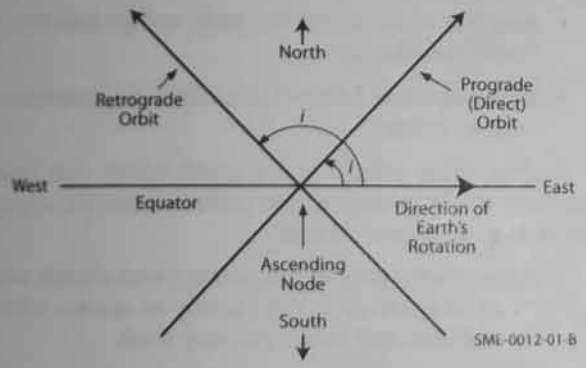


Fig. 9-10. Inclination, *i*, for Prograde and Retrograde Orbits. Satellite is moving from south to north.

The *trajectory* of a spacecraft is its path through space; if this path is closed (i.e., elliptical), then the trajectory is formally an *orbit*. Thus, correct usage would refer to a satellite in elliptical orbit or a probe on a hyperbolic trajectory. Although this distinction is maintained at times, orbit and trajectory are often used interchangeably. Throughout this book we will use "orbit" as the generic term for the path of a spacecraft.

For satellites it is frequently convenient to number the orbits so that we can refer to "a maneuver on the 17th orbit." In standard NASA practice, that portion of the orbit preceding the first ascending node is referred to as *orbit 0*, or *revolution 0*. *Orbit 1* or *revolution 1* goes from the first ascending node to the second ascending node, and so forth. (Note that *revolution* refers to one object moving about another in an orbit, whereas *rotation* refers to an object spinning about an axis.)

When spacecraft are launched, the initial stages of the launch vehicle are typically jettisoned or returned to Earth. However, the final stage may remain inactive and attached to the spacecraft during the coasting phase or parking orbit. The final stage is then ignited or reignited to *inject* the spacecraft, or place it into its proper orbit. The orbit in which the satellite conducts normal operations is referred to as the *mission orbit*, and is typically either *Earth-referenced*, if the principal purpose of the satellite is to observe or communicate with the Earth, or *space-referenced* if its principal purpose is to be or look somewhere in space. A *transfer orbit* is one that is used to transport a spacecraft from one orbit or location to another, as in the case of an Apollo transfer orbit to the Moon or the Mars Rovers orbit to Mars. Finally, at the end of its useful life, a satellite is typically *deorbited* or brought back to Earth, usually burning up in the atmosphere. If the satellite is too high to de-orbit, then an alternative at end of life is to raise the satellite into a parking orbit above the mission orbit such that it will not interfere with other satellites.

A satellite which revolves above the Earth in the same direction that the planet rotates on its axis is in a *prograde* or *direct orbit*. If it revolves in a direction opposite to the rotation, the orbit is *retrograde*. As shown in Fig. 9-10, the inclination of an Earth satellite orbit is measured from east toward north. Therefore the inclination of a prograde satellite is less than 90 deg, and the inclination of a retrograde satellite is greater than 90 deg. In a *polar orbit* $i = 90$ deg. Most satellites are launched in a prograde direction, because the rotational velocity of the Earth provides a part of the orbital velocity. Although this effect is not large (0.46 km/s for the Earth's rotational velocity at the equator compared with a circular velocity of around 7.5 km/s), the available energy is typically the limiting feature of a space mission. Thus, all factors that change the energy which must be supplied by the launch vehicle are important.

9.4 Orbit Perturbations, Geopotential Models, and Satellite Decay

Recall that Kepler's laws are based on a spherically symmetric mass distribution and do not take into account non-gravitational forces or the gravity of other bodies. Consequently, real orbits never follow Kepler's laws precisely, although at times they come very close. Keplerian orbits provide a convenient analytic approximation to a true orbit. In contrast, a *definitive orbit* is the best estimate that can be obtained with all available data of the actual path of a satellite. Because formal analytic solutions are almost never available for real orbit problems with multiple forces, observed orbits are generated numerically based on both orbit theory and observations of the spacecraft. Thus, definitive orbits are only generated for times that are past, although the information from the definitive orbit is frequently extrapolated to the future to produce a predicted or *propagated* orbit.

A *reference orbit* is a relatively simple, precisely defined orbit (usually, though not necessarily, Keplerian) which is used as an initial approximation to the spacecraft's motion. *Orbit perturbations* are the deviations of the true orbit from the reference orbit and are typically classified according to the cause, e.g., perturbations due to the Earth's oblateness, atmospheric drag, or the grav-

Table 9-7◀, Fig. 9-10, Eq. 9-20b◀

Table 9-8. Summary of Forces Acting on a Spacecraft in Earth Orbit.

Source	Regime	Effect in LEO
Earth (point mass)	Dominant force for orbits between Earth and Moon*	Results in Keplerian orbit of satellite about the Earth's center of mass
Earth (higher-order geopotential)	Significant perturbing force at GEO and below; declines rapidly with increasing altitude	Depends on specific orbit; orbit plane rotation of up to 14 deg/day is possible
Sun/Moon (point mass)	Dominant force for interplanetary flight; minor perturbing force in Earth orbit	Low level perturbations; orbit plane rotation of up to 0.007 deg/day
Atmosphere	LEO only, decays exponentially with altitude	Reentry occurs rapidly below 120 km Atmosphere negligible above 1,000 km
Solar Radiation Pressure	Minor perturbing force for normal spacecraft	Small eccentricity growth
Relativistic Effects	Near very massive objects	Negligible
Incidental Forces (leaks, RF, explosive bolts)	Very minor perturbing force for normal spacecraft	Negligible in most circumstances

*The Sun's perturbing force is less than 0.01 times the central force below 370,000 km; the Earth perturbing force is less than 0.01 times the central solar force above 2.5×10^6 km.

itational force of the Moon. In this section we describe the causes of orbit perturbations, the principal effects caused by them, and, where possible, formulas to determine the approximate magnitude of specific effects. More detailed methods for the numerical treatment of perturbations may be found in any of the references in the annotated bibliography on the book website.

Effects which modify simple Keplerian orbits are shown in Table 9-8. These may be divided into four classes: non-gravitational forces (principally atmospheric drag and solar radiation pressure), third body interactions, non-spherical mass distributions, and relativistic mechanics. For some orbits the first two effects dominate the motion of the spacecraft, as in satellite re-entry into the atmosphere or motion about both the Earth and the Moon. Although the effects of non-spherical mass distributions never dominate spacecraft motion, they do provide the major perturbation relative to Keplerian orbits for most intermediate altitude satellites, i.e., those above the atmosphere and below where effects due to the Moon and Sun become important. Additionally, the nonspherical mass distribution becomes much more important for orbits about small bodies such as comets or asteroids where the basic object itself may be fundamentally nonspherical. Finally, relativistic mechanics may be completely neglected in essentially all applications except those specifically intended to test elements of relativity. The largest orbit perturbation in the Solar System due to general relativity is the rotation of the perihelion of Mercury's orbit by about 0.012 deg/century or 3×10^{-5} deg/orbit. Although a shift of this amount is measurable, it is well below the magnitude of other effects which we will consider.

The relative importance of the various orbit perturbations will depend upon the construction of the spacecraft, the specific orbit it's in, and even the level of solar activity. Nonetheless, the Keplerian orbit is an excellent approximation for most astrodynamic problems, with a few important exceptions. For Earth orbits these are:

- Atmospheric drag in low-Earth orbit (less than 1,000 km)
- Rotation of the ascending node and perigee due to Earth's oblateness
- Resonance and Lunar-Solar effects at geosynchronous altitudes

Each of these effects is discussed below. For interplanetary orbits, places where perturbations are important or may dominate include:

- Atmospheric drag in low orbits about planets with an atmosphere or in the vicinity of comets where the gravitational forces are very weak
- Oblateness effects near rapidly rotating planets, such as Jupiter
- Higher order nonspherical effects near small bodies such as natural satellites or asteroids
- Multiple body effects in regions where gravitational forces approximately balance, e.g., between the Earth and the Moon or the Earth and the Sun (often called Lagrange point orbits, see Sec. 9.5)

The impact of each of the major perturbation is discussed in the following subsections.

9.4.1 Nonspherical Mass Distribution

The Earth is very nearly spherically symmetric. However, the rotation of the Earth causes it to assume approximately an equilibrium configuration of an oblate spheroid with an equatorial bulge and flattening at the poles. The difference between the equatorial radius, a, and the polar radius, c, is 21.4 km which produces a *flattening factor* or *ellipticity*, $f \equiv (a-c)/a$ of 1/298.257. As illustrated in Fig. 9-11 this is imperceptible as seen in almost any photograph or illustration.

In addition, the Earth has a slight pear shape at the equator (approximately 100 m out of round) and a variety of minor mass anomalies such as continents, mountain ranges, and San Francisco. Mathematically, this is

Lagrange Planetary Equations. As a practical matter, most current orbit computations are done by simply integrating the equations of motion to obtain a numerical approximation of the motion of the spacecraft. This process works extremely well for computer calculations but provides little physical insight into the effects which are occurring. We began the discussion at the front of the chapter by applying $\mathbf{F} = m\mathbf{a}$ and letting $\mathbf{F}$ be the gravitational force due to a spherically symmetric central body. We can generalize this approach to take additional forces into account by writing:

$$\begin{aligned} m\mathbf{a} &= \mathbf{F}_{\text{central body}} + \mathbf{F}_{\text{spherical harmonics}} + \mathbf{F}_{\text{Sun}} + \mathbf{F}_{\text{Moon}} + \cdots \\ &= -m\nabla U = -m\nabla(\text{central body} + \text{perturbations}) \\ &= -m\nabla(\mu/r) - m\nabla R \end{aligned} \tag{9-21a}$$

Therefore:

$$\mathbf{a} = -\left(\mu r^{-3}\right)\mathbf{r} - \nabla R \tag{9-21b}$$

where as usual, m is the mass of the spacecraft, $\mathbf{a}$ is the vector acceleration, U is the potential energy of the orbit, $\mu \equiv GM$ for the central body and R is the potential function of the perturbation, i.e., representing all of the forces other than the symmetric central body force. R is called the *disturbing potential* and provides a convenient mechanism for evaluating the impact of various forces acting on a spacecraft minus the central body effects.

We can use the above decomposition of the forces to derive *Lagrange's planetary equations* for the variation of the Keplerian orbital elements with time. (The derivation is given in most of the astrodynamics books listed in the bibliography.) The results are:

$$\frac{da}{dt} = \frac{2}{na}\left(\frac{\partial R}{\partial M}\right) \tag{9-22a}$$

$$\frac{de}{dt} = \frac{\left(1-e^2\right)^{1/2}}{na^2 e}\left\{\left(1-e^2\right)^{1/2}\left(\frac{\partial R}{\partial M}\right) - \left(\frac{\partial R}{\partial \omega}\right)\right\} \tag{9-22b}$$

$$\frac{di}{dt} = \frac{1}{na^2\left(1-e^2\right)^{1/2}\sin i}\left\{\cos i\left(\frac{\partial R}{\partial \omega}\right) - \left(\frac{\partial R}{\partial \Omega}\right)\right\} \tag{9-22c}$$

$$\frac{d\omega}{dt} = \frac{1}{na^2}\left\{-\frac{\cos i}{\left(1-e^2\right)^{1/2}\sin i}\left(\frac{\partial R}{\partial i}\right) + \frac{\left(1-e^2\right)^{1/2}}{e}\left(\frac{\partial R}{\partial e}\right)\right\} \tag{9-22d}$$

$$\frac{d\Omega}{dt} = \frac{1}{na^2\left(1-e^2\right)^{1/2}\sin i}\left(\frac{\partial R}{\partial i}\right) \tag{9-22e}$$

$$\frac{dM}{dt} = n - \frac{1-e^2}{na^2 e}\left(\frac{\partial R}{\partial e}\right) - \frac{2}{na}\left(\frac{\partial R}{\partial a}\right) \tag{9-22f}$$

where $R(a, e, i, \omega, \Omega, M)$ is the disturbing potential and n is the mean motion. Note that here a is the semimajor axis and not the acceleration, and M is the mean anomaly (measured relative to the moving perigee) and not the mass of the central body. In the absence of any orbit perturbations all of the partial derivatives are zero, so that only M changes with time as is appropriate for a Keplerian orbit. The advantage of Lagrange's planetary equations is that they allow us to analytically assess the effects of any disturbance on a Keplerian orbit.

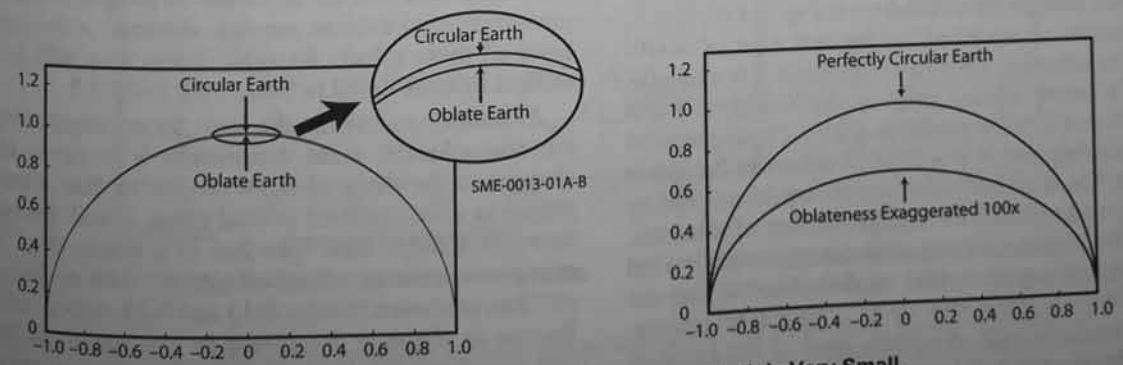
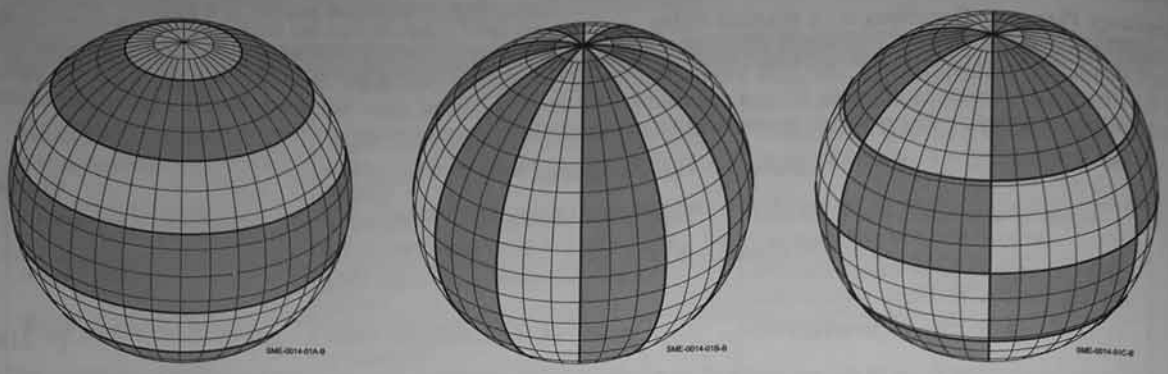


Fig. 9-11. The Difference Between a Spherical Earth and an Oblate Earth is Very Small.

Table 9-8◀, Fig. 9-11◀, Eq. 9-22f



(A) Zonal Harmonics J_n (M = 0, n = 6)

(B) Sectoral Harmonics (m = n = 12)

(C) Tesseral Harmonics (m = 6, n = 3)

Fig. 9-12. Basis of Spherical Harmonic Expansions. Boundaries between shaded regions are where the expansion changes sign. Thus, J_2 represents a change of sign on either side of the equator and, therefore, is the principal term representing the equatorial bulge.

dealt with by expanding the geopotential function at the point (r, θ, ϕ) in a series of spherical harmonics:

$$U(r,\theta,\phi) = \sum_{n=0}^{\infty}\left(\frac{r}{R_E}\right)^{n+1} J_n P_{n0}(\cos\theta) + \sum_{n=1}^{\infty}\sum_{m=1}^{n}\left(\frac{r}{R_E}\right)^{n+1} \times \left[C_{nm}\cos m\phi + S_{nm}\sin m\phi\right] P_{nm}(\cos\theta) \quad (9\text{-}23)$$

where J_n is defined to be C_{n0}, $\mu = GM$ is the Earth's gravitational constant, R_e is the Earth's equatorial radius, P_{nm} are Legendre polynomials, r is the geocentric distance, θ is the latitude, and ϕ is the longitude. Terms with $m = 0$ are called *zonal harmonics* and J_n are the zonal harmonic coefficients. The zonal harmonics are independent of the longitude and have n sign changes over the full range of latitude on the Earth. As shown in Fig. 9-12A, they divide the Earth into a series of longitude independent zones analogous to the temperate and tropic zones. Terms with $n = m$ are called *sectoral harmonics* since they divide the Earth into sectors which are independent of latitude as shown in Fig. 9-12B. Finally, terms with $m \neq 0$ and $m \neq n$ are called *tesseral harmonics* from the Latin *tessera* for tiles, since these divide the world into a tiled pattern as shown in Fig. 9-12C.

A *geopotential model* of the Earth is a set of coefficients in the spherical harmonic expansion. For example, one of the more widely used is the *Goddard Earth Model 10b* or *GEM10b* which is a 21 × 21 geopotential model, meaning that it is a 21 × 21 spherical harmonic expansion. These geopotential models are determined by analyzing tracking data for a large number of satellites. The need for higher accuracy orbit determination has led to more complex geopotential models. Models that are 50 × 50, 70 × 70, 100 × 100, up to 2160 × 2160 are currently in use, although of course most of the terms in higher order models will be extremely small.

Table 9-8◀, Fig. 9-12, Eq. 9-24

The J_0 term in the spherical harmonic expansion represents the point mass or spherically symmetric distribution for which the potential falls off as $1/r$. The J_1 term changes sign at the equator and therefore represents the difference in mass between the northern and southern hemispheres. The J_2 term has two changes in sign between the north and south poles and represents the mass distribution of the equatorial bulge. This is, by far, the largest of the geopotential terms. (The words "J_2" and "Earth oblateness" are frequently used interchangeably.) Beyond J_2, coefficients become small quickly as can be seen by looking at the first several terms:

$$\begin{aligned} J_0 &= 1 \\ J_1 &= 0 \\ J_2 &= 0.001\,082\,635\,9 \\ J_3 &= -0.000\,002\,54 \\ J_4 &= -0.000\,001\,61 \end{aligned} \quad (9\text{-}24)$$

$J_0 = 1$ by definition. In addition, because the coordinate frame is defined as going through the center of mass, $J_1 \equiv 0$. The oblateness term, J_2, is much larger than any of the other perturbations and has important effects for the orbit in that it causes both the right ascension of the ascending node and the argument of perigee to rotate at rates of several degrees per day. Because of the importance of these effects for orbit design they will be described in more detail in Secs. 9.5.2 and 9.5.3.

In addition to rotating the orbit, the principal effect of the non-spherical mass distribution is to cause small changes in the shape of the orbit. Rather than a perfect ellipse in a well-defined inertial plane, actual spacecraft move in a shape more like that of a potato chip, with maximum deviations from a Keplerian orbit on the order of 3 km, as shown in Figs. 9-13 and 9-14. Again this difference in shape is dominated by the equatorial bulge and, consequently, has a pattern in the various components that is nearly sinusoidal.

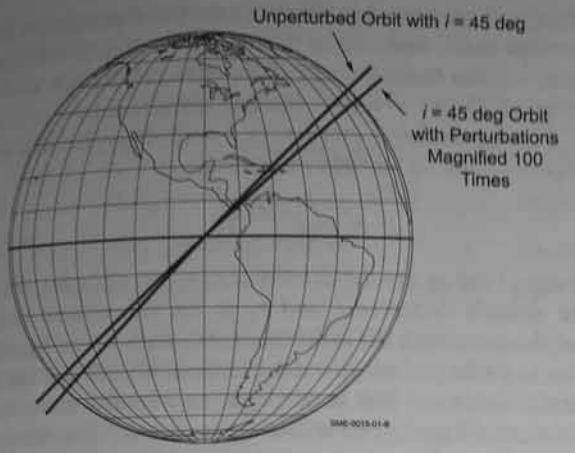


Fig. 9-13. Greatly Exaggerated Shape of a "Circular" Low-Earth Orbit.

Typically the effects of the higher order harmonics are extremely small, except for special circumstances. For example, in geosynchronous orbit, a satellite remains largely over one location on the Earth; therefore, the sectoral harmonics are felt continuously and pull the satellite in an east-west direction. Similarly, in a repeating ground-track orbit, the satellite flies over the same locations on the ground, day after day, and it is possible for resonances to build up when the satellite continuously passes, for example, to the right of the Himalayas, such that the effects of tesseral terms can be substantially magnified. As central bodies become less spherically symmetric the impact of the higher order harmonics becomes more important. For example, the Moon has a variety of mass concentrations or *mascons* which cause large variations in the orbits of lunar spacecraft and make it difficult to predict the orbits even though atmospheric drag plays no role. Orbits around asteroids or other very small objects will also be dominated by higher order harmonics both because of the distinctly nonspherical shape and the smallness of the point mass gravitational forces.

9.4.2 Third Body Interactions

The gravitational forces of the Sun and the Moon cause periodic variations in all of the orbit elements. The effect is generally similar to that of the Earth's equatorial bulge as described in Sec. 9.5.3, i.e., the Sun and the Moon apply an external torque to the orbits and cause the angular momentum vector to rotate. This effect is extremely small and in LEO is dominated by the orbit rotations caused by the Earth's oblateness. However, as described in Sec. 9.5.1 the effect becomes important in geosynchronous orbit and is the dominant source of the need for stationkeeping in GEO.

For nearly circular orbits, the Lagrange planetary equations can be used to determine the approximate rotation rates for the ascending node, Ω, and the argument of perigee, ω, due to the Sun and the Moon. These are:

$$\dot{\Omega}_{MOON} = -0.003\,38(\cos i)/n \tag{9-25}$$

$$\dot{\Omega}_{SUN} = -0.001\,54(\cos i)/n \tag{9-26}$$

$$\dot{\omega}_{MOON} = 0.001\,69\left(4 - 5\sin^2 i/n\right) \tag{9-27}$$

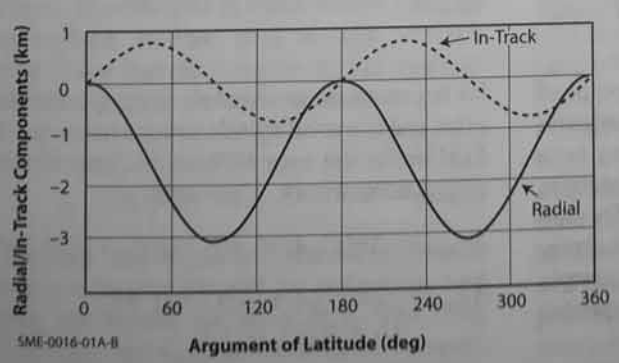
$$\dot{\omega}_{SUN} = 0.000\,77\left(4 - 5\sin^2 i/n\right) \tag{9-28}$$

where i is the orbit inclination, n is the number of orbit revolutions per day, and $\dot{\Omega}$ and $\dot{\omega}$ are in deg/day. The equations neglect the variations caused by the changing orientation of the orbital plane with respect to the Moon's orbit and the ecliptic plane and, therefore, are only average values.

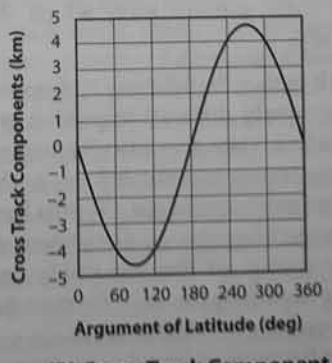
9.4.3 Solar Radiation Pressure

Solar radiation pressure causes periodic variations in all of the orbit elements. The effect is strongest for satellites which have low mass and large cross-sectional areas, such as balloons or spacecraft with a large solar sail. The magnitude of the acceleration a_r in m/s^2 due to solar radiation pressure is:

$$a_r = -4.5\times10^{-6}\,A/m \tag{9-29}$$



(A) Radial and In-Track Components in a Polar Orbit (*i*=90 deg)



(B) Cross-Track Component for a Polar Orbit (*i*=45 deg)

Fig. 9-14. Effect of Nonspherical Earth on the Satellite Orbit. The three curves are the difference between the position of a real satellite and a satellite in a perfectly circular orbit about a point-mass Earth. *In-track* is the direction of the velocity vector; *cross-track* is toward the normal to the orbit.

Table 9-8◀, Fig. 9-14, Eq. 9-29

where A is the cross-sectional area of the satellite normal to the Sun in m^2 and m is the mass of the satellite in kg. This formula applies to an absorbing surface. A mirror normal to the Sun would have twice the acceleration. For satellites below approximately 800 km altitude the acceleration from atmospheric drag is typically much greater than from solar radiation pressure. Above 800 km however, the acceleration from solar radiation pressure can become an important perturbing force. It is frequently the dominant perturbing force for interplanetary spacecraft simply because the other perturbations become dramatically small.

9.4.4 Atmospheric Drag and Satellite Decay

Atmospheric drag is the principal nongravitational force acting on most satellites in low-Earth orbit. Drag acts opposite the direction of the velocity vector, thus slowing the satellite and removing energy from the orbit. This reduction of energy causes the orbit to get smaller, leading to increases in the drag until eventually the altitude becomes so small that the satellite reenters the atmosphere. Below approximately 120 km, the satellite lifetime is extremely short and it will reenter quickly. Above 600 km the lifetimes due to drag will typically exceed spacecraft operational lifetime of approximately 10 years, although drag may still be important for orbit maintenance or maintaining the structure of a constellation.

Although the physics of atmospheric drag is very well understood, drag is nearly impossible to predict with even moderate precision and, therefore, is a major source of the unpredictability of satellite positions at future times and of when and where they will reenter the atmosphere. There are two reasons for this difficulty in predicting drag:

- Drag can vary by as much as an order of magnitude due to the attitude of the spacecraft and, particularly, the orientation of the solar arrays, with respect to the velocity vector.
- The atmospheric density at satellite altitudes varies by as much as two orders of magnitude depending upon the solar activity level which itself varies dramatically.

The net result is that the lifetime and impact point of satellites are extremely difficult to predict and satellite lifetimes themselves vary dramatically depending on a variety of circumstances and conditions. In this section, we provide a brief summary of the physical basis for drag and its effect on satellite orbits, a discussion of the atmosphere and the causes of the variability, and simple mechanisms for bounding the drag level to be expected for satellites in various low-Earth orbits.

The equation for acceleration due to drag, a_D, is:

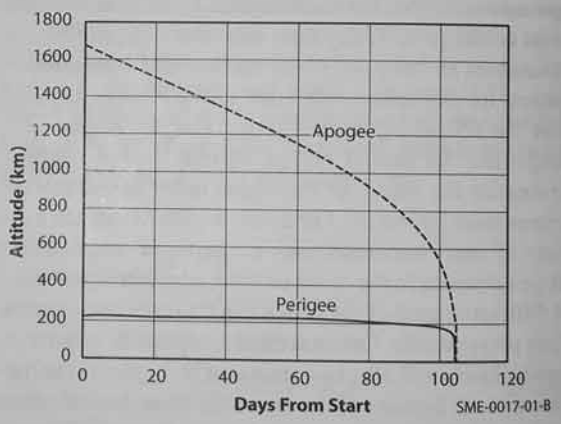
$$a_D = -(1/2)\,\rho(C_D A/m)V^2 \qquad (9\text{-}30)$$

where ρ is the atmospheric density, C_D is the dimensionless *drag coefficient* of the satellite (typically $C_D \approx 2.2$ as discussed below), A is the cross-sectional area, m is the satellite mass, and V is its velocity. The quantity $m/C_D A$ is called the *ballistic coefficient* and is typically in the range of 25 to 100 kg/m^2 for satellites.*

The atmospheric density, ρ, decreases approximately exponentially as altitude increases:

$$\rho \approx \rho_0 e^{-\Delta h/h_0} \qquad (9\text{-}31)$$

where ρ and ρ_0 are the density at any two altitudes, Δh is the altitude difference, and h_0 is the atmosphere *scale height*, by which the atmospheric density drops by 1/e. The scale height is typically between 50 and 100 km at satellite altitude. The scale height and density at various altitudes are given just inside the rear cover. For circular orbits, drag will act approximately continuously, and the orbit will spiral downward. Because of the exponential character of the atmosphere, drag for elliptical orbits will act predominantly at perigee and, therefore, will lower apogee while having relatively little effect on the perigee height. As shown in Fig. 9-15, this divides the drag effect on elliptical orbits into two distinct phases. In the first phase, the elliptical orbit is circularized by reducing apogee and having only a small effect on perigee. As the orbit becomes more circular, drag operates over the entire orbit and the orbit spirals inward more quickly. For elliptical orbits the approximate changes in semimajor axis, a, and eccentricity, e, per orbit are:



Live Calc

Fig. 9-15. Evolution of an Elliptical Orbit Due to Drag. Orbit began with a perigee height of 215 km and apogee height of 1,680 km.

$$\Delta a_{rev} = -2\pi(C_D A/m)\,a^2 \rho_p \exp(-c)\,[I_0 + 2eI_1] \qquad (9\text{-}32)$$

$$\Delta e_{rev} = -2\pi(C_D A/m)\,a\rho_p \exp(-c)\,[I_1 + e(I_0 + I_2)/2] \qquad (9\text{-}33)$$

where ρ_p is the atmospheric density at perigee, $c \equiv ae/h$,

* Note that some authors define the ballistic coefficient as $C_D A/m$. It's important to check the units on the relevant equations.

h is the atmospheric scale height, and I_j are the modified Bessel functions of order j and argument c.

For circular orbits, Eqs. (9-32) and (9-33) can be substantially simplified, to yield:

$$\Delta a_{rev} = -2\pi(C_D A/m)\rho a^2 \quad (9\text{-}34)$$

$$\Delta P_{rev} = -6\pi^2(C_D A/m)\rho a^2/V \quad (9\text{-}35)$$

$$\Delta V_{rev} = \pi(C_D A/m)\rho a V \quad (9\text{-}36)$$

$$\Delta e_{rev} = 0 \quad (9\text{-}37)$$

where P is the orbit period, and V is the satellite velocity. Thus, the computation of effective drag is straight forward with the minor complication that the key parameters (atmospheric density and ballistic coefficient) are largely unknown.

The spacecraft drag coefficient, C_D, depends primarily on four factors:*

- Shape
- Attitude
- Surface condition: smooth (*specular reflection*), rough (*diffuse reflection*), or sticky (*absorption* or *skin friction*)
- Multi-collision effects (*form drag*)

Form drag is very important at low speeds and low altitudes but is negligible for satellites. This is fortunate, because form drag is very difficult to compute.

Ignoring multi-collision effects, the coefficient of drag C_D can be computed using the following expressions:

$$C_D = \frac{4}{A}\int_S \cos^3\psi \, dS \qquad \text{Specular Reflection} \quad (9\text{-}38)$$

$$C_D = 2 + \frac{1}{A}\int_S \cos^2\psi \, dS \qquad \text{Diffuse Reflection} \quad (9\text{-}39)$$

$$C_D = 2 \qquad \text{Absorption} \quad (9\text{-}40)$$

where A is the cross-sectional area, S is the surface area, and ψ is the angle between the velocity vector and normal to the surface. The integral is taken over the entire forward-facing surface, so that ψ is always between 0 deg and 90 deg. Note that $dA = \cos\psi \, dS$. For any satellite, C_D can be divided into three components:

$$C_D = \alpha C_{DS} + \beta C_{DD} + \gamma C_{DA} \quad (9\text{-}41)$$

$$\alpha + \beta + \gamma = 1 \quad (9\text{-}42)$$

where C_{DS} is the drag coefficient for specular reflection, C_{DD} is the drag coefficient for diffuse reflection, and C_{DA} is the drag coefficient for absorption. The drag coefficients C_{DS}, C_{DD}, and C_{DA}, can usually be computed from the shape and attitude of the spacecraft. The component fractions, α, β, γ, however, must be determined by experiment. For most satellites, C_D is dominated by absorption or specular reflection. Table 9-9 gives values of C_D for various common shapes and attitudes.

Table 9-9. Drag Coefficients for Common Shapes, Attitudes, and Surface Conditions. See text for discussion. (Formulas due to Leo Early.)

Shape	Surface	Drag Coefficient
Flat Plate, Face First	Specular Diffuse	4 3
Flat Plate, Angle of Incidence α	Specular Diffuse	$4\sin^2\alpha$ $2 + \sin\alpha$
Sphere	Specular Diffuse	2 8/3 = 2.667
Circular Cylinder, Side First	Specular Diffuse	8/3 = 2.667 $2 + \pi/4 = 2.785$
Octagonal Cylinder, Side Face First	Specular	$2\sqrt{2} = 2.828$
Octagonal Cylinder, Side Edge First α = cos 22.5° β = cos 67.5°	Specular	$4\frac{\alpha^3+\beta^3}{\alpha+\beta} = 2.586$
Octagonal Cylinder, Side First, Average Over All Side-first Orientations	Specular	$C_D A = 2.812\, A_{\text{face first}}$
Circular Cone, Half-angle α, Point First	Specular Diffuse	$4\sin^2\alpha$ $2 + \sin\alpha$

The mass of the satellite is ordinarily very well known. However, the ballistic coefficient depends on both C_D and the cross-sectional area, A, perpendicular to the velocity vector. The fundamental difficulty is that most spacecraft have large deployable solar arrays which are very thin and thus have a high cross-sectional area perpendicular to their face and very low cross-sectional area when flying edge-on. Typically the solar arrays rotate to track the Sun and antennas, instruments, and other appendages rotate to track ground targets, geosynchronous relay satellites, or targets in space. Consequently, the cross-sectional area "into the wind" can vary by approximately a factor of ten, as shown in Table 9-10 for a number of representative low-Earth orbit satellites.

The next fundamental problem in calculating atmospheric drag is the extreme variation in atmospheric density. Variations in solar activity, particularly over the course of a solar cycle, cause both a variety of geomagnetic effects and heating of the upper atmosphere due primarily to extreme ultraviolet radiation which is not monitored from the Earth. When the atmosphere is heated, it expands and becomes denser at any given altitude. This seems counter-intuitive because heating a gas causes it to expand and to become less dense. The explanation here is that heating the atmosphere causes it to expand and rise, such that portion of the atmosphere which was previously at 200 km is now at 250 km and represents a much denser atmosphere for a satellite at that altitude.

Figures 9-16 through 9-19 provide a means of estimating satellite lifetimes based on a general evaluation of atmospheric densities. Fig. 9-16 shows the historical record of solar activity. What is plotted is the mean daily F10.7 index of radio flux from the Sun. The *F10.7 index*

* The discussion of drag coefficients is due to Leo Early.

Table 9-10. Typical Ballistic Coefficients for Low-Earth Orbit Satellites. Values for cross-sectional area and drag coefficients are estimated from the approximate shape, size, and orientation of the satellite and solar arrays. [XA = cross-sectional area].

Satellite	Mass (kg)	Shape	Max. XA (m^2)	Min. XA (m^2)	Max. Drag Coef.	Min. Drag Coef.	Max. Ballistic Coef. (kg/m^2)	Min. Ballistic Coef. (kg/m^2)	Type of Mission
Oscar-1	5	box	0.075	0.0584	4	2	42.8	16.7	Comm.
Intercos.-16	550	cylind.	3.16	2.7	2.67	2.1	82.9	76.3	Scientific
Viking	277	octag.	2.25	0.833	4	2.6	128	30.8	Scientific
Explorer-11	37	octag.	0.18	0.07	2.83	2.6	203	72.6	Astronomy
Hubble	11,000	cylind.*	112	14.3	3.33	4	192	29.5	Astronomy
OSO-7	634	9-sided	1.05	0.5	3.67	2.9	437	165	Solar Physics
Pegasus-3	10,500	cylind.*	264	14.5	3.3	4	181	12.1	Scientific
Landsat-1	891	cylind.*	10.4	1.81	3.4	4	123	25.2	Remote Sensing
ERS-1	2,160	box*	45.1	4	4	4	135	12.0	Remote Sensing
LDEF-1	9,695	12-face	39	14.3	2.67	4	169	93.1	Environment
Vanguard-2	9.39	sphere	0.2	0.2	2	2	23.5	23.5	Scientific
SkyLab	76,136	cylind.*	462	46.4	3.5	4	410	47.1	Scientific
Echo-1	75.3	sphere	731	731	2	2	0.515	0.515	Communications
Extrema							437	0.515	

* With solar arrays

is simply a measure of the radio flux at a wavelength of 10.7 cm measured in units of 10^{-22} watts/m^2/Hz. This is a convenient measure largely because data for the F10.7 index is available for historical periods beginning in 1945. The peaks in the F10.7 index correspond approximately to solar maxima, to levels of high solar activity, and, therefore, to levels of high atmospheric density. Several things are immediately clear from the plot. First, there is an 11-year RF solar cycle corresponding essentially to the sun spot cycles which have been observed for centuries. In addition, it is clear that there are very large month-to-month variations and that while the overall solar cycle can be predicted, the level at any specific time cannot be predicted with any reasonable accuracy. In addition, some solar cycles have levels which are much higher than other solar cycles. Consequently, while the average F10.7 index over an extended period of time is well known, predicting the level at any specific future time is highly uncertain.

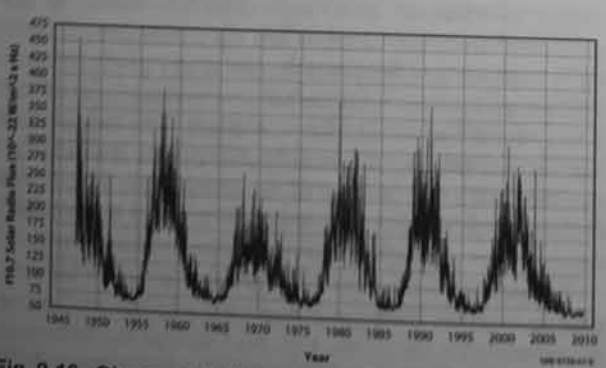


Fig. 9-16. Observed Daily Mean Radio Flux at 10.7 cm. This flux is called the F10.7 index.

Table 9-10, Fig. 9-16, Eq. 9-42◀

Figure 9-17 shows atmospheric density versus altitude corresponding to various values of the F10.7 index. These densities were obtained from the MSIS atmospheric model. Below approximately 150 km, the density is not strongly affected by solar activity. However, at satellite altitudes in the range of 400–800 km, the density variations between solar maximum and solar minimum are approximately two orders of magnitude. Because of these large density variations, satellites will decay far more rapidly during periods of solar maximum and more slowly during solar minimum.

To demonstrate the wide fluctuations in spacecraft lifetimes, we "de-orbited" nine satellites in a computer model, with the results shown in Fig. 9-18. All of the satellites were started at an initial altitude of 700 km in circular orbits. Three were started in 1956 at the beginning of solar maximum, three were started in 1959 near the peak of solar activity, and three were started in 1962 at the beginning of the subsequent solar minimum. Within each set, one satellite had a ballistic coefficient of 20 kg/m^2, one was at 65 kg/m^2, and one at 200 kg/m^2, corresponding roughly to light, moderate and dense satellites respectively. The histories of the nine satellites are shown on the graph. The bars at approximately 400 km altitude mark the periods of solar maxima where the F10.7 index was above 150.

Several characteristics of satellite decay are easily seen in Fig. 9-17. Satellites decay very little during solar minimum and then rapidly during solar maximum. For any one satellite, each solar maximum period will generally produce greater decay than the previous maximum because the satellite is lower. It will also depend on the level of the particular solar maximum. The effect of the

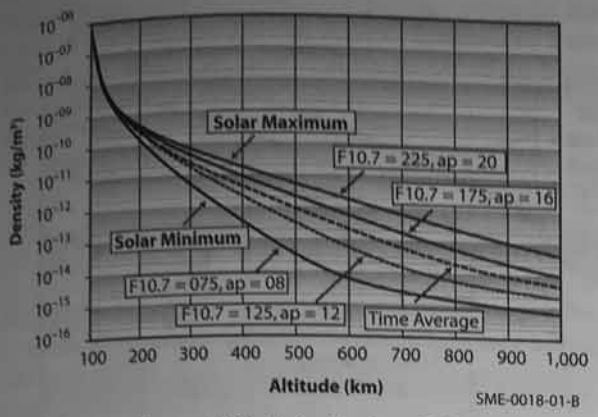


Fig. 9-17. Density vs. Altitude for Various F10.7 Values.

solar maximum also depends on the satellite ballistic coefficient. Those with a low ballistic coefficient respond quickly to the atmosphere and tend to decay promptly. Those with high ballistic coefficients push through a larger number of solar cycles and decay much more slowly. In the end the time for satellite decay is generally measured better in solar cycles than in years. All 9 satellites reentered during periods of solar maxima. For the range of ballistic coefficients shown, the lifetimes varied from approximately one-half of a solar cycle (5 years) to 17 solar cycles (190 years). Predicting in advance where any of the satellites come down would be remarkably difficult.

For a specific satellite whose parameters and current or expected orbit are known, the best way to predict the orbit lifetime is to integrate the equations of motion in an orbit propagator with realistic atmospheric models and assumptions about the satellite cross section. Alternatively, if the altitude is high such that orbit propagation over the satellite lifetime will take excessive computer time, we can simply integrate the equations for the change in altitude to provide approximate lifetime predictions, as was done for Fig. 9-18.

A third alternative, shown in Fig. 9-19, is to estimate a range of lifetimes based on the satellite ballistic coefficient and the approximate time of launch with respect to the solar cycle. The three sets of curves show satellite lifetime as a function of initial circular altitude for satellites with three different ballistic coefficients. The spread of the curves near the center represents the difference between solar maximum and solar minimum. At high altitudes and long lifetimes the curves come together because the satellite will see a large number of solar cycles and it will make very little difference when the satellite starts. Satellites in the midrange go through only a few solar maxima, so the lifetimes will depend strongly on when in the solar cycle the satellite is launched. Notice the steep bends in the curves at multiples of the 11 year solar cycle. As the satellite starts out lower and lower, the atmospheric variations are less, so the curves tend to come together again. Thus, a satellite launched at 100–200 km will decay very quickly irrespective of solar maximum or solar minimum. This chart provides a convenient mechanism for estimating the satellite lifetime range. The actual lifetime will depend on detailed design parameters, the details of the actual orbit, and the launch date relative to the solar cycle.

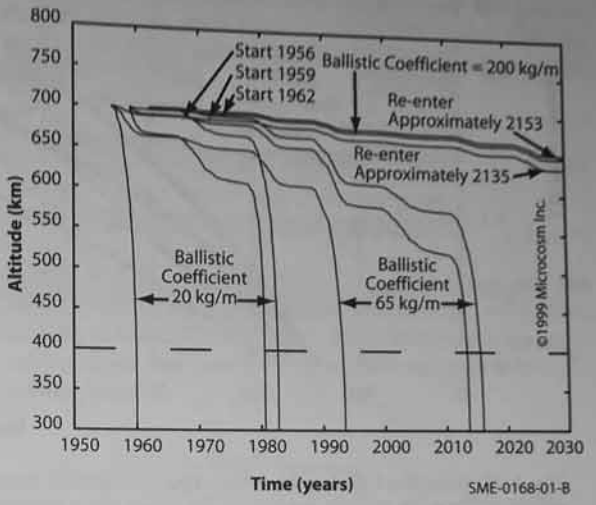


Fig. 9-18. Altitude as a Function of Date for 9 Hypothetical Satellites Launched Over a 6-year Period.

9.5 Specialized Orbits

Spacecraft orbits fall into two broad groups. First, there is a continuum of orbits with varying eccentricities and altitudes ranging from low-Earth to higher orbits, to eventually interplanetary and interstellar orbits. Typically, most such orbits are nearly Keplerian and are conveniently described and analyzed by the techniques in Sec. 9.2. In addition, superimposed on this orbit continuum is a set of specialized orbits summarized in Table 9-11. Within this group, geosynchronous orbits and the repeating ground track come about simply because of the relationship between the orbit period and the rotation period of the Earth. All the rest of the specialized orbits are the result of special effects arising from one of the major orbit perturbations described in Sec. 9.4. This section describes each of these specialized orbits. For each we discuss the physical effect which produces it, the basic equations which govern it, and why these orbits are important. Specific applications of these orbits are discussed in Secs. 10.4 and 10.5.

Whenever possible, expressions for the characteristics of specialized orbits are given in three forms:

- Analytic expression applicable to any central body
- First order approximation applicable to any central body
- Expression where constants have been evaluated for satellites in Earth orbit

The purpose is to allow the equations to be used for any planet or other central body and to allow them to

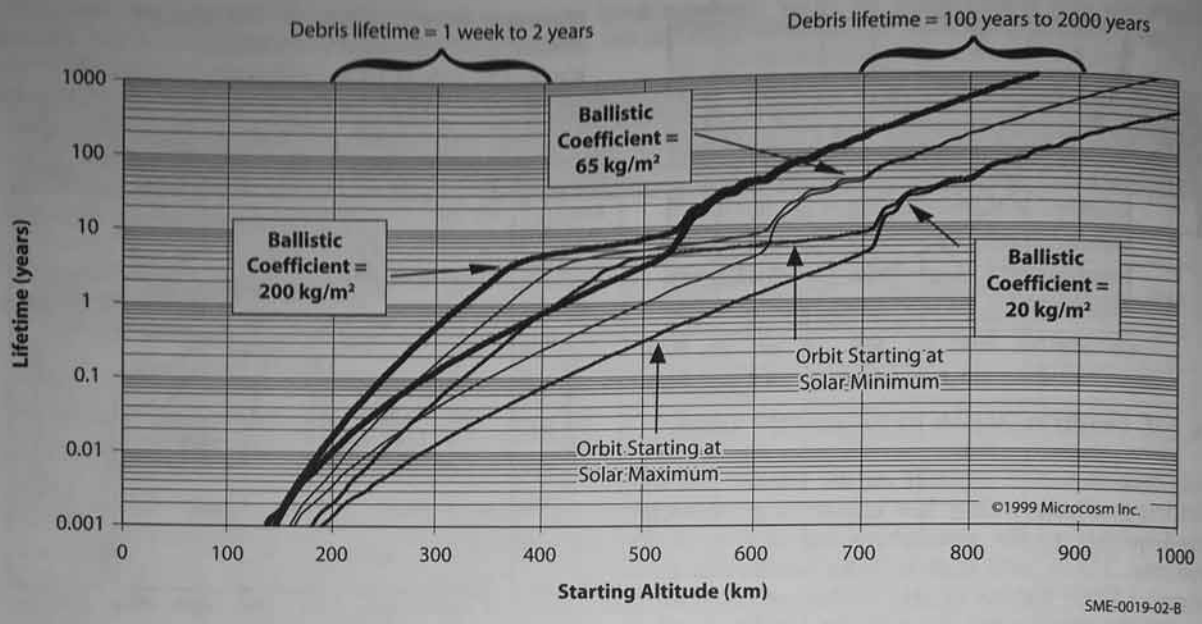


Fig. 9-19. Satellite Lifetime as a Function of Altitude, Relationship to the Solar Cycle, and Representative Ballistic Coefficient. For each ballistic coefficient, the upper curve represents launch during solar minimum and the lower curve represents launch during solar maximum. Note that below approximately 500 km essentially everything will decay and re-enter at the next solar maximum.

Table 9-11. Specialized Orbits and Their Cause.

Specialized Orbit	Characteristic	Cause	Physics Discussed	Applications Discussed
Geostationary	Maintains nearly fixed position with respect to the surface of the Earth	Orbit period = 1 day	Sec. 9.5.1	Sec. 10.4.2.1
Repeating Ground Track	Ground trace repeats after *k* orbits in *n* days	Orbit period = *n* / *k* days, where *n* and *k* are integers	Sec. 9.5.2	Sec. 10.4.2.4
Sun Synchronous	Orbit plane maintains approximately fixed orientation relative to the Sun	Torque due to equatorial bulge causes orbit to rotate 360 deg in 1 year	Sec. 9.5.3	Sec. 10.4.2.2
Repeat Coverage Orbit	Provides coverage of a single location on successive orbits	Inclination chosen to be several deg higher than the target latitude	Sec. 10.4.2.6	Sec. 10.4.2.6
Molniya	Apogee and perigee don't rotate at *i* = 63.4 deg	Greater equatorial force balanced by lesser polar force so that net in-plane; 0 net rotation	Sec. 9.5.4	Sec. 10.4.2.3
Lagrange Point	Maintains fixed position with respect to 2 bodies (e.g., Earth-Moon, Earth-Sun)	Forces between the 2 bodies balance to cause motion with their common period about the barycenter	Sec. 9.5.5	Sec. 10.5
Frozen Orbit	Orbit with fixed, low eccentricity for which apogee doesn't rotate	Balance of higher order harmonics	Sec. 9.5.6 Vallado [2007]	Sec. 10.4.2.5
Statite	Maintains fixed position relative to the Earth and Sun	Gravity balanced by solar radiation pressure	Sec. 18.7.2, Forward [1989]	Sec. 18.7.2, Forward [1989]

be evaluated quickly for Earth orbits. In addition, a complete numerical evaluation for circular Earth orbits at various altitudes for most of these expressions are contained in the appendices.

9.5.1 Geosynchronous Orbit

Geosynchronous orbit or *GEO* is both the simplest and most used of the specialized orbits. Approximately half of the spacecraft launched are geosynchronous spacecraft intended principally for Earth observation and communications. The period of satellites in low-Earth orbit is approximately 90 min. As the altitude increases, the orbit period also increases, according to Kepler's third law. At an altitude of 35,786 km, the period becomes 1,436 min, which is just the rotation period of the Earth relative to the fixed stars, called the *sidereal day*. Consequently, a satellite in a circular orbit at 0 inclination at this altitude will remain indefinitely fixed over a point on the Earth's equator.* In practice, of course, the orbit is never exactly circular nor at zero inclination and,

consequently, will move relative to the surface of the Earth. For a perfectly circular orbit of inclination, i, the path of the satellite as seen from the center of the Earth will be a narrow *analemma* or figure 8 with the following dimensions:

$$\text{height}_{inc} = \pm i \tag{9-43}$$

$$\text{width}_{inc} = \pm\left(\frac{i}{4}\right)\sin i \tag{9-44}$$

This is illustrated in Fig. 9-20A. A small eccentricity, e, in geosynchronous orbit with $i = 0$ will produce a back and forth motion with 1 cycle per orbit. The amplitude of this, in radians, will be:

$$\text{width}_{ecc} = \pm 2e \tag{9-45}$$

In most cases the sidewise component of the eccentricity effect will be larger than that of the inclination effect such that the resulting figure will be an approximately oval shape of height $\pm i$ and width $\pm 2e$ (Fig. 9-20B).

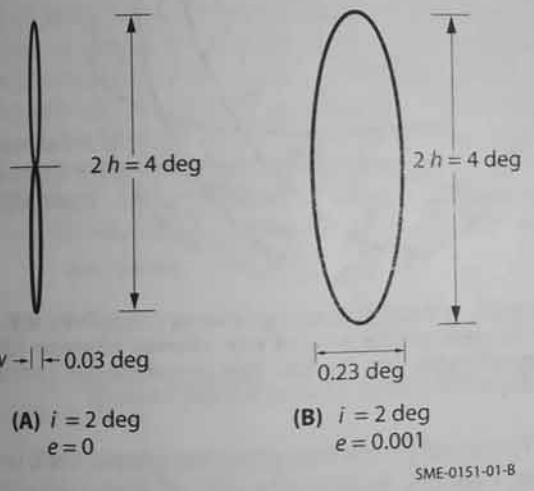


Fig. 9-20. Apparent Daily Motion of a Satellite in Geosynchronous Orbit. Horizontal scale is exaggerated to show the shape.

In addition to the daily oscillations due to small variations in the orbit elements, perturbations continuously act to change the orbit. The largest perturbation in GEO is a north-south drift or inclination change caused by the gravitational interaction with the Sun and the Moon. If not corrected, this perturbation will cause the inclination to vary between 0 and 15 deg over a period of approximately 55 years. As shown in Fig. 9-21, the general motion of the orbit pole under the effect of solar lunar perturbations will be a circle of approximately 7.5 deg radius centered at a point between the ecliptic pole and the equatorial pole. Because the Sun and the Moon are effectively point gravitational masses and are moving continuously with respect to the Earth and the geostationary satellite, the motion is a complex spiral. The approximate worst case changes in velocity per year are given by:

$$\Delta V_{Moon} = 102.67 \cos\alpha \sin\alpha \text{ (m/s per year)} \approx 36.93 \text{ m/s per year, for } i = 0 \tag{9-46}$$

$$\Delta V_{Sun} = 40.17 \cos\gamma \sin\gamma \text{ (m/s per year)} \approx 14.45 \text{ m/s per year, for } i = 0 \tag{9-47}$$

where α is the angle between the orbit plane and the Moon's orbit, and γ is the angle between the orbit plane and the ecliptic. The net effect of these forces is an inclination drift for satellites in GEO of approximately 0.9 deg/year.

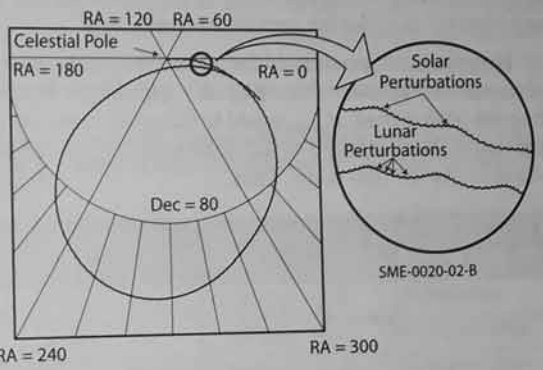


Fig. 9-21. Motion of the Orbit Pole Over 60 Years for a Satellite in Geosynchronous Orbit. The celestial pole is at the coordinate intersection near the top of the plot in the center. The long term motion is roughly a circle of 15 deg radius centered on a right ascension of 270 deg and a declination of 82.5 deg. As can be seen in the figure, the real motion is substantially more complex because of the motion of the Sun and Moon relative to the spacecraft. The large "waves" are due to the Sun and have a 6 month periodicity. The smaller ones (best seen in the enlarged view) are semimonthly "waves" due to the Moon. In general, geosynchronous satellites drift about 0.9 deg per year in inclination. RA = right ascension, Dec = declination.

In addition, there is an east-west drift caused by the out of roundness of the Earth's equator (i.e., the J_{22} sectoral term in the spherical harmonic expansion of the geopotential). If uncorrected, this transverse acceleration would result in a sinusoidal motion about either of 2 stable longitudes located at approximately 75 deg and 255 deg East longitude. The ΔV in m/s/year required to compensate for the East-West drift is given by:

$$\Delta V = 1.715 \sin(2|l_D - l_S|) \tag{9-48}$$

where l_D is the desired longitude and l_S is the closest stable longitude. Typically, the ΔV required for East-West stationkeeping is approximately 10% of that required for North-South stationkeeping.

Table 9-11◀, Fig. 9-21, Eq. 9-48

* Geosynchronous orbit was first described mathematically in 1928 by a Slovene rocket engineer, Herman Potocnik. The potential for GEO communications satellites was first described by the popular science and science fiction writer, Arthur C. Clarke in the October 1945 issue of *Wireless World* [Clarke, 1945]. Clarke often commented on the potential impact had he chosen to patent GEO communications satellites (For more details, see, for example, Gruntman [2004].)

None of the above perturbations serve to change the radius of the geostationary orbit. Whereas atmospheric drag provides a housekeeping function in low-Earth orbit, there is no natural perturbation which provides a similar process for removing satellites from the geostationary ring. Consequently, a satellite in GEO which dies will drift in inclination and drift back-and-forth in longitude among the other geosynchronous satellites. Similarly, if one of the spacecraft in GEO explodes or collides with another satellite, the resulting debris cloud will also remain in geosynchronous orbit. Thus, geosynchronous orbit is both a unique resource and a fragile environment because of the potential for debris to accumulate. The key to this problem is to raise spacecraft 500 km above geosynchronous orbit when they have finished their useful life. At this altitude, they will remain essentially indefinitely with no perturbations to bring them back into the geostationary arc. Such a maneuver takes an extremely small amount of propellant. The largest difficulty with spacecraft disposal is that it must be done while the spacecraft itself is still operating so that thrusters can be commanded to make the appropriate maneuver.

The synchronous altitude for any central body can be obtained immediately from Kepler's Third Law by setting the orbit period, P_{sync}, equal to the rotation period of the central body relative to the fixed stars, D^*, called the *sidereal day* for that body. Thus:

$$P_{sync} = D^* \tag{9-49}$$

$$a_{sync} = (\mu / 4\pi^2)^{1/3} (D^*)^{2/3} \tag{9-50}$$

$$H_{sync} = a_{sync} - R_{planet} \tag{9-51}$$

where as usual $\mu \equiv GM$ for the planet, a_{sync} is the semi-major axis of the synchronous orbit, R_{planet} is the equatorial radius of the planet, and H_{sync} is the synchronous altitude. Values of the synchronous altitude for principal bodies in the Solar System are listed in Table 9-12.

Live Calc

Table 9-12. Synchronous Orbits for Major Solar System Objects.

Central Object	Sidereal Period (days)	Equatorial Radius (km)	Synchronous Altitude (km)
Earth	0.99726	6,378.136 6	35,786
Moon	27.322	1,738	86,715
Sun	25.38	695,990	24,588,000
Mercury	58.6462	2,439.7	240,453
Venus	243.01	6,051.9	1,530,500
Mars	1.02596	3,397	17,031
Jupiter	0.41354	71,492	88,565
Saturn	0.44401	60,268	51,981

Table 9-12, Fig. 9-22, Eq. 9-52d

9.5.2 Repeating Ground Track Orbits

In a *repeating ground track orbit*, the spacecraft returns after one or more days to the same location with respect to the surface of the Earth such that the path of the spacecraft then repeats itself. For this to occur, an integral number of orbit periods, j, must equal an integral number of days, k. For example, if a spacecraft makes exactly 15.25 orbits per day, then it will make $j = 61$ orbits in $k = 4$ days and will have a 4 day repeating ground track. Every 4th day, the satellite will follow the same path and, therefore, will see the same geometry relative to features on the ground and will move across the sky in the same apparent path as seen by a ground observer. An example of a 2-day repeating ground track orbit is shown in Fig. 9-22.

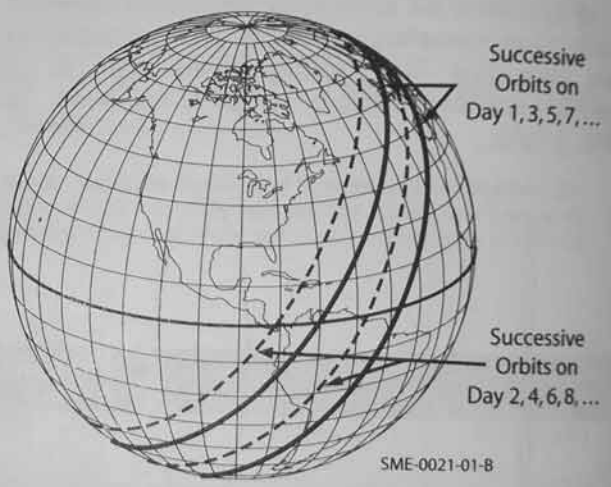


Fig. 9-22. A 2-day Repeating Ground Track Orbit will cross the Equator on the second day midway between where it crossed on the first day and then repeat the day 1 crossings on day 3. The pattern repeats at 2-day intervals.

The actual period for a repeating ground track is made more complex by the presence of orbit perturbations which cause the orbit itself to rotate. We want the orbit period to be a harmonic rotation of the Earth relative to the orbit plane, rather than the fixed stars. Consequently, we need to take into account the rotation of the orbit (in node, perigee, and mean anomaly) due to the oblateness of the Earth. Because the oblateness effects are a function of the altitude, solving the altitude for a given ground track repeat pattern is an iterative process. Our initial estimate H_0, although inaccurate, is the repeating ground track altitude ignoring the effects of oblateness:

$$H_0 = \mu^{1/3}\left(\frac{2\pi j}{D^* k}\right)^{-2/3} - R_E \tag{9-52a}$$

$$\equiv k_1 (j/k)^{-2/3} - R_E \tag{9-52b}$$

$$k_1 \equiv \mu^{1/3}\left(2\pi / D^*\right)^{-2/3} \tag{9-52c}$$

$$\approx 42{,}164.173 \text{ km} \tag{9-52d}$$

where $D^* \approx 86{,}164.10035$ sec is the sidereal rotation period of the Earth (i.e., relative to the fixed stars), R_E is equatorial radius of the Earth, and $\mu = GM$ is the Earth's gravitational constant. We then determine the rotation rate of the Earth, $\dot{L}$, and the rotation rate of the three orbit parameters due to oblateness as follows:

$$\dot{L} = 360 \text{ deg/day}^* \tag{9-53}$$

$$\dot{\Omega} = -2k_2 a^{-7/2} \cos i \left(1 - e^2\right)^{-2} \tag{9-54}$$

$$\dot{\omega} = k_2 a^{-7/2} \left(5\cos^2 i - 1\right)\left(1 - e^2\right)^{-2} \tag{9-55}$$

$$\dot{M} = k_2 a^{-7/2} \left(3\cos^2 i - 1\right)\left(1 - e^2\right)^{-3/2} \tag{9-56}$$

where

$$k_2 = 0.75\left(\frac{360}{2\pi}\right) J_2 \mu^{1/2} R_E^2 \tag{9-57a}$$

$$\approx 1.029\,553\,297 \times 10^{14} \text{ km}^{3.5} \text{ deg/day}^* \tag{9-57b}$$

where $\dot{\Omega}$ is the rate of change of the ascending node, $\dot{\omega}$ is the rate of change of perigee, $\dot{M}$ is rate of change of the mean anomaly, J_2 is the Earth's oblateness, e is the orbit eccentricity, i is the inclination, and a is the estimate of the semimajor axis. For the first iteration, we take $a \approx H_0 + R_E$. All of the angular rates are expressed in deg/sidereal day ≡ deg/day*. We next estimate the mean angular motion, n, for the repeating ground track as:

$$n = (j/k)\left(\dot{L} - \dot{\Omega}\right) - \left(\dot{\omega} + \dot{M}\right) \tag{9-58}$$

Finally, the revised estimate for the repeating ground track altitude, H, is:

$$H = \mu^{1/3}\left(\frac{n\pi}{180D^*}\right)^{-2/3} - R_E \tag{9-59}$$

If necessary, we can then recompute a and iterate again from Eq. (9-54) on. Note that for prograde orbits, $\cos i$ is positive, for retrograde orbits, $\cos i$ is negative, and that a must be in km if the value of k_2 in Eq. (9-57b) is used. Values for the above constants for different objects in the Solar System are given in App. B.

The principal advantage of this type of orbit is that the ground track and, therefore, the geometry relative to the surface of the Earth repeats. This in turn, means that we can view the surface of the Earth from the same viewpoint on a recurring basis and, therefore, compare one image or data set with another taken at a later time from the same orientation. This makes the interpretation of changes and variations far easier. In addition, the geometry of the satellite's path as seen by the ground also remains the same such that a ground station can follow the path of the satellite more easily since it will follow the same path on a recurring basis. The principal demerit of the repeating ground track is that it has the potential for resonance effects with the Earth's higher order harmonics. If we pass just to the left of a major mass concentration such as the Himalayas today, then we will do so again tomorrow, and the day after, and so on. Thus, perturbations have the potential to accumulate over time, and resonance effects can build up which will significantly distort the orbit. Consequently, repeating ground tracks are useful for observations, but should generally be avoided unless there is a specific reason for using them.

There are a very large number of potential repeating ground track orbits. For the Earth, there are 15 orbits per day at an altitude of 482.25 and 14 per day at an altitude of 817.14. However, in between these we have 14.33 orbits per day, 14.5 per day, 14.75 per day, 14.8 per day, and so on. This set of representative repeating ground track orbits is shown in Table 9-13.

An interesting subgroup of the repeating ground track orbits are those for which the inclination and, therefore, node rotation rate is adjusted such that the repeat occurs on precise multiples of the *civil orbits* day of 24 hours. For these orbits, the ground track not only repeats, but does so at the same time each day, every other day, or on a weekly basis depending upon the orbit parameters.

Live Calc

Table 9-13. Representative Repeating Ground Track Orbits. A large number of such orbits are possible.

Altitude (km)	Inclination (deg)	Period (min)	Orbits per Day	Repeat Pattern
817.14	28	101.24	14.0	14 orbits/day
701.34	28	98.80	14.33	43 orbits/3 days
645.06	28	97.63	14.50	29 orbits/2 days
562.55	28	95.91	14.75	59 orbits/4 days
546.31	28	95.57	14.8	74 orbits/5 days
482.25	28	94.25	15.0	15 orbits/day

9.5.3 Sun-Synchronous Orbits

Because the gravitational force is conservative, the total energy and mean values of the semimajor axis, apogee and perigee heights, and eccentricity do not change due to oblateness or other higher order terms in the Earth's geopotential. Nonetheless, secular variations in some of the orbit elements can occur. Perhaps the easiest of these to visualize is the rotation of the orbit due to the torque produced by the Earth's oblateness. In this context, it is easiest to visualize the Earth as a point mass plus a ring of additional mass along the equator representing the equatorial bulge (the J_2 term). The gravitational effect of the equatorial bulge is to continuously try to pull the orbit plane downward toward the equatorial plane. However, the result of this force is not a change in the inclination, but a rotation of the right ascension of the ascending node.

The reason for this effect is the angular momentum of the orbit. In this regard, we can think of the orbit as a

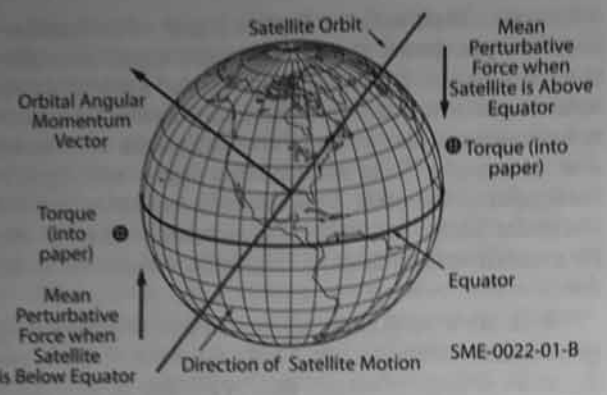


Fig. 9-23. Regression of the Nodes Due to Oblateness. The equatorial bulge of the Earth produces a torque on the orbit, causing prograde orbits to rotate from east to west (i.e. regress). Retrograde orbits rotate from west to east due to oblateness.

rotating wheel with its axis along the normal to the orbit plane. Recall from freshman physics that when a bicycle wheel is rotated rapidly and one end is allowed to drop, it does not fall, but rather *precesses*, i.e., the axis rotates in a horizontal direction. A similar result applies to the spacecraft orbit. As shown in Fig. 9-23, when the spacecraft is north of the equator, the oblate Earth is pulling it down thus producing a torque into the paper. Similarly, when the spacecraft is south of the equator, the equatorial bulge is pulling it up and, again, there is a torque into the paper. This continuous torque causes the angular momentum vector to move in the direction of the torque, i.e., to rotate about the Earth's pole. For a satellite in a prograde orbit, this rotation is in the westward direction opposite the direction of the Earth on its axis. Consequently, this motion is called the *regression of the nodes*. So long as the other orbit elements do not change, the orbit will rotate indefinitely in inertial space and the node will continue to change while the inclination remains constant.

To evaluate the rate of node rotation, we will return to the Lagrange planetary equations (Eq. 9-22e) and evaluate the impact of the J_2 term. When only the secular variations due to J_2 are considered, the right ascension of the ascending node, Ω, at time t is given by:

$$\Omega = \Omega_0 + \frac{d\Omega}{dt}(t - t_0) \tag{9-60a}$$

$$= \Omega_0 - \frac{3}{2} J_2 n \left(\frac{R_E}{a}\right)^2 \left(1 - e^2\right)^{-2} \times (\cos i)(t - t_0) \tag{9-60b}$$

$$\approx \Omega_0 - \frac{3}{2} J_2 \sqrt{\mu}\, R_E^2\, a^{-7/2} \left(1 - e^2\right)^{-2} \times (\cos i)(t - t_0) + \mathcal{O}\left(J_2^2\right) \tag{9-60c}$$

$$\approx \Omega_0 - 2.064\,74 \times 10^{14}\, a^{-7/2} \left(1 - e^2\right)^{-2} \times (\cos i)(t - t_0) + \mathcal{O}\left(J_2^2\right) \tag{9-60d}$$

Table 9-14, Fig. 9-23, Eq. 9-60d

where in the last form, $t - t_0$ is in days, Ω is in degrees, a is the semimajor axis in km, e is the eccentricity, i is the inclination, and Ω_0 is Ω at the epoch time t_0. In the final two forms, n is approximated by n_0 which may produce an error as large as 0.1% in the $d\Omega/dt$ term.

For example, in a circular low-Earth orbit at 500 km, we may further evaluate Eq. (9-60d) to yield $\dot{\Omega} = -7.652 \cos i$. Consequently, for low inclination orbits, the node can rotate up to 7.7 deg per day. This represents a significant motion of the orbit plane over the lifetime of most satellites, since the orbit could rotate through 360 deg in only about 45 days.

For a polar orbit, $i = 90$ deg, $\dot{\Omega} = 0$ and the orbit plane remains fixed in inertial space. For a polar orbit, if the Sun is in the orbit plane on any given date, then it will be in the vicinity of the orbit pole 3 months later and back in the orbit plane after 6 months when the Earth has moved 180 deg in its orbit.

If the orbit inclination is more than 90 deg, i.e., a retrograde orbit, then cos i will become negative and $\dot{\Omega}$ will become positive. In this case, the orbit progresses and rotates in the same direction as the Earth is rotating on its axis or as the Earth is going around the Sun. A particularly interesting case occurs whenever the product of the three terms in a, e, and i in Eq. (9-45) has the value of -4.7737×10^{-15} km$^{-3.5}$. In this case, the rotation rate of the node will be 0.9856 deg per day or 1 rotation per year. This is called a *Sun-synchronous orbit* because the orientation of the orbit plane will remain nearly fixed relative to the Sun as the Earth moves in its orbit. In a Sun-synchronous orbit, we are using the natural perturbation caused by the Earth's oblateness to pull the orbit around in inertial space at a rate of 1 rotation per year, so that we can maintain an approximately constant Sun angle. As shown in Table 9-14, the specific inclination in which this occurs will vary slightly depending upon the altitude of the satellite.

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Table 9-14. Inclination for Sun-Synchronous Orbits at Varying Altitudes.

Mean Altitude (km)	e = 0	e = 0.1		
	i (deg)	i (deg)	h_P (km)	h_A (km)
0	95.68	—	—	—
200	96.33	—	—	—
400	97.03	—	—	—
600	97.79	—	—	—
800	98.60	98.43	82	1,518
1,000	99.48	99.29	262	1,738
2,000	104.89	104.59	1,162	2,838
3,000	112.41	111.94	2,062	3,938
4,000	122.93	122.19	2,962	5,038
5,000	138.60	137.32	3,862	6,138
5,974	180.00	168.55	4,738	7,209

The effect of oblateness on the rotation of the orbit remains essentially constant throughout the year. Consequently, the orbit pole will move at a uniform rate at a fixed angle relative to the celestial pole. On the other hand, because of the eccentricity of the Earth's orbit, the Sun moves at a nonuniform rate going faster at perihelion and more slowly at aphelion. In addition, the Sun moves along the ecliptic inclined at 23.5 deg to the Earth's equator. This means that relative to the celestial pole, the Sun moves north and south through a total angle of 47 deg and additionally moves more rapidly along the equator at some times of the year than at other times. (The general character of this motion including the reasons for different speeds along the equator are explained in detail in Chap. 10 of Wertz [2009].) The net result is that throughout the year, the Sun moves in an analemma or figure 8 relative to the position of the orbit as shown in Fig. 9-24.

The general path of the Sun relative to the orbit plane will be an analemma with a half height equal to 23.5 deg. The width of the analemma will be 2.5 deg due simply to the obliquitity of the ecliptic. When the eccentricity of the Earth's orbit is taken into account, the analemma will be 3.8 deg across in the southern hemisphere and only 1.3 deg in the northern hemisphere.

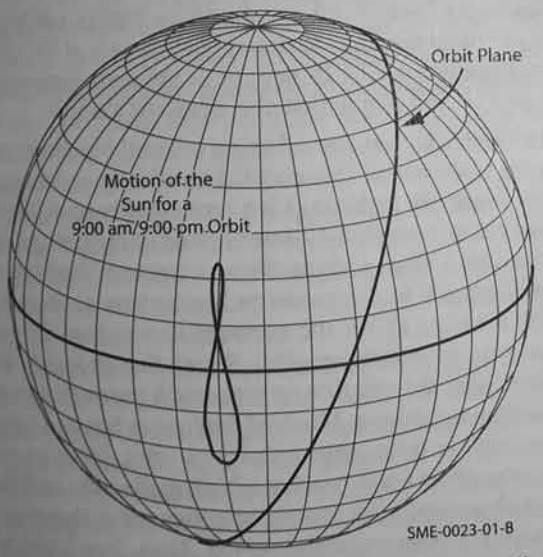


Fig. 9-24. Changing Position of the Sun Relative to the Orbit Plane in a Sun-Synchronous Orbit. The position of the Sun remains approximately fixed. However, since the J_2 perturbation is being used to balance the non-uniform revolution of the Earth about the Sun, and because the Sun isn't in the equatorial plane, the relative position will change somewhat throughout the year. For a detailed discussion, see Wertz [2009], Chap. 10.

Sun-synchronous orbits are typically labeled by the approximate orientation of the Sun relative to the orbit. If the Sun is approximately in the orbit plane, it is called a *noon-midnight orbit* because it will be either noon or midnight at the subsatellite point. If the Sun is near the orbit pole, it is called a *twilight orbit* or *6 am–6 pm orbit*, since the Sun will be nearly at right angles to the direction to the Earth. In twilight orbits, the Sun is always in the vicinity of the orbit pole and, therefore, there are almost no eclipses and the available power is at a maximum. In addition, the lighting on the ground below has very long shadows and, therefore, good contrast. The applications of the Sun-synchronous orbits are discussed in Sec. 10.6.

Finally, it is convenient to compute the local mean time* at the subsatellite point of the spacecraft. For any satellite in a circular orbit, the local mean time, T, at which the satellite is over latitude λ is given by:

$$T = \frac{1}{15}\left[-\alpha_s + \Omega + \arcsin\left(\frac{\tan\lambda}{\tan i}\right)\right] + 12 \qquad (9\text{-}61a)$$

$$\approx \frac{1}{15}\left[-\frac{360}{365.24}\Delta D + \Omega + \arcsin\left(\frac{\tan\lambda}{\tan i}\right)\right] + 12 \qquad (9\text{-}61b)$$

where α_s is the right ascension of the mean Sun, Ω is the right ascension of the ascending node of the orbit, i is the orbit inclination, ΔD is the number of days from the vernal equinox to the time Ω is evaluated, the angular quantities in square brackets are in degrees, and T is in hours. For Sun-synchronous orbits, $\alpha_s - \Omega$ is constant and, therefore, the local mean time is a function only of the latitude and does not change from orbit to orbit.

9.5.4 Molniya Orbits

In addition to causing rotation of the nodes, the oblateness of the Earth also causes the major axis of the orbit connecting apogee and perigee to rotate within the orbit plane. The reason for this is shown in Fig. 9-25. The $1/r^2$ force corresponding to a perfectly spherically symmetric mass distribution will cause the spacecraft to move in an elliptical Keplerian orbit such that the spacecraft would return exactly to its starting conditions and the orbit would repeat itself over and over in inertial space. If the force is stronger than $1/r^2$, the orbit will curve more strongly and the spacecraft will return to perigee sooner than it otherwise would have, thus causing perigee to rotate in a positive direction. Similarly, if the force is weaker than $1/r^2$, the orbit will rotate more slowly than a Keplerian orbit, the spacecraft will take longer than one period to return to perigee and the orbit will rotate opposite the direction of motion. For the case of the oblate Earth, the gravitational force over the equator is stronger than $1/r^2$ and weaker than $1/r^2$ in the vicinity of the poles. In a low inclination orbit, the spacecraft will feel a stronger force at all times and perigee will advance. In a polar orbit, the forces will be stronger over the equator and weaker when the spacecraft is over the pole. The

* The *local mean time* is the solar time that would be the case if the Sun moved at a uniform rate along the equator rather than at a varying rate along the ecliptic. Thus local mean time does not take into account either time zones or the eccentricity and obliquity of the ecliptic.

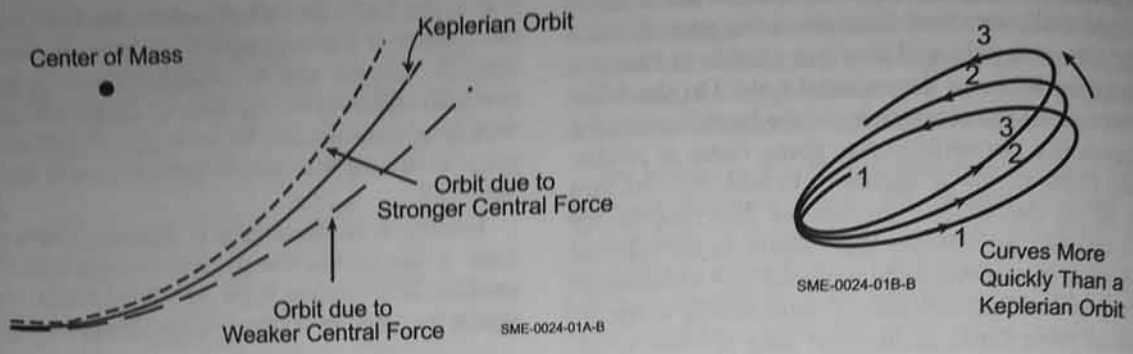


Fig. 9-25. Physical Basis for Rotation of Eccentric Orbits. Near the equator the force will be greater than the $1/r^2$ force from the Earth point mass and the orbit will rotate faster than the closed Keplerian orbit. Near the pole it will be less and the orbit will be more open.

net effect is that in a polar orbit perigee rotates backwards but not as rapidly as it rotates forward for an equatorial orbit.

Once again, we can use the Lagrange planetary Eqs. (9-22d) to evaluate the rate at which the argument of perigee, ω, rotates:

$$\omega = \omega_0 + \frac{d\omega}{dt}(t - t_0) \tag{9-62a}$$

$$= \omega_0 + \frac{3}{2} J_2 n \left(\frac{R_E}{a}\right)^2 \left(1 - e^2\right)^{-2} \times \left(2 - \frac{5}{2}\sin^2 i\right)(t - t_0) \tag{9-62b}$$

$$\approx \omega_0 + \frac{3}{2} J_2 \sqrt{\mu}\, R_E^2\, a^{-7/2} \left(1 - e^2\right)^{-2} \times \left(2 - \frac{5}{2}\sin^2 i\right)(t - t_0) + \mathcal{O}\left(J_2^2\right) \tag{9-62c}$$

$$\approx \omega_0 + 2.064\,74 \times 10^{14}\; a^{-7/2} \left(1 - e^2\right)^{-2} \times \left(2 - \frac{5}{2}\sin^2 i\right)(t - t_0) + \mathcal{O}\left(J_2^2\right) \tag{9-62d}$$

where in the last term $t - t_0$ is in days, ω is in degrees, a is the semimajor axis in kilometers, e is the eccentricity, i is the inclination, and ω_0 is the value of ω at the epoch t_0. For example, we can evaluate the expression for a circular orbit at 500 km to obtain $\dot{\omega} = 3.826\ (4 - 5\ \sin^2 i)$. This implies that for an equatorial orbit, the line of apsides or major axis will rotate in the direction of motion by up to 15.3 deg per day and for polar orbits it will rotate opposite the direction of motion by as much as –3.8 deg per day.

The rotation of perigee can be even more rapid than the rotation of the line of nodes. For example, in a low eccentricity, low inclination orbit that begins with apogee over the northern most latitude, apogee will have rotated to the equator after only a week, and in 2 weeks will be at the southern most latitude. At this high rate of rotation, it would be difficult to find good utility for highly elliptical orbits except perhaps to successively sample different regions of space. Notice, however, that there is a term in the expansion $(4 - 5\ \sin^2 i)$ that goes to 0 at an inclination of 63.435 deg. For an elliptical orbit at this *critical inclination*, the line of apsides does not rotate. Consequently, nearly all spacecraft in highly elliptical orbits intended for long-term applications are at the critical inclination of 63.435 deg, or at the complementary angle of 116.565 deg which also does not rotate.

The Soviet Union started using critically inclined orbits very early in the space age. Specifically, a *Molniya* orbit (Russian for lightning) is a semisynchronous, highly elliptical orbit with a 12-hour period at the critical inclination such that perigee does not rotate. Perigee is typically placed in the southern hemisphere so that the satellite remains above the northern hemisphere for approximately 11 hours per orbit. Recall that in highly elliptical orbits, the satellite moves much more slowly at apogee than at perigee. Much of the former Soviet Union was not visible from geosynchronous orbit. Molniya orbits, on the other hand, were at a high inclination and the spacecraft moved very slowly at apogee and, therefore, was in view for an extended period. Thus, 2 or 3 spacecraft in a Molniya orbit could provide communications for northern latitude regions well beyond the reach of geosynchronous satellites. Consequently, most of the Soviet communications satellites were placed in Molniya orbits. The basic parameters of representative Molniya orbits are shown in Table 9-15.

The rate of perigee rotation depends on the value of J_2 and, therefore, will be different for different central bodies. Note, however, that the critical inclination of 63.4 deg is a function only of the fact that the expansion is in terms of spherical harmonics and thus will be the same for all planets. Consequently, Molniya-like orbits about any of the planets will be inclined at 63.4 deg to the planet's equator.

Table 9-14◀, Fig. 9-25, Eq. 9-62d

9.5.5 Lagrange Point Orbits

Most specialized orbits are defined with respect to a single central body such as the Earth or the Moon. *Lagrange point orbits,** however, are defined with respect to 2 large co-orbiting bodies such as the Earth and the Moon or the Sun and the Earth. This is often called the *circular restricted three-body problem*, which refers to the two large bodies being in circular orbits about each other and the mass of the spacecraft or small body being much less than the mass of the other two objects. For convenience, we will define the Lagrange points for the Earth-Moon system. However, they can be applied to any pair of co-orbiting large bodies.

Table 9-15. Characteristics of Representative Molniya Orbits.

Rev Per Day	Period (sec)	Semimajor Axis (km)	Inc (deg)	Alt at Apogee (km)	Alt at Perigee (km)	Eccentricity
1	86,164	42,164	63.4	53,622	17,951	0.423
2	43,082	26,562	63.4	39,863	504	0.740
3	28,721	20,270	63.4	33,659	504	0.660

As shown in Fig. 9-26, the *Lagrange points*† or *libration points* for the Earth and the Moon are the five points such that an object placed at one of them will remain there indefinitely relative to the Earth-Moon system. All five of the Lagrange points are in the plane of the Moon's orbit. The 3 Lagrange points on the Earth-Moon line are positions of unstable equilibrium, i.e., any small change causes the object to drift away. However, L_4 and L_5 which form equilateral triangles with the Earth and the Moon in the plane of the Moon's orbit are positions of stable equilibrium. A satellite placed in the vicinity of L_4 or L_5 with an appropriate velocity will continue to orbit in that vicinity indefinitely.

The Moon orbits the Earth with a period of one month. If the Moon were not present, a satellite placed closer to the Earth, say at L_1, would rotate with a period shorter than a month and one placed at L_2 would rotate with a period longer than a month. For the satellite at L_1, "putting the Moon back" decreases the central force and, therefore, increases the period. Similarly, the satellite at L_2, the Moon increases the central force and decreases the period. L_1 and L_2 are simply the points at which the change is just sufficient to make the period exactly equal

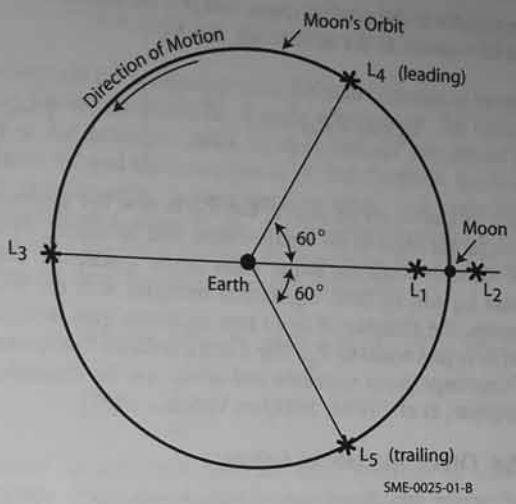


Fig. 9-26. Earth-Moon Lagrange Point Orbits. In the general literature, the L_4 and L_5 locations are always regarded as the ones off the Earth-Moon line; however, the specific nomenclature for each individual point (i.e., which one is which) varies widely among authors. The numbering scheme here is the one used by NASA and, therefore, applied to space missions associated with the Lagrange points. A similar set of points apply to the Sun and Earth and any other sets of two co-orbiting bodies.*

to the Moon's orbital period. Similarly, at L_3 the sum of the forces due to the Earth and the Moon are just enough to cause the satellite to rotate in a period of one month. The source of the stable equilibria at L_4 and L_5 is a bit more difficult to conceptualize. Although we usually think in terms of the Moon revolving around the Earth, in fact, both the Earth and the Moon revolve around their common *barycenter*, or mutual center of mass. (For the Earth-Moon system the center of mass is displaced from the center of the Earth approximately 4,700 km toward the Moon, i.e., still below the Earth's surface.) The effect of the Moon's gravitational force on the L_4 and L_5 Lagrange points is to displace the net gravitational force away from being directed toward the center of the Earth, changing the direction just enough so that the total force on the small satellite is directed toward the barycenter of the Earth-Moon system and, again, produce a period of revolution equal to that of the Moon.

Ehricke [1960] provides the following approximations for the locations of the Lagrange points for a spacecraft of negligible mass. For L_1 and L_2:

$$L_1 : \rho_1 = R_B \left[1 - \left(\frac{m}{3M} \right)^{1/3} \right] \tag{9-63}$$

$$L_2 : \rho_2 = R_B \left[1 + \left(\frac{m}{3M} \right)^{1/3} \right] \tag{9-64}$$

where M is the larger mass (Earth in our example), m is the smaller mass (the Moon), R_B is the distance from the

* Named after the 18th century French mathematician and astronomer Joseph Lagrange, who was also the principal inventor of the metric system.

† The naming convention for the Lagrange points is particularly bad. First, the numbering scheme varies widely between authors. Fig. 9-26 shows the scheme adopted by NASA. Second, the two central bodies are often left unstated. L_1 and L_2 missions most often refer to the Sun-Earth Lagrange points, while L_4 and L_5 most often refer to the Earth-Moon ones. The best approach is to be unambiguous, i.e., "the L_1 Lagrange point between the Earth and the Sun."

barycenter to the smaller mass, and ρ is the distance from the barycenter to the spacecraft. For L_3:

$$L_3 : \rho_3 = R_M \left[1 + \left(\frac{7m}{12M} \right)^{1/3} \right] \qquad (9\text{-}65)$$

Here M and m are as above, but R_M is now the distance from larger mass to the smaller mass and, similarly, ρ_3 is the distance from the larger mass to the spacecraft. Because L_4 and L_5 form equilateral triangles with the two masses, the distance of these two locations from both M and m is just equal to R_M. For a more detailed discussion of Lagrange point missions and orbits, see for example, Farquhar, et al. [1994, 1998] or Vallado [2007].

9.5.6 Other Specialized Orbits

Circular low-Earth orbits are not inherently stable. Higher order harmonics cause the eccentricity to oscillate about a small nonzero value as shown in Fig. 9-27. To avoid this, we can put a spacecraft in an orbit with a low eccentricity and an argument of perigee of 90 deg or 270 deg and the orbit will be stable. This is called a

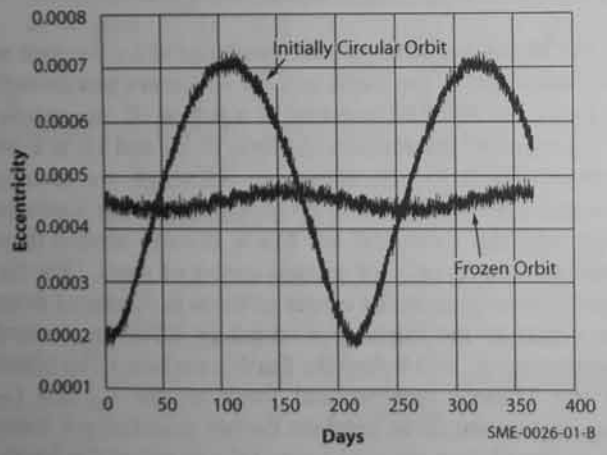


Fig. 9-27. Eccentricity Oscillation in an Initially Circular LEO Orbit. Central line shows the eccentricity in a comparable frozen orbit. Orbit shown is at 650 km at an inclination of 57 deg.

frozen orbit. Frozen orbits can be used with any low-Earth orbit. They are not so much a specialized orbit as a mechanism for maintaining a more stable orbit and avoiding a low level oscillation. The argument of perigee, ω_f, and eccentricity, e_f, for frozen orbits are as follows:

$$\omega_f = 90 \text{ deg} \quad \text{or} \quad 270 \text{ deg} \qquad (9\text{-}66)$$

and

$$e_f \approx -\frac{J_3}{2J_2}\left(\frac{R_E}{a}\right) \sin i \sin \omega_f \qquad (9\text{-}67a)$$

$$\approx +7.459\,335 \left(\frac{\sin i \sin \omega_f}{a} \right) \qquad (9\text{-}67b)$$

Table 9-15◀, Fig. 9-27, Eq. 9-67b

where, as usual, i is the inclination, R_E is the equatorial radius of the Earth, J_2 and J_3 are the second and third terms in the spherical harmonic expansion of the geopotential, and a is the semimajor axis, expressed in km in Eq. (9-67b). For a polar orbit of 500 km, $e_f \approx 0.00108$. (Including higher order terms will add about 0.00010 to this value.) In addition, both the eccentricity and argument of perigee will be frozen at the critical inclination of 63.4 deg or 116.6 deg and any values of e or ω.

In 1989, Bob Forward invented and patented the *statite* [Forward, 1989]. It is not called a satellite because it is not in an orbit but is an object fixed with respect to the Earth and Sun using solar radiation pressure to balance gravity. Because solar radiation pressure tends to be remarkably small, the statite is required to be very light weight, have an extremely large cross-sectional area, and be at a substantial distance from the Earth or other gravitating bodies. Thus far, no spacecraft have been placed in statite "orbits."

In 2005, Microcosm introduced the *Repeat Coverage Orbit* (RCO), as a more appropriate choice than Sun synchronous orbits for short or moderate duration missions or coverage of specific events or regions of the world [Wertz, 2005]. The RCO gives up global coverage in exchange for much better coverage of a specific region. (The RCO can provide 4 or 5 successive orbits of coverage every day for a particular latitude region vs. typically less than once a day for Sun synchronous orbits.) Because it best discussed as a part of Earth coverage, the RCO is described in more detail in Sec. 10.4.2.6.

9.6 Orbit Maneuvers

Thus far we have discussed primarily orbits which are either constant or slowly varying with time. In this section we discuss the physics and provide basic equations for orbit maneuvers intended to move the spacecraft from one orbit to another, typically for the purpose of getting somewhere, such as transfer orbits to a geosynchronous orbit or to the Moon or Mars. (Orbit maintenance and stationkeeping are discussed in Sec. 18.4.) This section discusses the basic physics of such orbits. Chapter 10 discusses the principal advantages and disadvantages of the various alternatives and reasons for selecting them.

9.6.1 Transfer Orbits

One of the fundamental goals of space missions is to get from one place to another—from the Earth to Mars, from low-Earth orbit to a higher orbit, or from the Earth to the Moon and back. The basic mechanism for doing this is the *transfer orbit*. The simplest and most common orbit transfer is between co-planar circular orbits. While most orbit transfers are not precisely co-planar, it is a good approximation to many transfer problems, and gets us started on the analysis and design of transfer orbits.

Putting mass and, therefore, propellant into space is remarkably expensive. Consequently, nearly all transfer orbits are intended to minimize the propellant required to

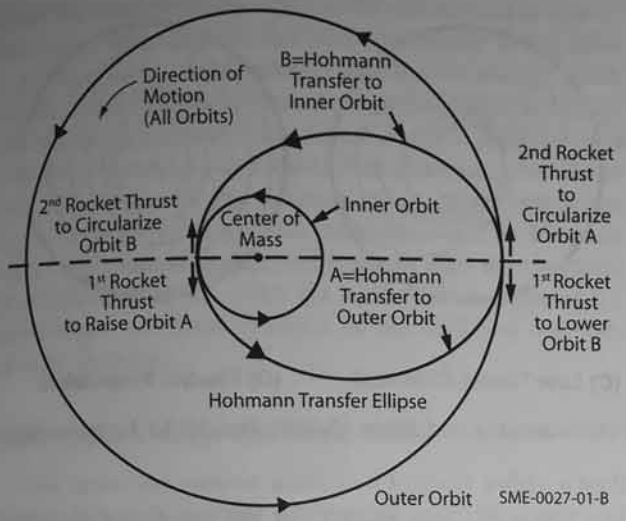


Fig. 9-28. The Hohmann Transfer Trajectory. This is normally the lowest energy approach for going between 2 co-planar circular orbits.

achieve the desired end orbit. The most fundamental and most often used is the *Hohmann transfer trajectory*, which is normally the lowest energy transfer between two co-planar circular orbits. This approach is based on the analytical results published in 1925 by Walter Hohmann, who was at the time the city architect of Essen, Germany. As shown in Fig. 9-28, the Hohmann transfer is an elliptical orbit with perigee tangent to the inner orbit and apogee tangent to the outer orbit, 180 deg away. If we're starting in a low orbit and wish to transfer to a higher one, we need to add energy. The most efficient way to do this is to increase the velocity of the spacecraft without changing its direction. Consequently, we point our rocket in the direction of motion (i.e., with the rocket exhaust going out the back, opposite the direction of motion). We then fire the rocket engine 180 deg from where we wish to end up and simply add ΔV to the current circular velocity. This creates an elliptical orbit with perigee where the engine fired. At this first firing, we provide just enough ΔV to raise apogee to the radius of our higher circular target orbit. When the spacecraft arrives at apogee, 180 deg later, we again fire the rocket engine in the direction of motion in order to raise perigee to the value of the apogee altitude, circularizing the orbit at the target altitude. The fundamental equations for the Hohmann transfer are as follows:

$$a_T = \frac{1}{2}(a_L + a_H) \tag{9-68}$$

$$\Delta V_L = \sqrt{\mu}\left[\left(\frac{2}{a_L} - \frac{1}{a_T}\right)^{1/2} - \left(\frac{1}{a_L}\right)^{1/2}\right] \tag{9-69}$$

$$\Delta V_H = \sqrt{\mu}\left[\left(\frac{1}{a_H}\right)^{1/2} - \left(\frac{2}{a_H} - \frac{1}{a_T}\right)^{1/2}\right] \tag{9-70}$$

$$T = \frac{1}{2}P_T = \pi\, a_T^{3/2}\mu^{-1/2} \tag{9-71}$$

where a is the semimajor axis, ΔV is the change in velocity, T is the transfer time, P is the orbit period, the values of μ for principal Solar System objects are given in Table 9-2 and the subscripts L, H, and T refer to the lower circular orbit, higher circular orbit, and Hohmann transfer orbit, respectively. For transfers between orbits at nearly the same altitude, the equations for the ΔV and transfer time may be further simplified to:

$$\Delta V_L \approx \Delta V_H \approx \frac{1}{2}(V_L - V_H) \tag{9-72}$$

$$T \approx \frac{1}{4}(P_L + P_H) \tag{9-73}$$

where the subscripts have the same meaning as above. Finally, if the initial orbit is circular and the magnitude of a single ΔV burn is known, then:

$$\Delta a = 2a\frac{\Delta V}{V} = 2\Delta V a^{3/2}\,\mu^{-1/2} \tag{9-74}$$

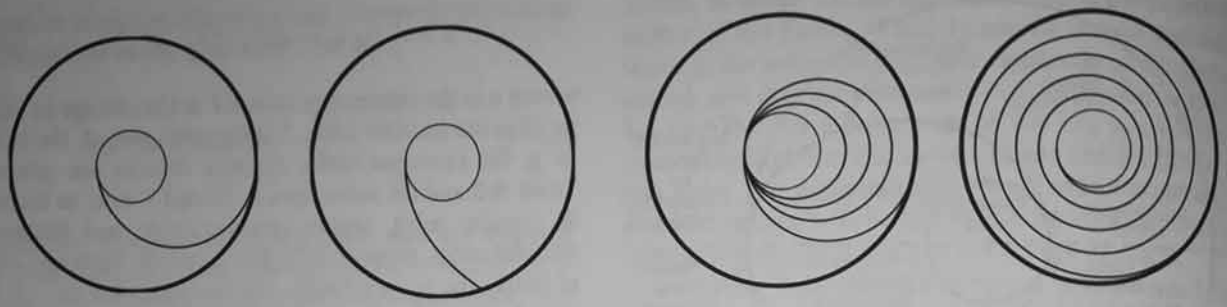
$$\Delta H = 2\Delta a \tag{9-75}$$

where Δa is the change in semimajor axis due to the single burn and ΔH is the corresponding change in altitude which will occur 180 deg away from where the burn was made.

A Hohmann transfer orbit from a higher orbit to a lower orbit is simply the reverse of the above process. The first burn is opposite the direction of motion and lowers perigee to the altitude of the inner orbit. When perigee is reached, 180 deg later, a second ΔV is applied, also opposite the direction of motion, to lower apogee and circularize the orbit. Thus it takes the same total ΔV to go from a high orbit to a low one as from a low orbit to a high one. Although we may at times use mechanisms such as atmospheric drag to slow the spacecraft, the fundamental problem of going to a lower energy orbit is just as difficult as going to a higher energy orbit.

Another interesting characteristic can be seen from Eqs. (9-68) to (9-73). If we start in a circular orbit with a velocity of 7 km/s and **add** 1 km/s in a two-burn Hohmann transfer then we will end up in a higher orbit with a velocity of 6 km/s. By adding energy to the orbit, more of the energy has been transferred into potential energy, and the total kinetic energy has been reduced. Similarly, if we go from a high orbit to a lower one, we will reduce the velocity and the total energy, but will end up in a lower orbit at a higher velocity and higher kinetic energy.

Most practical orbit transfers are relatively close to a Hohmann transfer because of the need to maximize efficiency. Nonetheless, as shown in Fig. 9-29 and listed in Table 9-16, alternative direct transfer orbits are used from time to time. Fig. 9-29A shows the classical Hohmann transfer. Fig. 9-29B shows a high-energy transfer which takes less time, but requires a higher ΔV. In the first burn, the ΔV will be higher because apogee will be



(A) Hohmann Transfer (B) High Energy (C) Low Thrust Chemical (D) Electric Propulsion

Fig. 9-29. Alternative Direct Transfer Orbits. See Table 9-16 for characteristics and Table 10-23 in Sec. 10.5.1 for advantages and disadvantages.

above the higher orbit. In the second burn, the ΔV will also be higher because we will need to change both the magnitude and the direction of the velocity. There are two practical applications for the high-energy transfer. One is for simply hitting a target in the higher orbit rather than matching velocities with it, since the second ΔV is not necessary in this case. The other application is when transfer time is critical and more rapid transfers are necessary. (See the discussion of rapid round trip transfers in Sec. 10.7.3.)

Both the Hohmann transfer and high-energy transfer assume impulsive burns, i.e., rocket engines which are large enough that the ΔV is applied over a short period of time. Alternatively, as described in Sec. 10.5, there can be significant advantages to using smaller rocket engines and applying the ΔV over a longer period such that the burn time is no longer short relative to the orbit period. Two specific cases are of practical interest. Fig. 9-29C shows the orbit path for a low-thrust chemical transfer in which rocket engines from the size of a few pounds to a few hundred pounds of thrust are used. In this case, the ΔV for a large orbit transfer cannot be applied in a single orbit, but must instead be applied over several orbits. First, a series of burns is made at successive perigees, such that the perigee height remains essentially unchanged, but the apogee height is raised by each maneuver until it ultimately reaches the desired final apogee. Then one or two burns are used at apogee to raise perigee to the circular orbit height. (Fewer burns are needed at apogee than at perigee because the spacecraft is moving at a lower velocity there.) For the low-thrust chemical transfer, the time for the transfer covers several orbits, and can be determined by adding up the orbit periods using Eqs. (9-4) and (9-5). The total ΔV that is required is identical to the ΔV used in the Hohmann transfer. Consequently the low-thrust chemical approach has the same optimal transfer efficiency as the Hohmann transfer, but a significantly lower thrust and, therefore, lower acceleration using a smaller engine and providing smaller disturbance forces on the spacecraft.

Finally, Fig. 9-29D shows the characteristics for electric propulsion transfer. Electric propulsion thrusters are much smaller than chemical ones, with thrust levels that are only a small fraction of a pound. Consequently, it is not practical for any large transfer to be done with burns only at perigee because the transfer time would be excessively long. In this case, the ΔV is applied continuously for the entire period of the orbit transfer. Because the thrust is very low, transfer from low-Earth orbit to geosynchronous orbit still takes many months rather than the few hours required for a Hohmann transfer. In addition, because apogee and perigee are being raised simultaneously, the electric propulsion transfer is significantly less efficient than the Hohmann transfer for large maneuvers, such as transfer to GEO. Specifically, the total ΔV required to transfer between coplanar circular orbits via spiral electric propulsion is given by:

$$V_T = V_L - V_H \tag{9-76}$$

Table 9-16. Physical Characteristics of the Alternative Transfer Trajectories Shown in Fig. 9-29. (See Table 10-23 in Sec. 10.5.1 for advantages and disadvantages.)

Transfer Type	Orbit Type	Typical Acceleration	ΔV	Typical Transfer Time
A. Hohmann (minimum energy)	Hohmann transfer	1 to 5 g	Eqs. (9-69), (9-70)	T= 1/2 orbit period, P is given by Eqs. (9-71) and (9-73)
B. High Energy	Elliptical or hyperbolic	1–10 g	> Hohmann	<T
C. Low Thrust Chemical	Hohmann transfer segments	0.02 to 0.5 g	Same as Hohmann	3 to 4 times P = 6 to 8 times T
D. Electric Propulsion	Spiral transfer	0.0001 to 0.001g	Eq. (9-76)	60 to 120 times P

where the parameters have the same definitions as above. Note once again that we are adding velocity to the orbit to get to a lower final velocity but higher energy orbit. For the ultra low-thrust transfer, transfer time is longer and the ΔV is greater. Nonetheless electric propulsion is inherently much more efficient than chemical propulsion and, therefore, can use a much lower propellant mass. (I_{sp}, a measure of propellant efficiency, is 5 to 10 times higher for electric propulsion systems than for chemical propulsion.) See Sec. 10.5 for further discussion of the advantages and disadvantages of the different methods of orbit transfer.

9.6.2 Plane Change and Phase Shifts

The previous section described transfer orbits which changed the total energy of the orbit itself. However, we may wish to change the orbit without changing the semimajor axis or the total energy. The two most common transfers in which this is done are changing the orbit plane (i.e., the inclination or the right ascension of the ascending node) or leaving the orbit the same and simply changing where we are in the orbit (i.e., moving forward or backward relative to an imaginary satellite which remains in the original orbit position).

Changing the orbit plane is required when the desired orbit plane is different from where we are initially. For example, in going from the Earth to Mars, we need to change the plane of the spacecraft from the Earth's orbital plane, i.e., the ecliptic, to the orbit plane of Mars. Similarly, for a spacecraft launched from anywhere not on the equator and, therefore, at a nonzero inclination, we need to make a plane change in order to put the spacecraft into geosynchronous orbit. In both of these examples, as in many plane change maneuvers, the need for the plane change occurs at the same time that the semimajor axis is to be changed. Consequently, combined plane-change and orbit-raising maneuvers are common.

As shown in Fig. 9-30, plane change maneuvers are simply a matter of vector addition. We have a velocity vector, $\mathbf{V}_i$, which defines the initial orbit plane, and need to obtain a different velocity, $\mathbf{V}_f$, to define a different orbit plane and potentially a different semimajor axis.

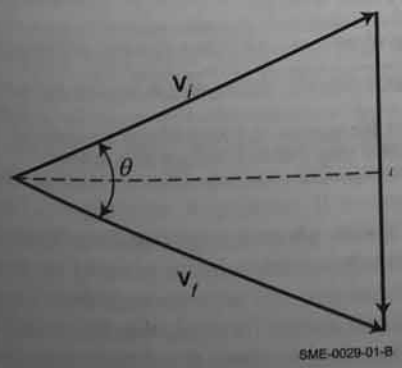


Fig. 9-30. Plane Change ΔV. Plane change maneuvers are simply problems in vector addition.

Combining a plane change with an altitude change is more efficient in terms of the total ΔV than doing the two maneuvers separately. (See Fig. 9-31.) Also, the ΔV required to undertake a plane change will be proportional to the velocity itself. Consequently, because spacecraft velocities are very large, plane change maneuvers typically require a very large ΔV, sometimes as much or more than the original orbital velocity. This implies that plane change maneuvers are highly energy intensive and will usually be undertaken only when absolutely necessary. Normally plane changes are done only once in the course of a mission and everything possible is done to reduce the ΔV required for the plane change. For example, the bielliptic transfer uses less ΔV than a standard two-burn transfer when launching to GEO from a latitude greater than about 35 deg.

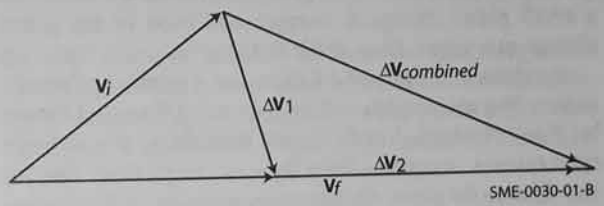


Fig. 9-31. Combining Plane Change and Orbit Raising Reduces the Total ΔV Required. The savings will be $|\Delta V_1| + |\Delta V_2| - |\Delta V_{combined}|$.

Because the plane change is simply a vector sum, we can determine the total ΔV required as a function of the initial and final velocities as:

$$\Delta V = \left(V_i^2 + V_f^2 - 2V_i V_f \cos\theta\right)^{1/2} \qquad (9\text{-}77)$$

where V_i is the initial velocity, V_f is the final velocity, and θ is the required angular change. If $V_i = V_f$, then the magnitude of the required ΔV reduces to:

$$\Delta V = 2V_i \sin\left(\frac{\theta}{2}\right) \qquad (9\text{-}78)$$

As an example, we find the total change in velocity to transfer from a 250 km parking orbit at 28.5 deg inclination to a geosynchronous equatorial orbit as follows:

$r_i = 6{,}628$ km	$r_f = 42{,}159$ km
$i_i = 28.5$ deg	$i_f = 0$ deg
$V_i = 7.75$ km/s	$V_f = 3.07$ km/s
$\Delta V_A = 2.44$ km/s	$\Delta V_B = 1.83$ km/s
$\Delta V_{total} = 4.27$ km/s	

Completing a Hohmann transfer followed by a simple plane change would require a velocity change of 5.425 km/s, so the Hohmann transfer with a combined plane change of apogee of the transfer orbit represents a savings of 1.15 km/s. As we see from Eq. (9-77), a small plane change ($\theta \geq 0$) can be combined with an energy change for almost no cost in ΔV or propellant. Conse-

Table 9-16◀, Fig. 9-31, Eq. 9-78

Table 9-17. Methods for Reducing the ΔV Required for Orbit Plane Change. It may be possible to combine techniques to increase the amount of plane change. See Table 10-22 in Sec. 10.5.1 for a discussion of the advantages and disadvantages of each. See text for discussion.

Method	Mechanism for Reducing ΔV	Computations
Do ΔV at Lowest Velocity (Highest Altitude)	Smallest velocity is easiest to change.	Eqs. (9-77), (9-78)
Combine ΔV with Orbit Raising	Vector sum is less than sum of components	Eqs. (9-77), (9-78)
3-Burn Transfer	Go to high altitude where plane change ΔV is low and then lower orbit	Eqs. (9-77), (9-78)
Use Differential Node Rotation	Allow natural perturbations to change plane	Eq. (9-61a)
Aero-Assist Trajectory	Use aerodynamic forces to change direction	[Albee et al., 1998]
Planetary Fly-By	Elastic "collision" with a planet changes direction	Sec. 9.6.3

quently, in practice, we do geosynchronous transfer with a small plane change at perigee and most of the plane change at apogee. (See Table 9-18).

As shown in Table 9-17 there are a number of mechanisms that are used to reduce the total ΔV needed for orbit plane changes. In addition to combining the ΔV with orbit raising, it is clear from the above equations that we should also do plane change maneuvers at the lowest velocity and, therefore, at the highest altitude in the orbit. Consequently, plane change maneuvers are typically done predominately at apogee.* If the plane change is very large, then it may be more efficient to use a *bi-elliptic* or *three burn transfer* rather than a two burn Hohmann transfer with plane change. In this case, we use the first burn to take the spacecraft to a very high altitude above the desired final orbit where the apogee velocity is very low. Here the plane change ΔV is small. At the high altitude we do a combined maneuver to raise perigee to the end altitude and change the plane. We then do a third and final burn to circularize the orbit at the final altitude. In this process we have used more than the minimum required energy to change altitude, but have still saved propellant by substantially reducing the ΔV required for the plane change.

Another technique for changing the ascending node of the orbit is to make use of the node rotation caused by the Earth's oblateness. If satellites are at different altitudes the node rotation rate will be different; the orbits will slide relative to each other in an East-West sense. Consequently, we can change that ascending node by changing the altitude, allowing differential node rotation to work, and then returning the altitude to its original value after the node has shifted as far as needed. Depending on the speed of the desired node rotation, we can use this mechanism to change the node with very small amounts of propellant.

We can also use the atmospheres of the Earth or other planets as a means of changing the orbit plane, using *aero-assist trajectories* in which aerodynamic forces on the spacecraft are used to provide a turn maneuver. Such maneuvers can be efficient, but also place additional stress on the spacecraft by applying significant aerodynamic forces. For example, Mars Global Surveyor used an aero-assist trajectory to adjust its orbit around Mars, but larger than anticipated aerodynamic forces caused the mission to be delayed to reduce further stress on the solar arrays [Albee, et al., 1998].

If we are going to change the altitude of the spacecraft then a small amount of plane change can be obtained at very low cost. This implies that when combining orbit raising and plane change it is most efficient to do some plane change at both apogee and perigee with most done at apogee. The process is optimized by iterating on the amount of plane change done at apogee and perigee to determine the total ΔV. An example of the potential savings is shown in Table 9-18.

Table 9-18. Alternative Approaches for Going from a 250 km Circular Orbit at 28.5 Deg Inclination to an Equatorial Geosynchronous Orbit. Because the last alternative uses the least propellant, it is essentially always used in practice.

Approach	Total ΔV (km/s)
Plane Change at 250 km Followed by Hohmann Transfer to GEO	7.696
Hohmann Transfer to GEO Altitude Followed by Separate Plane Change	5.425
Combined Maneuver, All Plane Change at Perigee	6.483
Combined Maneuver, All Plane Change at Apogee	4.273
Combined Maneuver, 4.0 deg Plane Change at Perigee, 24.5 deg Plane Change at Apogee	4.265

In addition to plane changes, spacecraft are frequently required to adjust their relative phasing in the orbit. Unlike plane changes, this requires very little ΔV and can be accomplished easily. To make the spacecraft drift relative to its nominal position we simply raise or lower the orbit to a slightly different altitude and, therefore, change the orbit period and in-track velocity. When we are done

* Part of the plane change will be done at perigee because a small amount of plane change takes very little ΔV. The correct ratio is determined by trial and error with Eq. (9-77) evaluated at apogee and perigee.

shifting it as far as we would like, we return the spacecraft to its original altitude. The transfer into a *drift orbit* can be done either using a two burn Hohmann transfer, or a single burn so that the drift orbit is elliptical, with apogee or perigee remaining at the initial altitude. In either case, the drift rate, ω_{drift}, in deg/orbit, is given by:

$$\omega_{drift} = 1{,}080\,\frac{\Delta V}{V} \tag{9-79}$$

where ΔV is the applied velocity change (in either one burn or both burns) and V is the nominal orbit velocity. The total ΔV required to shift phase depends only on the drift velocity and, therefore, the time for making the phase change. Thus in total:

$$\Delta V_{total} = \Delta V_{start} + \Delta V_{stop} \tag{9-80a}$$

$$= \left(\omega_{drift}\, V\right)/540 \tag{9-80b}$$

where ω_{drift} is in deg/orbit. For a satellite in geosynchronous orbit, $V = 3.0747$ km/s and consequently:

$$\Delta V_{total} = 5.6939\,\omega_{drift} \tag{9-81}$$

where ΔV is in m/s and ω_{drift} is in deg/day. Sample calculations of the ΔV required as a function of time are given in Table 9-19.

Live Calc

Table 9-19. Drift Rate and Total ΔV Required to Shift a Geosynchronous Satellite 60 deg in Longitude. In general, rapid maneuver requirements correspond to high propellant usage.

ω_{drift} (deg/day)	ΔV (m/s)	Time to Shift 60 deg (days)
1	5.7	60
2	11.4	30
6	34.2	10
10	56.9	6
30	170.8	2
60	341.6	1

9.6.3 Planetary Assist Trajectories

For interplanetary spacecraft, we can perform both plane change and velocity change maneuvers by one or more interactions with other planets or with the Earth. The general process of a planetary fly-by is shown in Fig. 9-32 [Minovitch, 1963]. We assume the spacecraft is initially in an orbit about the Sun a long way from any planet. With respect to the planet, the spacecraft approaches on a hyperbolic trajectory. It swings by the planet at "periplanet" and leaves on the other leg of the hyperbolic trajectory at the same velocity **with respect to the planet** at which it came in. The total angle through which the spacecraft orbit velocity has changed is known as the *turn angle*, ψ, and is the angle between the two asymptotes of the hyperbola. Because the orbit has undergone an angular change, this can be used to either change

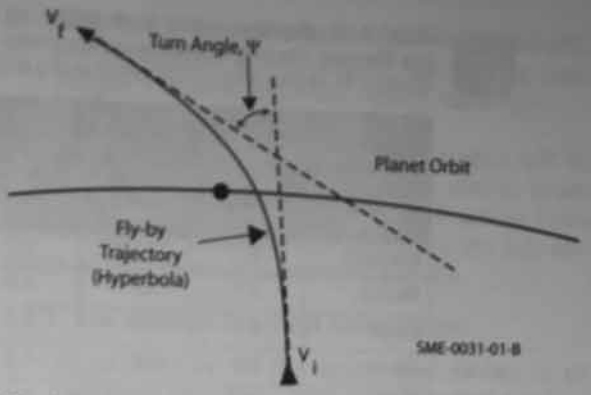


Fig. 9-32. Geometry of a Planetary Fly-By. The spacecraft velocity with respect to the planet remains the same incoming as outgoing. However, the vector sum of the planet and spacecraft velocities can be used to increase energy, decrease energy, or change direction.

the orbit plane or increase or decrease the energy. Because Jupiter has a very large mass and, consequently, can produce a large turn angle, it is the planet most commonly used for fly-bys. Depending upon our objective, we can use a fly-by of Jupiter to increase the spacecraft velocity and therefore increase the mass or reduce the transfer time to the outer planets, to decrease the velocity and allow the spacecraft to go to the vicinity of the Sun, or to change the plane of the spacecraft's orbit to allow us to explore space in the direction of the ecliptic poles.

A planetary fly-by is equivalent to an elastic collision between a baseball and a bat. The velocity of the baseball with respect to the bat is the same coming from the pitcher as leaving toward the outfield. The difference is that the baseball has picked up the velocity of the bat in the interaction, or in the case of a pop up, has principally changed direction.* Conceptually, we can think of the spacecraft as bouncing off the planet without having to undergo the excessive acceleration associated with actually hitting the surface.

The turn angle, ψ, obtained in a planetary fly-by is:

$$\sin\left(\psi/2\right) = 1/e \tag{9-82}$$

where e is the eccentricity of the hyperbolic trajectory. This may also be expressed in a form more convenient for analysis as:

$$\sin\left(\psi/2\right) = \left(1+\frac{V_\infty^2 q}{\mu}\right)^{-1} \tag{9-83}$$

where q is the perifocal distance or distance of closest approach to the planet, V_∞ is the hyperbolic velocity or the velocity of the spacecraft with respect to the planet when

* Hans Meissinger [1970] was the first to point out that the Earth could be used for a planetary fly-by. [Hollenbeck, 1975.] Continuing our baseball analogy, this is similar (though not mathematically equivalent) to throwing the baseball up in the air and then hitting it with the bat when it comes down.

Live Calc

Table 9-20. Representative Turn Angle, ψ, and Velocity Change, ΔV, for Fly-by Missions to the Planets. The assumed hyperbolic verocity, V_∞, is the velocity of approach for a Hohmann transfer orbit. R_p is the radius of the planet and q is the perifocal distance or distance of closest approach.

	Assumed	$q = R_p$ (Grazing)		$q = 2R_p$		$q = 10\ R_p$	
Planet	V_∞ (km/s)	ψ (deg)	$\lvert\Delta V\rvert$ (km/s)	ψ (deg)	$\lvert\Delta V\rvert$ (km/s)	ψ (deg)	$\lvert\Delta V\rvert$ (km/s)
Mercury	9.6	10.2	1.7	5.4	0.9	1.1	0.2
Venus	2.7	123	4.8	104	4.2	50	2.3
Earth	3.5	114	5.8	92	5.0	40	2.4
Moon	0.8	109	1.3	87	1.1	36	0.5
Mars	2.6	81	3.4	58	2.5	18	0.8
Jupiter	5.6	159	11.0	150	10.8	116	9.5
Saturn	5.4	146	10.3	133	9.9	86	7.4
Uranus	4.7	132	8.6	114	7.9	61	4.8
Neptune	4.1	141	7.7	127	7.3	78	5.1
Pluto	3.7	5.57	0.36	2.86	0.18	0.58	0.04

it is infinitely far away, and as usual $\mu \equiv GM$ for the planet. The magnitude of the velocity change, ΔV, resulting from the fly-by is a function of the turn angle between the two asymptotes and is given by:

$$\left|\Delta V\right| = 2V_\infty \sin(\psi/2) \tag{9-84}$$

Table 9-20 lists the values of the turn angle and velocity change for fly-bys of the principal objects of interest in the Solar System.

9.6.4 Spacecraft Disposal

Orbital debris is potentially a major problem for space missions, particularly in regions of high spacecraft density such as low-Earth orbit or geosynchronous orbit. Consequently, spacecraft disposal is becoming a key element of mission design. NASA guidelines require that spacecraft in low-Earth orbit be removed from orbit within 25 years of the end of their useful life. Of course it is desirable to remove them from orbit much more rapidly, but it may require more propellant to do so.

There are two basic ways to deorbit a satellite in low-Earth orbit. The first approach is to lower perigee to a low enough value, approximately 75 km, that the orbit will decay and the spacecraft will reenter essentially immediately. The second approach is to reduce perigee to the point where atmospheric drag will cause the satellite to decay within an appropriate period of time, though not immediately. With this approach we will have an approximate idea of how long reentry will take, but have no way of knowing or controlling when or where it will occur. Rough estimates of the re-entry time required for representative orbits and spacecraft parameters are given in Table 9-21.

The ΔV required for a satellite to drop from an initial circular altitude H_i to a reentry perigee altitude H_e is given by:

$$\Delta V_{deorbit} \approx V\left[1 - \sqrt{\frac{2(R_E + H_e)}{2R_E + H_e + H_i}}\right] \tag{9-85a}$$

$$\approx V\left[\frac{0.5(H_i - H_e)}{2R_E + H_e + H_i}\right] \tag{9-85b}$$

where V is the initial satellite velocity and R_E is the radius of the Earth.

Table 9-21. Spacecraft Disposal Options. Lifetime assumes a ballistic coefficient of 100 kg/m^2.

	From 600 km Circular		From 1,200 km Circular	
Perigee	ΔV (m/s)	Approx. Lifetime (days)	ΔV (m/s)	Approx. Lifetime (days)
50	157	0.0	304	0.0
75	149	0.0	297	0.0
100	142	0.1	290	0.3
125	134	0.7	282	2.6
150	127	7.2	275	26.4
200	112	47	261	174
250	97.8	152	247	615

For high altitude orbits deorbiting the spacecraft is impractical because of the large ΔV required. Nonetheless, there is still a strong desire to remove spent spacecraft from their operational orbit, particularly if they are part of a constellation with many satellites at that altitude or part of an orbit region, such as GEO, where there are a large number of different satellites. In this case the preferred option is to raise the orbit by several hundred kilometers above the nominal value. In the regime where atmospheric drag is no longer relevant, the spacecraft will be left in an orbit where the orbit elements oscillate.

but where the spacecraft will remain for thousands or hundreds of thousands of years, effectively eliminating it as a threat to other spacecraft. The disposal orbit is placed above the operational orbit so that spent spacecraft do not interfere with new operational spacecraft being brought up from the Earth. Spacecraft in disposal orbits still represent a theoretical threat to other spacecraft passing through that regime. However, this threat is dramatically reduced relative to the threat that is present if they remain in the operational orbit. In the future it may be possible to retrieve or destroy old spacecraft, but this is not a practical option at the present time.

The ΔV needed to raise a spacecraft to a higher altitude is typically very low. For example, in geosynchronous orbit the ΔV requirement is given by:

$$\Delta V_{disposal} = 0.0365\,(\text{m/s})/\text{km} \qquad (9\text{-}86)$$

Thus, an increase of 500 km above geosynchronous altitude requires a ΔV of only 18 m/s, as opposed to a deorbit ΔV of 4,000 m/s. The principal problem with any of the spacecraft disposal options is that they must be done at a time when both commanding and thruster firings can still be achieved by the spacecraft, and, therefore, at least critical spacecraft subsystems are still operating and propellant remains available.

9.7 Summary—The Rules of Practical Astronautics

9.7.1 Important Characteristics of Spacecraft Orbits

1. **Spacecraft in an elliptical orbit will always return to the location in space where the last maneuver was made.** Because the elliptical orbit is a closed figure, the spacecraft will always return to the location in space where we gave it the last ΔV. Thus, we cannot lower perigee by firing at perigee, unless we apply sufficient ΔV to change perigee into apogee. Similarly, we can't launch directly from the Earth's surface into an Earth orbit. We must first get into space and then apply sufficient ΔV to put the spacecraft into an orbit.
2. **Lower orbits have less energy, but greater velocity than higher orbits.** Higher orbits have a greater potential energy, but less kinetic energy than lower ones.
3. **The semimajor axis depends only on the total energy of the orbit.** This means that to change the size of the orbit, at least a portion of the applied ΔV must be along the velocity vector.
4. **For orbits to stay synchronized, they must be at the same altitude and inclination.** Because of the node rotation due to J_2, orbits at different inclinations or altitudes will drift with respect to each other, potentially quite rapidly. The exception here is a circular, equatorial orbit, for which the drift of the orbit doesn't matter.
5. **Elliptical orbits will almost always be at the critical inclination of 63.4 deg.** If this isn't the case, perigee will rotate relatively rapidly due to J_2.

As shown in Chap. 10, these characteristics will be important for both orbit and constellation design. In addition, they lead to what I call "Wertz's Practical Rules of Astronautics." Summarized below are the key elements of orbit maneuvers and orbit transfer.

9.7.2 Ten Rules of Practical Astronautics

1. **To go forward in the orbit, slow down; to go backward speed up.** That is the practical result of characteristic 2 above. When the spacecraft slows down, it goes into a lower orbit with a shorter period and, therefore, moves faster than when it was higher. The spacecraft can drift forward as far as desired and then be raised back to the previous altitude (and slowed down) by speeding up.
2. **Add ΔV deep in a gravity well when the velocity is highest.** The energy increases for a given ΔV is directly proportional to the velocity, so adding ΔV is most efficient when the spacecraft is going fast.
3. **Change direction when the velocity is lowest.** An inclination or plane change "at infinity" costs essentially no ΔV. An exception to this rule is number 4.
4. **Let Mother Nature help with plane changes.** Changing the plane typically takes a large ΔV, potentially even more than launching the spacecraft. With patience, Mother Nature can do quite a bit for free with planetary fly-bys, aerodynamic forces (in the upper atmosphere), or differential node rotation.
5. **Changes well in advance are far cheaper.** If you want to fly over a particular location at a particular future time or move to a different location in GEO, it takes almost no ΔV if done far in advance of when you want to get there and a huge ΔV if you want to do it quickly. If you want to deflect an asteroid from hitting Earth, it's essentially impossible just before impact and pretty easy years in advance.
6. **Use high thrust to get off a planet; use low thrust in space.** To get to orbit, you need large engines and high thrust to overcome gravity losses. However, once you're in orbit, high thrust is of little or no value and is potentially damaging to the payload (due to the high forces), expensive to control, hard to measure, and hard to fine tune for precision adjustments. In addition, big engines are heavy, expensive, and known to blow up from time to time, as in the case of Clementine.
7. **Stage early and often.** It's remarkably inefficient to carry the rocket stage itself to Jupiter, to GEO, or even to a high LEO orbit. To the extent possible, get rid of the launch vehicle and upper stages as early as possible and let the spacecraft fly itself around.

Table 9-21◀, Fig. 9-32◀, Eq. 9-86

8. **For round trip travel, there are major discontinuities in the round trip time as a function of the transfer ΔV.** This will be discussed in Sec. 10.8.2, but under the right conditions, a small increase in ΔV can reduce interplanetary round trip times by more than a year and time can be an important factor in the success of round trip missions.
9. **Design both orbits and constellations with orbit debris in mind.** As discussed in Sec. 7.5, orbit debris can be a major long-term hazard, particularly in GEO or high LEO orbits. The right initial orbit design and operations can do a great deal to mitigate this problem.
10. **Clean up after yourself.** To keep debris from bothering other spacecraft, dispose of stuff when you're done with it—both spacecraft and stages. Burn it up, retrieve it, or put it in an orbit where it won't bother anyone for 10,000 years or so. Even better, find a way to use spent vehicles to look for water on the Moon, find out the composition of asteroids and comets, or build habitats out of external tanks. Just don't leave it lying around in orbit.

Using astronautics wisely can dramatically reduce launch cost and propellant usage and keep space available for future missions.

Applicable Standards:

ANSI/AIAA S-131-2010: Astrodynamics Standard—Propagation Specifications, Technical Definitions, and Recommended Practices

See website for reference.

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For annotated bibliography of orbits and astrodynamics, see the book website.

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10 Orbit and Constellation Design—Selecting the Right Orbit

James R. Wertz, *Microcosm, Inc./USC*

10.1 The Orbit Selection and Design Process

10.2 Orbit Performance—Evaluating Earth Coverage and Payload Performance
Evaluating Earth Coverage; Payload Performance vs. Altitude

10.3 Orbit Cost—Delta V Budget and the Orbit Cost Function
The Delta V (ΔV) Budget; The Orbit Cost Function; Estimating Launch Costs

10.4 Selecting Earth-Referenced Orbits
Mission Requirements for Earth-Referenced Orbits; Specialized Earth-Referenced Orbits

10.5 Selecting Transfer, Parking, and Space-Referenced Orbits
Design of Transfer and Parking Orbits

10.6 Summary of Constellation Design
The Constellation Design Process; Coverage and Constellation Structure; Overview; The Rules for Constellation Design

10.7 Design of Interplanetary Orbits—*Ron Noomen*
Design Process; One-Way Missions; Round-Trip Missions

Chapters 8 and 9 introduced the geometry and physics of spacecraft orbits as well as formulas for computing viewing and orbit parameters. Those chapters are fundamentally computational in that there is a well defined answer to how the satellite will move in response to a given set of forces or what portion of the Earth I can see with a camera oriented in a specific direction. In contrast, orbit selection and design is a much fuzzier process. The question that we are trying to deal with is not how the spacecraft will move, but what orbit should we put it in. This is, of course, a mix of mission objectives, cost, available launch vehicles, and operational requirements to support the mission. There is no precise answer or correct solution.

In most cases, orbit design consists of choosing among a continuum of orbits and some specialized orbits, such as GEO or Sun synchronous, that come about because of specific orbit characteristics. In addition, the orbit selection process itself is typically complex, involving trades between many parameters. The orbit normally defines the space mission lifetime, cost, environment, viewing geometry, and, frequently, the payload performance. Nonetheless, the single most common trade is between the velocity required to achieve an orbit as a measure of cost vs. the coverage to be achieved as a measure of performance.

Section 10.1 addresses the overall orbit selection and design process. Section 10.2 discusses ways of evaluating orbit performance and Sec. 10.3 discusses the idea of orbit cost and introduces the orbit cost function as a comparative measure of what it costs for different orbits. Sections 10.4 and 10.5 then describe the design of Earth-referenced orbits and space-referenced orbits, respectively. Section 10.6 provides a summary of constellation design. Finally, Sec. 10.7 describes the design of interplanetary orbits. For a discussion of interstellar travel and interstellar orbits, see Sec. 12.8 of Wertz [2009].

There are relatively very few references on orbit and constellation design, per se. By far the most extensive is Wertz [2009], which forms the basis for much of the material here.

10.1 The Orbit Selection and Design Process

Orbit and constellation selection and design is a process rather than a set of specific computations. Throughout the chapter, we will use a series of process tables to summarize the fundamental steps and how to undertake them. These should be treated as guides to the issues involved rather than complete definitive recipes. There are a wide variety of mission types, each of which will be unique in the orbit selection process. Nonetheless, the tables here should be used as a starting point to create a process appropriate for your particular mission.

Table 10-1 summarizes the steps in the overall orbit selection and design process. Each step is discussed below, and later with greater detail in the sections listed in the table. Effective orbit design requires clearly identifying the reasons for orbit selection, reviewing these reasons regularly as mission requirements change or mission definition improves, and continuing to remain open to alternatives. Several different designs may be

Table 10-1. The Single Satellite Orbit Selection Process. See Table 10-24 in Sec. 10.6.1 for Constellation Design.

Step	Where Discussed
1. Establish Orbit Types	Sec. 10.1
2. Determine Orbit-Related Mission Requirements	Secs. 10.4.1, 10.5.1
3. Evaluate Orbit Performance	
A. Earth coverage	Sec. 10.2.1
Specialized orbits	Secs. 9.5, 10.4.2
Single satellite vs. constellation	Sec. 10.1
B. Other ways the orbit impacts performance	Sec. 10.2.2
C. Do mission orbit design trades	Secs. 10.4.3, 10.5.3, 10.5.4
4. Evaluate Orbit Cost	
A. Evaluate launch options and launch cost	Secs. 10.3.3, 11.3.6, 26.1
B. Look at low-cost options, if possible	Sec. 13.5.7
C. Add disposal options	Secs. 30.1, 30.2
D. Create delta V budget	Sec. 10.3.1
E. Evaluate the orbit cost function—launch cost vs. available on orbit mass	Sec. 10.3.2
5. Document Selection Criteria, Key Orbit Trades, Selected Orbit Parameters, and Allowed Ranges	Sec. 10.1
6. Iterate as Needed	Sec. 10.1

credible. Thus, communication systems may work effectively through a single large satellite in geosynchronous orbit, or a constellation of small satellites in low-Earth orbit. We may need to keep both options for some time before selecting one for a particular mission.

Step 1. Establish Orbit Types

To design orbits we first divide the space mission into segments and classify each segment by its overall function. In terms of orbit requirements, each segment falls into one of four basic orbit types, listed in Table 10-2. The first two entries are operational orbits in which the spacecraft is intended to spend a large fraction of its operational life, and to do most of the useful work of the mission. In contrast, the bottom two entries are simply means of getting the spacecraft where we want it, when we want it there. Typically, the requirements for the latter two are much less stringent, with the exception that it may be necessary to get the spacecraft where we want it with very precise timing, such as the requirement to attempt to land a probe at a particular location on the surface of Mars.

An example of the changing orbit types throughout the mission would be a geosynchronous communication satellite initially launched along with a transfer stage into a low-Earth orbit. (See Fig. 10-1.) Once ejected from the launch vehicle, the spacecraft stays briefly in a parking orbit to provide test and checkout of the spacecraft and transfer vehicle subsystems. The next mission segment is a transfer orbit that moves the spacecraft from the parking orbit to a geosynchronous equatorial orbit. Frequently, to preserve propellant, the spacecraft is initially put into a drift orbit near GEO such that any errors in the transfer process can be taken out by small adjustments associated with achieving the desired geosynchronous station location. The spacecraft then enters its operational orbit in the geostationary ring where it will spend the rest of its active life. While the spacecraft itself is still functional and before its stationkeeping propellant runs out, we must move it out of the geostationary ring to avoid possible collisions with other satellites and to free the orbital slot for a replacement. Putting the nearly dead



Fig. 10-1. Typical Orbit Phases. At end-of-life, the spacecraft should be either de-orbited or moved into a higher orbit where it will not interfere with other spacecraft.

spacecraft into a final disposal orbit a few hundred kilometers above the geostationary ring requires a relatively small amount of propellant (but not 0). Going above the geostationary ring rather than below avoids the potential for collisions with other satellites during subsequent geosynchronous transfers.

Table 10-2. Principal Orbit Types. For orbit design it is convenient to divide the mission into a series of orbit types based on the varying requirements as the mission progresses.

Type	Definition	Examples	Where Discussed
Earth-Referenced Orbit	An operational orbit which provides the necessary coverage of the surface of the Earth or near-Earth space (also applicable to the Moon and planets)	Geosynchronous orbit, low-Earth communications or observation constellations	Sec. 10.4, Sec. 13.5.6
Space-Referenced Orbit	An operational orbit whose principal characteristic is being or pointing somewhere in space, such that specific orbit parameters may not be critical	Hubble Space Telescope orbit, Chandra X-Ray Observatory, space manufacturing orbit	Sec. 10.5.3
Transfer Orbit	An orbit used for getting from place to place	Geosynchronous transfer, inter-planetary transfer to Mars	Sec. 10.5.4
Parking Orbit	A temporary orbit providing a convenient location for satellite check-out, storage between activity or at end-of-life, or used to match conditions between phases	After launch, while awaiting proper conditions for orbit transfer; end-of-life 500 km above geosynchronous	Sec. 10.5.4

Step 2. Establish Orbit-Related Mission Requirements

For each mission segment defined in Step 1, we next define the orbit-related requirements. They may include orbital limits, individual requirements such as the altitude needed for specific observations, or a range of values constraining any of the orbit parameters. Secs. 10.4.1 and 10.5.1 discuss in detail the requirements for designing operational orbits. Ordinarily, these multiple requirements drive the orbit in different directions. For example, resolution or launch capability tend to drive the orbit to low altitudes, but coverage, lifetime, and survivability drive the spacecraft to higher altitudes. Consequently, the operational orbit trades are typically complex, involving the evaluation of multiple parameters in selecting a reasonable compromise that meets mission objectives at minimum cost and risk.

Selecting parking, transfer, and space-referenced orbits is normally conceptually much simpler, although it may be mathematically complex. Here, the normal trade is meeting the desired constraints on a mission, such as lifetime, thermal, or radiation environments, at the lowest possible propellant cost. Section 10.5.4 discusses these types of orbits in more detail.

Step 3. Evaluate Orbit Performance

For all orbits, we are interested in the performance we can achieve (Step 3) vs. the cost of achieving it (Step 4). As discussed in Sec. 10.4, most missions are Earth-referenced, i.e., looking back at the Earth and here there is an immediate conflict. The best coverage occurs as we move up in altitude and can see more of the Earth at any one time (and, therefore need less satellites), but the best resolution occurs as we move down in altitude. It's like taking pictures of a facility or people. We like views from a distance to get a broad perspective, but like close-ups for detail. This suggests an elliptical orbit that provides both near and far views, but this comes with its own set of problems. Orbits that aren't Earth-referenced typically have fewer conflicting requirements, as discussed in Sec. 10.5, although there still may be a need for high precision, such as landing on a distant planet or rendezvous with a spacecraft or celestial object.

Step 3A. Earth Coverage

Coverage is a key requirement for any Earth or planet referenced mission and can be challenging to evaluate. As discussed in Sec. 10.2.1, coverage is not uniform but is also not a random variable and, therefore, statistical results that are often used can prove very misleading. Section 10.2.1 discusses some of the "traps" of evaluating coverage and relatively simple approaches to resolving these issues.

Specialized Orbits. In selecting the orbit for any mission phase, we must first determine if a specialized orbit applies. *Specialized orbits* (Sec. 9.5) are those with unique characteristics, which set them apart from the broad continuum of orbit parameters. (See Table 10-12 in Sec. 10.4.2 and Table 10-18 in Sec. 10.5.) Geosynchronous orbits, Sun-synchronous orbits, Molniya orbits, and Lagrange point orbits are typical examples. We examine each of these specialized orbits to see if its unique characteristics are worth its cost. This examination precedes the more detailed trades, because specialized orbits constrain parameters such as altitude or inclination, and, therefore, often lead to very different solutions for a given mission problem. Consequently, we may need to carry more than one orbit into more detailed design trades, such as keeping open a geosynchronous option and a low-Earth orbit option as possibilities for further trade studies.

Single Satellite vs. Constellation. The principal advantage of a single satellite is that it reduces cost by minimizing the mission overhead. One satellite will have one power system, one attitude control system, one telemetry system, and require only a single launch vehicle. A constellation or multiple spacecraft, on the other hand, may provide better coverage, higher reliability if a satellite is lost, and greater survivability. We may also need a constellation to provide the multiple conditions needed to carry out a mission, such as varying lighting conditions for observations, varying geometries for navigation, or continuous coverage of all or part of the Earth for communications.

Table 10-2, Fig. 10-1◀, Eq. 10-0

To meet budget limits, we often trade a single large satellite with larger and more complex instruments against a constellation of smaller, simpler satellites. This decision may depend on the technology available at the time of system design. As discussed in detail in Sec. 13.1, Wertz and Larson [1996], and Wertz et al. [2010], small satellites have become more capable through miniaturized electronics and on-board processing. Consequently, we may be able to construct constellations of small, low-cost satellites, frequently called *SmallSats* or *LightSats*, that weren't previously economically feasible. Another issue for large constellations is the operational problem of providing continuous navigation and orbit control. The introduction of low-cost, GPS-based navigation and autonomous orbit control (Sec. 19.2) can resolve many of these problems and promote larger constellations of small satellites in the future.

Step 3B. Other Ways the Orbit Impacts Performance

The orbit is one of the principal drivers of mission performance, as discussed in Secs. 10.2.2 and 10.4.3. The orbit impacts nearly every aspect of the system performance, including aperture and power requirements, drag and stationkeeping, the radiation environment, and the viewing angles and distance to various targets. As discussed in Sec. 9.6, orbit changes are very expensive in most cases, such that the spacecraft will typically remain very nearly in its initial operational orbit for the entire mission life. (There are some exceptions; such as changing phase in an operational orbit which has a very modest delta V cost.)

Step 3C. Do Mission Orbit Design Trades

The next step is to select the mission orbit by evaluating how orbit parameters affect each of the mission requirements defined in Chap. 5 and Sec. 10.4.1. As shown in Table 10-11 in Sec. 10.4.1, orbit design in Earth orbit depends principally on altitude. The easiest way to begin is by assuming a circular orbit and then conducting altitude and inclination trades. This process establishes a range of potential altitudes and inclinations, from which we can select one or more alternatives. Documenting the results of this key trade is particularly important, so we can revisit the trade from time to time as mission requirements and conditions change. We then evaluate the use of eccentric orbits as discussed in Sec. 10.4.3.

A relatively new criterion for orbit design is the problem of orbital debris and collision avoidance. As discussed in Sec. 7.5, orbital debris is a potential major problem for spacecraft. One of the principal solutions to this problem is to fly the spacecraft in a relatively low orbit. Below roughly 500 km, the life of any object is limited to a solar cycle of 11 years or less, as can be seen from Fig. 9-18 in Sec. 9.4.4. This means that debris cannot accumulate below this altitude. Traditionally satellites have been placed at higher altitudes specifically so that they will have a long orbit life. Satellites at lower altitudes will need some orbit maintenance, but have much better resolution in a smaller instrument and cannot contribute to the orbit debris problem simply because drag will remove any possible debris at the time of the next solar maximum.

Table 10-2◀, Fig. 10-1◀, Eq. 10-0

Step 4. Evaluate Orbit Cost

Step 3 gives us a sense of how well the orbit is doing at performing the fundamental needs of the mission. However, often lost in the discussion of orbits is that there is a significant cost to the choice of orbits as well. For example, putting spacecraft in GEO requires launching approximately 5 times the spacecraft mass into LEO and putting spacecraft on the surface of the Moon requires launching approximately 10 times that mass into LEO. It may be that we have no choice but to launch into a specialized orbit such as GEO so that we can provide direct broadcast TV into some part of the world. But if we do have a choice, then we need to ask ourselves, will we do better with a 1,000 kg satellite in GEO or a 5,000 kg satellite in LEO? Unfortunately, that question is rarely asked.

Step 4A. Evaluate Launch Options and Launch Cost

Section 26.6 discusses current satellite launch options, including selection issues associated with different alternatives. In addition, Isakowitz et al. [2004] provides detailed numerical summaries of currently available launch vehicles worldwide. The launch vehicle contributes strongly to mission cost, and ultimately limits the amount of mass that can be placed in any given orbit. During early mission definition, we must provide enough launch margin to allow for later changes in launch vehicles or spacecraft mass. New designs require more margin than existing ones, with 20% being typical for new missions. In order to keep cost down and maintain schedule, it is highly desirable to design our spacecraft so as to be compatible with more than one launch vehicle, as discussed in Sec. 13.5.7.

Step 4B. Look at Low-Cost Launch Options, if Possible

While launch is not the major cost element for most space missions, it tends to drive mission cost, and often the schedule as well. Therefore, if possible, it is desirable to look at low-cost options to a dedicated launch, as discussed in Sec. 13.5.7. In addition, some low-cost alternatives may offer observation opportunities that are simply not available with orbiting spacecraft. At the very beginning of X-Ray astronomy, which has to done above the atmosphere, it was clear that there were many discrete sources of X-Rays in the sky, but no way of knowing what they were due to the very poor angular resolution of detectors prior to the Chandra X-Ray telescope. A suborbital flight with an X-Ray detector on board was timed such the Crab Nebula would be occulted by the Moon during the very brief flight. As soon as the Crab Nebula disappeared behind the Moon, the brightest X-Ray source in the sky turned off and first source of extraterrestrial X-Rays was identified.

Step 4C. Add Disposal Options

Although given little consideration in many early missions, spacecraft disposal has become critical to mission design because of the orbital debris problem. This represents potentially major legal, political, and economic issues. Spacecraft disposal options are discussed in detail in Chap. 30. Spacecraft that will reenter the atmosphere

must either do a controlled reentry over the ocean or break up into pieces harmless to the Earth's surface.

If the spacecraft will not reenter the atmosphere in a reasonable time, we must still dispose of it at the end of its useful life so it isn't hazardous to other spacecraft. This problem is particularly acute in geosynchronous orbit where missions compete strongly for orbit slots. (See Cefola [1987] for an excellent analysis of the requirements for removing satellites from geosynchronous orbit.) As pointed out in Sec. 7.5, a collision between two spacecraft not only destroys them but also causes a debris cloud that is dangerous to their entire orbital area. Consequently, this is a major concern for satellite constellations as well.

Spacecraft beyond geosynchronous orbit are ordinarily allowed to continue to drift in interplanetary space at the end of their useful life. If a spacecraft is not specifically designed to encounter a planetary surface, then normal practice calls for preventing collisions with planets or moons in order to avoid the potential of biological contamination. At the end of their normal mission, Pioneers 10 and 11 became the first spacecraft to begin the exploration of interstellar space and continued to send back signals for an extended time on the environment that they encountered. (See Sec. 10.7.)

Step 4D. Create a Delta V Budget

To numerically evaluate the cost of an orbit, we first create a ΔV budget for the orbit, as described in Sec. 10.3.1. This then becomes the basis for the spacecraft propellant budget and a component of the Orbit Cost Function as described in Step 4E.

Step 4E. Evaluate the Orbit Cost Function

Section 10.3.2 discusses the *orbit cost function,* which is a mechanism for defining the approximate cost of orbit transfer in terms of the mass that must be put in low-Earth orbit to achieve the end mission orbit. Consequently, it provides a cost multiplier to be used in conjunction with launch cost estimates from other sources.

Steps 5 and 6. Document and Iterate

A key component of orbit or constellation design is documenting the mission requirements used to define the orbit, the reasons for selecting the orbit, and the numerical values of the selected orbit parameters. With this documentation, the baseline selection can be reevaluated from time to time as mission conditions change. Because mission design nearly always requires many iterations, we must make the iteration process as straightforward as possible and readdress orbit parameters throughout the mission design to ensure that they continue to meet all of the mission objectives and requirements at minimum cost and risk.

10.2 Orbit Performance—Evaluating Earth Coverage and Payload Performance

For a large number of spacecraft, Earth coverage is the fundamental element of performance. The number of satellites we need or the utility of the data depends on coverage. Thus, coverage is often a key parameter in orbit and constellation design. Most typically, this is done by computer simulation and statistical analysis of the results. However, one of the most important coverage characteristics is that

> ***Earth coverage is not a Gaussian parameter and statistical data can give very misleading results.***

I would like to give a specific example of this problem. Several years ago, a systems engineering group that I was a part of was asked to do a constellation coverage analysis. Continuous coverage was not necessary, but the customer was concerned with minimizing the duration of the gaps in coverage, consistent with keeping down the number of satellites in the constellation, and, therefore, the cost. Mindful of the fact that maxima are often a very poor measure of system performance, we chose instead to use as our most relevant measure of performance, the *mean gap duration*, i.e., the average length of coverage gaps under a variety of circumstances and constellation designs. We collected, analyzed, and plotted a variety of mean gap statistics as well as some additional statistical data based on our simulations of coverage performance. In one particular constellation design, we had a mean gap of approximately 40 min, which we determined was probably longer than the customer wanted. Consequently, we added a satellite to each of the orbit planes in the constellation configuration, rephased the satellites to provide the best coverage possible, and re-ran our statistical analysis. The result, with 50% more satellites, was that the mean gap went from approximately 40 min to more than an hour. Budgets were not quite so tight in those days when rockets were still pulled by horses; nonetheless, it seemed highly unlikely that our customer would be willing to pay for 50% more satellites in order to have longer gaps in coverage.

Although our computations were correct, it was clear that something had gone terribly wrong with our analysis. What had happened was actually straightforward. Our original constellation had a number of large and small gaps with an average duration of 40 min. We had improved the coverage by adding satellites and had actually filled in quite a few of the smaller gaps. What was left was a small number of larger gaps such that the average gap duration was longer, even though the total of all of the gaps was significantly reduced.

When obvious errors such as the above occur, it becomes clear that something has gone wrong and the problem can be identified and corrected. Unfortunately, that rarely happens. We collect statistical data on constellation coverage not because it has any type of a Gaussian distribution, but because we **always** collect statistical data from simulations. Often, the results look "good" and do not have any manifest problems. Nonetheless, statistical analysis of coverage is based on an assumption that is simply not true, and, therefore, must be approached with substantial care.

I would not propose that you avoid statistical data in coverage analysis. In any case, that is almost impossible to do. What is important is to use several of the other coverage analysis tools described in this section to provide substantial physical insight into the nature of the coverage that is being provided. It is this insight into what is occurring that provides the checks and balances to be able to draw conclusions that are truly valid.

10.2.1 Evaluating Earth Coverage

Earth coverage refers to the part of the Earth that a spacecraft instrument* or antenna can see at one instant or over an extended period. As Fig. 10-2 shows, the *instantaneous field of view*, typically called the *FOV* or *footprint*, is the actual area the instrument or antenna can see at any moment. In contrast, the *access area* is the total area on the ground that could potentially be seen at that moment by turning the spacecraft or instrument. In the case of an instrument that covers the entire visible surface of the Earth, these two would always be the same. For most operational instruments they are not.

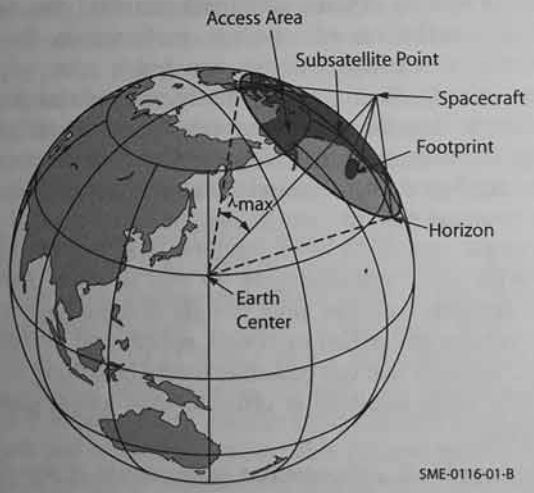


Fig. 10-2. Coverage of the Earth by a Satellite in Low-Earth Orbit. The instrument footprint (also called the field of view or FOV) is the instantaneous area of the ground being covered. The instantaneous access area is the total observable area at that time.

The second important distinction is between the area that can be seen at any one instant versus the rate at which new land comes into view as the spacecraft or instrument moves. Both are important, and either can be vital to mission success. In geosynchronous orbit, the instantaneous area is typically most important because the spacecraft is nearly stationary relative to the Earth's surface. In low-Earth orbit, satellites are moving rapidly over the surface, so the rate at which new land appears is often critical.

The two distinctions above lead to four parameters for Earth coverage:

- *Footprint Area* (*FA*, also *FOV area* or *instantaneous coverage area*) = area that a specific instrument or antenna is viewing at any instant.
- *Instantaneous Access Area* (*IAA*) = all the area that the instrument or antenna could potentially see at any instant.
- *Area Coverage Rate* (*ACR*) = the rate at which the instrument or antenna is sensing or accessing new land.
- *Area Access Rate* (*AAR*) = the rate at which new land is coming into the spacecraft's access area.

For an instrument that covers all of the area available to it as the spacecraft moves along, the coverage rate and access rate will be the same. For instruments operating only part of the time or continuously selecting the region to be examined, the coverage rate and access rate may be dramatically different. Generally the access area and access rate depend only on the orbit and limiting geometry of the system, so we can easily compute them with only a minimal knowledge of the detailed system design. On the other hand, the actual area coverage rate during spacecraft operations may well depend on the spacecraft control system, power, and management systems, as well as the details of mission operations.

Coverage assessment conveniently divides into two areas: first, an analytic assessment to provide approximate formulas for coverage parameters as a function of mission variables as described in Secs. 10.2.1.1 to 10.2.1.3; second, numerical simulations to provide coverage Figures of Merit for more detailed studies. These are discussed in Sec. 10.2.1.4.

10.2.1.1 Analytic Approximations

In this section, we present analytic approximations for various Earth coverage parameters. All of the formulas here take into account the spherical surface of the Earth, but do not account for oblateness, orbit eccentricity, or the rotation of the Earth underneath the orbit. These effects, plus coverage by multiple satellites, are ordinarily accounted for in numerical simulations as described in Sec. 10.2.1.4. Analytic expressions for coverage taking the Earth's rotation into account are given in Table 8-11 in Sec. 8.4.1 and Wertz [2009]. Analytic approximations appropriate for elliptical orbits are given in Table 8-12 in Sec. 8.4.2 and Wertz [2009].

All of the formulas here are derived directly from the single-satellite geometry described in Sec. 8.3. In particular, we will use the notation developed there and summarized in Fig. 8-28 in Sec. 8.3.1. In this section we will parameterize coverage in terms of the Earth Central Angle, λ.† However, we can use Eqs. (8-25) to (8-29) to

* Throughout this section we will use *instrument* to refer to any spacecraft sensor or antenna for which we want to compute coverage.

† λ may be thought of either as an angle at the Earth's center or as a distance measured along the Earth's surface and, therefore, may be expressed in either degrees or kilometers. We will use these two views interchangeably as convenient for the problem at hand.

transform each of the formulas below into one for either the spacecraft-centered nadir angle, η, or spacecraft elevation angle, ε, seen from the ground.

As Fig. 10-2 shows, the instrument footprint is normally a beam projected onto the Earth's surface with a circular cross section substantially smaller than the access area. The nomenclature and computational geometry for the footprint are in Fig. 10-3. (For instruments that see very large portions of the Earth, we can use the access area formulas below. For the projection of rectangular FOVs, corresponding to a typical focal plane array, onto the surface of the Earth, see Eqs. (8-59) to (8-62) in Sec. 8.3.4.)

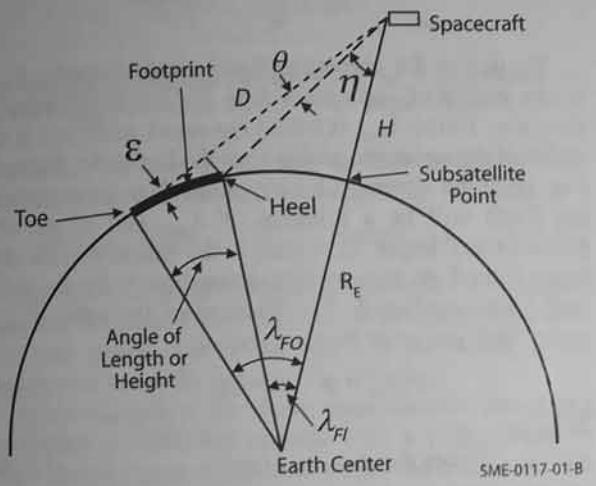


Fig. 10-3. Computational Geometry for Footprint Parameters. Note that ε is always measured at the toe because that is where performance is the worst.

The length (also called the height) of the footprint, L_F, is given by:

$$L_F = K_L(\lambda_{FO} - \lambda_{FI}) \tag{10-1a}$$

$$\approx D(\sin\theta / \sin\varepsilon) \tag{10-1b}$$

where the variables are defined in Fig. 10-3 and, for λ expressed in degrees, the conversion factor, K_L, is

$K_L = 1$	for length in deg
$K_L = 111.319\,543$	for length in km
$K_L = 60.107\,744\,7$	for length in nmi

Note: the linear approximation given in Eq. (10-1b) is computationally convenient but can be very inaccurate, particularly near the horizon where ε is small. (For a satellite at an altitude of 1,000 km with a 1 deg diameter beam, the error in Eq. (10-1b) is 400% at $\varepsilon = 1$ deg, 10% at $\varepsilon = 15$ deg, and 1% at $\varepsilon = 60$ deg). However, the alternative computation of Eq. (10-1a) is much less convenient. To find the footprint length for a given spacecraft elevation angle at the toe of the beam, ε, we begin by computing η and λ at the toe, then subtract the beam width, θ, from η to determine η at the heel, compute λ at the heel, and finally subtract to get the footprint length from Eq. (10-1a). An alternative that improves the approximation somewhat is to the center rather than the toe of the beam. Because the toe represents the worst-case link budget, it is most often used for performance computations and therefore is commonly used for geometry calculations as well.

The *footprint width*, W_F, is given by:

$$W_F = R_E \operatorname{asin}(D \sin\theta / R_E) \tag{10-2a}$$

$$\approx D \sin\theta \tag{10-2b}$$

where $R_E = 6{,}378.136$ km is the radius of the Earth, θ is the beam width, and D is the distance from the spacecraft to the toe of the footprint.* Here the error in the approximation in Eq. (10-2b) is proportional to $1 - (W_F / \sin W_F)$† and is generally small relative to other errors. Thus, Eq. (10-2b) is adequate for most practical applications.

Finally, if we assume that the projection on the ground is an ellipse, then the footprint area, F_A, is given by:

$$F_A \approx (\pi/4) L_F W_F \tag{10-3}$$

Assuming that L_F was computed by Eq. (10-1a), the error in ignoring the curvature of the Earth in Eq. (10-3) is again proportional to $1 - (W_F / \sin W_F)$ and is negligible for most applications.

The *instantaneous area* coverage rate for the beam is defined by:

$$ACR_{instantaneous} \equiv F_A / T \tag{10-4}$$

where T is the exposure time or dwell time for the instrument. The *average area coverage rate*, ACR_{avg}, will also be a function of the *duty cycle*, DC, which is the fraction of the total time that the instrument is operating, and the average between the footprint, O_{avg}, which is the amount by which two successive footprints cover the same area (typically about 20%):

$$ACR_{avg} = \mathrm{DC}\,(1 - O_{avg})\, F_A / T \tag{10-5}$$

Computing the instantaneous access area, IAA, will depend on the shape of the potential coverage area on the ground. Figure 10-4 shows several typical shapes. The most common of these is Fig. 10-4A, which assumes that the instrument can work at any point on the Earth for which the spacecraft elevation is above ε. This corresponds to a small circle on the Earth of radius λ centered on the current subsatellite point. However, some instruments, such as radar, cannot work too close to the subsatellite point. As Fig. 10-4B shows, these instruments have both an outer horizon, λ_1, and an inner horizon, λ_2.

For instruments with an access pattern as shown in Fig. 10-4A, the instantaneous access area, IAA, will be just the area of the small circle, that is,

$$IAA = K_A(1 - \cos\lambda) \tag{10-6}$$

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* See Sec. 8.3.4 for a discussion of the projection of non-circular fields of view onto the Earth.

† Here W_F should be in radians as seen from the center of the Earth.

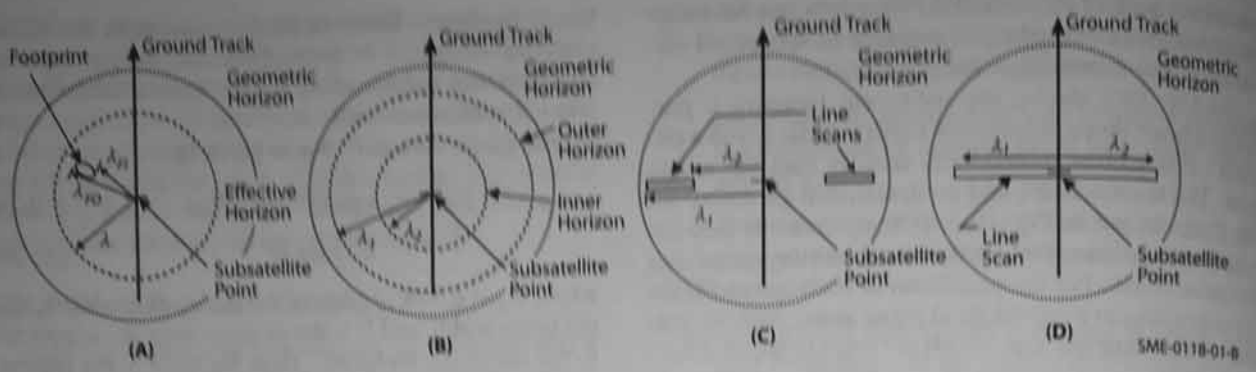


Fig. 10-4. Typical Spacecraft Access Areas. See Table 10-3 for formulas. See Sec. 8.3.4 for sensor FOV analysis.

where

$K_A = 2\pi \approx 6.283\,185\,311$ for area in steradians
$K_A = 20{,}626.4806$ for area in deg^2
$K_A = 2.556\,041\,87 \times 10^8$ for area in km^2
$K_A = 7.452\,225\,69 \times 10^7$ for area in nmi^2

The instantaneous access areas or access lengths for the other patterns in Fig. 10-4 are given in Table 10-3, which also summarizes all of the coverage formulas for these patterns. These access area formulas take into account the curved surface of the Earth and are accurate for any access area size or satellite altitude to within very small corrections for the Earth's oblateness.

We now wish to determine the length of time a particular point on the Earth is within the satellite access area and the access area rate at which land enters or leaves the access area. Consider a satellite in a circular orbit at altitude H. The orbit period, P, in minutes is given by:

$$P = 1.658\,669 \times 10^{-4} \times (6{,}378.14 + H)^{3/2}, \; H \text{ in km} \quad (10\text{-}7)$$

$$P = 4.180\,432 \times 10^{-4} \times (3{,}443.9 + H)^{3/2}, \; H \text{ in nmi} \quad (10\text{-}8)$$

We define the *maximum Earth Central Angle*, λ_{max}, as the radius of the access area for the observation in question. Twice λ_{max} is called the *swath width* and is the width of the coverage path across the Earth. As shown in Fig. 10-5, the coverage for any point **P** on the surface of the Earth will be a function of λ_{max} and of the off-ground-track angle, λ, which is the perpendicular distance from **P** to the satellite ground track for the orbit pass being evaluated. The fraction of the orbit, F_{view}, over which the point **P** is in view is:

$$F_{view} = \phi / 180 \text{ deg} \quad (10\text{-}9)$$

where:

$$\cos \phi = \cos \lambda_{max} / \cos \lambda \quad (10\text{-}10)$$

Therefore the *time in view*, T, for the point **P** will be

$$T = PF_{view} = \left(\frac{P}{180 \text{ deg}}\right) \text{acos}\left(\frac{\cos \lambda_{max}}{\cos \lambda}\right) \quad (10\text{-}11)$$

See Sec. 8.4 for a detailed discussion of the computation of all relevant parameters for a single target or ground station pass, including the corrections that take into account the rotation of the Earth during the pass.

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Table 10-3. Coverage Formulas for Patterns Shown in Fig. 10-4. See text for definition of variables. In pattern D, the minus sign applies if λ_2 is on the same side of the ground track as λ_1. The approximation for footprint area is invalid when $\varepsilon \approx 0$. The ACR formulas for patterns C and D assume that the instrument is side-looking. P is the orbit period.

Pattern	Typical Application	Footprint Area (*FA*)	Instantaneous Access Area (*IAA*) or Length (*IAL*)	Area Coverage Rate (*ACR*)	Area Access Rate (*AAR*)
A	Omni-antenna, Ground-station Coverage, General Sensing	$(\pi D K_L / 4) \sin\theta \times (\lambda_{FO} - \lambda_{FI}) \approx (\pi D^2 / 4) \times \sin^2 \theta / \sin \varepsilon$	$IAA = K_A (1 - \cos \lambda)$	$\frac{F_A(1 - O_{avg})DC}{T}$	$\frac{2K_A \sin\lambda}{P}$
B	Doppler Radar	As above	$IAA = K_A (\cos \lambda_2 - \cos \lambda_1)$	As above	$\frac{2K_A \sin\lambda}{P}$
C	Synthetic Aperture Radar	As above	$IAL = 2 K_L (\lambda_1 - \lambda_2) \approx 2K_L D \frac{\sin\theta}{\sin\varepsilon}$	$\frac{2K_A(\sin\lambda_1 - \sin\lambda_2)}{P}$	$\frac{2K_A(\sin\lambda_1 - \sin\lambda_2)}{P}$
D	Scanning Sensor	As above	$IAL = K_L (\lambda_1 \pm \lambda_2)$	$\frac{K_A(\sin\lambda_1 \pm \sin\lambda_2)}{P}$	$\frac{K_A(\sin\lambda_1 \pm \sin\lambda_2)}{P}$



Fig. 10-5. Earth Coverage Geometry. λ is the *off ground-track angle* and $2\lambda_{max}$ is the *swath width*. **P** is the target or ground station.

Finally, the area access rate as the satellite sweeps over the ground for the access area of Fig. 10-4A is:

$$AAR = (2\,K_A \sin\lambda)\,/\,P \qquad \text{(Pattern A)} \qquad (10\text{-}12)$$

Formulas for other patterns are in Table 10-3. Again note that because of the curvature of the Earth's surface, this area access rate is **not** equal to the width of the access area times the subsatellite velocity.

As an example of the above computations, consider a spacecraft at 2,000 km altitude with a 1 deg diameter beam looking perpendicular to the ground track at an elevation angle of 10 deg as seen from the ground. Our linear estimate of the footprint height is 446 km from Eq. (10-1b).* However, from Eq. (10-1a) we see that the true height is 355 km. From Eqs. (8-26), (8-27), and (8-30) in Sec. 8.3.1 we determine λ_{FO} = 31.43 deg and λ_{FI} = 28.24 deg. The footprint width from Eq. (10-2b) is 77 km. From Eq. (10-3) the footprint area is 21,470 km^2. The accuracy of the area is proportional to 1 (77/6,378)/sin (77/6,378) = 0.002%. The ground track velocity is the circumference of the Earth divided by the orbit period [from Eq. (10-7)] = 40,075 km/127 min = 315.6 km/min = 5.26 km/s. Multiplying this by the footprint height of 355 km gives a crude estimate of the area coverage rate of 1,867 km^2/s. Using the more accurate formula in Table 10-3 (Pattern D) and the values of λ above, we obtain a more accurate value of ACR = $2.556 \times 10^8 \times$ (sin 31.43 deg – sin 28.24 deg)/(127 × 60) = 1,620 km^2/s which implies an error of 15% in the less accurate approximation.

10.2.1.2 Identifying Coverage Patterns

As previously indicated, Earth coverage is not a statistically distributed parameter. Coverage tends to come in "chunks." Thus, at any given time, whether there is Earth coverage at some point on the Earth's surface is very strongly correlated with whether there was coverage at the last time step. Consequently, we need not only statistical data generated from simulations but also other types of analyses that can begin to identify coverage patterns and how coverage works in both a qualitative sense and analytical approximations.

One of the most useful approaches for analyzing coverage patterns is simply the *ground track plot* or *ground trace*, i.e., the trace of the subsatellite point over time. (For a detailed discussion of ground track plots and relevant formulas and examples, see Wertz [2009], Sec. 9.3.) A typical ground track plot is shown in Fig. 10-6 for a satellite in an orbit at an altitude of 1,000 km and inclination of 60 deg. In previous plots, we have drawn the access area on the surface of the Earth corresponding to the area that is visible for a specific satellite. In this figure, we draw a circle of the same size centered not on the position of the satellite, but on a specific target or ground station, in this case, Miami. For a satellite at 1,000 km and minimum elevation angle of 5 deg, the maximum Earth Central Angle is 25 deg, i.e., any point on the ground within 25 deg of the subsatellite point is within view. Conversely, whenever the subsatellite point for a spacecraft is within 25 deg of a specific target or ground station, then that target can be seen by the spacecraft. Therefore, the small circle of 25 deg radius centered on Miami encloses all possible subsatellite points for which a satellite at 1,000 km altitude can see Miami with a minimum elevation angle at the target greater than 5 deg.†

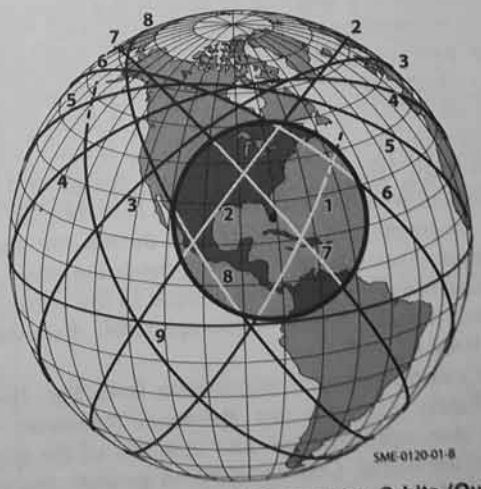


Fig. 10-6. Ground Track for 8 Successive Orbits (Out of 14 per Day) for a Satellite in a 1,000 km Circular Orbit. The heavy circle covers subsatellite points for which the satellite will be at an elevation angle, ε, of 5 deg or higher.

* As indicated previously, this estimate would be substantially improved if the 10 deg elevation angle was at the center of the beam. However, we would then need to keep track of beam-center parameters for the geometry and beam-edge parameters for performance estimates.

† Another way to think of the construction of this plot is to consider a cone of 85 deg radius centered on Miami, with its axis toward the zenith. The intersection of this cone with a sphere 1,000 km above the Earth produces the small circle shown in the figure. In this equivalent representation, the small circle represents all possible locations of the satellite that can see Miami with the appropriate conditions.

By simply looking at the plot, it is clear what the nature of the coverage will be. Notice that the orbits on the plot are numbered consecutively and that coverage of Miami is provided on orbits 1 and 2 and is then not available on orbits 3, 4, and 5 but is available again on orbits 6, 7, and 8. For this example, there will be 2 or 3 orbits of coverage followed by 3 or 4 orbits with no coverage and then another 2 or 3 orbits of coverage after which the process repeats. The specific coverage will depend on how the orbit lies with respect to the particular target. The orbit period at this altitude is approximately 105 min. A pass that goes directly over Miami will provide coverage for approximately (50/360) × 105, or a bit less than 15 min. Thus, an "orbit of coverage" will consist of somewhat less than 15 min of coverage, followed by a bit more than 90 min of no coverage.

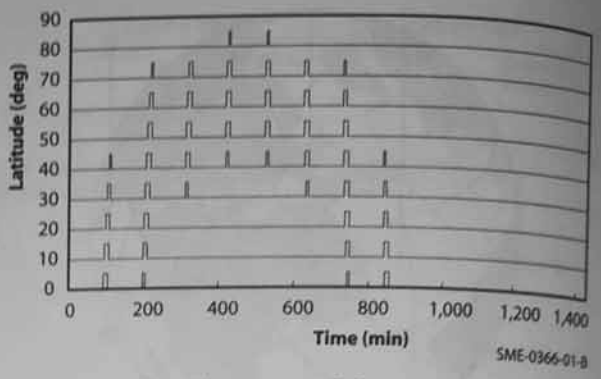
What happens if we move our ground station either north or south? As the ground station moves toward the equator, the coverage patterns become more symmetric, with coverage occurring twice per day at times centered 12 hrs apart. As we move toward the pole, coverage becomes more "one-sided" in terms of its daily distribution. There will be a larger number of orbits in a row that provide coverage, and then a longer gap with no coverage, with the process repeating on a daily basis. If we get close enough to the pole, there will be either no coverage if the radius of the coverage region is less than 90 deg minus the inclination, or a short region of coverage on every orbit.

We can learn a great deal about the nature of coverage and how it is distributed by simply looking carefully at a ground track plot for the orbit. One way to demonstrate this coverage as a function of latitude is a *coverage histogram* or *tombstone plot* as shown in Fig. 10-7. The horizontal axis is time and the various lines which look like a strip chart recorder are the coverage for points at the longitude of Miami and at 10 deg steps in latitude. When the line is up, there is coverage; when it is down, there is no coverage. Here you can see clearly the general characteristic of what we have discussed, including the symmetry of coverage that occurs near the equator and the asymmetry that occurs as we approach the poles. The extensive coverage at 40 deg latitude in this example is the basis for the Repeat Coverage Orbit discussed in Sec. 10.4.2.6.

We can also use the information from Fig. 10-6 to immediately calculate one statistical parameter. The access area represents just less than 5% of the surface area of the world. Consequently, in general, the mean coverage for any single point on the surface of the Earth will be 5%. Five percent of the Earth is covered at any one time, and over the course of time a random point will be covered approximately 5% of the time. While this is true, it does remarkably little to reflect the reality of the curves in Fig 10-7 or the understanding we can gain by looking at Fig. 10-6.

10.2.1.3 Analytic Coverage Computations

Figure 10-5 in Sec. 10.2.1.1 shows the geometry for a single ground station or target pass. Note that this is the same computational geometry that was done in detail in Sec. 8.4.1. Consequently, all of the tables and computations in that section are applicable to Earth coverage for a single point. These computations were summarized in Table 8-11 in Sec. 8.4.1 and are available on the book website for that table.



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Fig. 10-7. Histograms of Coverage Over a Day for Ground Stations at the Same Longitude Shown in Fig. 10-6, but Varying Latitudes. Note the asymmetry as the ground station moves toward the pole. See text for discussion.

It is also of interest to compute the coverage as a function of latitude, L. Conceptually, this is most conveniently done by thinking of the orbit and coverage pattern as fixed, and the Earth as rotating underneath as shown in Fig. 10-8.

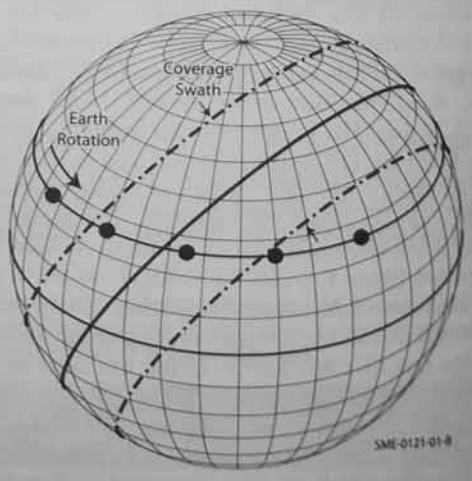


Fig. 10-8. Alternative View of a Rotating Ground Track. For some computations it is convenient to think of the orbit coverage pattern as fixed in inertial space and the Earth rotating under it. The large dots represent locations of a ground target on successive passes.

The large dots show the position of a ground station or target on successive orbits. Here we can see both the discrete nature of coverage and the derivation of the continuous analytic formulas. If the ground swath is thinner then the spacing between successive dots, then we may or may not get coverage on a given ascending or descending

pass. If it is wider than one space but less then two spaces, then we will get coverage on either 1 or 2 passes at each ascending and descending node. Thus, we could divide the world into latitude bands with each band having a discrete set of possible coverage patterns, corresponding approximately to the histogram of Fig. 10-7.

To obtain a continuous analytic approximation, we note that the arc length along the small circle connecting the dots is directly proportional to the percent coverage at that latitude.* Specifically, we assume that there is a satellite in a circular orbit at inclination i and that observations can be made at any off-ground-track angle less then or equal to λ_{max} on either side of the ground track. We also assume that L is positive, i.e., in the northern hemisphere. (The extensions are straightforward for the southern hemisphere or nonsymmetric observations.) As shown in Fig 10-9 there will be either no coverage, a single long region of coverage, or two shorter regions of coverage depending on the latitude and inclination as shown in Table 10-4.

Table 10-4. Coverage vs. Latitude.

Latitude Range	No. of Coverage Regions	Percent Coverage
$L > \lambda_{max} + i$	0	0
$i + \lambda_{max} > L > i - \lambda_{max}$	1	$\phi_1/180$
$i - \lambda_{max} > L > 0$	2	$(\phi_1 - \phi_2)/180$

where:

$$\cos\phi_{1 \text{ or } 2} = \frac{\pm \sin\lambda_{max} + \cos i \sin L}{\sin i \cos L} \qquad (10\text{-}13)$$

where the minus sign applies for ϕ_1 and the plus sign for ϕ_2. Here ϕ is one-half the longitude range over which coverage occurs. The formula in the third column above represents the fraction of all points at a given latitude in view of the satellite during one orbit. This is approximately equal to the fraction of orbits that will cover a given point at that latitude.

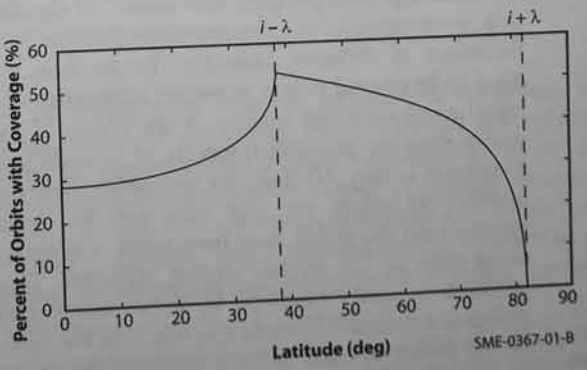
As an example, consider a satellite in a 62.5 deg inclined orbit which can see to an off-ground-track angle, $\lambda_{max} = 20$ deg. At a ground station latitude of 50 deg, the percent coverage will be 49.3%. On any orbit, 49% of the points at a latitude of 50 deg will be within view of the satellite at some time. Conversely, a given point at 50 deg latitude will be covered at some time on approximately 49% of the satellite orbits.



Fig. 10-9. Single Orbit Coverage is a Function of Latitude, Orbit Inclination, and Swath Width. See text for formulas.

Because there is only one coverage region, the covered orbits will occur successively during the day. If the satellite orbit period is 2 hrs, then our hypothetical ground station at 50 deg latitude will typically see the satellite on six successive orbits followed by six orbits of no coverage. The number and duration of coverage passes on a given day will depend on where the ground station is located with respect to the orbit node.

Figure 10-10 shows a representative plot of Earth coverage as a function of latitude, based on the above equations. For latitudes below $i - \lambda$, there will be two distinct coverage regions. In this area, the coverage increases somewhat as we go further from the equator, because the orbit is tilting more and therefore covering more of a given latitude line. At a latitude of $i - \lambda$, we transition from two coverage regions to a single coverage region which is extremely long. This can be seen in Fig. 10-9 and occurs where the underlying latitude line is just tangent to the southernmost edge of the coverage



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Fig. 10-10. Representative Plot of Earth Coverage as a Function of Latitude. Note that the chart gives the percentage of orbits which provide coverage, but does not reflect the coverage distribution, which is equally important. The satellite is assumed to be in a circular orbit at 1,000 km, an inclination of 60 deg and a minimum elevation angle of 10 deg.

* We need to emphasize once again that coverage is not a statistical variable. If the orbit is a 1-day repeating ground track, then the targets will appear in the same locations with respect to the orbit every day. In this case, coverage will be a function of both latitude and longitude and each Earth location will have its own fixed coverage pattern. If the orbit is not a repeating ground track, then the "percent coverage" rules are applicable.

band. This is the latitude with the highest coverage and is frequently missed in histograms such as those described above. As we proceed further north, coverage drops as the single coverage region gets smaller until coverage disappears entirely at a latitude of $i + \lambda$.

There are several conclusions that can be drawn from this curve. First of all, coverage varies significantly with latitude and we can design our orbit inclination to provide the greatest coverage where it is most needed. For example, a communications constellation might choose to use an inclination such that the coverage is maximized over the mid northern latitudes, where the principal population densities occur. Our second conclusion is that the coverage curve from Fig. 10-10 is interesting and informative but does not tell the whole story. It does not provide the information on the distribution of coverage that we get from either the ground track plot or the coverage histogram. While useful, it needs to be combined with other types of information in order to provide a clear understanding of coverage patterns.

10.2.1.4 Numerical Simulations

The analytic formulas presented above provide an easy and rapid way to evaluate Earth coverage and provide good physical insight into the nature of that coverage. However, to provide additional numerical details, simulations are nearly always required. The numerical simulations themselves are relatively straightforward. Simply set up representative points you are interested in or a grid of points covering the Earth, "fly" the spacecraft over the grid for one or more days, and collect the results. The difficult part is interpreting the meaning of the results and avoiding both computational problems and interpretations which misrepresent the real coverage.

There are four principal numerical interpretation problems that the user should be aware of in developing and using point coverage simulations. These are:

Spacing of Grid Points:

If we use a grid of points in our numerical simulation, we frequently want to compare coverage performance at different latitudes. To do this, we need to have grid points covering approximately equal areas over the surface of the globe. If the grid points are at equal intervals of latitude and longitude (every 10 deg, for example) then the number of points per unit area will be much greater near the poles, thus incorrectly weighting polar data in the overall global statistics. This problem is easily resolved by either using grid points at a constant latitude spacing with the number of points at each latitude proportional to the cosine of the latitude, or by adjusting the weighting of the points in creating the global statistics.

Gaps at the Beginning and End of the Simulation:

If the system being simulated is not intended for continuous global coverage, then there will be gaps in coverage for which the statistics are usually important. However, these statistics can be biased by gaps which occur at the beginning and end of the simulation run. For example, if the simulation covers exactly 24 hr and there is a 4-hr gap in coverage during the period when the simulation begins, then this may be misinterpreted as two 2-hr gaps. The easiest solution to this problem is to run the simulation long enough that start and end data have minimal impact on the overall statistics.

Incorrect Step Size:

In a time stepped simulation, the simulation itself does not see a continuous coverage distribution but rather a series of steps. If the steps are far apart in order to make the simulation easy to run it is possible to step "over" some of the grid points such that coverage which actually exists is missed. The principal requirement to avoid this happening too much is to keep the simulation step small relative to the dimensions of the coverage regions.

Incorrect Interpretation of Results:

As we have discussed previously, Earth coverage is not a statistically distributed phenomenon. Consequently, statistical figures of merit can prove extremely misleading. The principal solution to this problem is twofold. First, we should use more than one figure of merit for evaluating coverage so that multiple pieces of information are available and can be compared. The second approach is to use any of the more analytic approaches to understand the coverage such that we can be sure that the statistical data reflects a true assessment of the nature of the coverage.

Our most important step in creating a coverage simulation is to find the appropriate way to accumulate coverage statistics and to evaluate the quality of coverage. This is done by way of *coverage figures of merit* which are the numerical mechanisms for comparing the coverage of satellites and constellations. We would like to find a figure of merit which is physically meaningful, easy to compute in a numerical simulation, and fair in comparing alternative orbits and constellations. In the end, no one coverage figure of merit will prove successful by itself and a combination of techniques is most appropriate. The most common coverage figures of merit are:

Percent Coverage:

The *percent coverage* for any point on the grid is simply the number of times that point was covered by one or more satellites divided by the total number of simulation time steps. The advantage of percent coverage is that it shows directly how much of the time a given point or region on the ground is covered. However, it does not provide any information about the distribution of gaps in that coverage.

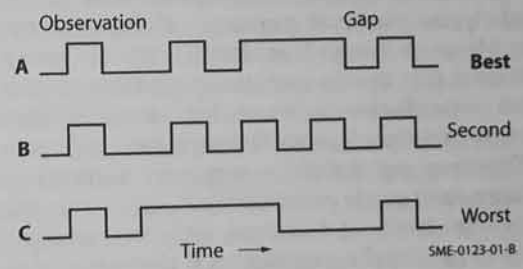
Table 10-4◀, Fig. 10-10, Eq. 10-13◀

Maximum Coverage Gap (= Maximum Response Time):
The *maximum coverage gap* is simply the longest coverage gap encountered for an individual point. When looking at statistics over more than one point, we can either average the maximum gaps or take their maximum value. Thus, the worldwide *mean maximum gap* would be the average value of the maximum gap for all the individual points, and the worldwide *maximum gap* would be the largest of any of the individual gaps. This statistic conveys some worst-case information, but it incorrectly ranks constellations because a single point or a small number of points determine the results. Thus, the maximum coverage gap, or maximum response time, is a poor figure of merit.

Mean Coverage Gap:
The *mean coverage gap* is the average length of breaks in coverage for a given point on the simulation grid. To compute mean gap statistics, we must have three counters for each point on the simulation grid. One counter tracks the number of gaps. A second tracks total gap duration. The third tracks the duration of the current gap and is reset as needed. During the simulation, if no satellite covers a given point on the grid, we increment the gap length counter (3) by one time step. If the point is covered but was not covered the previous time (indicated by a value of the gap length counter greater than 0), then we have reached the end of an individual gap. We increment the counter for the number of gaps (1) by one and add the gap duration to the total gap counter (2) or incorporate it in other statistics we want to collect. The final mean coverage gap is computed by dividing the total gap length by the number of gaps. As noted above, what happens at the beginning and end of the simulation influences all statistics relating to gap distribution.

Time Average Gap:
The *time average gap* is the mean gap duration averaged over time. Alternatively, it is the average length of the gap we would find if we randomly sampled the system. To compute the time average gap, two counters are required, one for the current gap length, and one for the sum of the squares of gap lengths. During the simulation, if no satellite covers a given point on the grid, add one to the current gap length counter. If the point is covered, square the current gap length, add the results to the sum of the squares counter, and reset the current gap length counter to zero. (If the current gap length counter was previously 0, then no change will occur in either counter.) The time average gap is computed at the end of the simulation by dividing the sum of the squares of the gaps by the duration of the simulation.

Mean Response Time:
The *mean response time* is the average time from when we receive a random request to observe a point until we can observe it. If a satellite is within view of the point at a given time step, the response time at that step will be 0.* If the point in question is in a coverage gap, then the response time would be the time until the end of the coverage gap. In principle, response time should be computed from a given time step to the end of a gap. But by symmetry we could also count the time from the beginning of the gap—a computationally convenient method with the same results. Thus, the response time counter will be set to 0 if a point is covered at the current time step. We advance the response time counter by one time step if the point is not now covered. The mean response time will then be the average value of all response times for all time steps. This figure of merit takes into account both coverage and gap statistics in trying to determine the whole system's responsiveness. As shown below, the mean response time is most often the best coverage figure of merit for evaluating overall responsiveness.



	Percent Cov.	Max Gap	Mean Gap	Time Average Gap	Mean Response Time
A	60	2	1.33	0.6	0.5
B	50	2	1.25	0.7	0.6
C	60	3	2.00	1.0	0.7

Fig. 10-11. Simulation of Coverage Figures of Merit. See text for explanation.

To illustrate the meaning and relative advantages of these figures of merit, Fig. 10-11 diagrams a simplified coverage simulation for 3 satellite systems: A, B, and C. These could, for example, be 3 sample FireSat II constellations. Our goal is to see events as quickly as possible, and therefore, minimize gaps. Constellation B is identical to A, except for one added gap, which makes B clearly a worse

* One advantage of response time as a figure of merit is that delays in processing or communications (for both data requests and responses) can be directly added to the coverage response time. This results in a *total response time*, which measures the total time from when users request data until they receive it. We can also evaluate minimum, mean, and maximum total response times which have much more operational meaning than simple gap statistics and are still easy to compute.

Table 10-5. Advantages and Disadvantages of Different Approaches to Evaluating Coverage. See text.

Approach	Advantages	Disadvantages	Example
Ground Trace Plot	Very simple to do; provides excellent physical insight	Does not provide quantitative results; hard to compare systems	Fig. 10-6
Coverage Histogram	Simple to calculate; gives excellent insight into multiple coverage characteristics; best overall evaluation approach	Difficult to compare systems; can not be reduced to a single number; may miss key features at locations not evaluated	Fig 10-7
Analytic Coverage Functions	Simple to calculate; gives good insight into percent coverage	Provides no information about gaps or coverage distribution	Fig. 10-10
Numerical Simulations and Coverage Figures of Merit	Relatively easy to determine via simulation; provides simple numerical answers that can be easily compared (but may be misleading)	Requires coverage simulation; many parameters to select; provides statistical information on inherently nonstatistical data; can lead to very misleading results and conclusions	Fig. 10-11

solution than A. C has the same overall percent coverage as A, but the gaps are redistributed to create a rather long gap, making C the worst constellation for regular coverage.

The table at the bottom of Fig. 10-11 shows the numerical values of the figures of merit defined above. The percent coverage correctly ranks constellation A better than B, but because it does not take gap statistics into account, it cannot distinguish between A and C. Similarly, the maximum gap cannot distinguish between A and B, even though B is clearly worse by having an additional gap. In this case, the maximum gap tells us which constellation is worse but cannot distinguish between two constellations that are clearly different.

The mean gap statistic is even more misleading. By adding a short gap to constellation B, the average length of the gaps has been decreased, and consequently, this figure of merit ranks constellation B **above** constellation A. (As discussed in the introduction to Sec. 10.2 this can happen in real constellation statistics. By adding satellites we may eliminate some of the very small gaps, thus **increasing** the average gap length, even though more satellites provide more and better coverage.)

Finally, the time average gap and mean response time in the fourth and fifth columns correctly rank the three constellations in order of preference by taking into account both the percent coverage and gap statistics. Consequently, both of these are better figures of merit than the other three. I believe the mean response time is the stronger figure of merit because it provides a more useful measure of the end performance of the system and because it can be easily extended to include delays due to processing, communications, decision-making, or the initiation of action. However, because each figure of merit represents different characteristics we should evaluate more than one. Specifically, I recommend evaluating mean response time, percent coverage, and maximum gap, and qualitatively (not quantitatively) weighing the results in that order, keeping strongly in mind the caveat at the beginning of Sec. 10.2.

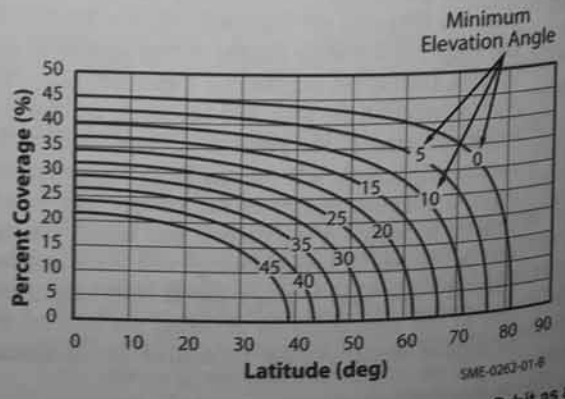
In summary, we have seen that there are four broad techniques available for evaluating Earth coverage:

- Ground track plots
- Coverage histograms
- Analytic coverage assessment (single location or vs. latitude)
- Numerical figures of merit

Table 10-5 summarizes some of the relative advantages and disadvantages of each of these approaches. In the end, there is no substitute for developing physical insight into the nature of the coverage and finding the most effective way to express that for each individual problem.

10.2.1.5 Earth Coverage for Geosynchronous Orbits

By definition, geostationary orbits maintain an approximately fixed location over the Earth. Small amounts of motion due to a low inclination, small eccentricity, or stationkeeping maneuvers have essentially no impact on the coverage area of the satellite. Consequently, the area of Earth covered is essentially fixed and is a function only of the minimum working elevation angle of the system. Figure 10-12 is a plot of the percent coverage of the surface of the Earth as a function of latitude for various minimum elevation angles. These coverage regions are drawn on the Earth in Fig. 10-13. For inclined geosynchronous orbits, all of the mechanisms of the low-Earth orbit analysis are applicable as discussed in Secs. 10.2.1.1 to 10.2.1.4. In addition, the motion of an inclined geosynchronous satellite relative to the surface of the Earth is a relative motion analemma as described in Sec. 8.5.



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Fig. 10-12. Coverage from Geostationary Orbit as a Function of Minimum Elevation Angle.

Table 10-5, Fig. 10-12, Eq. 10-13◀

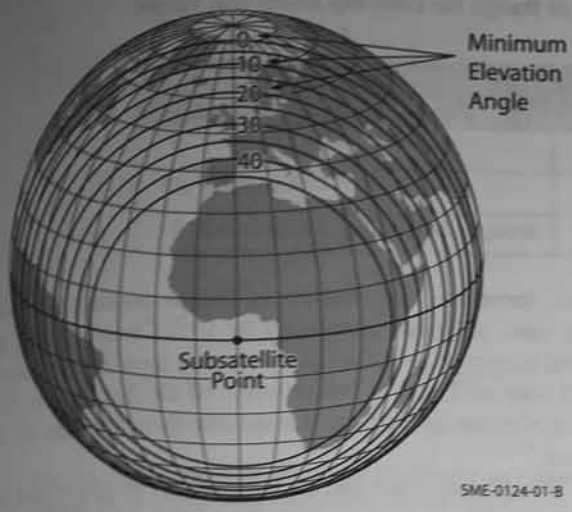


Fig. 10-13. Coverage from Geostationary Orbit Projected onto the Earth.

10.2.1.6 Coverage Analysis Example

Get More

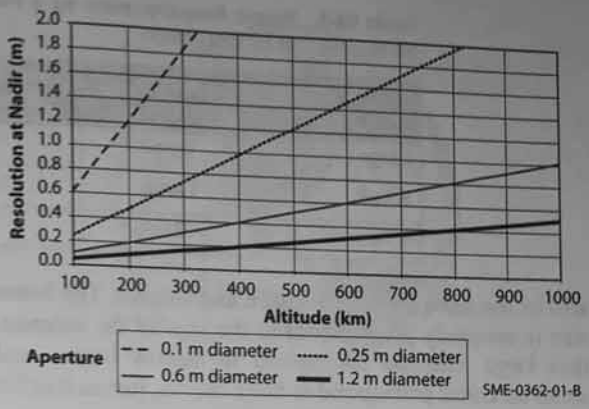
See the book website for a detailed coverage example and simplified method of analyzing coverage.

10.2.2 Payload Performance vs. Altitude

As discussed in Sec. 5.3, the orbit, and particularly the altitude, affect a great many system parameters. (See Table 5-4 in Sec. 5.3.) The effect of the altitude of the satellite on the performance of the payload depends largely on whether we are looking down (observation payloads discussed in Sec. 10.2.2.1), illuminating the surface and looking at the return (radar and laser payloads, discussed in Sec. 10.2.2.2), or communicating (communications payloads, discussed in Sec. 10.2.2.3).

10.2.2.1 Observation Systems

Most observation payloads looking back at the Earth operate in visible or IR regions of the spectrum because the atmosphere is largely transparent at these wavelengths and the resolution is good. (See Chap. 17.) The key issue for most observation systems will be the *resolution*, i.e., the fineness of the detail that we can see on the Earth's surface, which depends on the size of the instrument and the altitude, as shown in Fig. 10-14. It also depends on the quality of the optical instrument, but space instruments are sufficiently expensive that they are essentially all diffraction-limited, i.e., effectively perfect optics. (The effects of diffraction on the image are discussed in Chap. 17.) Another critical characteristic is that the *brightness* (the amount of light per square degree) of an object on the Earth depends on the Sun angle and cloud cover, but does **not** depend on the distance. The amount of light from any object arriving at a camera lens, falls off as the square of the distance. (At twice the distance, the light fills a sphere which is twice as large and, therefore, has 4 times the surface area so the light per square meter is 1/4th as much.) But the angular size of the



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Fig. 10-14. Resolution vs. Altitude at Visible Wavelengths for Satellites in Low-Earth Orbit. The resolution in the IR will be 4 to 6 times worse because of the longer wavelength.

object is only 1/2 what it was and the angular area it subtends is also 1/4th as much. Therefore, the light per unit of angular area, or brightness, is independent of distance. This is clear when we take photos or simply look at the world around us. People that are twice as far away are half as high, but just as bright as those that are closer. Therefore, as the satellite gets higher, we don't need more light gathering power, but the resolution, i.e., the size of smallest object that we can see, gets worse.

Note that atmospheric attenuation, described in Chap. 17, is very strong, but does not change as the altitude changes. Spacecraft are above essentially all of the atmosphere, so that the amount of air that the light goes through depends on the elevation angle as seen from the ground, but not on the range to the spacecraft. Therefore, atmospheric attenuation depends on the elevation angle and the wavelength, but not on the distance or altitude.

10.2.2.2 Radar and Laser Systems

Space-based radar and space-based lasers are generally very high power devices because of the large distances involved in space missions. Nonetheless, they have been used throughout the space program to serve a variety of purposes. There are a number of advantages to the use of both radar and lasers in space, as discussed in Chap. 17. First, we can control both the amount of illumination and the spectral characteristics in order to get as much information as possible. In addition, we can adjust the wavelength to provide cloud penetration as in the case of using radar to map the surface of Venus. The largest detriment to this approach is that we must carry our own source of radar or laser illumination, which typically requires very high levels of power.

The dependence of radar power on distance depends on how the radar is used, as shown in Table 10-6. Space-based radar works in a series of *dwells*, or looks, in which a burst of energy is sent out by the radar antenna and typically received by the same antenna after bouncing off the target. Since the same antenna is used for both transmit and receive, the beam size or angular diameter of the field of view of the radar,

Table 10-6. Power Requirements as a Function of Range for Differing Beam and Target Sizes. See text for discussion.

Beam Size	Target Size	# of Dwells Required	Power per Dwell	Power for Whole Target
Large	Small	1	r^4	r^4
Small	Large	$1/r^2$	r^2	Independent of r
= Target	= Beam	1	Independent of r	Independent of r

will be the same for both transmit and receive. The beam size is inversely proportional to the size of the antenna, thus large antennas correspond to narrow beams and small antennas correspond to large beams, just as it is for parabolic radio antennas.

Ordinarily, we think of the beam size as being much larger than the target we are looking at. For example, in using radar to search for spacecraft, we will use a large beam to search for spacecraft with a small angular diameter (row 1 in Table 10-6). Assuming that we hit the target with the radar beam, a single dwell or look is sufficient to see the target. However, the required power per dwell increases as r^4, where r is the distance between the radar and the target. As the target gets farther away, the area covered by the radar beam increases as r^2, which means we must increase the power as r^2 to provide a given level of illumination at the target. With a given level of illumination, the surface brightness of the target in terms of the reflected radar energy will be independent of distance. However, the angular diameter of the target as seen from the radar antenna will fall off as $1/r^2$ such that the total intensity will fall off as $1/r^2$. This implies a need to further increase the power by r^2 to account for the smaller angular size. The net effect is that the power required with a large radar beam and a small target increases as r^4. Because the distances involved for a space-based radar are typically very large, they are ordinarily very high-power devices.

Next, consider what happens if the target is much larger than the beam size (row 2 in Table 10-6). This would occur, for example, if we want to make a radar map of the state of Texas. As we increase the distance between the target and the radar, the area covered on the ground becomes larger by r^2. In order to maintain a fixed level of illumination per square meter, we must increase the power per dwell proportional to r^2. However, we do not have the same problem in the receive antenna. The size of the receive beam is the same as the size of the area being illuminated. The surface brightness is constant, therefore, the energy return will also be constant. From the point of view of the radar, it is very much like looking at the surface of the Earth illuminated by the Sun. There is a constant illumination level irrespective of distance (due to the increased transmitter power in proportion to r^2); therefore, the surface brightness remains the same and the exposure time or received energy also remains the same. Consequently, the total power required per dwell increases only as r^2 rather than as r^4. What's more, is that the area covered by each dwell also increases as r^2 and, therefore, the number of dwells required to cover the state of Texas is reduced by $1/r^2$. Consequently, the total power required to make a radar map from space of the state of Texas is independent of the distance so long as the radar beam remains smaller than the size of the state.*

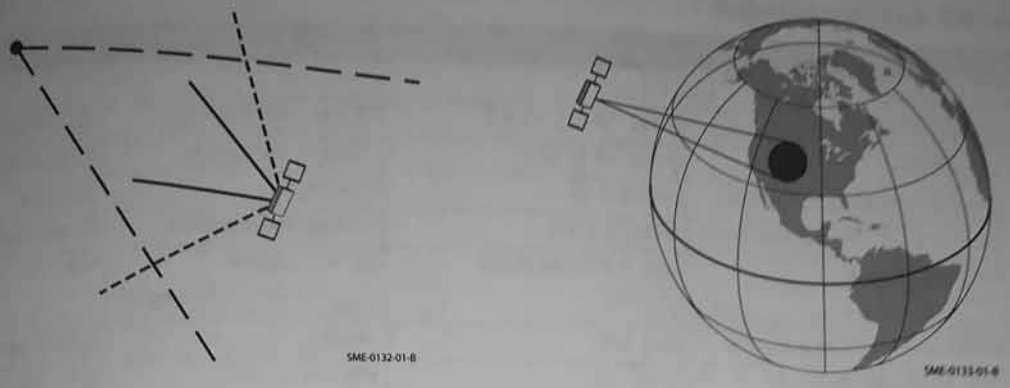
Finally, consider the case in which the target and the beam are the same size (row 3 in Table 10-6). Note that for a target of fixed size, this means that the antenna must get bigger as I get farther away in order to make the beam smaller so that it remains the same angular size as the target. In this case, the power per dwell required to provide a given level of illumination is independent of distance. All of the energy transmitted by the beam is falling on the target and none is being wasted. Consequently, the energy per square meter at the target is independent of distance (it is simply the energy in the beam divided by cross-sectional area of the target in m^2). As in the case above, I'm looking at a uniformly illuminated target, so once again, the required power per dwell will be independent of distance. Because the target and the beam are the same size, I require only a single look to see it and, consequently, the power required to view the entire target is independent of distance. Of course the practical limitation here is that as I get farther and farther away, I must use larger and larger diameter antennas in order to make the beam as small as the size of the target. If I do this, however, I can create a radar for which the power requirement is independent of distance.

The above results seem strange at first glance. The physical reason that the power required is independent of distance is that I am being efficient in using all of the energy that I am transmitting. None of it is wasted by spilling out into space. This is illustrated conceptually in Fig. 10-15. The same logic applies to laser illumination or the use of a flashlight and a pair of binoculars on the surface of the Earth.

10.2.2.3 Communications Systems

Communications systems follow essentially the same logic as radar systems, but without the need for observ-

* Another way to think of this is as follows. I need to illuminate the whole state with so many W/m^2 and, therefore, so many Watts. (There is a reasonably well-defined number of m^2 in Texas, even though that number is large.) I am being efficient in only taking a "picture" of those m^2 that are currently being illuminated. Therefore, I only need the prescribed total number of Watts, irrespective of how far away the "camera" and the "light source" are.



(A) Power Proportional to r^4 (B) Power Independent of r

Fig. 10-15. Power Requirements for Radar or Laser Illumination of Surfaces. (A) A broad illuminator and a small target. (B) A narrow illuminator and a large target. See text for discussion.

ing the reflected return signal. Normally, we have a large beam and a small target, as when we're trying to send signals to a spacecraft or a spacecraft at Jupiter is trying to send signals to Earth. In this case, the power required increases as r^2, where r is the distance to the target. However, there are circumstances where this is not the case. For a direct broadcast satellite in geosynchronous orbit, we design the size and shape of the antenna to just cover the region of interest, say the United States and southern Canada. If we were able to hold the spacecraft fixed at half the geosynchronous altitude, or we didn't mind the motion, then if we reduced the diameter of the antenna by half, the coverage area would be the same and it would take the same amount of power on the spacecraft.

The same logic holds for the GPS constellation. Here we want to provide a given number of W/m^2 from 4 non-collinear spacecraft over the entire surface of the Earth. If we change the constellation altitude, the number of satellites required will change, but, if the in coverage remains about the same, the total power required from the whole constellation will not change. The total number of Watts to cover the Earth does not change, no power is lost in space itself, and, therefore, the power needed by the constellation will be the same irrespective of altitude, although the size of the spacecraft antenna will change so as to just cover the Earth.

10.3 Orbit Cost—Delta V Budget and the Orbit Cost Function

The key characteristic of any orbit is performance vs. cost. What is the performance we can obtain vs. the cost required to put the payload into that orbit? In these terms, GEO is an "expensive" orbit in that approximately 5 kg must be launched into low-Earth orbit for every kilogram that ultimately arrives in geosynchronous orbit. The purpose of this section is to look at alternative launch and orbit options and quantify the assessment of cost as much as possible.

10.3.1 The Delta V (ΔV*) Budget

As discussed in Sec. 10.1 (Table 10-2), a space mission is a series of different orbits. For example, a satellite may be released in a low-Earth parking orbit, transferred to some mission orbit, go through a series of rephasings or alternative mission orbits, and then move to some final orbit at the end of its useful life. Each of these orbit changes requires energy. The ΔV *budget* is traditionally used to account for this energy. It is the sum of the velocity changes, or ΔVs, which must be imparted to the spacecraft throughout mission life. In a broad sense the ΔV budget represents the cost for each mission orbit scenario. In designing orbits and constellations, we must balance this cost against the utility achieved.

We use the ΔV budget to create a propellant budget and estimate the propellant mass, m_p, required for the space mission. (See Table 10-7 and Sec. 14.5.2.) For preliminary design, we estimate the propellant mass, m_p, needed for a space mission by using the *rocket equation* to determine the total required spacecraft plus propellant mass, $m_i \equiv m_0 + m_p$, in terms of the dry mass of the spacecraft, m_0, the total required ΔV, and the propellant exhaust velocity, V_0:

$$m_i = m_0 e^{(\Delta V / V_0)} \equiv m_0 e^{(\Delta V / I_{sp} g)} \qquad (10\text{-}14)$$

where the *specific impulse*, $I_{sp} \equiv V_0 / g$, and g is the acceleration of gravity at the Earth's surface.† Typical exhaust velocities for chemical propellants are in the range of 2 to 4 km/s and up to 30 km/s for electric propulsion systems. We can see from Eq. (10-14) that a total ΔV much smaller than the exhaust velocity (i.e. a few hundred meters per second) will require a propellant mass that is a small fraction of the total mass. If the total ΔV required

* ΔV is also often written as delta V. Both notations are used in the book.

† Note that g in the definition of I_{sp} is best thought of as a units conversion factor. It does **not** depend on the actual acceleration of gravity where the rocket happens to be operating.

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Table 10-7. Creating a ΔV Budget.

Item	Where Discussed	FireSat II Example	SCS Example
Basic Data Required			
Initial Conditions	Secs. 10.5.1, 13.5.7	200 km, 55 deg inc	20,500 km, 0 deg inc
Mission Orbit	Secs. 10.4, 10.5	700 km, 55 deg inc	21,000 km, 0 deg inc
Mission Duration (each phase)	Sec. 3.4	8 years	10 years
Orbit Maintenance Requirements	Secs. 9.5, 9.6	Altitude maintenance	Phase maintenance
Drag Parameters	Sec. 9.4.4, Table 9-9	BC = 25 kg/m^2 $\rho_{max} = 1.47 \times 10^{-13}$ kg/m^3	N/A
Orbit Maneuver Requirements	Sec. 9.6	None	Initial rephasing
End-of-Life Requirements	Chap. 30	Positive deorbit	raise 300 km
ΔV Budget (m/s)			
Orbit Transfer	Sec. 9.6, Eqs. (9-68)–(9-75), Fig. 10-16		
1st Burn	Eq. (9-69)	280 m/s	18 m/s
2nd Burn	Eq. (9-70)	280 m/s	17 m/s
Altitude maintenance			
LEO	Sec. 9.4.4, Fig. 10-17, Eq. (9-36)	21 m/s at solar max	N/A
GEO North/South	Eqs. (9-46, 9-47)*	N/A	N/A
GEO East/West	Eq. (9-48)*	N/A	N/A
Orbit Maneuvers			
Rephasing, Rendezvous	Eqs. (9-79)–(9-81)	N/A	5 m/s**
Node or Plane Change	Eqs. (9-77), (9-78)	N/A	N/A
Spacecraft Disposal	Eqs. (9-85)–(9-86), Fig. 10-17	183 m/s	20 m/s
Total ΔV		764 m/s	60 m/s
ΔV Savings	Tables (9-17)–(9-19), Sec. 9.6.3, Also see text.	N/A	N/A
Margin	Secs. 14.3.2, 26.1.2	In propellant budget	In propellant budget

* For GEO missions, N/S stationkeeping represents about 50 (m/s)/yr and E/W stationkeeping is at most about 10% of the N/S amount.
** Assigned value based on continuous adjustment of in-track phase

is equal to the exhaust velocity, then we will need a total propellant mass equal to $(e-1) \approx 1.7$ times the mass of the spacecraft. Propulsion systems require additional structure such as tanks, so a ΔV much greater than the exhaust velocity is difficult to achieve. It may make the mission effectively impossible, or require some alternative, such as staging or refueling. The fractional mass of the spacecraft that must be devoted to propellant depends directly on the ΔV that must be supplied during the life of the mission. Consequently, calculating ΔVs is a basic step in assessing the cost of a particular orbit.

The process for constructing a ΔV budget is summarized in Table 10-7. First we assemble the needed data: the launch vehicle's initial conditions, the mission orbit or orbits, the mission duration, required orbit maneuvers or maintenance, and the mechanism for spacecraft disposal. We then transform each item into an equivalent ΔV requirement using the formulas given in Sec. 9.6 and listed in Table 10-7. Each of these formulas is discussed in more detail in the location referenced.

What is the ΔV cost of putting a satellite into orbit at varying altitudes? Figure 10-16 shows representative orbit acquisition and de-orbit ΔV requirements for a satellite dropped off at 185 km altitude. As can be seen from the figure, a plane change of 30 deg tends to dominate all of the other characteristics. Figure 10-17 shows the altitude maintenance ΔV for spacecraft at varying altitudes and with a ballistic coefficient of 100 kg/m^2. Recall from Eq. (9-36) that the ΔV requirement is inversely proportional to the ballistic coefficient such that ΔVs required for spacecraft with different ballistic coefficients can be easily estimated from the figure.

The higher the mission orbit, the more propellant it takes for both orbit acquisition and de-orbit, but the less it takes for orbit maintenance. This represents the ΔV cost which must be compared to the overall utility of the orbit to make an intelligent selection decision. Figure 10-18 shows the total ΔV budget as a function of altitude for representative LEO orbits. Depending on the desired lifetime of the mission and the phase during the solar cycle, there exists an optimal altitude that balances orbit acquisition and orbit maintenance. However, this may or may not be useful for orbit design because of the dramatically high variability in the atmospheric density

Table 10-7, Fig. 10-15◀, Eq. 10-14◀

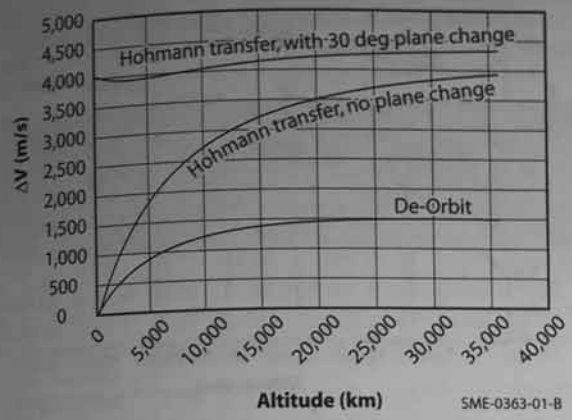


Fig. 10-16. Representative Orbit Transfer and De-Orbit ΔV Requirements. Assumes satellite is dropped off at 185 km altitude and raised via a 2-burn Hohmann transfer. De-orbited is assumed to be an elliptical orbit with a perigee altitude of 50 km.

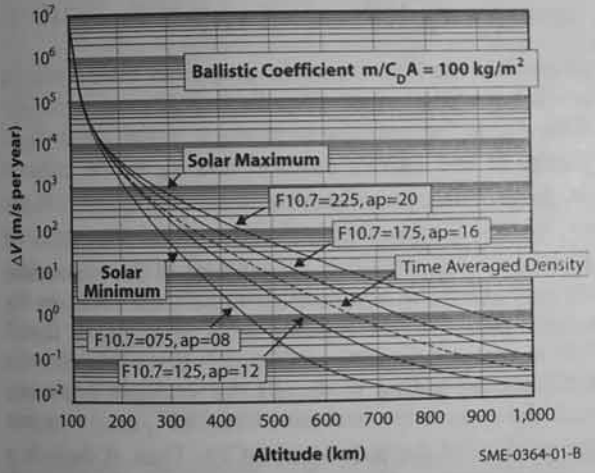


Fig. 10-17. Altitude Maintenance ΔV for a Ballistic Coefficient of 100 kg/m^2. The required ΔV will be inversely proportional to the ballistic coefficient. See Sec. 9.4.4 for a discussion of ballistic coefficients, solar activity, and atmospheric density.

(Sec. 9.4.4 and Fig. 10-17) and the other requirements on the orbit altitude. Beyond plane change requirements, the principal factors which will increase the ΔV cost in low-Earth orbit are a low altitude and long mission lifetime. The atmosphere increases in density exponentially as the altitude decreases, thus dramatically increasing drag as altitude comes down. In addition, large solar arrays or other appendages represent a large cross sectional area that will increase drag. A low mass per unit area, i.e., a large, lightweight satellite will also have higher drag than a small, dense satellite. Note that the shape and surface characteristics typically play only a secondary role (See Sec. 9.4.4). They impact the drag coefficient somewhat but usually play a relatively minor role in overall ΔV requirements, except at low altitude where it may become important.

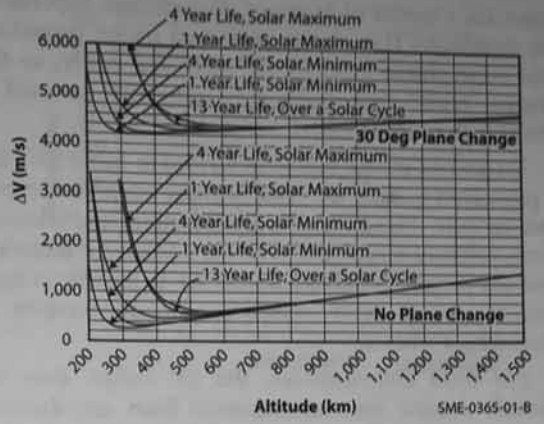


Fig. 10-18. ΔV Budget as a Function of Altitude for Low-Earth Orbits.

The ΔV budget has a strong impact on the propulsion requirements and, therefore, on the final cost and achievability of a space mission. Nonetheless, other factors can vary the propellant requirements relative to a nominal ΔV budget. For example, although rocket propulsion usually provides the ΔV, we can obtain very large ΔVs from a flyby of the Moon, other planets, or even the Earth itself [Kaufman et al. 1966; Meissinger, 1970]. As discussed in more detail in Sec. 9.6.3, a spacecraft in a planetary flyby leaves the vicinity of some celestial body with the same velocity relative to the body as when it approached, but in a different direction. This phenomenon is like the elastic collision between a baseball and a bat, in which the velocity of the ball relative to the bat is nearly the same, but its velocity relative to the surrounding baseball park can change dramatically. We can use flybys to change direction, to provide increased heliocentric energy for solar system exploration, or to reduce the amount of energy the satellite has in inertial space. For example, one of the most energy-efficient ways to send a space probe near the Sun is to use a flyby of Jupiter to reduce the intrinsic heliocentric orbital velocity of the Earth associated with any spacecraft launched from here.

A second way to produce a large ΔV without burning propellant is to use the atmosphere of the Earth or other planets to change the spacecraft's direction or reduce its energy relative to the planet. The manned flight program has used this method from the beginning to dissipate spacecraft energy for return to the Earth's surface. It can also be used to produce a major plane change through an *aeroassist trajectory* [Austin et al., 1982; Mease, 1988]. This has been used, for example, on Mars orbiter missions.

The *solar sail* is a third way to avoid using propellant. A large, lightweight sail uses solar radiation pressure to slowly push a satellite the way the wind pushes a sailboat. Of course, the low-pressure sunlight produces very low acceleration. Hence, very large solar sail areas are required as discussed in Sec. 18.7.

The aerospace literature discusses many alternatives for providing spaceflight energy because of its impor-

tance for a variety of missions. Nonetheless, experimental techniques (i.e., those other than rocket propulsion and atmospheric braking) are risky and costly, so that normal rocket propulsion will ordinarily be used to develop the needed ΔV, if this is at all feasible.

To convert the ΔV budget described in Table 10-7 into a propellant budget, we must consider other characteristics as well. These include, for example, inefficiencies from thrusters misaligned with the desired ΔV direction, and any propulsion diverted from ΔV to provide attitude control during orbit maneuvers or desaturation of momentum wheels.

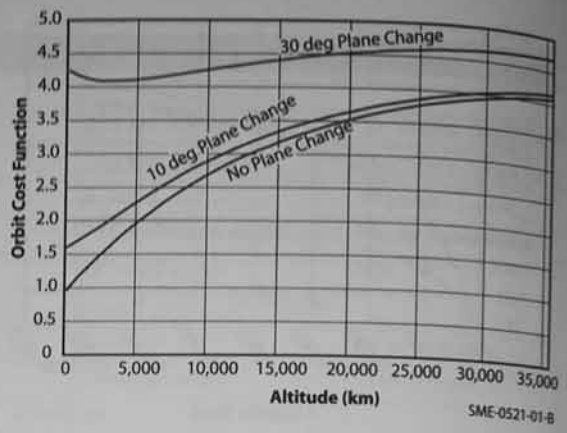
For most circumstances, the ΔV budget does not include margin because it results from astrodynamic equations with very little error relative to most other errors in system design. Instead, we maintain the margin in the propellant budget itself, where we can reflect other elements such as residual propellant. (See Sec. 14.5.) An exception is the use of ΔV to overcome atmospheric drag. Here the ΔV depends upon the density of the atmosphere, which is both highly variable and difficult to predict. Consequently, we must either conservatively estimate atmospheric density or incorporate a ΔV margin for low-Earth satellites to compensate for atmospheric variability.

10.3.2 The Orbit Cost Function

Given that we can estimate the total launch cost or specific launch cost for putting mass into low-Earth orbit, how can we translate this into the mass available in a mission orbit. Equivalently, what is the ultimate cost per kilogram to get where we want to go and achieve our mission objectives? To address this, we define the *Orbit Cost Function*, *OCF*, as the ratio of the mass available in a 185-km circular orbit due East from the launch site to that available in the mission orbit, or at the end of mission life. This can be thought of either as a multiplier on the cost of putting a spacecraft into its mission orbit, or, for a given launch vehicle, an inverse multiplier for the amount of payload that can be put into that mission orbit. Section 10.3.3 will look at the practical applications of using the *OCF* as a means of estimating launch costs and available mass.

The *OCF* depends on how the spacecraft is launched. For example, direct injection into Jupiter transfer orbit has a total *OCF* on the order of 10. Using a Venus swingby followed by an Earth swingby can reduce this to approximately 6. In this case, we can use the *OCF* to measure the performance gain associated with the swingbys and compare that with the time lost to determine whether the swingbys are worthwhile for a particular mission. Thus, the use of a Venus swingby for the Jupiter mission represents an increase in the available mass at Jupiter or a decrease in launch cost by a factor of approximately $10/6 \approx 1.7$. The question for the mission designer is then whether the added complexity and added time associated with these swingbys is worth the savings that are obtained.

Table 10-7◀, Fig. 10-19, Eq. 10-16b



Live Calc

Fig. 10-19. Orbit Cost Function for Various Earth Orbiting Missions. The curves assume starting at a 185 km circular orbit and a circular orbit at the final mission altitude. The curves are for orbit acquisition only.

The orbit cost function is related to the required ΔV by:

$$OCF = (1+K)e^{\Delta V/V_0} - K = (1+K)e^{\Delta V/gI_{sp}} - K \quad (10\text{-}15)$$

where K is the fraction of the propellant mass assigned to tankage and other propellant hardware, typically 10%. I_{sp} is the specific impulse, g is the constant of gravity at the Earth's surface, and V_0 is the exhaust velocity of the propulsion system. An I_{sp} of 300 sec (typical for a bipropellant system) gives a V_0 value of 2.94 km/s. Depending on the purpose of the evaluation, we may choose to compare the Orbit Cost Function for simply getting to the mission orbit, through the useful life, or through de-orbit or disposal. If we calculate the *OCF* corresponding to the ΔV for multiple segments, then the cumulative cost function, $OCF_{A+B+C...}$, is just the product of the individual *OCF*s. Thus, if there is a series of mission phases or maneuvers *A, B, C...*, with OCF_A corresponding to ΔV_A, and so on, then

$$\Delta V_{A+B+C...} = \Delta V_A + \Delta V_B + \Delta V_C + \ldots \quad (10\text{-}16a)$$

and

$$OCF_{A+B+C...} = OCF_A \times OCF_B \times OCF_C \times \ldots \quad (10\text{-}16b)$$

Figure 10-19 shows the Orbit Cost Functions for typical Earth orbiting missions as a function of altitude and plane change. For more complex orbits, or interplanetary missions, the *OCF* is computed by first calculating the required ΔV, and then accounting for added tank mass and any other additional requirements such as the extra power required for electric propulsion. The *OCF* is tabulated for various Earth-oriented and interplanetary missions in Table 10-8.

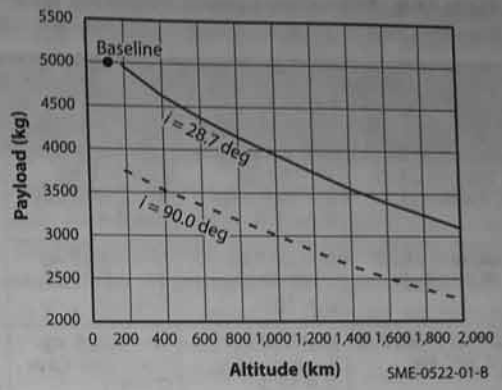
We can also construct the Orbit Cost Function for a launch vehicle alone from the data supplied by the manufacturers. For example, Fig. 10-20A shows the mass as a function of orbit parameters as provided by the launch vehicle manufacturer for the Delta 7925 launch vehicle.

Table 10-8. Typical Orbit Cost Functions for Various Earth-Orbiting and Interplanetary Missions. Interplanetary missions assume a Hohmann transfer. *OCF* assumes I_{sp} of 300 sec and 10% tank mass.

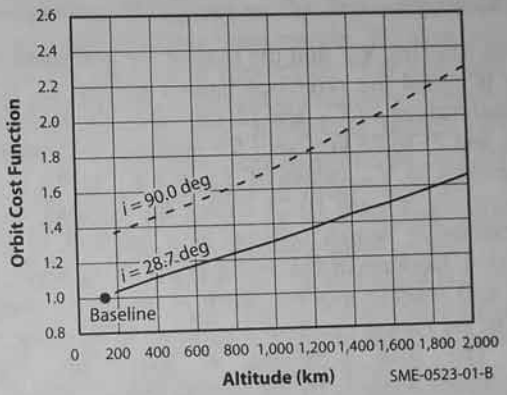
Mission	Incremental ΔV			Cumulative Orbit Cost Function		
	Getting There	Mission Life	Disposal	Getting There	Mission Life	Disposal
200 km @ 90°	0.75	12.66	0.045	1.32	107.01	108.83
407 km @ 51.6°	0.39	0.192 0	0.104	1.16	1.24	1.29
500 km @ 28.7°	0.24	0.43 2	0.130	1.47	1.49	1.57
500 km Sun-synchronous	0.89	0.43 2	0.130	1.39	1.41	1.48
500 km @ 90°	0.89	0.043 2	0.130	1.39	1.41	1.48
1,000 km @ 28.7°	0.51	0.000 2	0.257	1.21	1.21	1.33
1,000 km Sun-synchronous	1.38	0.000 2	0.257	1.66	1.66	1.82
GTO (200 × 35,786 km)	2.80	0.411 4	0.024	2.75	3.20	3.23
GEO	4.95	0.521 5	0.018	5.82	7.06	7.11
Lunar Fly-by	3.14			3.10		
Lunar Orbit	3.23			3.20		
Lunar Lander	7.07			12.06		
Mars Fly-by	3.61			3.65		
Mars Orbit	5.69			7.51		
Mars Lander	9.30			25.86		
Mission Life ΔV = ΔV required for 10 years						

The Orbit Cost Function is the ratio of the mass that can be launched due East to 185 km low-Earth orbit to the masses shown in figure. This is plotted in Fig. 10-20B. As a specific example, the Delta II 7920 can put 5,140 kg into a 185 km circular orbit at 28.7 deg inclination and 3,220 kg into a Sun synchronous orbit at 800 km and 98.6 deg. Thus, the *OCF* for the Sun-synchronous orbit is 5,140/3,220 ≈ 1.60.

Typically, the highest mass to the final orbit can be achieved by using on-board propulsion. The reason for this is that we can then avoid putting the vehicle upper stage into the final spacecraft orbit. For example, if we launch a spacecraft to Jupiter using the traditional approach of a large upper stage, then the upper stage vehicle itself will also be put in the Jupiter transfer orbit. Using onboard propulsion, we will put only the spacecraft and some additional tankage into this very high energy orbit. An example of a high cost approach to orbit transfer is the Hubble Space Telescope, which was launched on board the Space Shuttle and is re-boosted from time to time by the Orbiter at the same time instruments are changed out. The reasons for this approach were to avoid entirely an on-board propulsion system and the potential for contamination from thruster firings as well as to avoid the possibility of propellant slosh providing even a low level of vibration to the telescope. In this case, a higher cost approach to orbit acquisition and maintenance was used to meet the mission requirements for near-zero disturbances and absolute minimization of potential contamination.



(A) Payload Mass



(B) Orbit Cost Function

Fig. 10-20. Performance Data for the Delta 7925 Launch Vehicle (3 m fairing). (A) Mass vs. orbit as provided by the launch vehicle manufacturer. (B) The Orbit Cost Function is simply the ratio of the mass that can be launched due East to a 185 km LEO orbit to the mass that can be put in the mission orbit. See text for discussion.

Table 10-9 compares alternative scenarios for a low-cost Jupiter mission based on (a) the traditional approach of using the launch vehicle upper stage for final injection to the transfer orbit, (b) using a Venus-Earth-Earth swing-by as was done for the Galileo mission, and (c) using on-board propulsion to provide direct transfer to the target planet in a modified launch mode as proposed by Meissinger, et al. [1997, 1998]. The planetary swing-by approach provides 4 to 5 times the payload mass at Jupiter, but at the cost of increasing the transfer time from 2.3 years to over 6 years. The on-board propulsion approach increases the payload available from the traditional approach by 50% to 100%, but with no increase in transfer time.

Finally, the selection criteria for launch vehicles for interplanetary missions are similar to those for Earth orbit missions, but use a different set of parameters to characterize vehicle performance. As discussed further in Sec. 10.7, launch requirements are defined in terms of the velocity or energy of the spacecraft at the point at which it has just escaped the gravitational pull of the Earth. The velocity at this time is called the *hyperbolic*

Table 10-9. Alternative Launch Scenarios for a Low-Cost Jupiter Mission. The number below each vehicle is the baseline mass in a due east orbit at 185 km. See text for discussion.

	At Jupiter Arrival			In Jupiter Orbit		
	Traditional Direct	Venus-Earth-Earth	MLM Direct*	Traditional Direct	Venus-Earth-Earth	MLM Direct*
Taurus XL/S (1,400 kg)	M = 62 kg T = 840 days C = 22.6	M = 316 kg T = 2,240 days C = 4.4	M = 143 kg T = 840 days C = 9.8	M = 49 kg T = 840 days C = 28.6	M = 249 kg T = 2,240 days C = 5.6	M = 109 kg T = 840 days C = 12.8
Delta II 7925 (5,089 kg)	M = 280 kg T = 840 days C = 18.2	M = 1,004 kg T = 2,240 days C = 5.1	M = 443 kg T = 840 days C = 11.7	M = 221 kg T = 840 days C = 23.0	M = 792 kg T = 2,240 days C = 6.4	M = 330 kg T = 840 days C = 15.4
Delta II 7925 H (6,107 kg)	M = 336 kg T = 840 days C = 18.2	M = 1,205 kg T = 2,240 days C = 5.1	M = 520 kg T = 840 days C = 11.7	M = 265 kg T = 840 days C = 23.0	M = 950 kg T = 2,240 days C = 6.4	M = 397 kg T = 840 days C = 15.4

M = mass at destination; T = transfer time; C = Orbit Cost Function; MLM = modifed launch mode
** US Patent No. 6,059,235, Microcosm Inc.*

excess velocity, V_∞, and the energy per unit mass at this time is called the reference launch energy, $C_3 \equiv V_\infty^2$. Typical values of C_3 are 12 km^2/sec^2 for missions to Mars and 75 to 85 km^2/sec^2 for missions to Jupiter, corresponding to V_∞-values of 3.5 and 8.7 to 9.2 km/sec, respectively. Figure 10-21 shows the payload mass of representative launch vehicles as a function of the C_3 provided. Because of the strong effect of the upper stage mass and staging process, there is no simple standard form of the rocket equation applicable to all of the vehicles. The data presented is based on detailed performance analysis by the vehicle manufacturers.

10.3.3. Estimating Launch Cost

What does it cost to put a payload in orbit? In general, there are two answers, depending on how we view the problem. On the one hand, having selected a launch vehicle and possibly an upper stage or orbit transfer mechanism, the cost is a fixed dollar value. Changing the mass of the spacecraft will have no impact unless we ultimately are forced to use a larger vehicle, or find that we can launch on a smaller one. Once we have selected a launch vehicle, there is little point in reducing weight beyond what is needed to accommodate that vehicle.

The second view is to look at the launch capacity per kilogram, or *specific launch cost*. Generally, as shown in Fig. 10-22 launching twice as much mass will cost twice as much. From this perspective, reducing the mass of some portion of the spacecraft can allow us to either reduce the launch cost by potentially using a smaller vehicle, or alternatively, to increase the spacecraft mass in some other area, i.e., by flying an additional payload instrument or more propellant to allow a longer mission at the final destination.

There are two factors which make the specific launch cost inappropriate or at least a poor measure of effectiveness for launch cost. First is the quantization in launch vehicles. Launchers are not available in an infinite variety of sizes, so that once we have chosen a vehicle, the cost per kilogram is immaterial. Second, there are economies of scale as vehicles get bigger, as shown in Fig. 10-22. In general, launching a spacecraft which

Table 10-9, Fig. 10-22, Eq. 10-16b◀

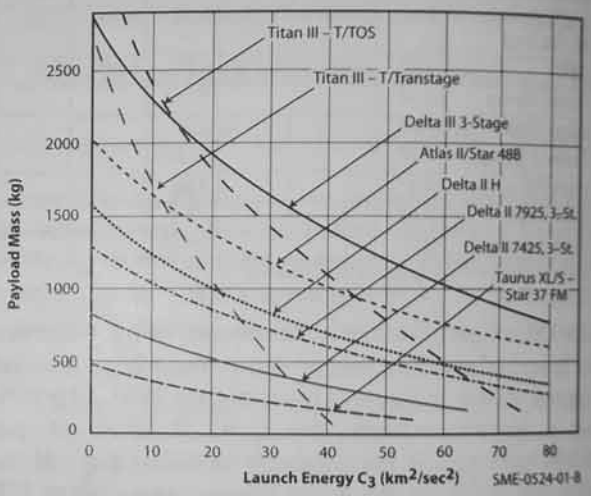


Fig. 10-21. Representative Range of US Launch Vehicle Escape Mission Performance. Data should be compared with mass to LEO from Table 11-23 to determine the Orbit Cost Function.

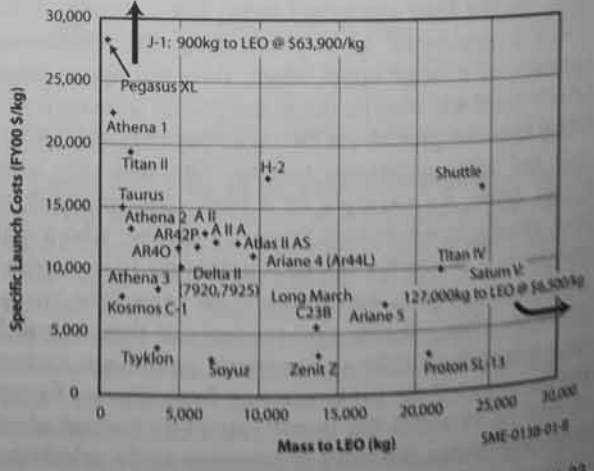


Fig. 10-22. Specific Launch Costs. Data from Tables 11-23 and 13-17. A II= Atlas II.

Table 10-10. Estimating Launch Cost. Specific launch costs will depend on additional factors, such as the vehicle selected and the negotiating leverage of the organization buying the launch. Thus, this process is more accurate at estimating relative cost than absolute cost.

Step	Where Discussed
1. *Establish orbit parameters*	Sec. 10.1–10.5
2. *Establish spacecraft mass*	Chap. 14
3. *Estimate Orbit Cost Function for direct ascent and on-board propulsion, if applicable*	Sec. 10.3.2
4. *Obtain corresponding LEO mass*	Result of (2) × (3)
5. *Obtain rough estimate of launch cost from specific launch costs in Table 11-23*	Secs. 10.3.3, 11.2.8
5A. *For dedicated launch, select vehicle, or vehicles based on LEO mass or direct injection*	Sec. 11.2.8, Chap. 26, Isakowitz et al. [2004]
6. *Determine launch cost and available mass margin*	See text.

is 10 times as massive will cost less than 10 times as much, because of the efficiency associated with larger vehicles. Many of the commercial low-Earth orbit communications constellations have chosen to use a variety of launch vehicles and typically launch several spacecraft on a single large vehicle. This allows them to take advantages of the economies of scale and, at the same time, implies that the cost per kilogram continues to be an important measure. As the mass of an individual spacecraft is reduced, this effect is multiplied by the number of spacecraft per vehicle and can allow either additional propellant on board each of the spacecraft, or potentially, allow an additional spacecraft to be put on a given launcher.

Despite the shortcomings, the specific launch cost is a useful concept in mission design. It allows us to get a rough estimate of the cost of getting our payload into low-Earth orbit at a time before launch vehicle options have been selected. The biggest shortfall of the specific launch cost model is that it only gives the cost of getting into low-Earth orbit. To use this concept effectively, we need to be able to apply it to multiple orbit types—to low-Earth orbit, geosynchronous orbit, Sun-synchronous orbits, interplanetary, and so on, so that we have an effective means of comparing orbit cost. This is the purpose of the Orbit Cost Function defined in Sec. 10.3.2. We can make use of this function to assess the relative costs between, for example, high-inclination posigrade orbits and Sun-synchronous orbits which are retrograde.

Table 10-10 summarizes the step-by-step process of estimating the launch cost for a particular mission. Overall, it is an iterative process of selecting launch parameters, spacecraft mass, and launch cost. We need to adjust all of these factors to achieve our mission objectives at minimum cost and risk.

We begin the launch cost estimation process by determining the orbit parameters for the mission. We then estimate the spacecraft dry mass consisting of the operational payload and the spacecraft bus exclusive of the tankage required for any propellant used in orbit transfer. Next, we establish the Orbit Cost Function based on the ΔV budget as defined in Sec. 10.3.1. If possible, we would like to do this for both direct ascent using the launch vehicle and for using on-board propulsion. This process allows us to compute a corresponding mass into low-Earth orbit for our particular mission. Given an amount of mass in low-Earth orbit, we can then estimate the launch cost from the specific launch cost given in Fig. 10-22 or using the costs for an appropriate vehicle or range of vehicles listed in Table 11-23 in Sec. 11.2.8. For dedicated launch, we can then do a preliminary selection of a specific vehicle or range of vehicles based upon either the mass in low-Earth orbit or direct injection. This gives us the first estimate of our launch cost and available mass margin. We then proceed to iterate the process as appropriate. For example, we may be able to reduce our mass estimates somewhat in order to obtain a smaller, lower-cost launch vehicle. Alternatively, if we have large margins, we may choose to increase the system mass in order to provide more payload or greater propellant for other mission activities.

There are a number of potential error sources in estimating launch costs. First, the cost of launch is a negotiated figure based in large part on the law of supply and demand. Thus, a user with a large number of payloads may be able to negotiate a lower price in exchange for a long-term contract. However, the capacity to negotiate a lower price depends on the existence of at least some level of competition within the launch vehicle arena. Thus, if the political decision is made to use only national launch vehicles, then typically, the competitive environment is substantially reduced and we can expect to pay significantly more for a given launch. Commercial launches, on the other hand, are usually not constrained to use vehicles from any particular nation or organization and, therefore, are in a better position to negotiate favorable prices based strictly on performance.

In addition to price negotiation, there is also the potential for dual payload launches. For example, we may choose to use a somewhat larger vehicle with substantial launch margin, and use that launch margin to include a secondary payload, which can cover a significant fraction of the cost of the launch vehicle. Third, there are overall evolutionary trends in launch costs that will impact the price which vehicle manufacturers can charge for vehicles of a given size. The net effect of these factors is to imply that the relative costs obtained by the above process will have a higher level of accuracy than the absolute costs.

For some missions, it is also useful to run the cost estimation process "backwards." To do this, we first select a representative launch vehicle, determine the mass available in low-Earth orbit, and use the Orbit Cost Function to estimate the mass available in the final mission orbit. This is a mechanism for sizing the payload to meet the capacity of a launch vehicle that falls within the budget constraints appropriate to the mission.

Table 10-10, Fig. 10-22, Eq. 10-16b ◀

In general, the Orbit Cost Function provides a much easier mechanism for including launch and orbit transfer costs in system trades than the traditional approach of selecting a vehicle and making the process work within this constraint. We would like to include the mission orbit itself as a part of the system trade process and determine the impact of the mission orbit on the overall mission cost. The Orbit Cost Function is an analytic mechanism doing this.

10.4 Selecting Earth-Referenced Orbits

Earth-referenced orbits are typically among the most complex in terms of mission design for two reasons. First, there are a variety of specialized orbits that are design options that need to be evaluated and assessed. Second, the orbit parameters impact a large number of requirements in varying ways, such that no one orbit is ideal for all aspects of a mission. For example, a higher orbit typically provides better coverage, but is more expensive to get to in terms of the Orbit Cost Function (OCF), has lower resolution for observations, requires more power for communications, and is in a more adverse radiation environment.

Most of the steps in the orbit design process for Earth-referenced missions are the generic ones, defined in Table 10-1 in Sec. 10.1. The key issues that are specific to Earth-referenced missions are the orbit-related mission requirements, specialized Earth-referenced orbits, and the potential use of eccentric orbits. These are discussed in the three subsections below.

10.4.1 Mission Requirements for Earth-Referenced Orbits

As listed previously in Table 10-1 in Sec. 10.1, the first step in the orbit selection process after dividing the mission into phases is to accumulate the orbit-related mission requirements. The mission requirements which normally have the greatest effect on Earth-referenced orbits are listed in Table 10-11. The principal conclusion here is that for Earth-orbiting missions, a large number of mission requirements are orbit-dependent. The second most important conclusion from the table is that the orbit parameter that is most important in orbit selection is the altitude, with the secondary parameter being the inclination. Altitude is the main determining factor in nearly all of the orbit-related mission requirements. The altitude is the primary determinant of most aspects of coverage, performance, spacecraft environment, launch cost, and even the end-of-life options available to the spacecraft. Consequently, the most common trade here is one of coverage as a measure of performance versus ΔV or OCF as a measure of cost.

Another key factor in altitude selection is the satellite's radiation environment. As described in Sec. 7.3, the radiation environment undergoes a dramatic change at an altitude of approximately 1,000 km (See Fig. 7-11). Below this altitude, the atmosphere quickly clears out

Table 10-11. Principal Mission Requirements that Normally Affect Earth-Referenced Orbit Design. See Sec. 10.1 for a discussion of the orbit design process.

Mission Requirement	Parameter Affected	Where Discussed
Coverage		
Continuity	Altitude	Chap. 8,
Frequency	Inclination	Sec. 10.2
Duration	Node (only	
Field Of View (Or Swath Width) Ground Track	relevant for some orbits)	
Area Coverage Rate		
Viewing Angles		
Earth Locations Of Interest		
Sensitivity or Performance		Sec. 10.2.2,
Exposure or Dwell Time	Altitude	Chaps. 16, 17
Resolution		
Aperture		
Environment and Survivability		Secs. 7.5,
Radiation Environment	Altitude	8.2.4
Lighting Conditions	(Inclination	
Hostile Action	usually secondary)	
Launch Capability		Sec. 10.3.3,
Launch Cost	Altitude	Chap. 26
On-Orbit Weight	Inclination	
Launch Site Limitations		
Ground Communications		Sec. 8.4,
Ground Station Locations	Altitude	Chaps. 28, 29
Use Of Relay Satellites	Inclination	
Data Timeliness		
Orbit Lifetime	Altitude	Fig. 9-18
	Eccentricity	(Sec. 9.4.4)
Legal or Political Constraints		Secs. 12.3
Treaties	Altitude	
Launch Safety Restrictions	Inclination	
International Allocation	Longitude in GEO	

charged particles such that the radiation density is relatively low. Above this altitude are the Van Allen Belts, where the high level of trapped radiation can greatly reduce the lifetime of spacecraft components. Consequently, most Earth-referenced mission orbits separate naturally into *low-Earth orbit* (LEO), below 1,000 km, and *geosynchronous orbit* (GEO), which is largely above the Van Allen Belts. Mid-range altitudes have coverage characteristics that may make them particularly valuable for some missions. However, the added shielding or reduced life stemming from this region's increased radiation environment also makes them more costly. Nonetheless, the increased coverage to be obtained at higher altitudes has tended to push up the minimum altitude defined as LEO. For example, LEO communications constellations have been considered at altitudes of up to 1,500 km. At these altitudes, not only is the radiation environment harsher, but the potential options for end-of-life are also more difficult. Low-Earth orbit spacecraft are often de-orbited or allowed to reenter at the end of their useful life. Above 1,000 km, this process occurs extremely slowly when driven only by drag and requires

Table 10-12. Specialized Orbits Used for Earth-Referenced Missions. Typically the orbit design process consists of deciding whether any of the specialized orbits should be used followed by a general trade on other orbits. Section 9.5 provides the physical basis for each orbit; this section discusses applications.

Orbit	Characteristic	Applications	Applications Discussed	Physics Discussed
Geosynchronous (GEO)	Maintains nearly fixed position over equator	Communications, weather, surveillance	Sec. 10.4.2.1	Sec. 9.5.1
Sun Synchronous	Orbit plane rotates so as to maintain approximately constant orientation with respect to Sun	Earth resources, weather	Sec. 10.4.2.2	Sec. 9.5.3
Molniya (also called HEO)	Apogee/perigee do not rotate	High latitude communications	Sec. 10.4.2.3	Sec. 9.5.4
Frozen Orbit	Minimizes changes in orbit parameters	Any orbit requiring stable conditions	Sec. 10.4.2.4	Sec. 9.5.6, Vallado [2007]
Repeating Ground Track	Subsatellite trace repeats	Any orbit where constant viewing angles are desirable	Sec. 10.4.2.4	Sec. 9.5.2
Repeat Coverage Orbit	Coverage of multiple successive orbits	Whenever multiple successive views are needed	Sec. 10.4.2.6	Sec. 10.4.2.6

Table 10-13. Advantages and Disadvantages of Specialized Earth-Referenced Orbits.

Orbit	Advantages	Disadvantages
Geosynchronous	Continuous viewing of one region; fixed ground antenna	Poor polar region coverage, very high cost, very long range
Sun-Synchronous	Maintains roughly constant Sun angle relative to orbit plane	High cost
Molniya (HEO)	Provides extended coverage of high latitude regions	High cost; widely varying ranges, Earth size, and rates drive up spacecraft cost; high radiation
Frozen Orbit	Maintains stable conditions for near-circular orbits	None
Repeating Ground Track	Coverage repeats	Orbit perturbations can resonate
Repeat Coverage Orbit	Best coverage of a specific latitude band	Doesn't provide whole Earth coverage; time of day of coverage changes

a very substantial ΔV if it is to be done by an on-board propulsion system. Consequently, in the regime above 1,000 km, the end-of-life option often regarded as the most desirable is to raise the spacecraft 50 to 100 km higher in order to remove it from the constellation structure. Note however, that this process creates one or more "graveyard" regimes in which a large number of spent non-functioning spacecraft could accumulate. Of course, one constellation's graveyard orbit could be another constellation's operational orbit.

10.4.2 Specialized Earth-Referenced Orbits

After having defined the orbit related mission requirements, the next step in finding an appropriate Earth-referenced orbit is to determine if any of the specialized orbits listed in Table 10-12 are appropriate. The advantages and disadvantages of these alternatives are summarized in Table 10-13 and discussed in more detail below. Chapter 9 provides a detailed discussion of the physical basis of each of these orbit types. For purposes of mission design, we need to examine each of these specialized orbits individually to see if its characteristics will meet the mission requirements at a reasonable cost. Space missions do not need to be in specialized orbits, but these orbits have come into common use because of their valuable characteristics for many missions. Because they do constrain orbit parameters such as altitude and inclination, we first determine whether or not to use one of the specialized orbits before doing the more detailed design trades in Table 10-1 in Sec. 10.1.

It is frequently the existence of specialized orbits which yields very different solutions for a given space mission problem. For example, a geosynchronous orbit may provide the best coverage characteristics but may demand too much propellant, instrument resolution, or power, or have too high an Orbit Cost Function. The trade of value vs. cost can lead to dramatically different solutions depending on mission needs. A Sun-synchronous orbit is typically 30% more expensive than a low-inclination prograde orbit. Nonetheless, for many obser-

Table 10-13, Fig. 10-22◀, Eq. 10-16b◀

vation satellites such as Earth resources or weather, the advantage of being able to see locations on the ground at the same time of day and under the same lighting conditions on a continuing basis outweighs the additional cost of this type of orbit.

10.4.2.1 Geosynchronous Orbits (GEO)

Prior to the introduction of low-Earth orbit communications constellations, geosynchronous orbit represented the single most used orbit in space. It accounted for approximately 50% of satellite launches during the 1990s. The physical properties of geosynchronous orbit and orbit perturbations are discussed in Sec. 9.5.1; in addition, Pocha [1987] and Soop [1994] provide very detailed discussions of geosynchronous orbit design and stationkeeping approaches. Chapter 16 discussed the design of geosynchronous communications satellites. GEO satellite applications and design are also discussed in detail by Gordon and Morgan [1993] and Maral and Bousquet [2010].

The principal applications for geosynchronous orbit are communications (both two-way and direct broadcast), weather, and Earth surveillance. However, as shown in Chap. 16, GEO is getting full. There are now relatively few orbital slots available over regions of particular interest, such as the mid-Atlantic or mid-Pacific areas used for communications between North America, Europe, and Japan. Similarly, slots over Europe and North America used for internal communications are also becoming full.

Overcrowding and the potential for satellite collisions represent major problems for geosynchronous orbit, as discussed in detail by Reijnen and De Graaff [1989]. In a sense, space is large and the potential for collisions between satellites is small. Nonetheless, the cost of getting spacecraft to GEO is extremely high and there is no current potential for being able to retrieve or actively recover either spent spacecraft or debris. In addition, there are essentially no natural perturbations that will remove debris from geosynchronous orbit. Consequently, any sequence of either spacecraft explosions or inadvertent collisions which produce a significant debris cloud in GEO may make the orbit effectively unusable for the foreseeable future. (An interesting potential alternative to GEO using solar radiation pressure for orbit maintenance is the cylindrical orbit discussed in Sec. 18.7.2.)

Because there is currently no mechanism for removing debris from GEO, the only approach for protecting it is to not allow actions which have the potential for generating debris. This means that spacecraft should be removed from geosynchronous orbit while they are still responding to commands and while there is still propellant available to do so. Thus, in order to preserve this unique resource, we must be willing to turn off operational spacecraft prior to the use of their last few grams of propellant. The graveyard orbit for geosynchronous spacecraft is to put them in an orbit approximately 200 km above GEO. At this altitude, natural perturbations will cause the inclination to oscillate as discussed in Sec. 9.5.1. However, these oscillations in orbit elements will not bring the spacecraft back into the geosynchronous ring in any reasonable amount of time, i.e., within several hundred thousand years. In addition, the ΔV required to raise the altitude is extremely small. Increasing the altitude by 200 km requires a ΔV of only 7.4 m/sec.

As discussed in Sec. 9.5.1, the principal perturbative forces in GEO are solar/lunar perturbations, which cause a general North/South drift or change in inclination, and the effect of the out-of-roundness of the Earth's equator, which cause a very slow drift in the East/West direction. In both cases, the effects are extremely predictable and essentially continuous. Consequently, this represents an excellent application for electric propulsion for orbit maintenance. A nearly continuous low level of thrust can be used to maintain the spacecraft in geosynchronous orbit. Typically, the North/South stationkeeping is required for operational purposes so that the spacecraft can keep its target regime on Earth in view, and equally important, so that Earth antennas can be pointed at the spacecraft without having to be continuously steered. The East/West stationkeeping, while requiring only 10% of the ΔV of North/South stationkeeping, is perhaps more important in that it prevents either physical or communications interference with satellites in a neighboring orbital slot. Without nearly continuous East/West stationkeeping, geosynchronous satellites would continuously slide through the orbit slots assigned to other spacecraft, thus providing both significant communications interference and even the potential for satellite collisions.

Synchronous orbits are possible around other planets and satellites as well, although, like the Earth, they may be significantly influenced by various perturbations. Both the formulas to determine these orbits and the characteristics of synchronous orbits for a variety of other central bodies are listed in Table 9-12 in Sec. 9.5.1.

10.4.2.2 Sun-Synchronous Orbits

As discussed in Sec. 9.5.3, a *Sun-synchronous orbit* is one in which the perturbation due to the Earth's oblateness causes the orbital plane to rotate in inertial space at a rate equal to the average rate of the Earth's rotation about the Sun. Consequently, in a Sun-synchronous orbit, the position to the Sun relative to the orbit plane remains approximately constant. The general motion of the Sun relative to the orbit plane is shown in Fig. 9-27 in Sec. 9.5.3. From an applications perspective, the Sun-synchronous orbit has the principal advantage of maintaining approximately constant Sun angles. This means that for Earth resources satellites, for example, the angle of illumination will be approximately constant as photographs of a given region are taken over time. Consequently, it is significantly easier to interpret the photos and look for changes over time. In addition, visual observations are best done when the shadows are long, i.e., in the morning or late afternoon. Thus, 9 am–9 pm and 3 am–3 pm orbits are often chosen for Earth observations.

Sun-synchronous orbits have also been used by some missions as a means of reducing the battery size or elim-

Table 10-13◀, Fig. 10-22◀, Eq. 10-16b◀

inating a second gimbal in the solar array drive because of the nearly constant Sun angle. Thus, rather than requiring a 2-axis gimbal, a single-axis gimbal is sufficient and the cost of the spacecraft can be marginally reduced. This is an example of an application of the Sun-synchronous orbit which is probably inappropriate in most circumstances. The Sun-synchronous orbit requires an additional cost of approximately 30% to obtain this orbit. This cost is far in excess of the benefit achieved by having only a single gimbal in the solar array drive. In addition, it also adds another element of risk to the mission by creating a new failure mode. If the launch vehicle or propulsion system are unable to achieve or to maintain a Sun-synchronous orbit, then a failure of the mission is possible. If the system could operate under non-Sun-synchronous conditions, then there would have been no reason for the orbit in the first place. Consequently, choosing a Sun-synchronous orbit for purely spacecraft bus functions is an incorrect engineering choice in most circumstances.

Live Calc

Table 10-14. Representative Sun-Synchronous Orbits for Other Central Bodies.

Central Body	Altitude (km)	Inclination (deg)	Period (min)
Mars	1,000	94.85	147.2
	5,000	144.52	388.4
Jupiter	1,000	90.08	181.5
	5,000	90.09	196.7
Saturn	1,000	90.04	257.8
	5,000	90.05	283.5
Uranus	1,000	90.01	183.3
	5,000	90.02	232.4
Neptune	1,000	90.02	166.0
	5,000	90.04	206.1

Sun-synchronous orbits are also possible around other central bodies. Representative parameters for these orbits are given in Table 10-14. The advantages of these orbits are similar to their advantages when used on the Earth. That is, a constant level of illumination allows us to compare photographs taken at different times in order to more accurately determine physical changes that may be taking place on the planet or moon that we are orbiting. For example, Sun-synchronous orbits would be appropriate for examining seasonal variations on Mars, or variations as a function of volcanic activity or electromagnetic storms in the vicinity of Jupiter's moon, Io. Note that in the case of other planets and satellites, Sun-synchronous orbits are significantly less expensive to achieve. Generally, the orbit plane in which the spacecraft arrives can be adjusted by very small ΔVs applied during the orbit transfer. Consequently, the high cost associated with Sun-synchronous orbits on the Earth is not necessarily a problem for other central bodies. However, the parameters associated with Sun-synchronous orbits for other central bodies, such as the required altitude or inclination may make them inappropriate for other reasons, such as viewing distance or excessive perturbations. Note that for the gas giants, the oblateness is large and the revolution rate about the Sun is slow such that Sun-synchronous orbits are essentially polar.

10.4.2.3 Molniya (HEO) Orbits

A *Molniya orbit* is a *Highly Elliptical Orbit* (HEO) inclined at the critical inclination of 63.4 deg (or 180 – 63.4 = 116.6 deg) in order to prevent apogee and perigee from rotating. The fundamental equations and physical properties for Molniya orbits were discussed in Sec. 9.5.4. The basic application of these orbits is for communications or observations at high northern latitudes where geosynchronous spacecraft are ineffective or inaccessible. Consequently, Molniya orbits have been used for communications in Russia and the former Soviet Union for an extended period of time.

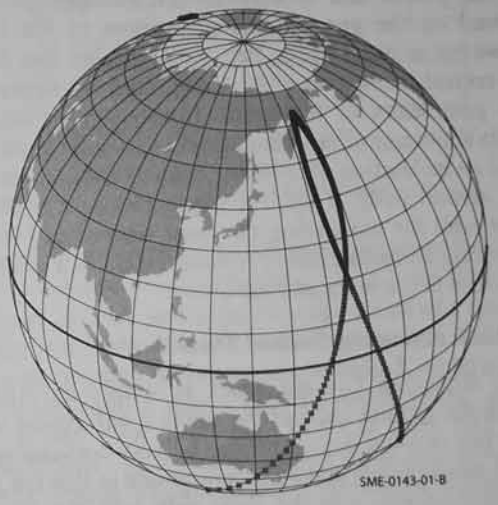


Fig. 10-23. A Molniya Orbit with a 12-Hour Period. With two satellites in this orbit, one of them will always be within 24 deg of apogee in true anomaly and with three satellites one will always be within 15 deg.

The properties of representative Molniya orbits for the Earth are summarized in Table 10-15 and a typical orbit ground track is shown in Fig. 10-23. Note that there are tick marks along the orbit at equal time intervals that clearly illustrates the long period of time that the spacecraft remains over high northern latitudes. Because each of the satellites tends to "hang" at apogee, it is straightforward to use two or three satellites in orbits of this type to provide good coverage of far northern or far southern regimes. Unfortunately, there are a number of disadvantages of the Molniya orbit, such as the non-stationary ground track with respect to the Earth, and the fact that it travels through the Van Allen radiation belts on a continuing basis, and consequently has a severe radiation environment. Nonetheless, the Molniya orbit provides the best mechanism for providing extended coverage in high latitude regions, and therefore, is appropriate for any type of communication, observation, or sampling system that requires coverage of this type. (The statite, a

Table 10-14, Fig. 10-23, Eq. 10-16b◀

Table 10-15. Properties of Representative Molniya Orbits. (See also Table 9-15 in Sec. 9.5.4.)

Period Hours	Eccentricity	Apogee Height (km)	Perigee Height (km)	Fraction of Orbit with 90° < True Anomaly < 270°	Fraction of Orbit with 120° < True Anomaly < 240°
2	0.1713	3,062	300	60.9%	43.4%
6	0.6016	20,471	300	85.8%	73.6%
12	0.7490	40,164	300	92.7%	84.9%
24	0.8419	71,426	300	96.3%	91.7%
36	0.8794	97,647	300	97.5%	94.3%
48	0.9004	121,051	300	98.1%	95.6%
72	0.9240	162,674	300	98.8%	97.0%

potential alternative using solar radiation pressure, is discussed in Sec. 18.7.2.)

The period and eccentricity of the Molniya orbit depend on the gravitational parameters of the Earth. However, as discussed in Sec. 9.5.4, the fact that it is at the critical inclination of 63.4 deg, does not depend on any parameters of the Earth. Consequently, Molniya orbits about any of the other planets or satellites will also be at an inclination of 63.4 deg relative to the equator of that central body. Thus, we might choose a Molniya-style orbit for monitoring the polar regions of Mars or providing communications or observations for scientific activities at the poles and the Moon.

10.4.2.4 Repeating Ground Track Orbits

A *Repeating Ground Track* is simply an orbit in which the ground trace of the satellite will repeat itself after one or more days. This can be very convenient for Earth observations and communications in that the general characteristics of the flight path with respect to a ground target or observer on the ground is a repeating pattern. Recall from Sec. 9.5.2 that a repeating ground track comes about when we go through an integral number of orbits in a sidereal day, taking into account the orbit rotation due to the oblateness of the Earth. Consequently, the fundamental requirement for a repeating ground track is:

$$H = \mu^{1/3}\left(\frac{n\pi}{180}\right)^{-2/3} - R \qquad (10\text{-}17)$$

$$n = (j/k)\left(360 - \dot{\Omega}\right) - \left(\dot{\omega} + \dot{M}\right) \qquad (10\text{-}18)$$

where

$$\dot{\Omega} = -2K_2 a^{-7/2}\cos i\left(1-e^2\right)^{-2} \qquad (10\text{-}19)$$

where the initial estimate of a for Eqs. (10-20 and 10-21)

$$\dot{\omega} = K_2 a^{-7/2}\left(5\cos^2 i - 1\right)\left(1-e^2\right)^{-2} \qquad (10\text{-}20)$$

$$\dot{M} = K_2 a^{-7/2}\left(3\cos^2 i - 1\right)\left(1-e^2\right)^{-3/2} \qquad (10\text{-}21)$$

is just the repeating ground track value of the semimajor axis in the absence of the oblateness correction, i.e., $a_{initial} = K_1(j/k)^{-2/3}$, where j and k are integers. As discussed in Sec. 9.5.2, $\dot{\Omega}$ is the node rotation rate $\dot{\omega}$ is the argument of perigee rotation rate, and $\dot{M}$ is the mean anomaly rotation rate due to oblateness. As usual, the parameters, a, e, and i are the semimajor axis, eccentricity, and inclination of the repeating ground track orbit and R is the radius of the central body. Values of K_1 and K_2 for various objects in the solar system are given in Table 10-16. (See Sec. 9.5.2 for the formulas for these constants.) Note that with the constants in the table, the rotation rates are all expressed in deg/sidereal day ≡ deg/day*, where "sidereal day" is the rotation period **of the body the satellite is orbiting about** relative to the fixed stars. The satellite completes j orbits in k sidereal days.

The principal merit of a repeating ground track is that we continue to sample the same region of the planet's surface on a repeating basis. This is useful, for example, in planet observations where we are interested in determining changes from one time to the next. For the Earth, this can be particularly important for evaluating changes in vegetation, water levels, pollution parameters, population growth, or deforestation. The principal demerit of a repeating ground track orbit is that it also flies through the variations in the planet's geopotential in the same fashion on successive orbits. Thus, if the spacecraft flies just to the left of the Himalayas on one day, it will do so on successive days as well. This allows the potential of resonance effects to build up, comparable to vibrations in a physical structure, which can greatly magnify the impact of the perturbations and cause significantly larger variations in the motion of the satellite. Consequently, a repeating ground track should be chosen only if there is an operational reason to do so, since it may require significant propellant utilization to counter the cumulative perturbative forces.

For the Earth, an interesting subgroup of the repeating ground track orbits are those for which the inclination, and therefore node rotation rate, is adjusted so that the repeat occurs on precise multiples of the civil day of

Table 10-16. Repeating Ground Track Parameters for Representative Central Bodies.

Body	Radius, R (km)	Sidereal Day (sec)	μ (km^3/sid day^2)	K_1 (km)	K_2 ($km^{3.5}$/sid day)
Mercury	2,439	$5.067\,032 \times 10^6$	$5.656\,674 \times 10^{17}$	$3.666\,292 \times 10^6$	$2.013\,338 \times 10^{11}$
Venus	6,052	$2.099\,606 \times 10^7$	$1.435\,799 \times 10^{20}$	$5.988\,628 \times 10^7$	$8.887\,296 \times 10^{12}$
Earth	6,378.136	$8.616\,410\,04 \times 10^4$	$2.959\,310\,16 \times 10^{15}$	$2.959\,310\,16 \times 10^{15}$	$1.029\,549\,648 \times 10^{14}$
Moon	1,738	$2.360\,591 \times 10^6$	$2.732\,032 \times 10^{16}$	$8.023\,799 \times 10^5$	$7.590\,274 \times 10^{10}$
Mars	3,397.2	$8.864\,266 \times 10^4$	$3.382\,663 \times 10^{14}$	$2.081\,545 \times 10^4$	$3.126\,615 \times 10^{11}$
Jupiter	71,492	$3.572\,986 \times 10^4$	$1.618\,757 \times 10^{17}$	$8.884\,214 \times 10^4$	$2.274\,883 \times 10^{16}$
Saturn	60,268	$3.780\,000 \times 10^4$	$5.421\,019 \times 10^{16}$	$6.405\,589 \times 10^4$	$1.043\,377 \times 10^{16}$
Uranus	25,559	$5.616\,000 \times 10^4$	$1.827\,396 \times 10^{16}$	$5.804\,523 \times 10^4$	$7.947\,793 \times 10^{14}$
Neptune	24,764	$6.635\,520 \times 10^4$	$2.998\,011 \times 10^{16}$	$7.651\,243 \times 10^4$	$3.185\,514 \times 10^{14}$
Pluto	1,151	$5.518\,109 \times 10^5$	$2.740\,457 \times 10^{14}$	$6.566\,573 \times 10^4$	N/A
Phobos	11.3	$2.755\,384 \times 10^4$	$5.385\,410 \times 10^5$	$1.115\,318 \times 10^1$	N/A
Deimos	6.3	$1.090\,749 \times 10^5$	$1.889\,503 \times 10^6$	$4.241\,016 \times 10^1$	N/A
Io	1,821.3	$1.528\,535 \times 10^5$	$1.392\,519 \times 10^{14}$	$2.226\,665 \times 10^4$	N/A
Europa	1,565	$3.068\,220 \times 10^5$	$3.015\,221 \times 10^{14}$	$4.583\,899 \times 10^4$	N/A

24 hours. For these orbits, the ground track not only repeats but does so at the same time each day, every other day, or on a weekly basis, depending on the orbit parameters chosen. We call an orbit of this type a *civil orbit* because it is one in which the spacecraft is keeping time with civil functions, rather than requiring the many users on the ground to keep track of a spacecraft ephemeris. Thus, for example, for relaying data to the home office, we could catch the "5:10 to London" on a daily basis such that we could transmit data at the same time each day and have it received at a fixed, later time.

10.4.2.5 Frozen Orbits

As discussed in Sec. 9.5.6, frozen orbits come about because circular orbits are not inherently stable. On the other hand, stable or frozen orbits are available with very low values of eccentricity, such that they are nearly circular for most practical applications. Frozen orbits can be used with essentially any low-Earth orbit and are largely used as a means of maintaining a more precise orbit and reducing the propellant requirements for orbit maintenance. For applications in which a circular orbit is desirable, a frozen orbit will somewhat reduce the propellant utilization and has no significant drawbacks. For additional details on creating a frozen orbit, for example, Vallado [2007].

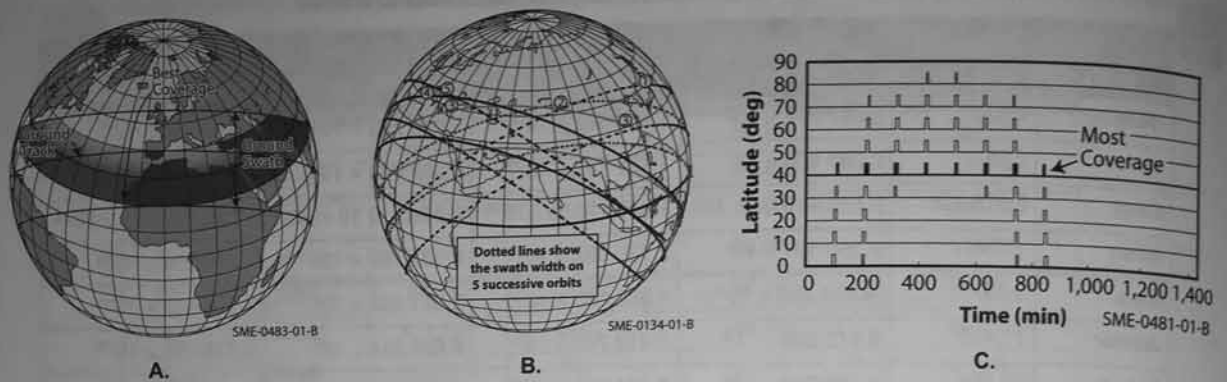
10.4.2.6 Repeat Coverage Orbits

A *Repeat Coverage Orbit* (RCO) is a prograde orbit that provides multiple successive orbits of coverage for a specific location on the Earth [Wertz, 2005]. Most of the other specialized orbits make use of particular orbit perturbations, such as J_2, to create unique conditions. In contrast, the RCO simply takes advantage of the coverage characteristics brought about by the Earth rotating under the orbit plane as discussed in detail in Secs. 10.2.1.2 and 10.2.1.3.

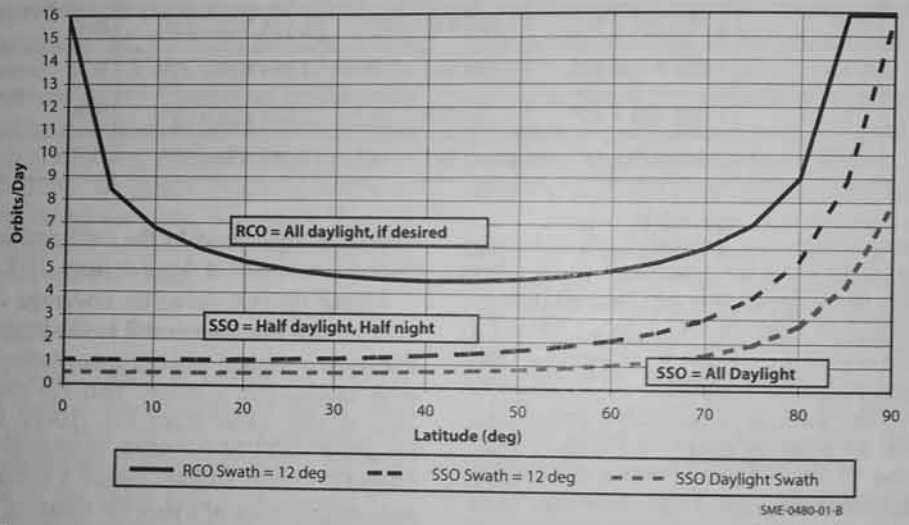
Figure 10-24A shows the coverage swath of a typical satellite in a low Earth orbit as the Earth rotates under it. The most coverage will occur at a latitude, L, equal to the orbit inclination, i, minus half the swath width, λ_{max}. This is also clear from Fig. 10-10 in Sec. 10.2.1.3. Equation 10-13 in the same section gives the coverage formula. Figure 10-24B shows 5 successive orbits that provide coverage of a specific location, shown as a small black square. Here, you can see what happens as the Earth rotates under the satellite or, equivalently, as the ground swath moves approximately 22 deg westward on each orbit. This is the same coverage pattern that was in the hypothetical HawaiiSat example in Sec. 5.4.2.2. The tombstone plots, equivalent to Fig. 10-7, for a slightly different example are shown in Fig. 10-24C. Notice that at 40 deg latitude, there are 8 successive orbits that provide coverage. This type of plot is particularly useful in designing a small constellation of RCO satellites.

The coverage vs. latitude for a *Repeat Coverage Orbit* is shown in Fig. 10-25 and compared with that of a Sun-synchronous orbit, SSO. Note that the RCO provides substantially more coverage than the SSO, particularly at mid and low latitudes. However, the RCO coverage is tuned to the specific latitude of interest while the SSO coverage is from one SSO orbit. Thus, with the RCO, we are giving up global coverage in order to provide better coverage of a specific latitude band of interest.

In the RCO, all of the revisits occur one after another. The time of day of the first pass will depend on the time of day when the satellite was launched and the difference in longitude between the launch site and the target. On



Live Calc **Fig. 10-24. The Repeat Coverage Orbit.** (A) The maximum coverage occurs at a latitude equal to the inclination minus half the swath width. (B) Five successive orbits provide coverage of a target at 32 deg N latitude. (C) A tombstone plot showing the region of maximum coverage. For additional plots and Excel calculation models, see Figs. 10-7, 10-9, and 10-10.



Live Calc **Fig. 10-25. Coverage vs. Latitude for Single Satellites in an RCO and an SSO.** The SSO is a single orbit intending to cover the entire world. The RCO is adjusted to each individual latitude.

successive days, the combination of node rotation due to J_2 and the revolution of the Earth around the Sun will move the sequence of passes over a given location forward in time by:

$$\Delta T \approx 3.9424 + 8.1896 \times 10^{14}\, a^{-7/2} (1 - e^2)^{-2} \cos i \qquad (10\text{-}22a)$$

where ΔT is the amount earlier that each pass occurs in min/day, a is the semimajor axis in km, e is the eccentricity, and i is the inclination. (The first term is due to the revolution of the Earth around the Sun and the second term is due to node rotation caused by J_2, as shown in Eq. (9-60d).) For a circular orbit at an altitude of 500 km, this becomes

$$\Delta T \approx 34.289 \cos i \qquad (10\text{-}22b)$$

Thus, for mid-latitude orbits, the time of day of the sequence of passes gets earlier by approximately 30 min/day.

Depending on the latitude, a constellation of 3 or 4 satellites can provide coverage of a given location every 90 min, 24 hours/day. Similarly 6 or 8 satellites could provide coverage every 45 min, 24 hrs/day, and 24 or 32 satellites could provide 24 hrs/day coverage approximately every 10 min.

To choose a Repeat Coverage Orbit, we first select the latitude, L, (or latitude band) of interest and determine the width of the ground swath width, $2\lambda_{max}$, for the satellite, as discussed in Sec. 10.2.1.3. We then select an inclination, i, somewhat less than $L + \lambda_{max}$. Recall from Fig. 10-10 that the coverage drops off rapidly below $i - \lambda_{max}$. Also note that most launch vehicles can put approximately 30% more payload into a prograde orbit due east from the launch site than into a Sun-synchronous orbit such that the launch cost will be less for the RCO than for the SSO or, equivalently, we can launch a somewhat larger satellite, or more of them, into an RCO with a given launch vehicle. The RCO is an excellent

Table 10-16, Fig. 10-25, Eq. 10-22b◀

option for observation satellites, if we are able to limit the latitude band of interest.

10.4.2.7 Eccentric Orbits for Earth Coverage

Eccentric orbits can be thought of in either of 2 ways. On one hand, they add a level of complexity that increases the design cost of the spacecraft and the mission by requiring the spacecraft to work at multiple altitudes and at differing rotation rates if one side is to remain facing the Earth. On the other hand, eccentricity adds a degree of freedom in the design that can be used to adjust the orbit to best meet specific mission needs. Eccentric orbits can be used to minimize the number of satellites, maximize the coverage with respect to areas of interest, or combinations of these purposes. The use of eccentric orbits has been studied at length by Draim [1997, 1998, 2004, 2008, 2009], who has made an extremely strong case for using eccentric orbits to achieve maximum mission performance with a minimum number of satellites [Draim et al., 1999, 2008].

For any specific mission, the question becomes whether this additional level of performance is worth the increased cost and complexity of the spacecraft and mission parameters. As discussed in Sec. 10.6.2, one of the most interesting examples of the use of eccentric orbits is using only 4 spacecraft to provide complete, continuous coverage of the surface of the Earth.* Other interesting examples include the proposed Ellipso and Cobra communications constellation which uses eccentric orbits, potentially in combination with a supplemental equatorial orbit in order to optimize a low-Earth orbit communications system with small numbers of satellites and optimized coverage for both geographic location and times of day [Draim, 1999, 2001, 2002]. As in the case of Molniya orbits, virtually all satellites in elliptical orbits will need to be at the critical inclination of 63.4 deg in order to prevent the line of apsides from rotating. Since this rotation rate can be up to 14 deg per day, it is important to maintain the critical inclination with good precision in order to maintain apogee and perigee over an extended mission life. This implies that orbit maintenance will typically be necessary for satellites in long term elliptical orbits. However, the ΔV requirements to maintain these parameters are very modest. For a discussion of elliptical orbits vs. circular orbits for communications, see Wertz [2007].

10.5 Selecting Transfer, Parking, and Space-Referenced Orbits

In a space-referenced orbit, our objective is typically to be in space for manufacturing, low-gravity or high-vacuum experiments, or to observe celestial objects from above the Earth's atmosphere. These orbits are used by celestial observatories such as Space Telescope, Chandra, or JWST. In these orbits we typically have only a minimal concern with our orientation relative to the Earth. Consequently, we select such orbits to use minimum energy, while maintaining the orbit altitude and possibly to gain an unobstructed view of whatever celestial objects we may be observing. As listed in Table 10-17, the mission requirements that normally affect the design of such orbits tend to be less stringent than for Earth-referenced orbits. We are looking for a reasonable environment, easy accessibility, and good communications, all of which can be satisfied by a wide range of orbits. Consequently, the orbit design process is typically much more straightforward than it is for Earth-referenced orbits. Of course, some space-referenced mission orbits are designed specifically to sample space, such as magnetic field or solar wind explorers. In these cases, the orbit design is chosen to sample the region of interest on a schedule appropriate to the science return.

Table 10-17. Principal Requirements that Normally Affect the Design of Space-Referenced Orbits. Typically none of the requirements is as stringent as for Earth-referenced orbits.

Requirement	Where Discussed
Accessibility (ΔV or OCF)	Sec. 10.3
Orbit Decay Rate and Long Term Stability	Sec. 9.4.4
Ground Station Communications, Especially for Maneuvers	Sec. 8.4, 10.2.1
Radiation Environment	Sec. 7.3
Thermal Environment (Sun angle and eclipse constraints)	Secs. 8.2.4, 22.2

As listed in Table 10-18, there are a few specialized space-referenced orbits that need to be considered in much the same fashion as specialized Earth-referenced orbits. The Sun-synchronous orbit was discussed previously in Sec. 10.4.2.2. This type of orbit may be appropriate for maintaining a constant Sun angle with respect to a satellite instrument such as an orbiting solar observatory.

Perhaps the most interesting and most widely used of the space-referenced orbits are the Lagrange point orbits, described in Sec. 9.5.5. These are orbits which use the gravitational attraction of two orbiting bodies such as the Earth and the Moon or the Earth and the Sun to maintain the spacecraft in a constant orientation relative to the two bodies. The locations of Lagrange point orbits are shown in Fig. 10-26 and Fig. 9-29 in Sec. 9.5.5. Because of their long-term stability, these orbits have a wide variety of potential applications as summarized in Table 10-20.

Recall from Sec. 9.5.5 that the L_4 and L_5 Lagrange points are stable, such that objects orbiting these points will remain there indefinitely. In contrast, L_1, L_2, and L_3 are unstable equilibrium points such that an object placed there will ultimately drift away. However, it takes a relatively small amount of propellant to maintain a spacecraft in a *halo orbit* about one of the Lagrange points. In

* This orbit has been patented by Draim, [1987a].

Table 10-18. Specialized Space-Referenced Orbits.

Orbit	Characteristic	Application	Where Discussed
Sun-Synchronous	Orbit rotates so as to maintain approximately constant position with respect to Sun	Solar observations; Missions concerned about Sun interference or uniform lighting	Secs. 9.5.3, 10.4.2.2
Lagrange Point Orbit	Maintains fixed position relative to Earth/Moon system or Earth/Sun system	Interplanetary monitoring; potential space manufacturing	Secs. 9.5.5, 10.5
Statite	Solar radiation pressure balances gravity	Interplanetary monitoring and communications	Forward [1989], Sec. 18.7.2

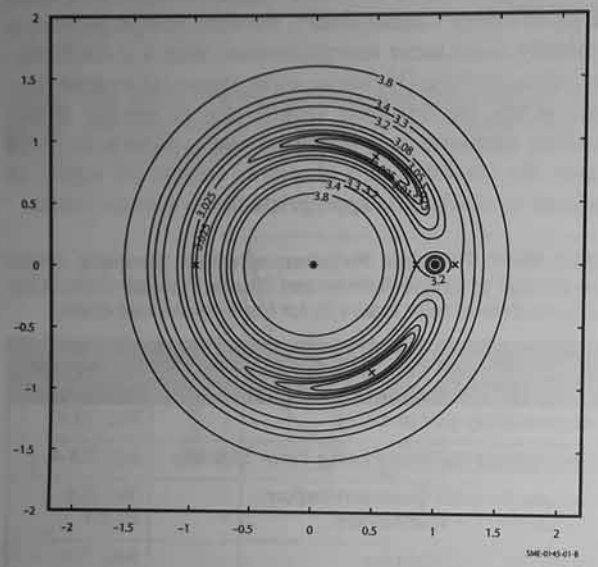


Fig. 10-26. Lagrange Point Orbits. These orbits are stable with respect to the 2-body system they are part of and are convenient locations for observations, communications, and manufacturing. (See also Fig. 9-29 in Sec. 9.5.5.)

Table 10-19. Principal Requirements that Normally Affect the Design of Transfer Orbits. Typically none of the requirements is as stringent as for Earth-referenced orbits.

Requirement	Where Discussed
Transfer ΔV	Sec. 9.6, Table 10-7
Transfer Time	Sec. 9.6
Departure and Arrival Conditions	Sec. 9.6.1
Ground Station Communications, Especially for Maneuvers	Secs. 8.4, 10.2.1
Radiation Environment	Sec. 7.3
Thermal Environment (Sun Angle and Eclipse Constraints)	Secs. 8.2.4, 22.2
Required Navigation and Control Accuracy	Sec. 19.2

any case, the halo orbit is necessary in most cases in order to provide communications. For example, a spacecraft at the L_2 Lagrange point on the far side of the Moon will be useful for communications with any lunar far side stations but would need to be in a halo orbit in order to provide line of sight communications back to the Earth.

Table 10-19, Fig. 10-26, Eq. 10-22b◀

Similarly, solar monitoring stations have been maintained for extended periods at the Earth-Sun L_1 Lagrange point in order to provide advance warning for solar storms. Again the halo orbit is required in order to keep the spacecraft away from a direct line with the Sun, which would prevent communications with the spacecraft due to the high level of radio noise from the solar disk.

The long term stability of the Earth-Moon L_4 and L_5 Lagrange points make these an ideal location for very large space colonies or space manufacturing facilities. Because they are at the distance of the Moon, the gravity-gradient forces will be extremely small such that a very large region of extremely low microgravity is possible. In addition, massive facilities are possible there, because no ΔV is necessary to maintain the orbit and there is no drag to provide orbit decay, as occurs for the International Space Station. Material for constructing a large facility at the L_4 and L_5 Lagrange points can be brought at relatively low ΔV cost from the surface of the Moon. The extensive work by Gerard O'Neill* and his colleagues in the 1960s and 1970s has clearly established the feasibility and potential utility of large colonies at these locations. [Heppenheimer, 1977; O'Neill, 1976a,b; O'Leary, 1982a,b; Faughnam et al., 1993]. Unfortunately, the continuing high cost of launch has prevented any of these from advancing beyond the mission study phase.

Lagrange point orbits are also possible with respect to other pairs of celestial bodies. For example, spacecraft at the L_4 or L_5 Lagrange point with respect to Jupiter or the Sun could be used for monitoring the asteroid belt, Similarly, a Lagrange point orbit between any of the major planets and their larger moons could be used as a stable orientation with which to observe that planetary system. Thus, long-term stable (or nearly stable) observation platforms could be created at any of several of the Lagrange points within the systems of Jupiter, Saturn, Uranus, or Neptune.

* O'Neill originated the concept as part of his undergraduate physics classes at Princeton in 1969. After several years working on technical details, he first published the concept of colonizing space itself at L_4 or L_5, rather than on a planetary surface, in the journal *Physics Today* in Sept., 1974 [O'Neill, 1974]. Substantial public enthusiasm led to the creation of the L-5 Society in 1975. In 1987 the L-5 Society merged with the National Space Institute to form the National Space Society.

Table 10-20. Potential Applications of Lagrange Point Orbits. E-S = Earth-Sun, E-M = Earth-Moon, H = Halo orbit.

Lagrange Point	Application
E-M L_4, L_5	Space Manufacturing
E-M L_4, L_5	Space Observatory
E-M L_4, L_5	Lunar Communications
E-M L_4, L_5	Large Space Colonies
E-M L_2 H	Lunar Far Side Communications
E-S L_1 H	Monitoring solar activity
E-S L_3 H	Observing events not visible from Earth

10.5.1 Design of Transfer and Parking Orbits

Typically, the *parking orbit*, or *storage orbit*, is a low-Earth orbit high enough to reduce atmospheric drag, but low enough to be easy to reach. We may store satellites in these orbits, referred to as *on-orbit spares*, for later transfer to a higher altitude, or may use them as a place for test and check-out following launch or waiting for the appropriate transfer conditions to occur. Consequently, the principal issues for parking orbits are accessibility and matching the orbit conditions on either end to what is needed. For example, a Mars probe may be launched into low-Earth orbit for test and check-out while awaiting the appropriate opportunity to begin the Mars transfer phase.

The fundamental purpose of a *transfer orbit* is to get the spacecraft where it wants to be, when it wants to be there. Ordinarily, the design process is straightforward, and we'll want to do this with a minimum cost in terms of ΔV or propellant utilization. The physics of transfer orbits and the equations that govern them were described in Sec. 9.6.1. The principal requirements that normally affect the design of transfer orbits are shown in Table 10-19. As with space-referenced orbits, these are typically less stringent requirements than for Earth-referenced orbits, although there is often a requirement to match the desired end conditions with high precision in order to minimize the propellant cost of any corrective ΔV.

A further requirement that can arise in manned flight is for orbit transfer options that are "forgiving." For example, the Apollo program used a *free return trajectory* such that if the burn required to go into lunar orbit did not occur on the backside of the Moon, then the spacecraft would return to the Earth with no further ΔV. This was critical for the successful rescue of Apollo 13, in which the propulsion system in the command module was effectively destroyed by an explosion and the free return was necessary in order to bring the astronauts back to the vicinity of the Earth such that the lunar module propulsion system could be used to provide the appropriate deorbit ΔV for recovery of the astronauts.

Similarly, in potential human Mars missions, there is a high premium for reducing the transfer time because of the high cost of maintaining people in orbit for extended periods. In addition, in any manned planetary mission, we need to plan the transfer time such that a return to Earth can be achieved in an appropriate time. Both of these requirements lead to somewhat higher energy transfers for human flight than the traditional Hohmann transfer.

As shown in Table 10-21, there are specialized transfer orbits that are available to the mission designer. The physics of these transfers was discussed in Chap. 9. For each of the specialized transfer orbits, the fundamental objective is to reduce the ΔV required to achieve a particular set of mission objectives. Lunar or planetary flybys, for example, are used to provide additional energy by providing effectively an elastic collision between the spacecraft and some other celestial object. Aeroassist trajectories are used principally to provide either a plane change or braking, as in the case of the Mars orbiter program.

As was discussed in Sec. 9.6.2, the ΔV required for simultaneous plane changing and orbit raising can be reduced by combining these maneuvers such that the total ΔV is the vector sum rather than the linear sum of the individual maneuvers. However, in cases where the plane change is extremely large, this ΔV can be further reduced by using a 3-burn transfer, in which the spacecraft is first raised to an altitude significantly higher than the final orbit altitude. At extremely high altitude, the velocity will be very low and consequently a plane change can be done with very small ΔV. Thus the plane change is accomplished at the high altitude and the orbit is then returned to the final altitude. Similar to the 3-burn transfer, Miller and Belbruno [1991], has evaluated and patented several trajectories which make use of the Moon to further reduce the ΔV required for plane change maneuvers. The principal demerit of this approach is that it requires a fairly extended time for the maneuver, although substantial ΔV savings are possible in the case of large plane changes. The principal advantages and disadvantages of alternative plane change mechanisms are given in Table 10-22.

Table 10-21. Specialized Transfer Orbits.

Orbit	Characteristic	Application	Where Discussed
Lunar or Planetary Fly-By	Same relative velocity approaching and leaving flyby body	Used to provide energy change or plane change	Sec. 9.6.3
Aeroassist Orbit	Use atmosphere for plane change or braking	Used for major energy savings for plane change or reentry	Sec. 9.6.2
3-Burn Transfer (= bi-elliptic transfer)	Go to altitude above final orbit to reduce plane change ΔV	Reduce total ΔV for orbits with large plane change	Sec. 9.6.2,

Table 10-22. Alternative Plane Change Mechanisms. See Table 9-17 in Sec. 9.6.2 for physical mechanisms and computations.

Orbit	Advantages	Disadvantages	Typical Application
Do ΔV at Lowest Velocity (highest altitude)	Smallest velocity is easiest to change	May not be a low-velocity orbit segment	Orbit transfer
Combine ΔV with Orbit Raising	Vector sum is less than sum of components	Works best when large in-track ΔV is required	Orbit transfer
3-Burn Transfer	Plane change can occur at essentially no ΔV cost	Requires going to a higher altitude and then returning	Large plane change with moderate to high altitude final orbit
Use Differential Node Rotation	No propellant cost	Time delay	Storage of satellites in a parking orbit
Aero-Assist Trajectories	No propellant cost	Requires precision maneuver and aerodynamic surfaces; may be heating	Planetary exploration
Planetary Fly-by	No propellant cost	Requires planetary swing-by and typically long delays	1-Outer planet exploration 2-Out of ecliptic mission (using Jupiter) 3-Trips to near the Sun (using Jupiter)

Table 10-23. Characteristics of the Alternative Direct Transfer Orbits Shown in Fig. 9-29 in Sec. 9.6.1. See Table 9-16 for transfer characteristics.

Type	Typical Accel.	Orbit Shape	Advantages	Disadvantages	Applications
High Energy	1g to 10g	Elliptical & hyperbolic	Rapid transfer	Uses more energy than necessary + Hohmann disadvantages	Human flight, military intercept
Minimum Energy, High Thrust (Hohmann)	1g to 5g	Hohmann transfer	Traditional High efficiency Rapid transfer Low radiation exposure	Rough environment Thermal problems Can't use S/C subsystems	Used for nearly all orbit transfers
Low Thrust Chemical	0.02 to 0.10g	Hohmann transfer segments	High efficiency Low engine weight Low orbit deployment & check-out Better failure recovery Can use spacecraft subsystems	Moderate radiation exposure 3 to 4 day transfer to GEO	Use when needed to reduce cost, weight, risk, or disturbances
Electric Propulsion	0.0001 to 0.001g	Spiral transfer	Can use very high I_{sp} engines = major weight reduction Low orbit deployment & check-out Can have reusable transfer vehicle	2 to 6 month transfer to GEO High radiation exposure Needs autonomous transfer for cost efficiency	Rarely used. Has been proposed for GEO and interplanetary transfer

The various alternatives for direct orbit transfers have been described in detail in Sec. 9.6.1. (See specifically Fig. 9-32 and Table 9-16.) Table 10-23 lists the relative advantages and disadvantages of using these alternative transfer methods and the principal applications to which they apply. High-energy direct transfers are useful principally when the transfer time is critical, as in human flight or some military missions. Typically, the ΔV penalty associated with high-energy transfer is small for minor increases in transfer time, but becomes very large for any major reduction in transfer time, such that very short transfers are prohibitively expensive in terms of ΔV. The other alternatives of successive burns using low-thrust chemical or a spiral transfer using electric propulsion both take significantly longer than the Hohmann transfer and are of use principally when transfer time is not a major constraint. The low thrust chemical transfer offers the significant advantage of providing a much more benign environment for the spacecraft, since the principal disturbance forces and maximum loads frequently occur during orbit transfer. In addition, low thrust transfer allows the spacecraft to be fully deployed prior to transfer, such that it can be tested and checked out before being put in a totally inaccessible orbit. Electric propulsion transfer has the merit of extremely high efficiency but can take a very long time. Consequently, electric propulsion is an extremely efficient approach for orbit maintenance but has not yet been widely used for orbit transfer.

In the design of orbit transfer, it is important to keep in mind the distinction between the basic requirements for launch vehicles and the requirements for in-orbit transfer. In launch vehicle design, high thrust is absolutely imperative in order to get to space as quickly as possible and minimize gravity losses that occur when thrusting is not done at right angles to the central body. In a number of launch attempts in the early space program, the thrust to weight ratio was very close to 1, such that the launch vehicle lifted a short distance off the pad,

got out of the ground effect, held itself briefly in the air, and then fell back to Earth. Thus, a driving characteristic for launch vehicles is the need for a high thrust system.

In contrast, once we have gotten into space, high thrust is no longer important and can be a major detriment to an in-orbit transfer system. A high thrust transfer stage frequently imparts the largest loads the spacecraft will ever see. Thus it has a strong potential for damaging the spacecraft. In addition, in any high thrust system, performance is critical and all maneuvers must be done with precision both in terms of the maneuver itself and the timing of the maneuver. Typically, any failures of high thrust transfer will lead directly to mission failure. Similarly, because the forces are extremely large, high thrust orbit transfer stages need their own separate control system to control the very large disturbance torques that are generated. Consequently, the transfer stage is nearly a separate spacecraft with a very large propulsion and control system. However, in the on-orbit environment, much of this performance is either wasted or detrimental. An on-board propulsion system, for example, can provide the ΔV appropriate to orbit transfer but do so at much lower thrust levels. For most orbit transfers, the time required for the burn is small relative to the transfer time and is unimportant in successfully completing the maneuver. If the orbit transfer is done by a sequence of low thrust burns, then individual burns become far less critical and an error or misfire on one burn may be able to be corrected on subsequent burns. Consequently, low thrust orbit transfer tends to be significantly more fail-safe than high thrust transfer. In addition, the disturbance torques imparted by a low thrust system are far less, such that it may be possible to control the transfer using the control system already on board the spacecraft itself. This can significantly reduce the amount of ancillary equipment that needs to be flown, thus reducing both the weight and cost of the mission. The use of on-board propulsion can have a dramatic impact on the mass available in the mission orbit, or, alternatively, on the launch cost because the entire mass of the transfer stage doesn't need to be sent all the way to the mission orbit.

The important point here is to make a clear distinction between the requirements for launch and the requirements for orbit transfer once we have achieved low-Earth orbit. These frequently are regarded as nearly the same process because they are often assigned to a single organization which has the responsibility for putting the spacecraft in its final mission orbit. While this has some possible organizational advantages, it typically provides a system which is less robust, higher risk, more expensive, and lower performing than would be the case using a low thrust system once we have achieved orbit. The critical issue here is to regard in-orbit transfer not as simply an extension of the launch process, but as a separate mission phase. As always, we want to achieve the overall mission objectives at minimum cost and risk. In many cases, we may be able to substantially reduce both cost and risk by making use of straightforward, low-cost, low-thrust technology for in-orbit transfer.

10.6 Summary of Constellation Design

A *constellation* is a set of satellites distributed over space intended to work together to achieve common objectives. If the satellites are close together, such as a flying interferometer, or spacecraft next to each other exchanging data or material, then it is called a *cluster* or *formation*, rather than a constellation. Because constellations are inherently very expensive, most are used for communications, navigation, or similar functions that take advantage of the global Earth coverage provided by satellites and difficult to achieve by other means.* The constellation design process tends to be complex with no agreed upon solution by the community as a whole. A thorough discussion of this process is beyond the scope of this volume and is covered in detail in Wertz [2009]. Here we provide a summary of the constellation design process and the broad rules for constellation design.

One of the most important results of decades of constellation studies has been that no absolute rules exist for constellation design. For example, the Multi-Satellite System Program (MSSP), was a proposed constellation of satellites in random low-Earth orbits in order to create a survivable communications system. (See Wertz [2009], Sec. 13.2.) One of the most interesting characteristics of the low-Earth orbit communications constellations is that constellation builders have invested billions of dollars and come up with distinctly different solutions. For example, a higher altitude means fewer satellites, but a much more severe radiation environment, such that the cost of each satellite will be higher, and the life potentially shorter. Both solutions are being used by commercial constellations. Similarly, circular orbits provide a simple, low-cost satellite design. However, elliptical orbits allow an additional degree of freedom, which allows the constellation to be optimized for multiple factors, but requires a more complex satellite operating over a range of altitudes and velocities, and passing through heavy radiation regimes. (See, for example, Draim et al. [1996, 1999].) Again, both solutions are being used by modern constellations. Because the constellations' size and structure strongly affect system cost and performance, we must carefully assess alternative designs and document the reasons for trades that are made. It is this list of reasons that allows the constellation design process to continue in order to achieve the system objectives at minimum cost and risk.

10.6.1 The Constellation Design Process

The overall process of constellation design is summarized in Table 10-24. The key requirement in this process is to understand the mission objectives and what is needed to fulfill the mission, particularly with respect to coverage. This means understanding both the coverage

* In contrast, the F6 program is intended to create a resource-sharing formation specifically to reduce cost and vulnerability and allow low-cost augmentation or replenishment.

Table 10-23◀, Fig. 10-26◀, Eq. 10-22◀

Table 10-24. The Constellation Design Process. See also Tables 10-25, 10-27, and 10-28. See Wertz [2009] for a more extensive discussion.

Step	Where Discussed Here	Where Discussed OCDM
1. *Establish constellation-related mission requirements, particularly* • Latitude-dependent coverage • Goals for growth and degradation plateaus • Requirements for different modes or sensors • Limits on system cost or number of satellites	Secs. 8.4, 10.2.1, 10.6.1	Chap.5 Secs. 12.1, 13.1
2. *Do all single satellite orbit trades except coverage*	Table 10-1 (Sec. 10.1), Sec. 10.2	Chap. 12, Sec. 13.3
3. *Do trades between swath width (or maximum Earth central angle), coverage, and number of satellites.* *Evaluate candidate constellations for:* • Coverage Figures of Merit vs. Latitude and mission mode • Coverage excess • Growth and degradation • Altitude plateaus • End-of-life options Consider the following orbit types: • Walker Delta pattern (GlobalStar, GPS) • Polar constellations with seam (Iridium) • Equatorial (TDRS, GEO Comm, MEO Comm [Wertz, 2007]) • Equatorial supplement (Ellipso) • Elliptical (Molniya, Ellipso, Cobra)	Secs. 8.4, 10.6.2 Sec. 10.2.1 Sec. 9.6.4, Chap. 30	Chap. 9, Secs. 13.3, 13.6, 13.7 Chap. 9, Secs. 13.1–13.3 Sec. 13.7 Sec. 13.6 Sec. 13.6 Secs. 2.6.4, 13.6
4. *Evaluate ground track plots for potential coverage holes or methods to reduce the number of satellites*	Sec. 10.2	Sec. 9.5.1.2
5. *Adjust inclination and in-plane phasing to maximize the intersatellite distances at plane crossings for collision avoidance*		Sec. 13.5
6. *Review the Rules of Constellation Design.*	Sec. 10.6.3 (Table 10-30)	Sec. 13.7
7. *Document reasons for choices and iterate.*		

needs for the mission, and the spacecraft needs to provide that coverage, such as swath width or constraints on lighting, power, or communications. Can the system operate in the Van Allen belts? A second key part of the requirements is the growth and degradation goals. Are outages acceptable if a satellite dies on orbit? How soon is a replacement required? How soon will the constellation as a whole be replaced and upgraded?

The principal factors to be defined during constellation design are listed in Table 10-25, along with the major selection criteria for each. These factors have been discussed in detail earlier in the chapter. A key point to keep in mind here is that constellation design consists of more than just orbit elements. The minimum working elevation angle is perhaps the single most critical parameter in defining constellation coverage. Collision avoidance parameters, including the method for stationkeeping and size of the stationkeeping box are key to insuring the integrity of the constellation over its lifetime. While we often think of a constellation as a single, coherent pattern, it may be that some combination of patterns will satisfy our mission objectives at a lower cost and risk. This approach has been adapted, for example, by Ellipso with a mix of inclined eccentric and circular equatorial orbits.

Typically the purpose of creating a constellation is to provide Earth coverage, or at times, coverage of near-Earth space. The demerit of a constellation is that you need multiple satellites and, therefore, multiple reaction wheels, communication systems, and so on. A typical goal of constellation design is to provide the needed coverage with a minimum number of satellites. Consequently, the principal system trade is most frequently coverage as a measure of performance vs. number of satellites as a measure of cost. Nonetheless, it is important to bear in mind that the number of satellites is not necessarily an accurate representation of system cost. For example, raising the altitude of the constellation will normally result in needing fewer satellites. However, as the altitude increases, the launch cost for each satellite increases, and the radiation hardening requirements increase dramatically as we enter the Van Allen radiation belts. Minimizing the number of satellites may or may not minimize the system cost and it is the cost which is typically of most interest to constellation sponsors, both government and commercial.

Specifying a constellation by defining all of the orbit elements for each satellite is complex, inconvenient, and overwhelming in its range of options. A reasonable way to begin is by looking at constellations with all satellites in circular orbits at a common altitude and inclination. This means that the period, angular velocity, and node rotation rate will be the same for all of the satellites. This leads to a series of trades on altitude, inclination, and constellation pattern involving principally the number of satellites, coverage, launch cost, and the environment

Table 10-25. Principal Factors to be Defined During Constellation Design.

Factor	Effect	Selection Criteria	Where Discussed Here	Where Discussed OCDM
		PRINCIPLE DESIGN VARIABLES:		
Number of Satellites	Principal determinant of cost and coverage	Minimize number consistent with meeting other criteria	Chap. 8, Sec. 10.6.2	Chap. 9 Sec. 13.3
Constellation Pattern	Determines coverage vs. latitude, plateaus	Select for best coverage	Sec. 10.6.2	Sec. 13.1
Minimum Elevation Angle	Principal determinant of single satellite coverage	Minimum value consistent with payload performance and constellation pattern	Secs. 8.4, 10.6.2	Secs. 9.1, 13.3
Altitude	Coverage, environment, launch, and transfer cost	System level trade of cost vs. performance	Secs. 10.2.2, 10.4.1	Chap. 9, Secs. 12.4, 13.3
Number of Orbit Planes	Determines coverage plateaus, growth and degradation	Minimize consistent with coverage needs	Secs. 10.6.2, 10.6.3	Sec. 13.6
Collision Avoidance Parameters	Key to preventing constellation self destruction	Maximize the intersatellite distances at plane crossings	Sec. 10.6.3	Sec. 13.5
		SECONDARY DESIGN VARIABLES:		
Inclination	Determines latitude distribution of coverage	Compare latitude coverage vs. launch costs*	Sec. 10.2, Fig. 10-10	Sec. 13.3
Between Plane Phasing	Determines coverage uniformity	Select best coverage among discrete phasing options*	Sec. 10.6.2	Sec. 13.1
Eccentricity	Mission complexity and coverage vs. cost	Normally zero; non-zero may reduce number of satellites needed	Table 8-12, Sec. 8.4.2, Sec. 10.4.2	Sec. 13.3
Stationkeeping Box Size	Coverage needed; cross-track pointing	Minimize consistent with low cost maintenance approach		Sec. 13.4
End-of-Life Strategy	Elimination of orbital debris	Any mechanism that allows you to clean up after yourself	Sec. 9.6.4, Chap. 30	Secs. 2.6.4, 13.6

* Fine tune for collision avoidance

(primarily drag and radiation). We then examine the potential of elliptical orbits and the addition of an equatorial ring. After exploring the consequences of some of the choices, we will summarize the key trades and Rules for Constellation Design in Sec. 10.6.3.

10.6.2 Coverage and Constellation Structure

For most constellations, Earth coverage is the key reason for using multiple satellites. A constellation can provide observations and communications more frequently than a single satellite can. Given this objective, the normal trade in constellation design is coverage as a measure of performance versus the number of satellites as a measure of cost. Thus, we normally assume that a 5-satellite constellation will be less expensive than a 6-satellite one, but this assumption may be wrong. The larger constellation may be at a lower altitude or inclination and, therefore, cost less to launch or have a less harsh radiation environment. Alternatively, we may be able to have a smaller constellation with elliptical orbits, for which increased spacecraft complexity could offset the lower cost due to the number of satellites.

A principal characteristic of any satellite constellation is the number of orbit planes in which the satellites reside. Symmetry in constellation structure requires an equal number of satellites in each orbit plane. This means that an eight-satellite constellation may have either one, two, four, or eight separate orbit planes. But because moving satellites between planes uses much more propellant than moving them within a plane, it is highly advantageous to place more satellites in a smaller number of planes. Moving satellites within an orbit plane requires only a slight change in the satellite altitude. This changes the period so we can slowly rephase the satellite within the constellation, and then return it to the proper altitude to maintain its position relative to the rest. Thus, we can rephase many times using relatively little propellant. If a satellite fails or a new satellite is added to a given orbit plane, we can rephase the remaining satellites so that they are uniformly spaced. The consequence of this is to provide a significant premium to constellations which contain more satellites in a smaller number of orbit planes.

The number of orbit planes relates strongly to a coverage issue often overlooked in constellation design: the need to provide the constellation performance plateaus and graceful degradation. Ideally one would like to achieve some performance level with the very first satellite launched and to raise that level of performance with each succeeding satellite. Generally, however, performance tends to come in plateaus as we put one more satellite into each orbit plane of the final constellation. If a constellation has 7 orbit planes, we will achieve some performance with the first satellite, but the next major

performance plateau may not come until one satellite is in each of the 7 planes. We would expect this constellation to have plateaus at 1, 7, 14, 21, (and so on) satellites. Again, constellations with a small number of orbit planes have a distinct advantage over many-plane ones. A single-plane constellation produces performance plateaus with each added satellite, whereas one with 2 planes would have plateaus at 1, 2, 4, 6, 8, (and so on) satellites. Thus, more complex constellations will require more satellites for each performance plateau.

Frequent performance plateaus have several advantages. First, because individual satellites are extremely expensive, we may want to build and launch one or two satellites to verify both the concept and the constellation's ultimate usefulness. If a constellation is highly useful with just 1 or 2 satellites, it offers a major advantage to the system developer.

Another advantage is that coverage requirements are rarely absolute. More coverage is better than less, but we may not know at the time the constellation is designed how useful added coverage will be. For example, we may design the FireSat II system for 30-min revisits, then later revise the response strategy so 45-min revisits can provide nearly equal performance. Communications constellations are normally thought of as having a very rigid requirement of continuous global coverage. Even here, however, they may want more coverage or greater redundancy over regions of high population density.

A constellation of 1 or 2 planes can be more responsive to changing user needs than a system with multiple planes. Because we often design constellations many years before they are launched, we may not be able to correctly balance performance vs. cost. Both needs and budgets follow political constraints and economic priorities over much shorter periods than a constellation's lifetime. Thus if an 8-satellite constellation is highly useful with only 6 satellites, budget constraints may delay the launch of the remaining 2. At the same time, the constellation may expand to 10 satellites if the first set generates substantial demand for more performance or greater capacity. This responsiveness to political and performance demands provides perhaps the largest advantage to constellations with a smaller number of orbit planes.

Finally, a smaller number of orbit planes leads to more graceful degradation. In an 8-satellite, 2-plane constellation, if one satellite is lost for any reason, we may rephase the constellation at little propellant cost and thereby maintain a high performance level corresponding to a 6-satellite plateau. This rephasing and graceful degradation may be impossible for constellations with a large number of orbit planes.

Another important characteristic is the orbit inclination. In principle, one could design satellite constellations with many different inclinations to get the best coverage. In practice this is extremely difficult because the rate of nodal regression for a satellite orbit is a function of both altitude and inclination. Consequently, satellites at a common altitude but different inclinations will regress at different rates, and a set of orbit planes which initially have a given geometric relationship with respect to each other will change that relationship with time. Otherwise, we would have to expend propellant to maintain the relative constellation spacing, a technique that is extremely expensive in terms of propellant and is achievable for only a short time or under unique circumstances. Thus, we usually design constellations to have all the satellites at the same inclination. A possible exception is to have all satellites at a single inclination except for a set of satellites in a 0 inclination (equatorial) orbit. Regression of the nodes is not meaningful for the equatorial orbit, so we can maintain constant relative phasing indefinitely between satellites in equatorial and inclined orbits. An example of such a constellation is satellites in three mutually perpendicular orbit plane—2 polar and 1 equatorial.

As shown in Fig. 10-27, the spacing between satellites in a single orbit plane determines whether coverage in that plane is continuous and the width of the continuous coverage region. Assume that λ_{max} is the maximum Earth central angle as defined in Sec. 5.3.1 and that there are N satellites equally spaced at $S = 360/N$ deg apart in a given orbit plane. There is intermittent coverage throughout a swath of half-width λ_{max}. If $S > 2\,\lambda_{max}$, the coverage is intermittent throughout the entire swath. If $S < 2\,\lambda_{max}$, there is a narrower swath, often called a *street of coverage*, centered on the ground trace and of width $2\,\lambda_{street}$, in which there is continuous coverage. This width is given by:

$$\cos\lambda_{street} = \cos\lambda_{max} / \cos(S/2) \qquad (10\text{-}23)$$

If the satellites in adjacent planes are going in the same direction, then the "bulge" in one orbit can be used to offset the "dip" in the adjacent orbit as shown in Fig. 10-28. In this case, the maximum perpendicular separation, D_{max}, between the orbit planes required for continuous coverage is,

$$D_{maxS} = \lambda_{street} + \lambda_{max} \qquad (10\text{-}24)$$

(moving in the same direction)

If the satellites are moving in opposite directions, then the bulge and dip cannot be made to line up continuously and, therefore,

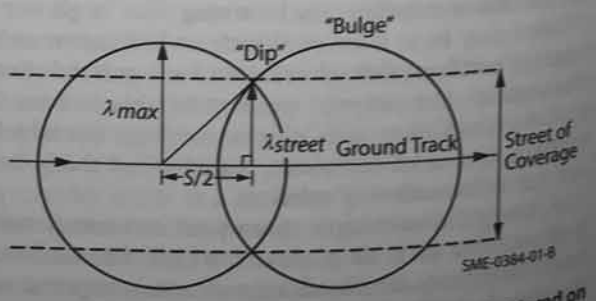


Fig. 10-27. The "Street of Coverage" is a Swath Centered on the Ground Track for which there is Continuous Coverage.

Table 10-25◀, Fig. 10-27, Eq. 10-24

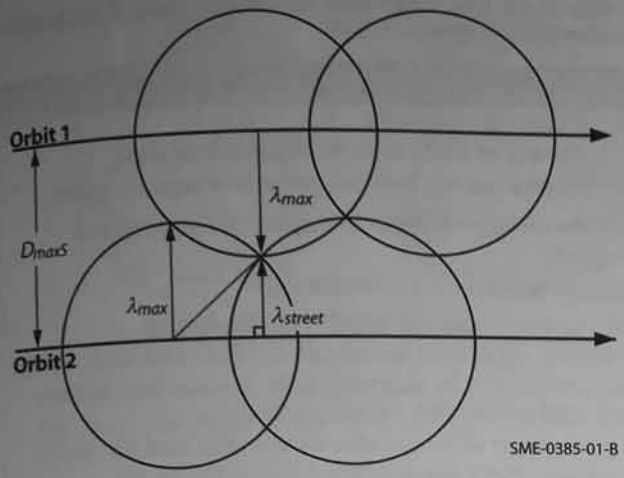


Fig. 10-28. Coverage in Adjacent Planes. If the satellites in the planes are moving in the same direction, the coverage pattern can be designed to provide maximum.

$$D_{maxO} = 2\lambda_{street} \quad \text{(moving opposite directions)} \qquad (10\text{-}25)$$

This leads to a polar constellation often called *Streets of Coverage*, illustrated in Fig. 10-29, in which M planes of N satellites are used to provide continuous global coverage. At any given time, satellites over half the world are going northward and satellites over the other half are going southward. Within both regions, the orbit planes are separated by D_{maxS}. Between the two halves there is a seam in which the satellites are going in opposite directions. Here the spacing between the planes must be reduced to D_{maxO} in order to maintain continuous coverage.

This pattern clearly shows another critical characteristic of constellations—**coverage does not vary continuously and smoothly with altitude**. There are discrete jumps in coverage that depend primarily on λ_{max} which, in turn, depends on the minimum elevation angle, ε_{min}, and the altitude. (See Table 8-11 in Sec. 8.4.1.) If we keep ε_{min} fixed and lower the constellation altitude, then we will reach an altitude plateau at which we will need to add another orbit plane, and N more satellites, to cover the Earth. The Iridium communications constellation was originally intended to have 77 satellites in a streets of coverage pattern. (The element iridium has an atomic number of 77.) By slightly increasing the altitude and decreasing the minimum elevation angle, the number of orbit planes was reduced by one, and the number of satellites required for continuous coverage was reduced to only 66. (Unfortunately, Dysprosium is not a compelling constellation name.)

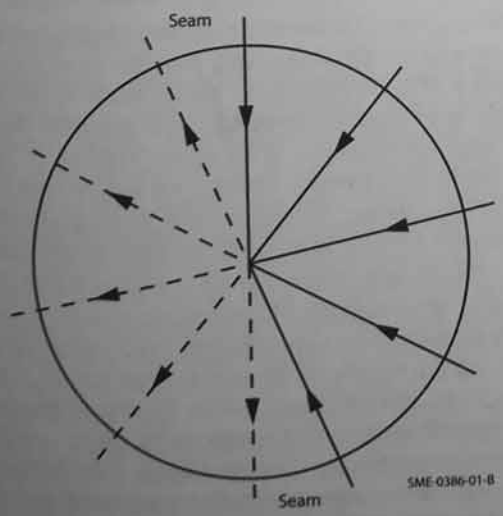


Fig. 10-29. "Streets of Coverage" Constellation Pattern. View seen from the north pole Northward portions of each orbit are drawn as solid lines and southward portions are dashed. To achieve full coverage, orbit planes on either side of the seam must be closer together than the others.

As the altitude changes, the fundamental constellation design changes and, consequently, the number of satellites and coverage characteristics change in steps. As a result, we cannot provide a meaningful chart, for example, the number of satellites vs. altitude without examining different constellation designs at different altitudes. While we may use this sort of chart to estimate constellation size, it would not provide realistic data for orbit design.

Requirements other than coverage can also be important in constellation design, but most are directly related to coverage issues. For example, we may need several satellites to cover a point on the ground or in space at the same time. Navigation with GPS requires that four satellites be in view with a reasonably large angular separation. A similar requirement is cross-link connectivity among satellites in the constellation. Cross-link connectivity is geometrically the same issue as overlapping coverage. At any time when the coverage of two satellites overlaps (that is, they can both see at least one common point on the ground), then the two satellites can see each other and we can establish a cross-link. Thus, forming cross-links is equivalent to the problem of multiple coverage.

Even apparently simple design problems can be very difficult, with solutions depending on various mission conditions. Perhaps the simplest constellation design problem is the question "What is the minimum number of satellites required to provide continuous coverage of the Earth?" In the late 1960s, Easton and Brescia [1969] of the United States Naval Research Laboratory analyzed coverage by satellites in two mutually perpendicular orbit planes and concluded we would need at least 6 satellites to provide complete Earth coverage. In the 1970s, J.G. Walker [1971, 1977, 1984] at the British Royal Aircraft Establishment expanded the types of constellations considered to include additional circular orbits at a common altitude and inclination. He concluded that continuous coverage of the Earth would require 5 satellites. Because of his extensive work, Walker constellations are a common set of constellations to evaluate for overall coverage. More recently in the 1980s, John Draim [1985, 1987a, 1987b] found and patented a constellation of 4 satellites in elliptical orbits which would provide continuous Earth coverage. A minimum of four satellites are required at any one instant to

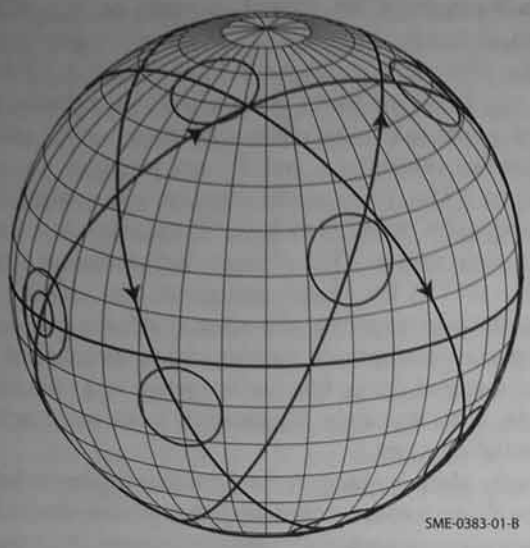


Fig. 10-30. A 15/5/1 Walker Constellation at 65 deg Inclination. Circles are centered on each of the 15 satellites. The double circle is on a satellite at its ascending node.

provide full coverage of the Earth. Consequently, while the above progression looks promising, the 1990s did not yield a 3-satellite full Earth coverage constellation or the 2000s a 2-satellite constellation.

While extensively studying regular, circular orbit patterns, Walker [1984] developed a notation for labeling orbits that is commonly used in the orbit design community and frequently used as a starting point for constellation design. Specifically, the *Walker delta pattern* contains a total of t satellites with s satellites evenly distributed in each of p orbit planes. All of the orbit planes are assumed to be at the same inclination, i, relative to a reference plane—typically the Earth's equator. (For constellation design purposes, this need not be the case. But orbit perturbations depend on the inclination relative to the equator and, therefore, the equator is the most practical standard reference plane.) Unlike the streets of coverage, the ascending nodes of the p orbit planes in a Walker pattern are uniformly distributed around the equator at intervals of 360 deg/p. Within each orbit plane the s satellites are uniformly distributed at intervals of 360 deg/s.

The only remaining issue is to specify the relative phase between the satellites in adjacent orbit planes. To do this we define the *phase difference*, $\Delta\phi$, in a constellation as the angle in the direction of motion from the ascending node to the nearest satellite at a time when a satellite in the next most westerly plane is at its ascending node. In order for all of the orbit planes to have the same relationship to each other, $\Delta\phi$ must be an integral multiple, f, of 360 deg/t, where f can be any integer from 0 to $p-1$. So long as this condition holds, each orbit will bear the same relationship to the next orbit in the pattern. The pattern is fully specified by giving the inclination and the three parameters, t, p, and f. Usually such a constellation will be written in the shorthand notation of i: $t/p/f$. For example, Fig. 10-30 illustrates a Walker constellation of 15/5/1 at $i = 65$ deg. Table 10-26 gives the general rules for Walker delta pattern parameters.

Table 10-26. Characteristics of a Walker Delta Pattern Constellation. See Walker [1984].

T/P/F—Walker Delta Patterns
t = Number of satellites.
p = Number of orbit planes evenly spaced in node.
f = Relative spacing between satellites in adjacent planes.
Define $s \equiv t/p$ = Number of satellites per plane (evenly spaced).
Define Pattern Unit, $PU \equiv 360$ deg/t
Planes are spaced at intervals of s PUs in node.
Satellites are spaced at intervals of p PUs within each plane.
If a satellite is at its ascending node, the next most easterly satellites will be f PUs past the node.
f is an integer which can take on any value from 0 to (p–1).
Example: 15/5/1 constellation shown in Fig. 10-30.
15 satellites in 5 planes ($t = 15$, $p = 5$).
3 satellites per plane ($s \equiv t/p = 3$).
$PU = 360/t = 360/15 = 24$ deg.
In-plane spacing between satellites = $PU \times p = 24 \times 5 = 120$ deg.
Node spacing = $PU \times s = 24 \times 3 = 72$ deg.
Phase difference between adjacent planes = $PU \times f = 24 \times 1 = 24$ deg.

While Walker constellations are important to constellation design, they are not the only appropriate options and do not necessarily provide the best characteristics for a given mission. Walker intended to provide continuous multiple coverage of all the Earth's surface with the smallest number of satellites. This plan may or may not meet all the goals of a particular program. For example, equally distributed coverage over the Earth's surface may not be the most beneficial. We may wish to provide global coverage with the best coverage at the poles, mid-latitude regions, or the equator. In these cases, we may want constellation types other than Walker orbits.

If the regions of interest do not include the poles, then an equatorial constellation may provide all of the coverage with a single orbit plane, which leads to flexibility, multiple performance plateaus, and graceful degradation. Thus, for example, if all of the regions of interest were within 50 deg of the equator, we would want to consider a constellation having several equatorial satellites with enough altitude to provide the appropriate coverage at the smallest spacecraft elevation angle [Wertz, 2007].

If all or most regions of interest are above a given latitude, a directly polar constellation would allow all satellites to see the region of the pole on every orbit. Thus, if all targets of interest were within 50 deg of the pole, a polar constellation with a single orbit plane could provide excellent coverage. If most targets were in the polar region with lesser interest in the equatorial regions, a two-plane polar constellation could provide continuous or nearly continuous coverage of the pole while providing reduced but good coverage of the equatorial regions. Conversely, if most of the targets were at lower latitudes an excellent choice would be a ring of equatorial satel-

Table 10-30, Fig. 10-26, Eq. 10-25◀

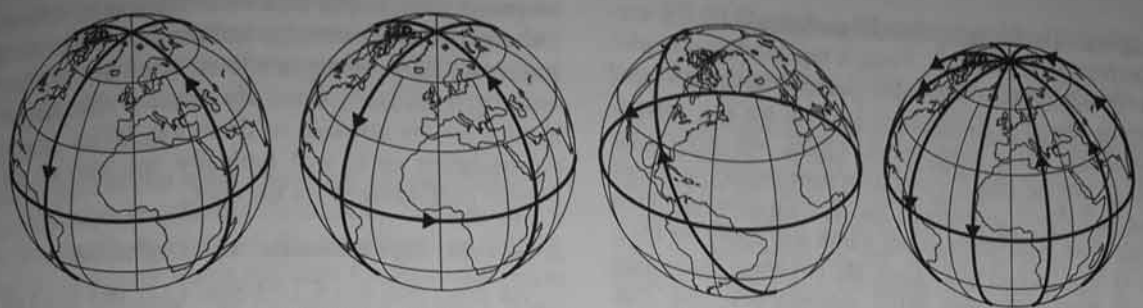


Fig. 10-31. Examples of Typical Non-Walker Constellations. All orbits are assumed to be circular.

lites, augmented by a ring with fewer satellites covering the polar regions if needed [Wertz, 2007]. One might also consider a mix of polar or high inclination satellites with some satellites at the equator to provide the added coverage needed there.

Another class of non-Walker constellations consists of two planes at right angles to each other. If both planes are perpendicular to the equator it will be a polar constellation. Although it will have substantial symmetry, it is not one of the Walker delta patterns. The two planes can also be tipped relative to the equator to achieve any inclination from 90 to 45 deg. Again the ascending nodes are such that they are not Walker constellations except when the inclination is 45 deg, in which case they reduce to a Walker two-plane configuration. Figure 10-31 shows examples of several non-Walker constellations.

A final example of non-Walker constellations is the Molniya orbits used for Russian communication satellites. Sections 9.5.4 and 10.4.2.3 describe them in more detail. These constellations can fully cover high northern latitudes while requiring less energy than circular high-altitude orbits.

10.6.3 Overview

Constellation design is complex, requiring us to assess many issues and orbit characteristics. When we pick a preliminary design, this complexity demands that we document the reasons for the design and remain aware of alternatives as orbit design continues. Unfortunately, systematic reassessments of constellation design are difficult under typical budget constraints and the constellation pattern is often locked in very early in mission design.

Perhaps the most important coverage issue to remember in terms of constellation design is that compiling statistical data on orbit coverage may not provide useful physical insight into the constellation design problem and can, in many cases, be dramatically misleading. The mission designer must always be aware that in constellation design, statistical data is being used to represent a dramatically non-Gaussian process. Examples of this problem have been given throughout the book—particularly at the beginning of Sec. 10.2.

There are two fundamentally different types of constellations —those that demand continuous coverage and those that do not. The former group represents the more straightforward design problem. What is the minimum number of satellites (or minimum total system cost) needed to provide the necessary single or multiple coverage? For example, GPS wants continuous coverage of the entire world by a minimum of four satellites. What is the minimum cost system to do this?

An interesting figure of merit for continuous coverage constellations is the *excess coverage*—i.e., the total instantaneous coverage available as a percentage of the total required. For N satellites in circular orbits at a common altitude with coverage defined as being within λ_{max} of the subsatellite point, the total instantaneous coverage, COV, is given by:

$$COV = N\,(1 - \cos\lambda_{max}) / 2K \qquad (10\text{-}26)$$

where K is the coverage multiplicity that is needed (i.e., K = 4 for GPS). Our goal is to drive COV toward 1.

For the 24-satellite GPS constellation, the satellites are all at half GEO (a = 26,562 km) and the assumed minimum elevation angle is 5 deg. From Eqs. (9-4) and (9-5), we find that λ_{max} = 71.2 deg and COV = 2.03. This means that it takes approximately twice as many satellites as would be needed if the coverage were static and could be perfectly distributed. Iridium is a communications constellation with 66 satellites at 780 km. Here $K = 1$ and the minimum working elevation angle is 9.5 deg corresponding to λ_{max} = 19.0 deg and COV = 1.80. Although the two constellations are very different in use and in coverage pattern, the excess coverage in the implemented constellations is nearly the same.

For constellations that do not demand continuous coverage, the most representative figure of merit is typically the mean and maximum response time as a function of the number of satellites (see Sec. 10.2.1.4). This is a substantially more difficult trade because it now needs to be worked iteratively with the basic objectives of the constellation and may require extended interaction with the end user to determine how those needs are best met.

With numerical simulation, we can evaluate some of the Figures of Merit defined in Sec. 10.2.1. (That section discusses how to lay out the simulation for unbiased results.) Generally the results of such a simulation are

best expressed as Figures of Merit vs. latitude for the various performance plateaus. Thus, a typical decision plot might include mean response, percent coverage, and maximum gap as a function of latitude for the various constellations being considered. Often, we must also evaluate coverage data for different instruments on board a spacecraft. Each instrument has its own coverage area and, therefore, a different swath width will apply for each principal observation type. Thus, the coverage associated with one instrument may differ dramatically from that associated with another instrument or operating mode. Alternative operating modes or instruments will likely lead us to prefer distinctly different constellation designs. We may then choose either different satellites or a compromise between the alternative instruments or modes.

Ultimately, constellations are expensive. Therefore, all of our system trades should be based on cost vs. performance. The principal issues that dominate constellation design are listed in Table 10-27. Generally, we evaluate each constellation design for four criteria:

1. Baseline Coverage vs. Latitude

The coverage associated with different instruments or operating modes is best expressed as coverage vs. latitude. (See Secs. 8.4 and 10.2.1.) I regard the mean response time as the best overall measure of coverage, although percent coverage and maximum gap can also be important in some applications. We must use the maximum gap measure carefully because this single point should not be allowed to drive the design of an entire constellation as it will typically not provide the best overall performance for the cost.

2. Growth, Replenishment, and Degradation

As described in OCDM Sec. 13.6, this is a key issue in practical constellation design. It will be different for each constellation type. In evaluating growth, replenishment, and degradation, we should assume that rephasing within the orbit plane can be done at very modest propellant cost, and that changing orbit planes is not feasible.* (See Sec. 9.6.2.)

3. Existence of Altitude Plateaus

We should evaluate each constellation to see if plateaus exist in which the number of orbit planes or other

* If we are willing to allow months for putting the satellites in place, differential node rotation due to a different altitude can be used to move the ascending node.

Table 10-27. Principal Issues that Dominate Constellation Design. See also Tables 10-24, 10-25, and 10-28.

Issue	Why Important	What Determines It	Principal Issues or Options	Where Discussed Here	Where Discussed OCDM
Coverage	Principal performance parameter	Altitude, minimum elevation angle, inclination, constellation pattern	Gap times for discontinuous coverage; Number of satellites simultaneously in view for continuous coverage	Secs. 8.4, 10.2.1	Chap. 9, Sec. 13.3
Number of Satellites	Principal cost driver	Altitude, minimum elevation angle, inclination, constellation pattern	Altitude, minimum elevation angle, inclination, constellation pattern	Chap. 8, Sec. 10.6.2	Sec. 13.3
Launch Options	Major cost driver	Altitude, inclination, spacecraft mass	Low altitude, low inclination costs less	Chap. 26	Secs. 12.3, 13.3
Environment	Radiation level and, therefore, lifetime and hardness requirements	Altitude	Options are below, in, or above Van Allen radiation belts	Chap. 7	Fig. 2-12, Sec. 12.4
Orbit Perturbations (Stationkeeping)	Causes constellations to disassociate over time	Altitude, inclination, eccentricity	Keep all satellites at common altitude and inclination to avoid drifting apart	Sec. 9.6	Secs. 2.4, 13.4
Collision Avoidance	Snowball effect can destroy entire constellation	Constellation pattern, orbit control	No option—must design entire system for collision avoidance	—	Sec. 13.5
Constellation Build-Up, Replenishment, and End-of-Life	Determines level of service over time and impact of outages	Altitude, constellation pattern, build-up and sparing philosophy	Sparing: on-orbit spares vs. launch on demand; End-of-life: Deorbit vs. raise to higher orbit	End-of-life: Sec. 9.6.4, Chap. 30	Secs. 2.6.4, 13.6
Number of Orbit Planes	Determines performance plateaus	Altitude, inclination	Fewer planes means more growth plateaus and more graceful degradation	Sec. 10.6.2	Secs. 2.6.2, 13.6
Legal or Political Constraints	Limits options, locations in GEO, frequency use	Law, policy, international treaties and agreements	Need to be sure all needed permissions can be secured and time needed to do so	Sec. 12.3	—
Viewing and Lighting Conditions	Determines visibility of targets and objects of interest	Distance, surface properties, Sun angle	May drive orbit selection, viewing direction, or observation timing	Viewing conditions: Sec. 8.3	Chap. 11

key characteristics make a discrete step. Plateaus may be different for different instruments and operating modes, but usually are functions of the swath width for each instrument or operating mode.

4. Collision Avoidance

Finally, one of the most important characteristics of any constellation is collision avoidance. The reason for this is not merely the loss of the satellites that collide, because we anticipate losing satellites for many reasons in any large constellation. The fundamental problem is the debris cloud that results from any satellite collision. The velocity imparted to the particles resulting from the collision is small relative to the orbital velocity. Consequently, the net effect of a collision is to take two trackable, possibly controllable satellites and transform them into thousands of untrackable particles that spread out with time in the same orbit regime as the original satellites. Because the energy is proportional to mv^2, even a small piece of a satellite carries an enormous amount of kinetic energy at orbital velocities. Because the debris cloud remains in the area of the constellation, it dramatically increases the potential for secondary collisions, which, in turn, continues to increase the amount of debris and the possibility of making the orbit "uninhabitable." The implication for constellation design is that we should go to great lengths to design the constellation and the spacecraft to avoid collisions, explosions, or generation of extraneous debris.

10.6.4 The Rules for Constellation Design

It is clear from the above discussion, and particularly the example of the MSSP program at the beginning of Sec. 10.6, that there are no absolute rules for constellation design. It is not a systematic process, where we can simply take, for example, all possible Walker constellations and enumerate how well they do, and then select an absolute winner for the best constellation. The constellation design process is much fuzzier than that. The key is to look at the fundamental mission objectives and determine how these objectives can best be met at minimum cost and risk.

Ordinarily, our goal would be to minimize the number of satellites while achieving the appropriate level of coverage. Nonetheless, this may not be the lowest cost approach. For example, we may be able to achieve our coverage objectives with 20 satellites in the Van Allen radiation belts, where 25 would be required below the belts. The key question then becomes whether the lower cost and longer life per satellite for the lower constellation is worth the additional number of satellites. The constellation design process per se cannot answer that question. It can only be answered within the broader context of mission, system, and spacecraft design.

While there are not absolute rules, there are broad guidelines that assist in the process of constellation design. These are summarized in Table 10-28. In the end, constellation design is coupled only moderately to astrodynamics, and strongly to the process of meeting mission objectives at minimum cost and risk.

10.7 Design of Interplanetary Orbits

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This section on interplanetary orbit design is application-oriented. Clearly, spacecraft on interplanetary missions obey the same universal laws of nature as spacecraft orbiting the Earth. This means that the types of trajectories of such vehicles are similar to those of missions restricted to the direct vicinity of the Earth; we can therefore similarly apply the well-known concepts of Kepler orbits and so on. As a matter of fact, the formulas that we use nowadays for the description of satellite orbits were based on observations of planets, i.e., motions on the scale of the Solar System: it was the works of people like Brahe, Kepler, and Newton which lead to our knowledge of orbital mechanics as it is now. Rather than going through a detailed discussion on the theory of orbital mechanics, this chapter will **"only"** focus on the process of designing interplanetary orbits. For more specific information, refer to Chap. 9 in this book and a range of excellent texts, e.g., Cornelisse et al. [1979], Battin, [1999], Chobotov, [2002], Vallado, [2007].

Although the title of this section might suggest differently, we will use the term "interplanetary" irrespective of whether our vehicle travels to another planet, the Moon, an asteroid, a comet, or a mathematical position in space, like a Lagrange libration point. Along the same lines, we will consider Pluto as a planet, although officially it lost this status in 2006 and should be referred to as a dwarf planet.

The design of interplanetary missions goes back a long time. The first man-made vehicle to orbit the Earth was Sputnik-1, launched on October 4, 1957. Since the extra energy needed to step from a closed orbit around the Earth (an ellipse) to an escape orbit (a parabola or hyperbola) is relatively modest (at least, compared to what is needed to achieve a stable Earth orbit), it comes as no surprise that the first attempts to fly missions beyond the Earth took place in the year after Sputnik-1. Table 10-29 gives a summary of milestones in interplanetary spaceflight. It clearly shows that it is a discipline with many new and exciting developments. A more detailed overview of interplanetary missions can be found on the website.

One of the key factors in spaceflight is propulsion, in particular the thrust needed to accelerate the vehicle such that it can overcome Earth's gravity and atmospheric drag, and achieve the required velocity to sustain a steady flight in a certain orbit. During the first four decades of its existence, spaceflight solely relied on chemical high-thrust propulsion, where the energy to provide this thrust is stored in and delivered by chemical rocket engines. Such rocket engines are known for their high reliability, large thrust range, but also their modest efficiency (see Chap. 18). The latter aspect is expressed by the so-called specific impulse, I_{sp}, which has typical values of 200–450 sec. for this kind of propulsion. Alternatively, more efficient low-thrust engines have been in development since many decades, but the very first real applica-

Table 10-27◀, Fig. 10-31◀, Eq. 10-26◀

Table 10-28. Rules for Constellation Design. While there are no absolute rules, these broad guidelines are applicable to most constellations. See also Tables 10-24, 10-25, 10-27.

Rule	Where Discussed Here	Where Discussed OCDM
1. ***To avoid differential node rotation, all satellites should be at the same inclination, except that an equatorial orbit can be added.***	Secs. 9.4.1, 9.5.3	Sec. 13.1
2. ***To avoid perigee rotation, all eccentric satellites should be at the critical inclination of 63.4 deg or (180–63.4)deg.***	Secs. 9.4.1, 9.5.4	Sec. 13.1
3. ***Collision avoidance is critical, even for dead satellites, and may be a driving characteristic for constellation design.***	Secs. 7.5, 10.6.3	Sec. 13.5
4. ***Symmetry is an important, but not critical element of constellation design.***	Sec. 10.6.2	Secs. 13.0, 13.1
5. ***Altitude is typically the most important of the orbit elements, followed by inclination. 0 eccentricity is the most common, although eccentric orbits can improve some coverage and sampling characteristics.***	Table 10-6, Sec. 10.4.1	Sec. 13.1
6. ***Minimum working elevation angle (which determines swath width) is as important as the altitude in determining coverage.***	Secs. 8.3.1, 10.2, Fig. 10-12	Sec. 9.1, 13.1
7. ***Two satellites can see each other if and only if they are able to see the same point on the ground.***	Fig. 8-46 (Sec. 8.5)	Sec. 10.1.1
8. ***Principal coverage Figures of Merit for constellations:*** • Percentage of time coverage goal is met • Number of satellites required to achieve the needed coverage • Mean and maximum response times (for non-continuous coverage) • Excess coverage percent • Excess coverage vs. latitude	Sec. 10.2.1.4	Secs. 9.5.1.4, 13.1
9. ***Size of stationkeeping box is determined by the mission objectives, the perturbations selected to be overcome, and the method of control.***	—	Sec. 13.2
10. ***For long-term constellations, absolute stationkeeping provides significant advantages and no disadvantages compared to relative stationkeeping.***	Sec. 19.2	Sec. 13.2
11. ***Orbit perturbations can be treated in 3 ways:*** • Negate the perturbing force (use only when necessary) • Control the perturbing force (best approach if control is required) • Leave perturbation uncompensated (best for cyclic perturbations)	—	Sec. 13.2
12. ***Performance plateaus and the number of orbit planes required are a function of the altitude.***	Sec. 10.6.2	Sec. 13.1
13. ***Changing position within the orbit plane is easy; changing orbit planes is hard; implies that a smaller number of orbit planes is better.***	Sec. 9.6.2	Sec. 13.4
14. ***Constellation build-up, graceful degradation, filling in for dead satellites, and end-of-life disposal are critical and should be addressed as part of constellation design.***	Sec. 9.6.4, 10.6, Chap. 30	Sec. 13.4
15. ***Taking satellites out of the constellation at end-of-life is critical for long-term success and risk avoidance. This is done by:*** • Deorbiting satellites in LEO • Raising them above the constellation above LEO (including GEO)	Sec. 9.6.4, Chap. 30	Sec. 13.4

Table 10-29. Most Prominent Successful "Firsts" in Interplanetary Spaceflight.

Satellite	Event Date	Achievement
Luna-2	September 14, 1959 (impact)	First Lunar mission
Mariner 2	December 14, 1962 (Venus flyby)	First planetary fly-by
Pioneer 10	July 1972	First to travel through asteroid belt
Mariner 10	February 5, 1974 (Venus GA)	First planetary gravity-assist
International Sun/Earth Explorer-3 (ISEE-3)	August 12, 1978	First mission to Lagrange point
Magellan	August 10, 1990 (Venus orbit insertion)	First use of aerobraking
Deep Space 1	October 1998	First mission with low-thrust propulsion as primary propulsion technique
Near Earth Asteroid Rendezvous (NEAR)	April 30, 2000 (asteroid Eros)	First to orbit and land on asteroid

tions as a primary propulsion technique were not flown until 1998 [NASA, 2010a] and 2003, [ESA, 2010a]. The ion engines used in these missions had I_{sp} values of 3,100 sec and 1,640 sec, respectively, which can still be considered as relatively conservative: the I_{sp} of (near) future low-thrust propulsion engines may range between 5,000 and 10,000 sec. This means that low thrust is to be considered as a very likely propulsion technique for virtually any future interplanetary mission. However, this section will restrict itself to high-thrust missions, if only since a treatment of the ins and outs of low-thrust interplanetary trajectory design would require quite a number of pages. For an introduction to low-thrust orbits and their design, refer to Petropoulos and Longuski [2004], and Izzo [2006], amongst others.

Another key element of interplanetary missions is the geometry of the problem: the location of the Earth and other celestial bodies involved, relative to the spacecraft. For first-order calculations, it is customary to consider the problem to be 2-dimensional, i.e., all bodies move about the Sun in one plane, the *ecliptic* (which is the orbital plane of the Earth). Realizing that the inclinations of the orbits of most other planets are quite small (apart from Mercury and Pluto, all planets have inclinations of a few degrees at most), this is a very reasonable approach. However, in order to achieve the most accurate trajectory, one should take one step further and work with a 3-dimensional problem formulation. This section will address both possibilities.

With the risk of being contradictory, the highest possible accuracy is not always what one is after in preliminary mission design. This would require including the most accurate models for the positions of the celestial bodies, covering all possible forces acting on the vehicle, use of accurate propagators, and so on. As a consequence, this would be a time-consuming activity. Rather, in first-order design, preference is given to fast, accurate-but-not-perfect algorithms. In doing so, a designer can gain several orders of magnitude on CPU-time, get a very reasonable first-order optimal solution, and later on, refine this with a fully detailed model of all aspects, only for a last iteration. In this section, we will mainly focus on fast, analytical techniques, and merely address some aspects that are relevant to "the real thing."

Finally, for those readers who have whetted their appetite after having read this section: an international competition is being organized on an almost yearly basis since 2005, in which teams are invited to design an optimum trajectory according to specific requirements: the Global Trajectory Optimization Competition (GTOC). The initial competition, GTOC1, was organized by the Advanced Concepts Team of the European Space Agency [ESA, 2010b], and the winner of each particular round is required to organize the competition for the year thereafter. Previous versions dealt, for example, with the design of an optimal trajectory through a series of asteroids, or a mission to such heavenly bodies followed by a return to the Earth. Experience teaches that these assignments are extremely difficult to solve, if only for the large number of unknowns in the problem and the fact that solutions have to be handed in within a month after release of the assignment. On the other hand they are extremely motivating for students, and very good projects to work on off-line, in order to study, develop, and fine-tune the various elements of interplanetary mission design.

10.7.1 Design Process

In order to structure the search for an optimal interplanetary trajectory, one should follow a series of well-defined steps. These are summarized in Table 10-30; a brief explanation of each element is given below.

Table 10-30. The Orbit Design Process. See text for discussion of each step.

Step	Where Discussed
1. Quantify Relevant Mission Parameters	Sec. 10.7.1
2. Obtain Model for Planetary Ephemerides	Sec. 10.7.1
3. Analyze Direct Hohmann Flight	Sec. 10.7.2
4. Analyze Alternative Direct Keplerian Orbit	Sec. 10.7.2
5. Assess (Round Trip) Travel Times	Secs. 10.7.2, 10.7.3
6. Identify Possible Multi-leg Scenarios	Sec. 10.7.2
7. Analyze Multi-leg Scenarios Using Lambert Targeting	Sec. 10.7 web article
8. Iterate	

Step 1. Quantify Relevant Mission Parameters

Any design process starts with an assessment of what is actually required. Questions to be answered are: what is the departure planet? What is the target? Do we go for a direct flight, or will we possibly allow fly-by's of planets in-between? Is the mission objective to arrive at the destination only, or do we want to return to the departure planet? Do we have specific constraints, like departure and arrival conditions (time, geometry), maximum allowable ΔV, maximum allowable time-of-flight? Usually, such aspects are set by the customer, but some aspects can also be traded during the design process. In this section, we are interested in the parameters which are related to orbital mechanics only; however, these may very well affect other mission parameters such as required propellant mass.

Step 2. Obtain Model for Planetary Ephemerides

One of the aspects that need to be fulfilled is the availability of a model that specifies the positions of the planets involved, as a function of time. Depending on what is actually required, one can do with a simple 2-dimensional model, where the orbits of the planets are assumed to lie within a single plane and described by simple Kepler orbits (i.e., coplanar orbits). This is a reasonable first-order assumption, since the inclinations of the orbits of most of the planets in our Solar System have a value of a few degrees at most; it can become a different story when wanting to study missions to and from asteroids. So one step further is to work with a 3-dimensional model with

Table 10-30, Fig. 10-31◀, Eq. 10-26◀

Kepler orbits, now also involving the orientation parameters inclination and the longitude of the ascending node (see App. C). To simulate reality as closely as possible, one can go for a fully detailed 3-dimensional representation, where perturbations acting on the trajectories of the planets are also taken into account. For the latter purpose, JPL's DE405 model [Standish, 1998] or later versions thereof can be used.

Step 3. Analyze Direct Hohmann Flight

The classical optimal solution, at least in terms of required ΔV, to a direct transfer problem between two coplanar orbits is the Hohmann transfer orbit. Such an orbit tangentially touches the orbits of the departure planet and that of the target planet (see Sec. 9.6). Here, it is assumed that the velocity changes are obtained instantaneously (i.e., the position of the spacecraft is unaffected, and the maneuver is achieved with an idealized rocket engine that can provide an infinite acceleration), and that the velocities directly before and after each maneuver are parallel. Although definitions may differ, it is also customary to assume that this transfer takes place between two circular orbits. The Hohmann transfer between two orbits around the Earth has already been discussed in Sec. 9.6; a similar concept can be applied for interplanetary purposes (although some elements do differ). Section 10.7.2 will provide more details.

Step 4. Analyze Alternative Direct Keplerian Orbits

Attractive as they may be from an energy point of view, Hohmann transfers are also reputed for having a long trip time. In order to overcome this problem, one can investigate whether faster transfers between two planets are possible, with limited consequences for the required ΔV. This is also covered in Sec. 10.7.2.

Step 5. Assess (Round Trip) Travel Times

Timing of events (launch, maneuvers, rendezvous) is already crucial in missions around the Earth, but for interplanetary missions it is even more stringent: the positions of the celestial bodies involved are prominent elements in the design, and dictate the geometry of potential transfers. For a trip to Mars, for example, you don't want to reach Mars' orbit without the planet being there! This problem becomes worse in the case of missions along a series of celestial objects, or round trips, in which we want to return to our original point of departure. The problem will be treated in Sec. 10.7.3.

Step 6. Identify Possible Multi-leg Scenarios

Direct transfers between two planets are attractive, if only for the ease with which the various options can be analyzed. However, the energy that can be provided by launchers, upper stages and on-board engines is restricted, meaning that we will be confronted with limitations as to what can be achieved with a direct (single-leg) flight. Fortunately, nature provides a free tool to increase the energy of a spacecraft without paying any cost: a planetary fly-by. This necessitates a more complex mission design, where a vehicle travels to one or more planets first before reaching its destination. In order to analyze the possible application of such a scenario, one first has to identify the candidate (series of) fly-by planets. The quantitative analysis of each of such proposed scenarios is dealt with in the next step. Both will be discussed in Sec. 10.7.2.

Step 7. Analyze Multi-leg Scenarios Using Lambert Targeting

When the trajectory proposal encompasses a series of legs between pairs of planets (such as Galileo's VEEGA trajectory: launch at the Earth, followed by fly-by's along Venus, Earth, and Earth again, before arriving at Jupiter), the analysis of the advantages and disadvantages of such an option cannot be done with straightforward Hohmann orbits anymore. In this case, the problem is tackled by means of solving a series of Lambert problems, which allows a straightforward evaluation of ΔV and time-of-flight, amongst others. This is the standard approach applied for any mission design; it can be used for a wide variety of missions: one-way, round trip, different departure and target conditions. The theory will be discussed briefly in the web article for this section.

Step 8. Iterate

It is very likely that the first result is not optimal: the required ΔV is too high, the trip takes too long. So, one may have to take a fresh look at the initial requirements (as defined under Step 1) and modify them such that a more satisfactory result can be obtained. When the techniques to analyze the interplanetary transfer problem have been coded in a general-purpose computer program, this should not be a problem at all.

10.7.2 One-Way Missions

The following sections will address most of the issues listed in Table 10-30. First, we will take a look at fundamental aspects that play a role for one-way missions, and in the next section we will treat round-trip missions.

In Sec. 9.6, the theory of the Hohmann transfer orbit was discussed, in particular for transfers between circular orbits around the Earth. Inspecting Fig. 9-28 in Sec. 9.6.1, the main characteristics of such a transfer are clearly visible: (1) it is a transfer between two circular, coplanar orbits (the circularity requirement is traditional, if only for providing unique solutions for the parameters of the Hohmann transfer orbit), (2) it is tangent to both the departure orbit and the target orbit, and (3) it is an ellipse. Its *pericenter* and *apocenter* exactly touch the departure and target orbits (note that the concept can be used for a transfer to an inner target orbit as well), so the value of the semi-major axis of the transfer orbit (and hence the corresponding energy) is minimal. This concept can also be applied to interplanetary spaceflight, albeit that the situation is slightly more complicated. Therefore, we take one step back and address the geometry for the most general trajectory between arbitrary objects around the Sun first. The situation is sketched in Fig. 10-32.

First of all, positions and velocities are given in a heliocentric system, i.e., with respect to an inertial reference frame with its center at the Sun. The spacecraft has

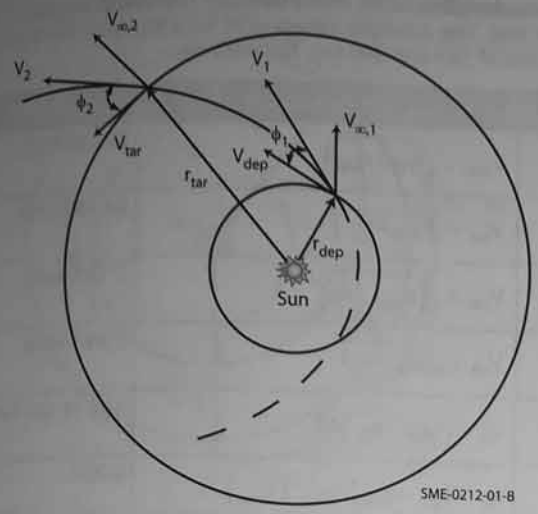


Fig. 10-32. Geometry of an Arbitrary Interplanetary Transfer Between Two Planets, and Definition of the Parameters Involved.

velocities V_1 and V_2 (at different epochs), whereas the departure planet and the target planet have velocities V_{dep} and V_{tar}, respectively. The relative velocities $V_{\infty,1}$ and $V_{\infty,2}$ of our spacecraft can be obtained by vectorially subtracting the heliocentric velocities of the spacecraft and the planet, respectively.

Secondly, Fig. 10-32 shows that we are dealing with (at least) 4 different objects: the central body (i.e., the Sun), the departure and target planets, and our spacecraft. However, when making a first-order design of an interplanetary trajectory, it is customary to describe this by a succession of 2-body problems: the so-called patched conics approach. According to this approach, the motion of our vehicle in each segment is considered to be dominated by the gravitational pull of a perfectly symmetrical central body, whereas all other forces (e.g., third-body attractions, solar radiation pressure, atmospheric drag) are ignored. So, each segment can be described by the equations of a Kepler orbit. In our interplanetary transfer problem (Fig. 10-32), we work with a series of 3 consecutive 2-body problems: (1) the satellite around the departure planet (planetocentric), (2) the satellite around the Sun (heliocentric), and (3) the satellite around the target planet (planetocentric again). The planetocentric phases take place within the *Sphere of Influence (SoI)* of the respective planets; these basically indicate the limit where the gravitational dominance of one celestial body (e.g., the departure planet, Earth) is taken over by another one (e.g., the Sun). Without derivation (which can be found in Cornelisse et al. [1979], for example), the average dimension of the *SoI* of a planet can be computed with the following equation:

$$r_{SoI} = r_{planet}\left(\frac{m_{planet}}{m_{Sun}}\right)^{0.4} \qquad (10\text{-}27)$$

where r_{planet} is the distance between the particular planet and the Sun (not to be confused with the radius of the former), and m_{planet} and m_{Sun} are the masses of the planet and the Sun, respectively. The resulting dimensions of the *SoI*'s for the planets in our Solar System are summarized in Table 10-31.

Table 10-31. Dimensions of the Sphere of Influence of the Planets in Our Solar System. The SoI values are based on the average distance between the planet and the Sun, and may vary slightly when evaluated at actual planetary positions.

Planet	Sphere of Influence	
	[AU]	[% of Average Distance from Sun]
Mercury	0.0008	0.2
Venus	0.004	0.6
Earth	0.006	0.6
Mars	0.004	0.3
Jupiter	0.322	6.2
Saturn	0.365	3.8
Uranus	0.346	1.8
Neptune	0.579	1.9
Pluto	0.021	0.1

It is clear that the size of any *SoI* is marginal compared to the size of our Solar System. Therefore, we can consider the heliocentric part of the transfer to (effectively) take place between the exact positions of the departure and target planets. In addition, the planetocentric part takes place on a local scale and involves a hyperbola around the planet of interest (since of course, the spacecraft has to escape/enter the gravity field of the departure/target planet).

Hohmann Transfers

The geometry of an interplanetary Hohmann transfer is a special case of the general geometry as sketched in Fig. 10-32. In the Hohmann case and evaluated at the departure epoch, the heliocentric velocity V_1 of the satellite is parallel to the velocity V_{dep} of the departure planet; at the arrival epoch, the same thing holds for the velocities V_2 and V_{tar} of the satellite and the target planet, respectively (note the vector notations). In essence we end up at the geometry for a Hohmann transfer between orbits around the Earth (see Fig. 9-28). In order to quantify the relevant characteristics of an interplanetary Hohmann transfer, we can apply the series of calculations summarized in Table 10-32. In this table, μ_S, μ_{dep} and μ_{tar} represent the gravitational parameter of the Sun, the departure planet and the target planet, respectively; r_{dep} and r_{tar} represent the radius of the orbits of the departure planet and target planet around the Sun, respectively (both assumed to be circular); and r_0 and r_3 represent the radius of the (circular) orbit around the departure planet and the target planet, respectively. The last column gives an example for the case of a Hohmann transfer from the Earth to Mars.

An important element of Table 10-32 are the maneuvers ΔV_0 and ΔV_3. The first one is responsible for the transition from a circular orbit around the departure planet to

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Table 10-32. Summary of the Procedure to Compute the Characteristics of an Interplanetary Hohmann Transfer. The meaning of the parameters is explained in the main body of the text. The example values hold for a trip from the Earth (at 1.0 AU) to Mars (at 1.52366 AU), with circular parking orbit altitudes of 185 and 500 km, respectively.

Step	Parameter	Expression	Example
1	V_{dep} (heliocentric velocity of departure planet)	$V_{dep} = \sqrt{(\mu_S / r_{dep})}$	29.785 km/s
2	V_{tar} (heliocentric velocity of target planet)	$V_{tar} = \sqrt{(\mu_S / r_{tar})}$	24.130 km/s
3	V_{c0} (circular velocity around departure planet)	$V_{c0} = \sqrt{(\mu_{dep} / r_0)}$	7.793 km/s
4	V_{c3} (circular velocity around target planet)	$V_{c3} = \sqrt{(\mu_{tar} / r_3)}$	3.315 km/s
5	a_{tr} (semi-major axis of transfer orbit)	$a_{tr} = (r_{dep} + r_{tar}) / 2$	188.77 10^6 km
6	e_{tr} (eccentricity of transfer orbit)	$e_{tr} = \lvert r_{tar} - r_{dep} \rvert / (r_{tar} + r_{dep})$	0.208
7	V_1 (heliocentric velocity at departure position)	$V_1 = \sqrt{[\mu_S (2 / r_{dep} - 1 / a_{tr})]}$	32.729 km/s
8	V_2 (heliocentric velocity at target position)	$V_2 = \sqrt{[\mu_S (2 / r_{tar} - 1 / a_{tr})]}$	21.481 km/s
9	$V_{\infty,1}$ (excess velocity at departure planet)	$V_{\infty,1} = \lvert V_1 - V_{dep} \rvert$	2.945 km/s
10	$V_{\infty,2}$ (excess velocity at target planet)	$V_{\infty,2} = \lvert V_2 - V_{tar} \rvert$	2.649 km/s
11	V_0 (velocity at pericenter of hyperbola around departure planet)	$V_0 = \sqrt{(2\mu_{dep} / r_0 + V_{\infty,1}^2)}$	11.408 km/s
12	V_3 (velocity at pericenter of hyperbola around target planet)	$V_3 = \sqrt{(2\mu_{tar} / r_3 + V_{\infty,2}^2)}$	5.385 km/s
13	ΔV_0 (maneuver at pericenter around departure planet)	$\Delta V_0 = \lvert V_0 - V_{c0} \rvert$	3.615 km/s
14	ΔV_3 (maneuver at pericenter around target planet)	$\Delta V_3 = \lvert V_{c3} - V_3 \rvert$	2.070 km/s
15	ΔV_{tot} (total velocity change)	$\Delta V_{tot} = \Delta V_0 + \Delta V_3$	5.684 km/s
16	T_{tr} (transfer time)	$T_{tr} = \pi \sqrt{(a_{tr}^3 / \mu_S)}$	0.709 yr

a hyperbola, whereas the second one causes the reverse, but now around the target planet. Knowing the characteristics of hyperbolas, the two maneuvers translate into excess velocities $V_{\infty,1}$ and $V_{\infty,2}$, respectively. Note that the required amount of propellant is determined by the magnitudes of ΔV_0 and ΔV_3, and not by the relative velocities $V_{\infty,1}$ and $V_{\infty,2}$. This represents a fundamental difference between Hohmann transfers on a heliocentric scale and Hohmann transfers between orbits around a particular planet (e.g., the Earth; see Fig. 9-28). Depending on the exact conditions, the difference between such a ΔV and the excess velocity V_∞ may in fact be easily in the order of a few kilometers per second, with significant consequences for the amount of propellant to be spent.

Timing

Of course, a Hohmann transfer has to physically connect 2 objects orbiting the Sun: the spacecraft leaves from a physical departure body, with a velocity tangent to the heliocentric velocity of that body, and should exactly arrive at the target object in its orbit around the Sun. In order to assess the timing of such a transfer, the geometry of the departure planet, an arbitrary (i.e., non-Hohmann) transfer orbit and the target planet is depicted in Fig. 10-33 (including the definition of the relevant parameters). Figure 10-33 refers to a trip from the Earth to Mars, but of course any combination of planets will be valid.

Here, the transfer starts at epoch t_1 whereas the vehicle is expected to arrive at epoch t_2. In addition, the angular positions of the 3 objects involved are expressed by a parameter ν, and an arbitrary epoch t_0 is taken as a reference to express the positions of the planets. In our case specifically, the connection between the 2 planets is a Hohmann transfer, with the time-of-flight denoted as T_{tr} and a transfer angle π (see Sec. 9.6.1). We then can write the positions at the departure epoch:

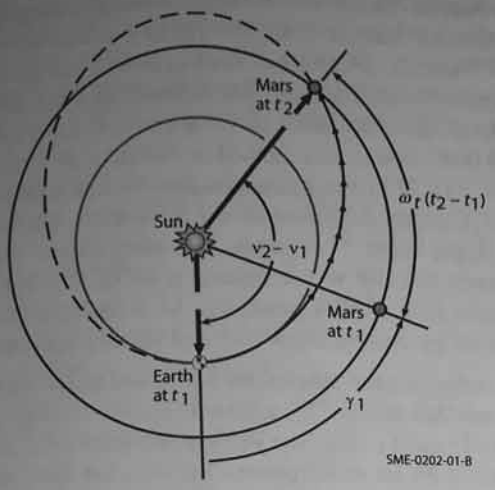


Fig. 10-33. The Geometry of Departure Planet, Arbitrary Transfer Orbit, and Target Planet [Wertz, 2009].

$$\begin{aligned} \nu_E(t_1) &= \nu_E(t_0) + \omega_E(t_1 - t_0) \\ \nu_M(t_1) &= \nu_M(t_0) + \omega_M(t_1 - t_0) \\ \nu_{spc}(t_1) &= \nu_E(t_1) = \nu_E(t_0) + \omega_E(t_1 - t_0) \end{aligned} \tag{10-28}$$

where ω_E and ω_M are the average orbital angular rates of the Earth and Mars, respectively.

And, similarly, the positions at the arrival epoch:

$$\begin{aligned} \nu_E(t_2) &= \nu_E(t_1) + \omega_E T_H \\ \nu_M(t_2) &= \nu_M(t_1) + \omega_M T_H \\ \nu_{spc}(t_2) &= \nu_{spc}(t_1) + \pi = \nu_M(t_2) \end{aligned} \tag{10-29}$$

From these 6 equations, it can easily be derived that

$$\nu_E(t_1) = \nu_M(t_2) - \pi = \nu_M(t_1) + \omega_M T_H - \pi \tag{10-30}$$

or

$$t_1 = t_0 + \frac{\nu_M(t_0) - \nu_E(t_0) + \omega_M T_H - \pi}{\omega_E - \omega_M} \tag{10-31}$$

and

$$t_2 = t_1 + T_H \tag{10-32}$$

As an example, take the situation Earth to Mars: at the epoch January 1, 2000, 12 hrs (i.e., J2000), the planets have a longitude of 100.46° and 355.45° in their orbits, respectively (ignoring any effect of differences in inclination, $\nu(t_0)$ can be taken identical to these values) [Cox, 2000], [Seidelmann, 1992]. Substituting these values in the equations above, with orbital periods of 1.0000 and 1.8807 years for the Earth and Mars, respectively, and a Hohmann transfer time of 0.709 years, gives as the first Hohmann transfer opportunity the epoch of departure, t_1, 456.4 days after January 1, 2000 (i.e., April 1, 2001), and the epoch of arrival, t_2, 715.2 days after January 1, 2000 (i.e., Dec 16, 2001). Of course, the same option repeats after exactly one synodic period (2.135 years, see Table 9-5). It must be emphasized that these considerations are valid for an ideal case where the planets move in circular, coplanar orbits; a more realistic simulation will be discussed in Table 10web-1.

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Faster Trajectories

Using the recipe given in Table 10-32, one can compute the parameters of a Hohmann transfer between any pair of planets in a straightforward manner. Attractive as it is from an energy and computational point of view, a Hohmann transfer does have a number of drawbacks: (1) the launch epoch is very strict, with little or no room for deviations (at least ideally; pericenter and apocenter are by definition on opposite sides of the central body, with consequences for the departure and arrival epochs), (2) the time-of-flight may be too long for some applications, and (3) although it represents the most energy-efficient direct transfer, the total ΔV required for a direct trip to distant planets may simply be too large to be delivered by even the most powerful combination of launcher, upper stage and on-board engine (an indirect trajectory, involving planetary flybys, could provide a solution; more on this later in this section).

In order to compensate for the first 2 drawbacks, one might consider to leave the departure planet (e.g., Earth) under conditions which are different from those of a Hohmann transfer; a similar flexibility can be introduced for the arrival geometry. This is the most general situation as sketched in Fig. 10-32. Clearly, our vehicle now follows an elliptical orbit with dimensions that exceed those of a Hohmann transfer orbit. As shown in Fig. 10-32, it can reach the orbit of the target planet on its way out, before reaching apocenter; in such a case we speak of a so-called Type-I transfer, which is clearly faster than a Hohmann transfer. In case the orbit of the target planet is reached after having passed the apocenter, we speak of a Type-II transfer, which has a longer transfer time compared to a Hohmann transfer. Both options (Type-I, Type-II) can have advantages and disadvantages (travel time, ΔV, departure, and arrival geometry, possibilities for continuation to another planet), so both should be considered.

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A summary of the recipe to analyze such options is given in Table 10web-2.

Multiple Legs

As mentioned earlier, a Hohmann trajectory is the most attractive transfer from one planet to another from an energy point of view. However, it may bring drawbacks in terms of flight time, geometry, and required total ΔV (still). An alternative could be to refrain from entering in a circular orbit around the target planet and instead make a simple pass by it; however, this may not be rewarding enough (scientifically, financially) and jeopardize the entire mission. As an alternative, it is possible to